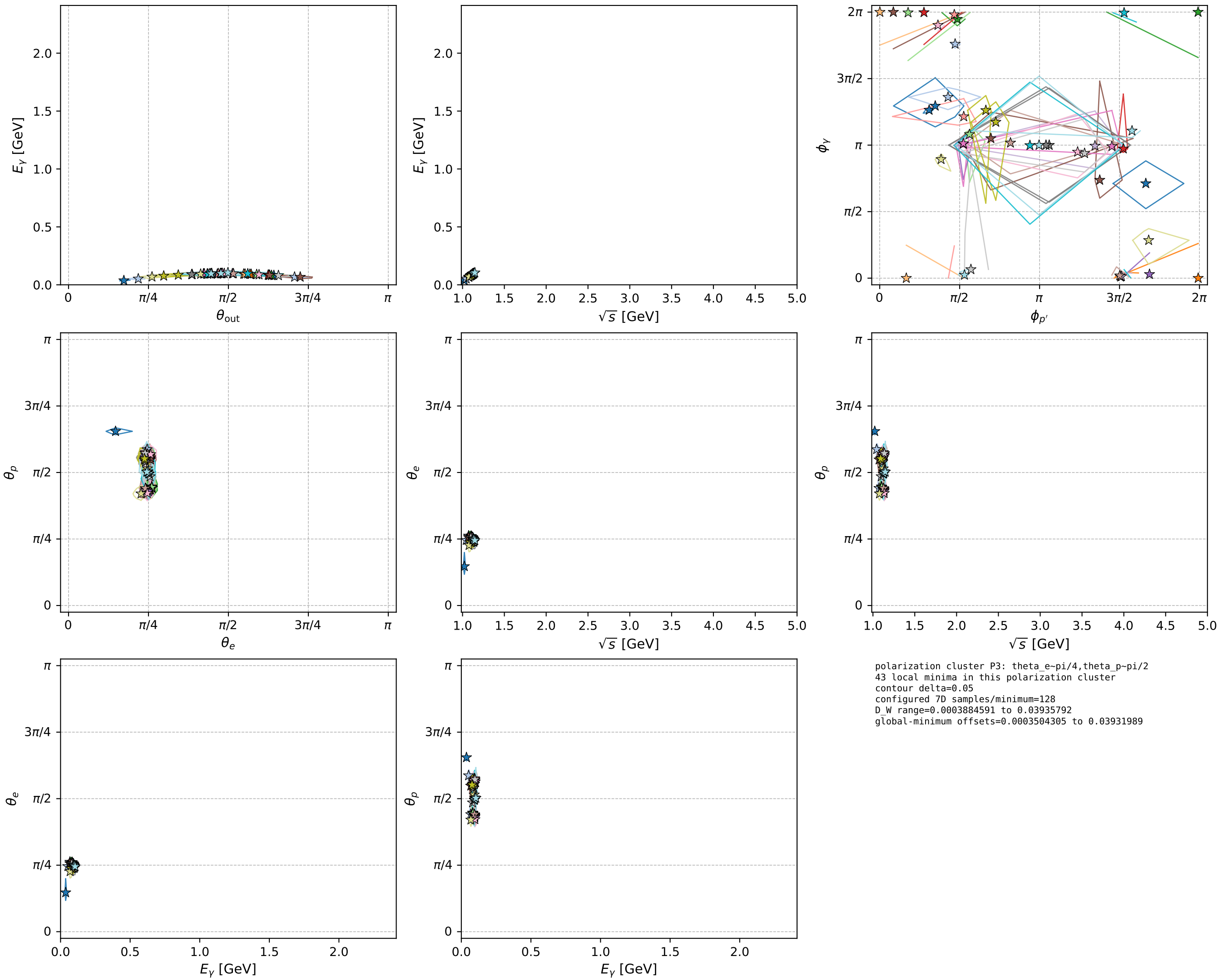


electron: polarization cluster P3; all local-minimum configurations and pairwise projections of their 7D contours

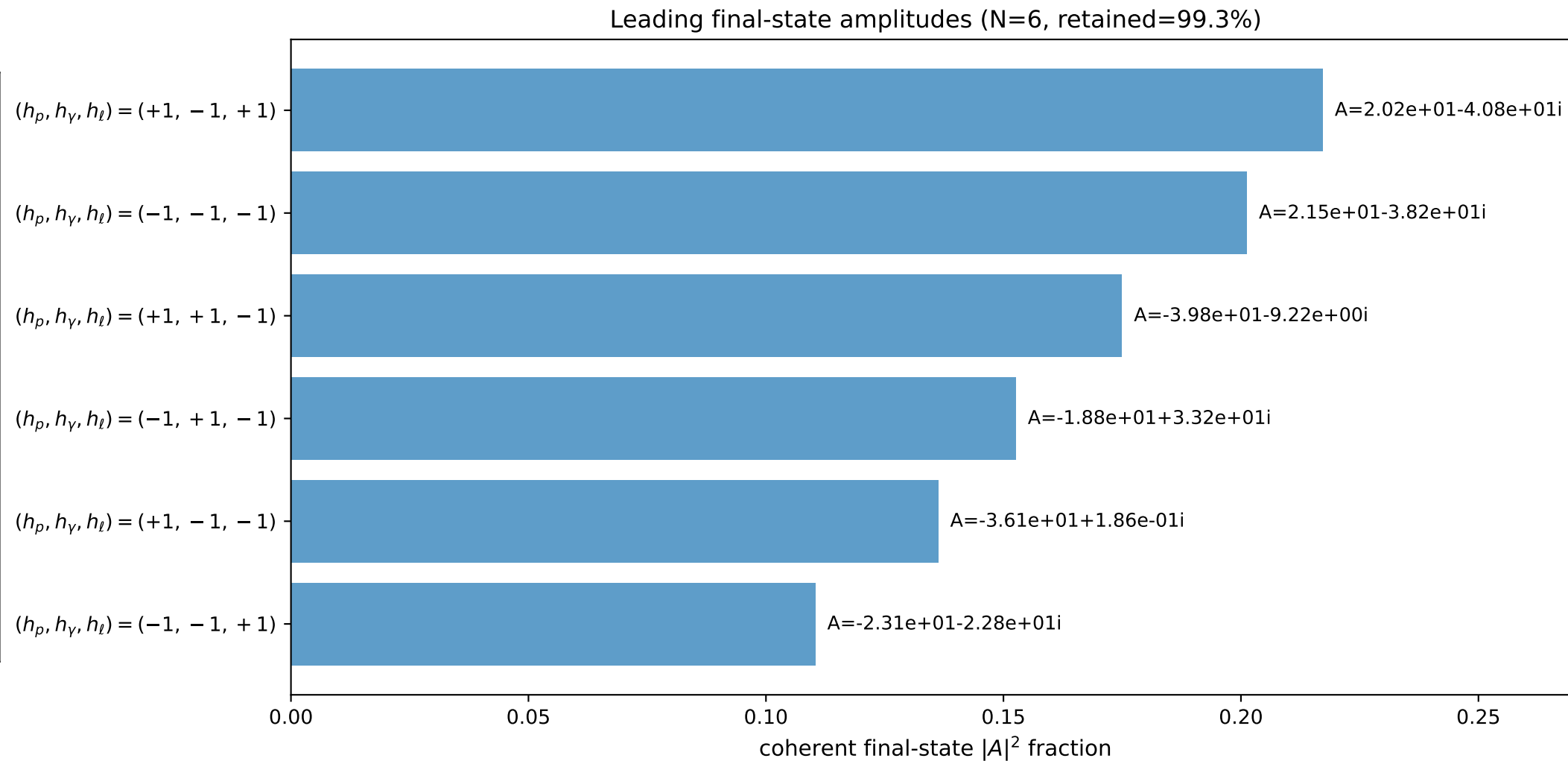
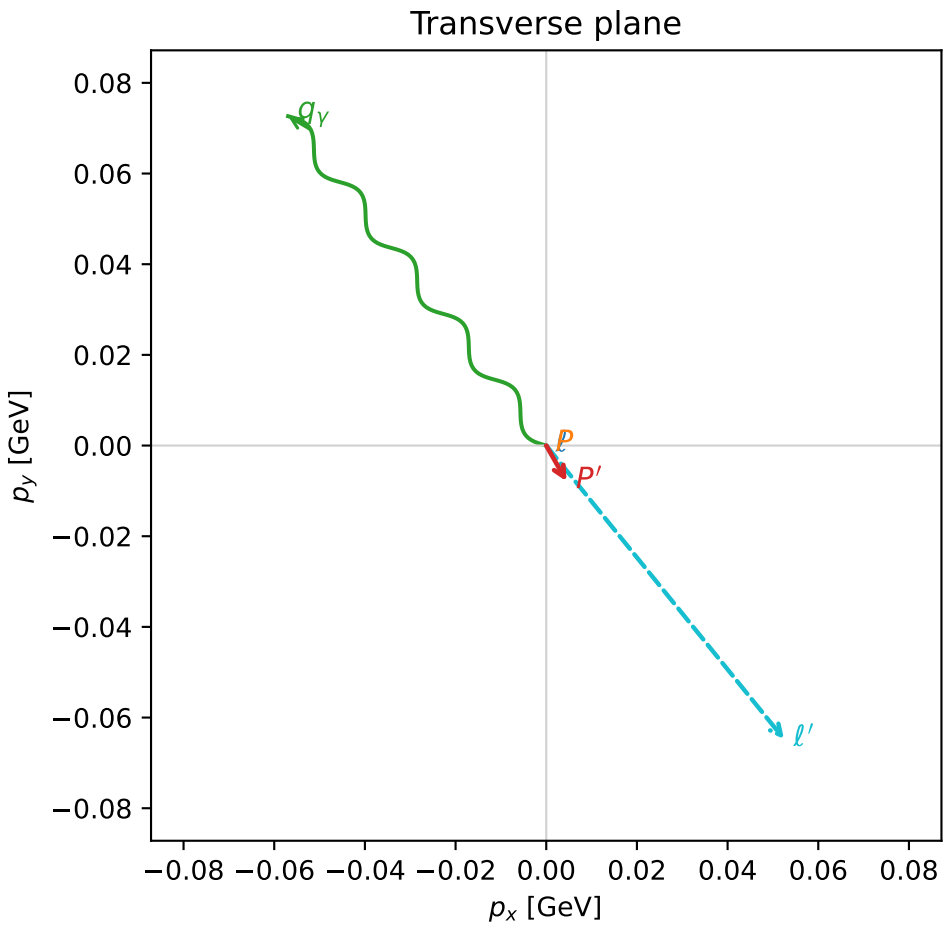
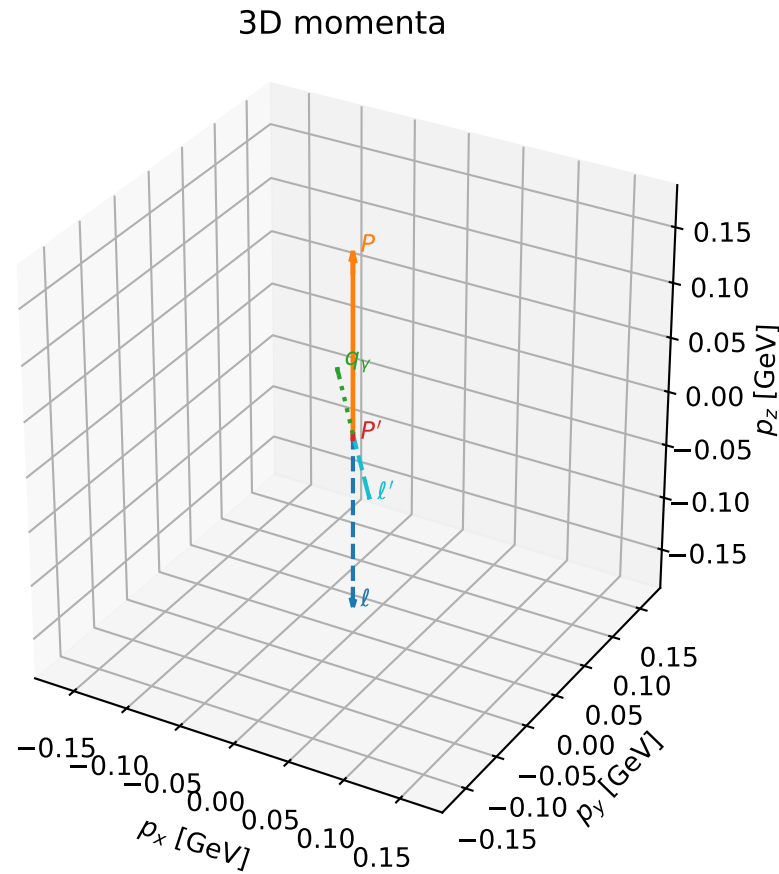


coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

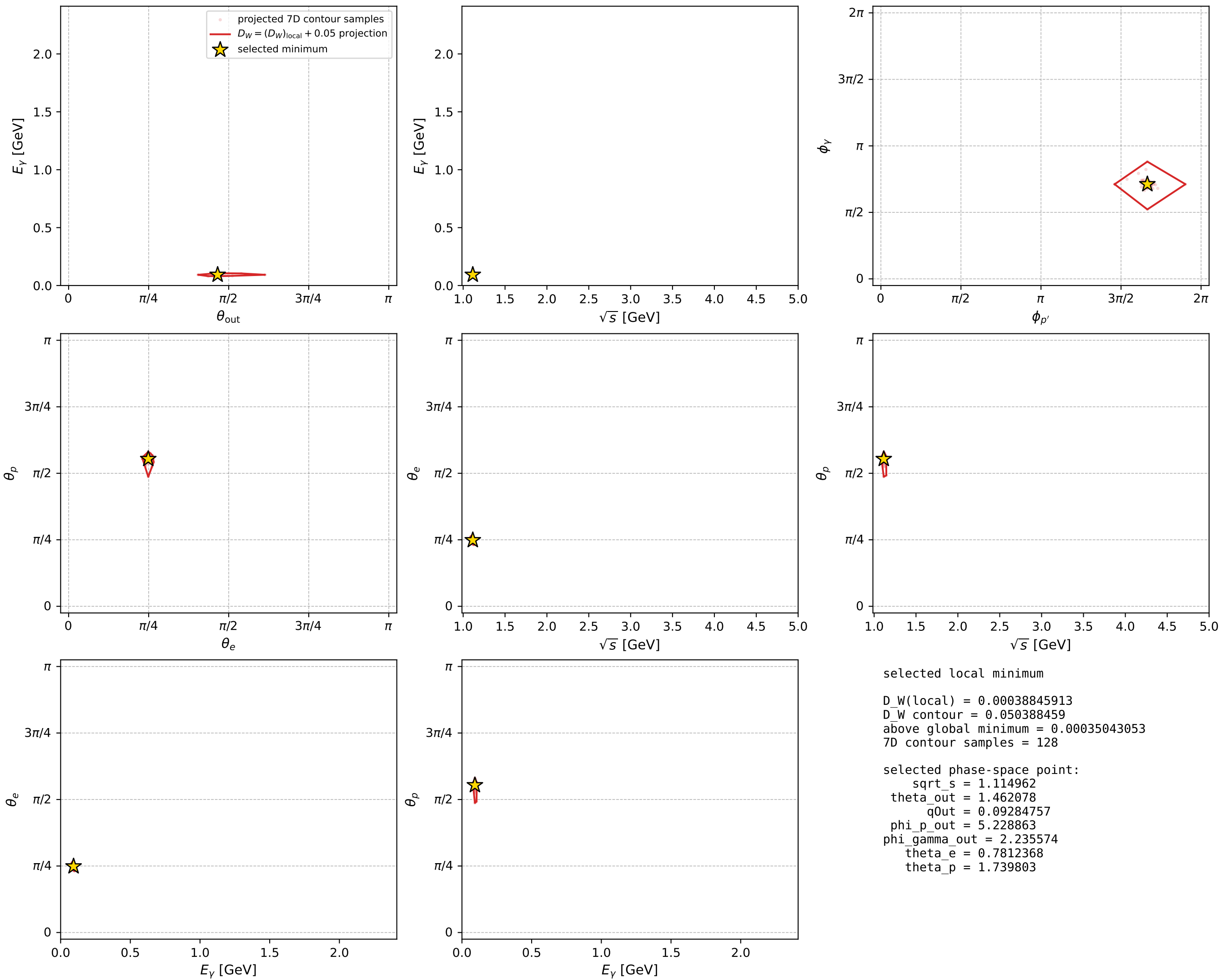
dw_mixing_angles_polarization_03_local_minimum_79 D_W=0.000388459
 region: polarization_cluster_03_local_minimum_79
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\theta_e=0.7812368$ rad
 incoming lepton: $0.710043|+\rangle + 0.704158|-\rangle$
 proton mixing angle: $\theta_p=1.7398032$ rad
 incoming proton: $-0.168203|+\rangle + 0.985752|-\rangle$

$s=1.24314$, $\sqrt{s}=1.11496$, $|\vec{P}|=0.162918$, $|\vec{P}'|=0.00912681$
 $\theta_{\text{out}}=1.46208$, $\phi_{P'}=5.22886$, $\phi_\gamma=2.23557$
 $E_\gamma=0.0928476$, $\theta_\ell=1.70279$, $\phi_{\ell'}=5.39325$
 $Q^2=0.0237873$, $x_B=0.119166$, $t=-0.0261068$, $W^2=1.05567$, $y=0.549454$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1629$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1629$
 P $E= 0.9520$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1629$
 ℓ' $E= 0.0841$ $p_x= 0.0525$ $p_y= -0.0648$ $p_z= -0.0111$
 P' $E= 0.9380$ $p_x= 0.0045$ $p_y= -0.0079$ $p_z= 0.0010$
 q_γ $E= 0.0928$ $p_x= -0.0569$ $p_y= 0.0726$ $p_z= 0.0101$



electron: polarization cluster P3, local minimum 79; pairwise projections of the 7D $D_W = (D_W)_{\text{local}} + 0.05$ contour



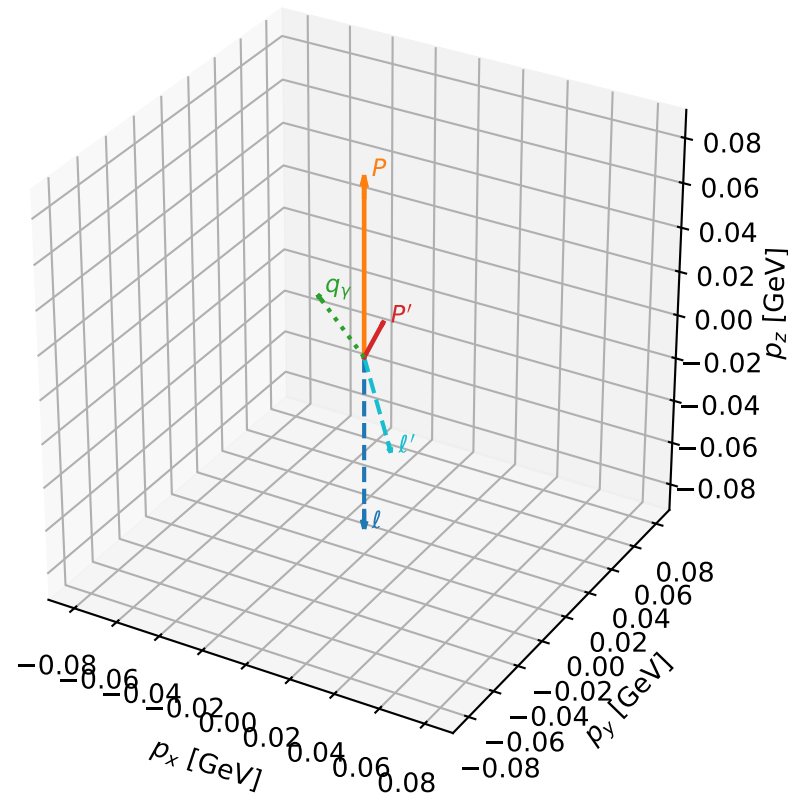
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

dw_mixing_angles_polarization_03_local_minimum_87 D_W=0.000428559
 region: polarization_cluster_03_local_minimum_87
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\theta_e=0.46191114$ rad
 incoming lepton: $0.895202|+\rangle +0.44566|-\rangle$
 proton mixing angle: $\theta_p=2.0573025$ rad
 incoming proton: $-0.46754|+\rangle +0.883972|-\rangle$

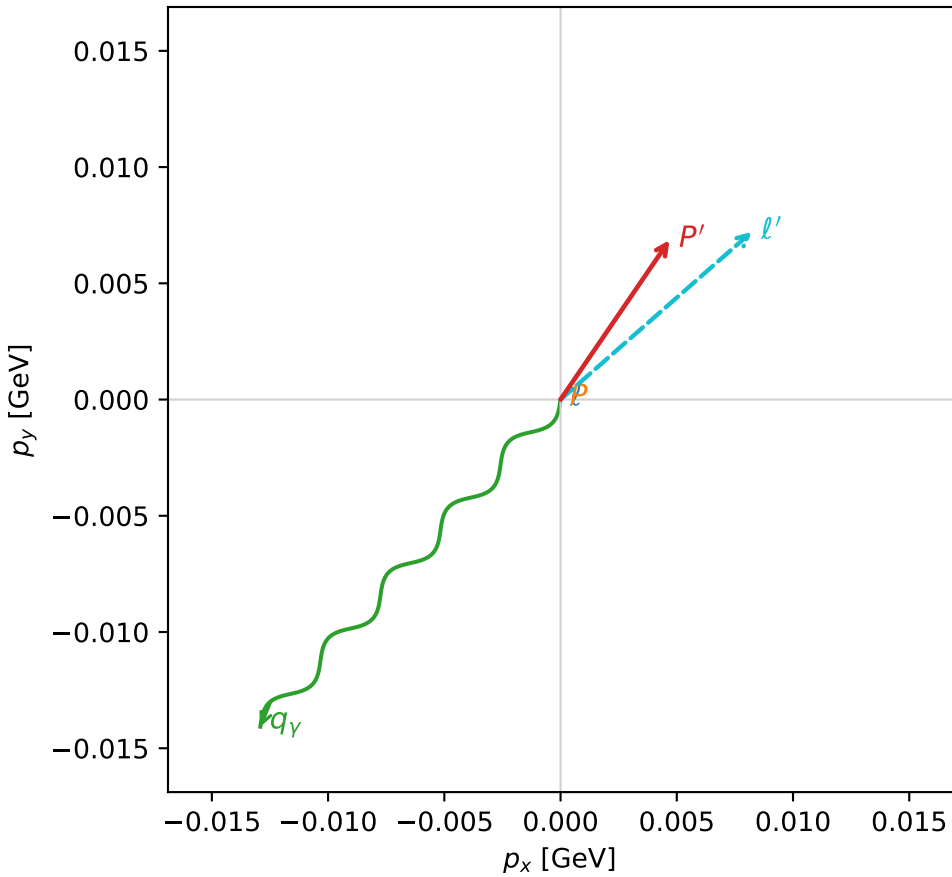
$s=1.04395$, $\sqrt{s}=1.02174$, $|\vec{P}|=0.080307$, $|\vec{P}'|=0.0160685$
 $\theta_{\text{out}}=0.543476$, $\phi_{p'}=0.971276$, $\phi_Y=3.9699$
 $E_Y=0.0369339$, $\theta_{\ell'}=2.90498$, $\phi_{\ell'}=0.719864$
 $Q^2=0.000208908$, $x_B=0.00302977$, $t=-0.0044876$, $W^2=0.948587$, $y=0.420159$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E=0.0803$ $p_x=0.0000$ $p_y=0.0000$ $p_z=-0.0803$
 P $E=0.9414$ $p_x=0.0000$ $p_y=0.0000$ $p_z=0.0803$
 ℓ' $E=0.0467$ $p_x=0.0082$ $p_y=0.0072$ $p_z=-0.0454$
 P' $E=0.9381$ $p_x=0.0047$ $p_y=0.0069$ $p_z=0.0138$
 q_Y $E=0.0369$ $p_x=-0.0129$ $p_y=-0.0141$ $p_z=0.0316$

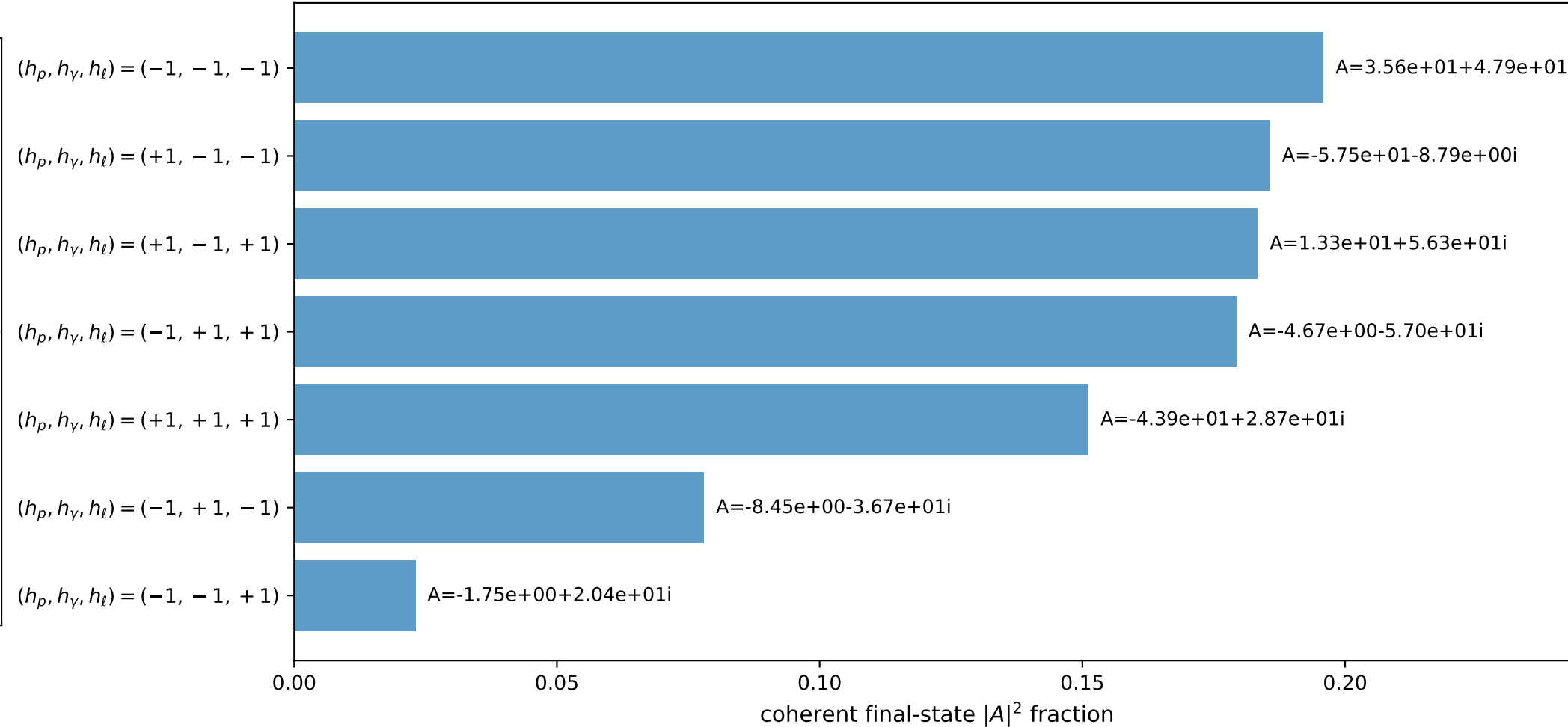
3D momenta



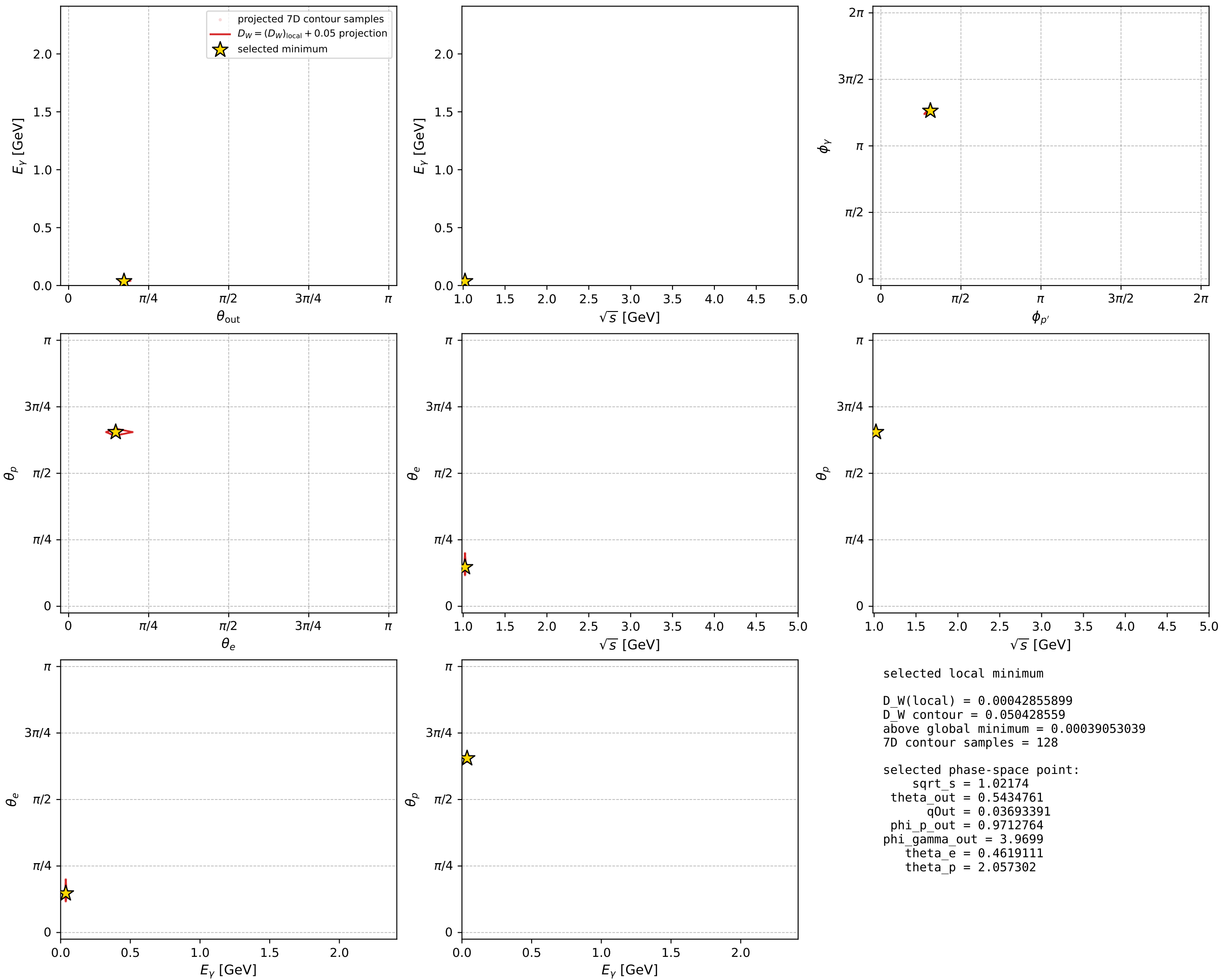
Transverse plane



Leading final-state amplitudes (N=7, retained=99.6%)



electron: polarization cluster P3, local minimum 87; pairwise projections of the 7D $D_W = (D_W)_{\text{local}} + 0.05$ contour

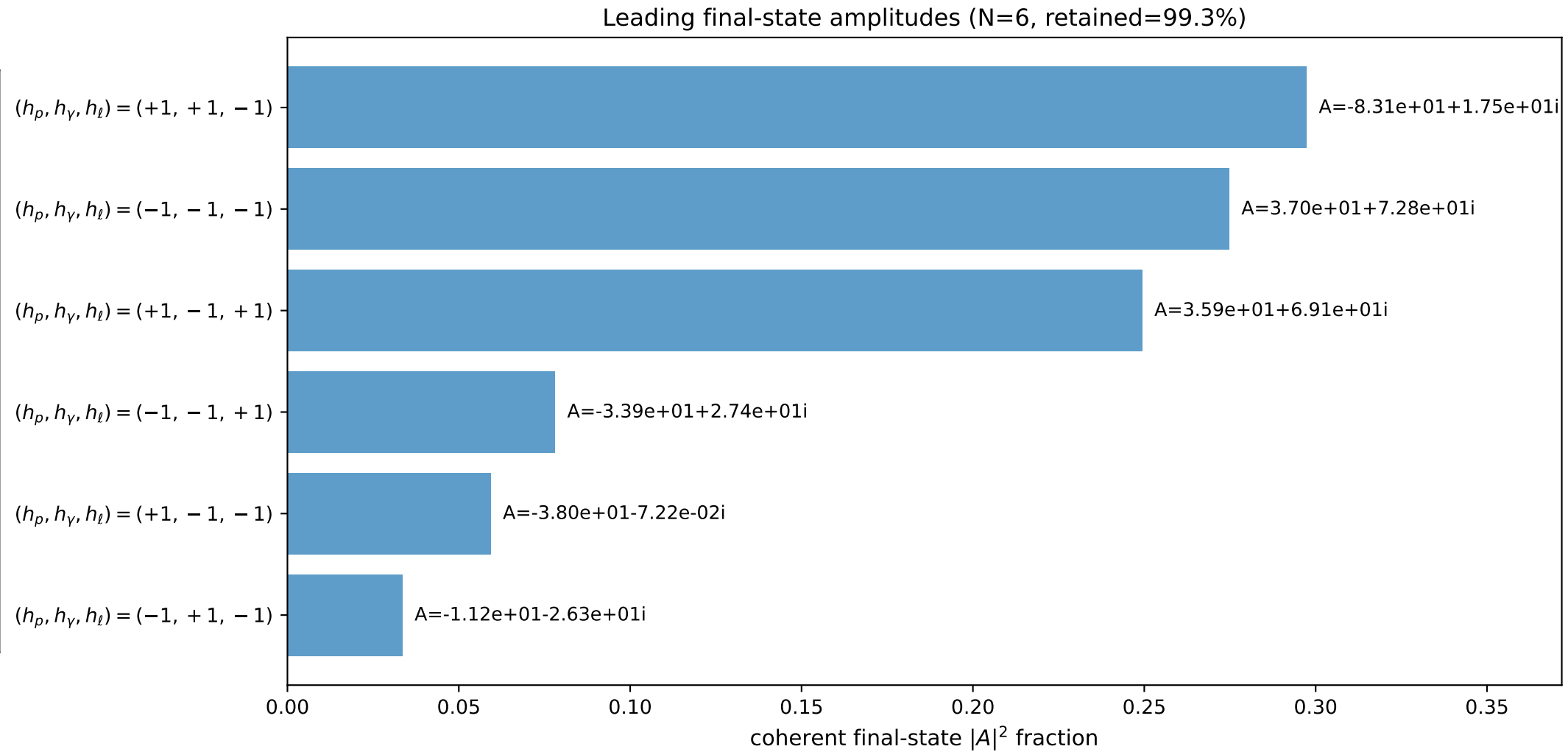
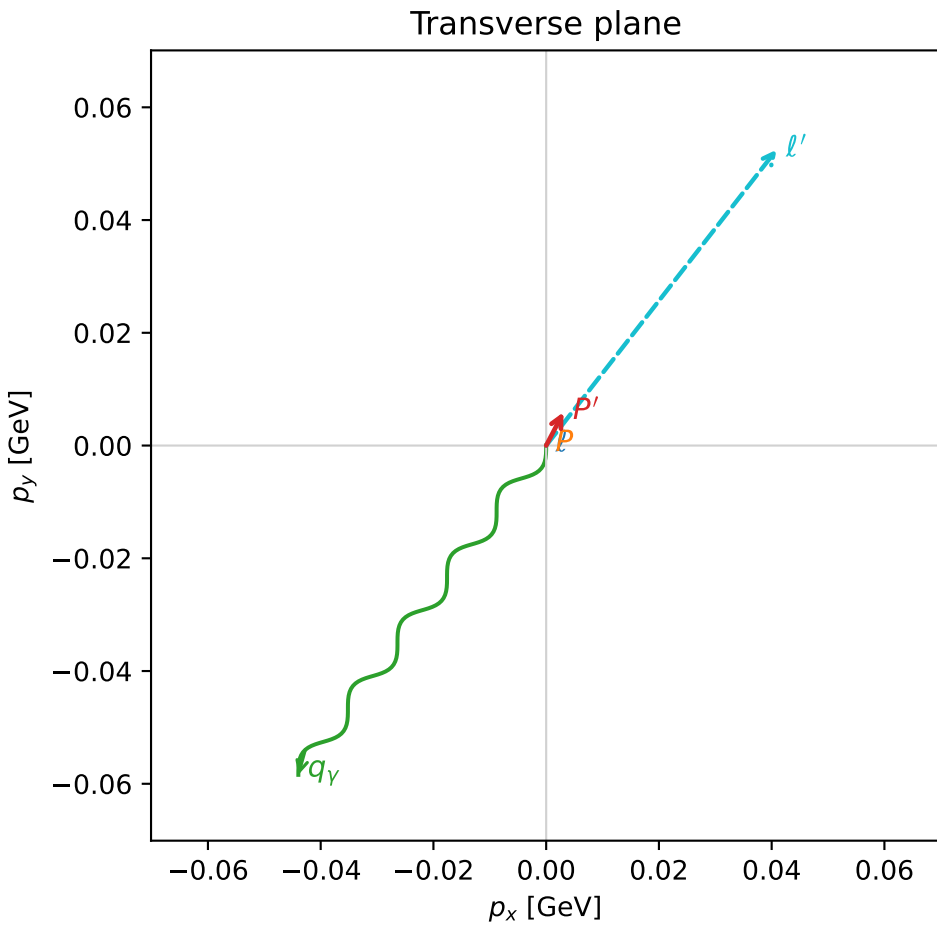
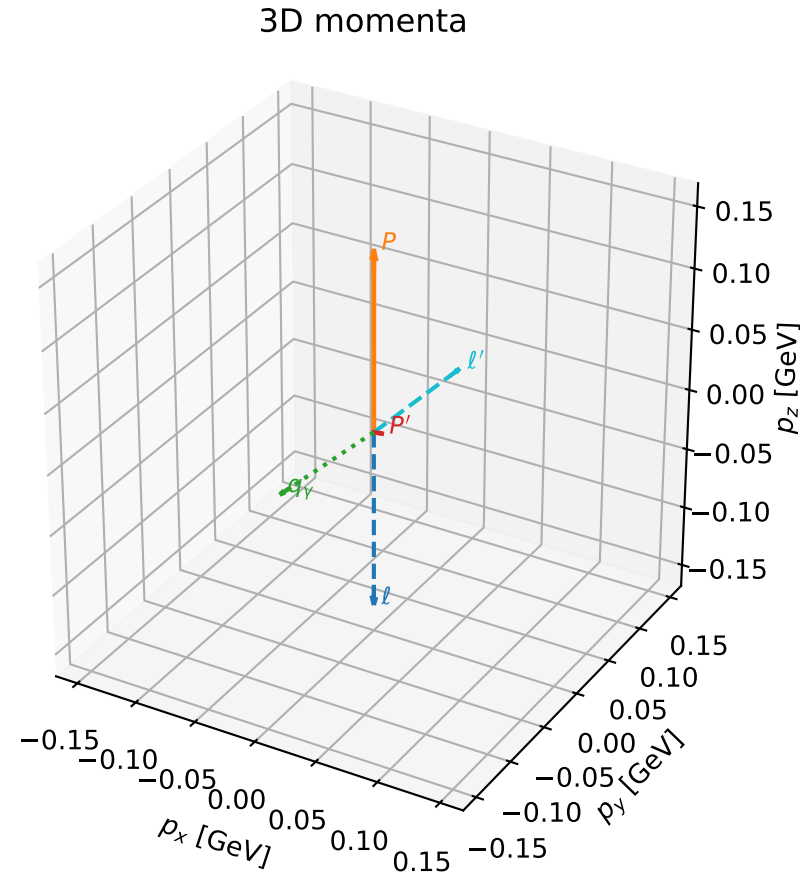


coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

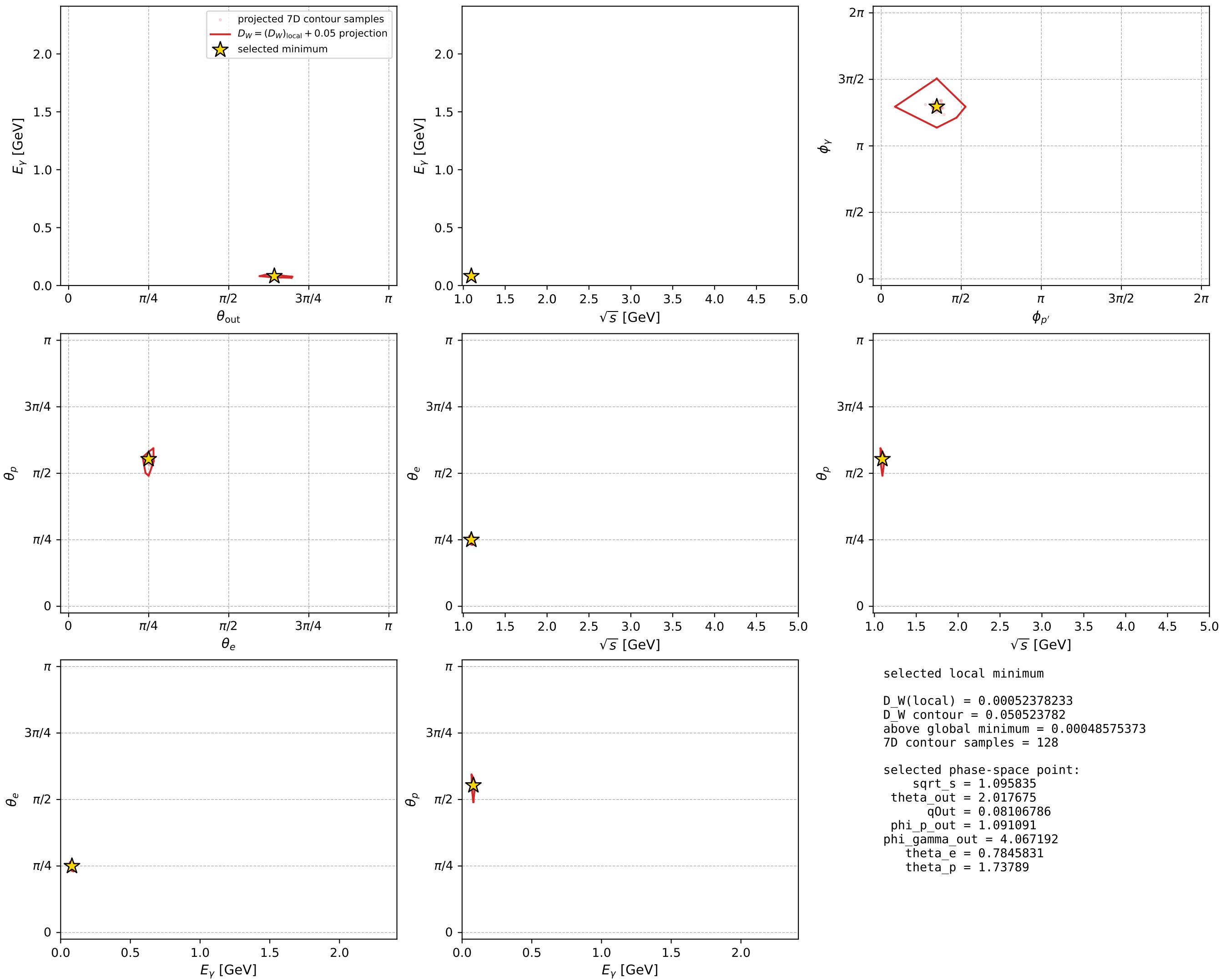
dw_mixing_angles_polarization_03_local_minimum_110 D_W=0.000523782
 region: polarization_cluster_03_local_minimum_110
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\theta_e=0.78458305$ rad
 incoming lepton: $0.707683|+\rangle + 0.70653|-\rangle$
 proton mixing angle: $\theta_p=1.7378903$ rad
 incoming proton: $-0.166317|+\rangle + 0.986072|-\rangle$

$s=1.20085$, $\sqrt{s}=1.09583$, $|\vec{P}|=0.146467$, $|\vec{P}'|=0.00740015$
 $\theta_{\text{out}}=2.01768$, $\phi_{P'}=1.09109$, $\phi_Y=4.06719$
 $E_Y=0.0810679$, $\theta_{\ell'}=1.04925$, $\phi_{\ell'}=0.909075$
 $Q^2=0.0336782$, $x_B=0.180576$, $t=-0.0223157$, $W^2=1.03267$, $y=0.580994$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1465$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1465$
 P $E= 0.9494$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1465$
 ℓ' $E= 0.0767$ $p_x= 0.0409$ $p_y= 0.0525$ $p_z= 0.0382$
 P' $E= 0.9380$ $p_x= 0.0031$ $p_y= 0.0059$ $p_z= -0.0032$
 q_Y $E= 0.0811$ $p_x= -0.0440$ $p_y= -0.0584$ $p_z= -0.0350$



electron: polarization cluster P3, local minimum 110; pairwise projections of the 7D $D_W = (D_W)_{\text{local}} + 0.05$ contour

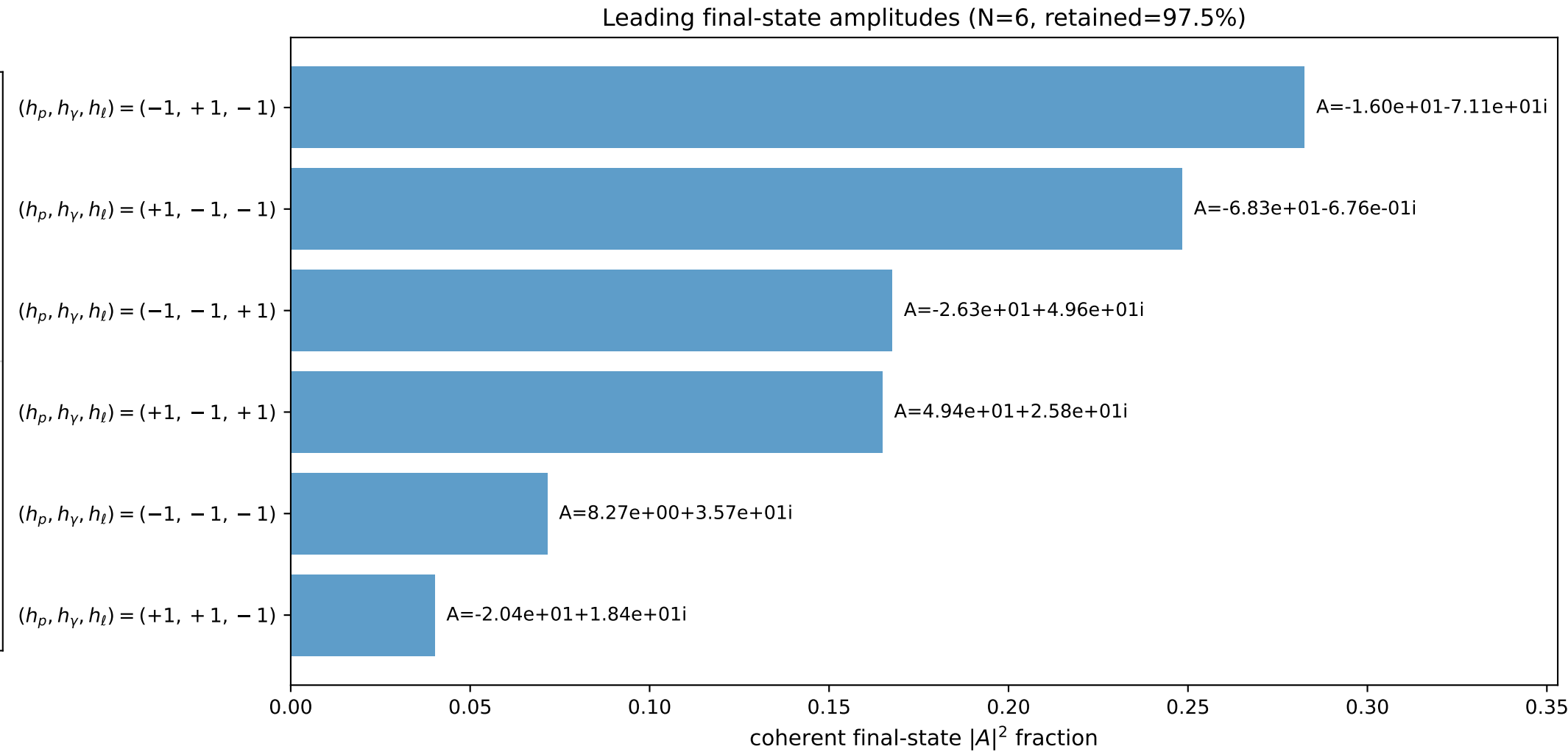
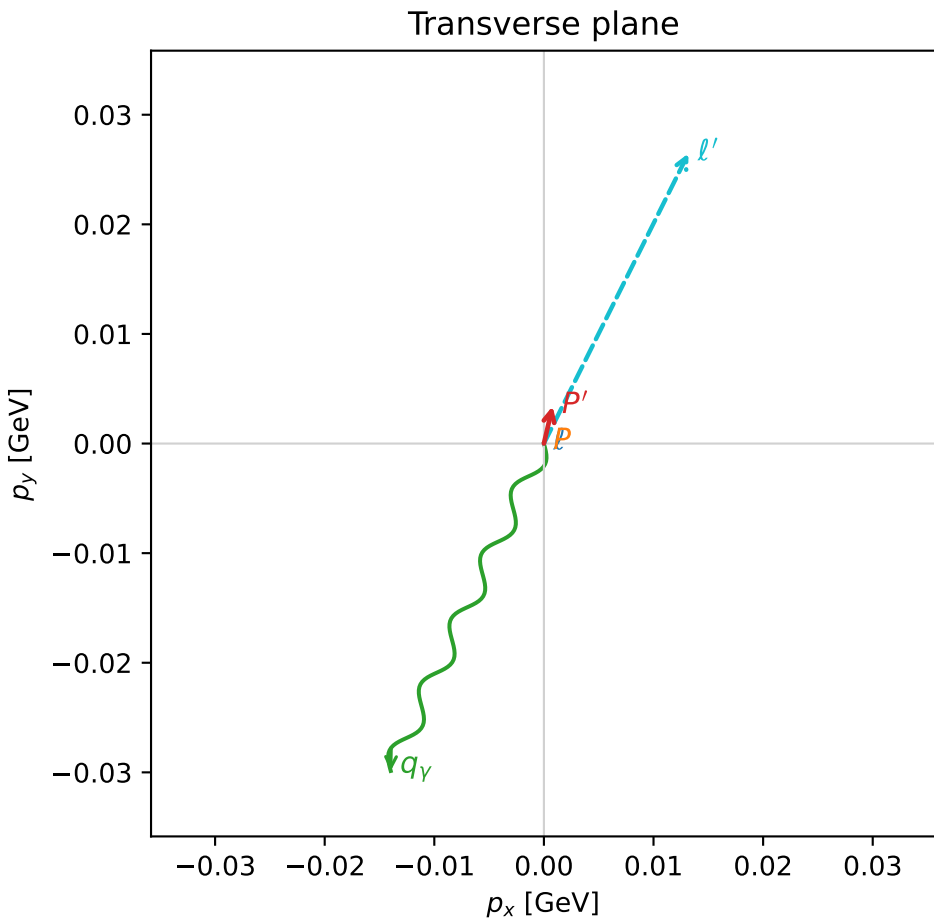
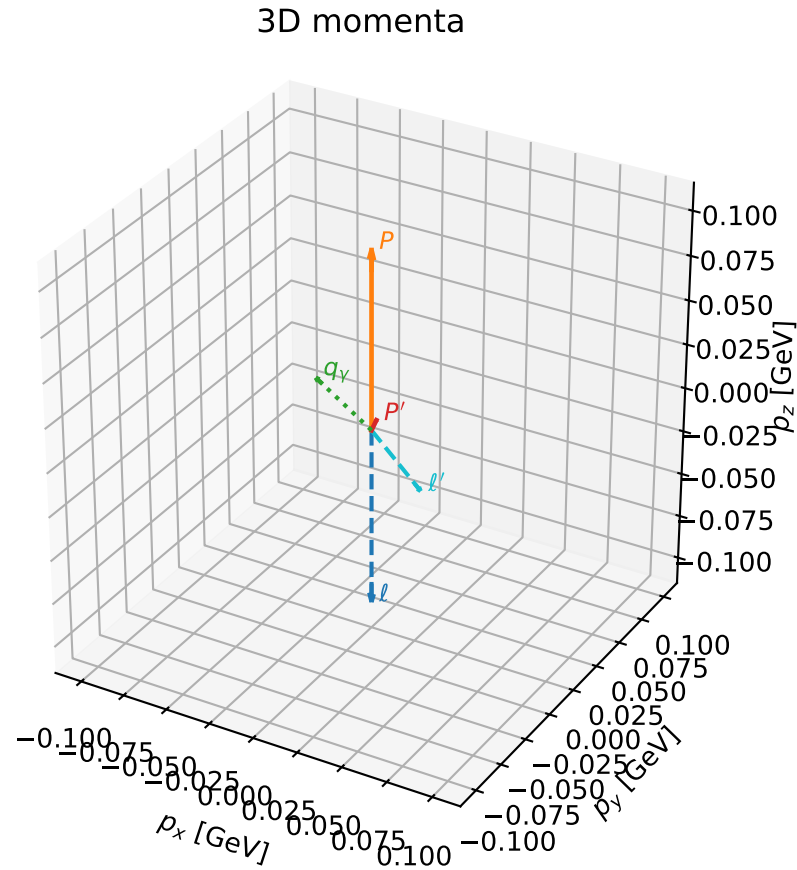


coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

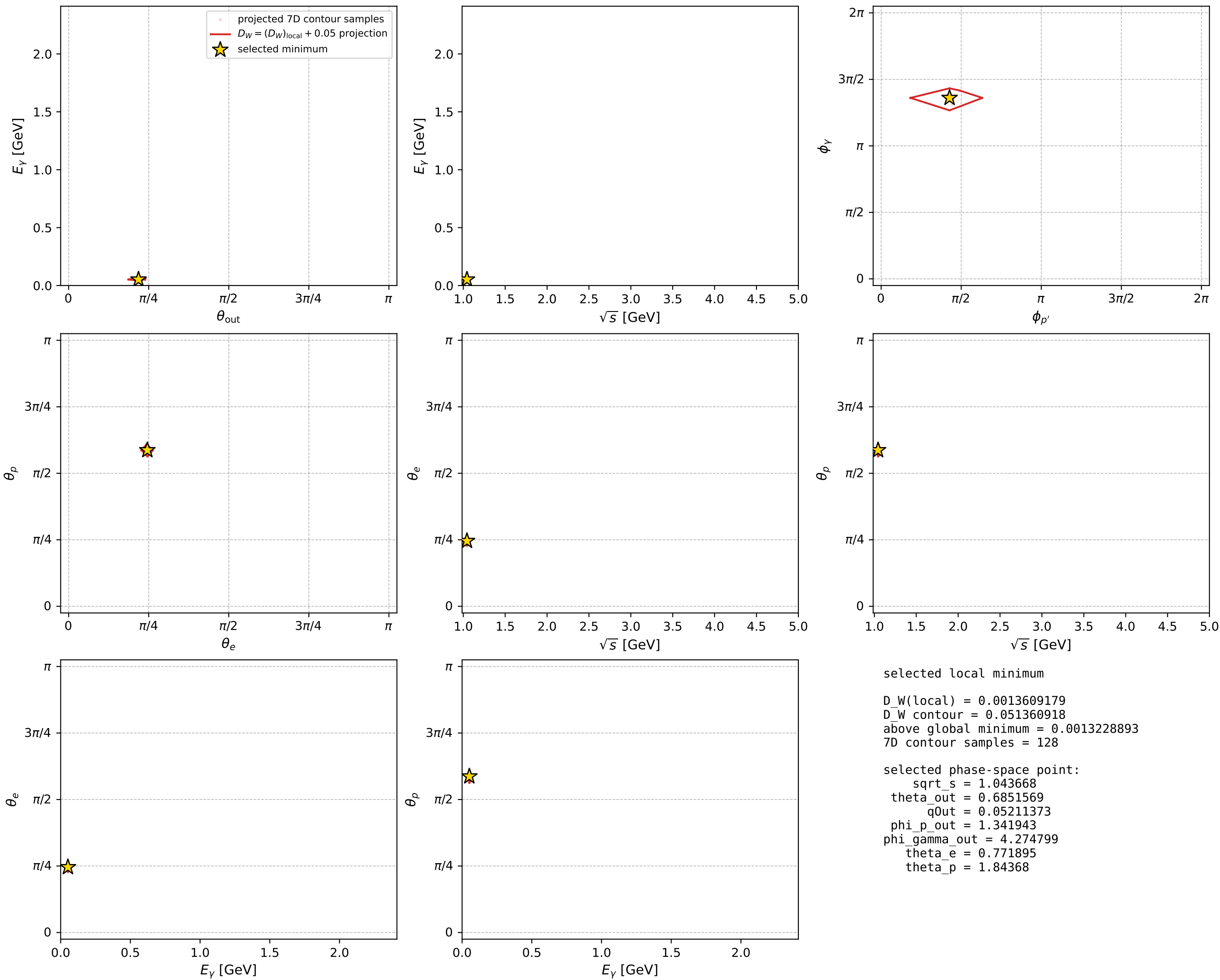
dw_mixing_angles_polarization_03_local_minimum_263 D_W=0.00136092
 region: polarization_cluster_03_local_minimum_263
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\theta_{\bar{e}}=0.77189495$ rad
 incoming lepton: $0.71659|+\rangle +0.697494|-\rangle$
 proton mixing angle: $\theta_p=1.84368$ rad
 incoming proton: $-0.26951|+\rangle +0.962998|-\rangle$

$s=1.08924$, $\sqrt{s}=1.04367$, $|\vec{P}|=0.100318$, $|\vec{P}'|=0.00552562$
 $\theta_{\text{out}}=0.685157$, $\phi_{P'}=1.34194$, $\phi_Y=4.2748$
 $E_Y=0.0521137$, $\theta_{\ell'}=2.55652$, $\phi_{\ell'}=1.1087$
 $Q^2=0.00178664$, $x_B=0.0179681$, $t=-0.00920729$, $W^2=0.977491$, $y=0.474854$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1003$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1003$
 P $E= 0.9433$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1003$
 ℓ' $E= 0.0535$ $p_x= 0.0132$ $p_y= 0.0265$ $p_z= -0.0446$
 P' $E= 0.9380$ $p_x= 0.0008$ $p_y= 0.0034$ $p_z= 0.0043$
 q_Y $E= 0.0521$ $p_x= -0.0140$ $p_y= -0.0299$ $p_z= 0.0404$



electron: polarization cluster P3, local minimum 263; pairwise projections of the 7D $D_W = (D_W)_{\text{local}} + 0.05$ contour

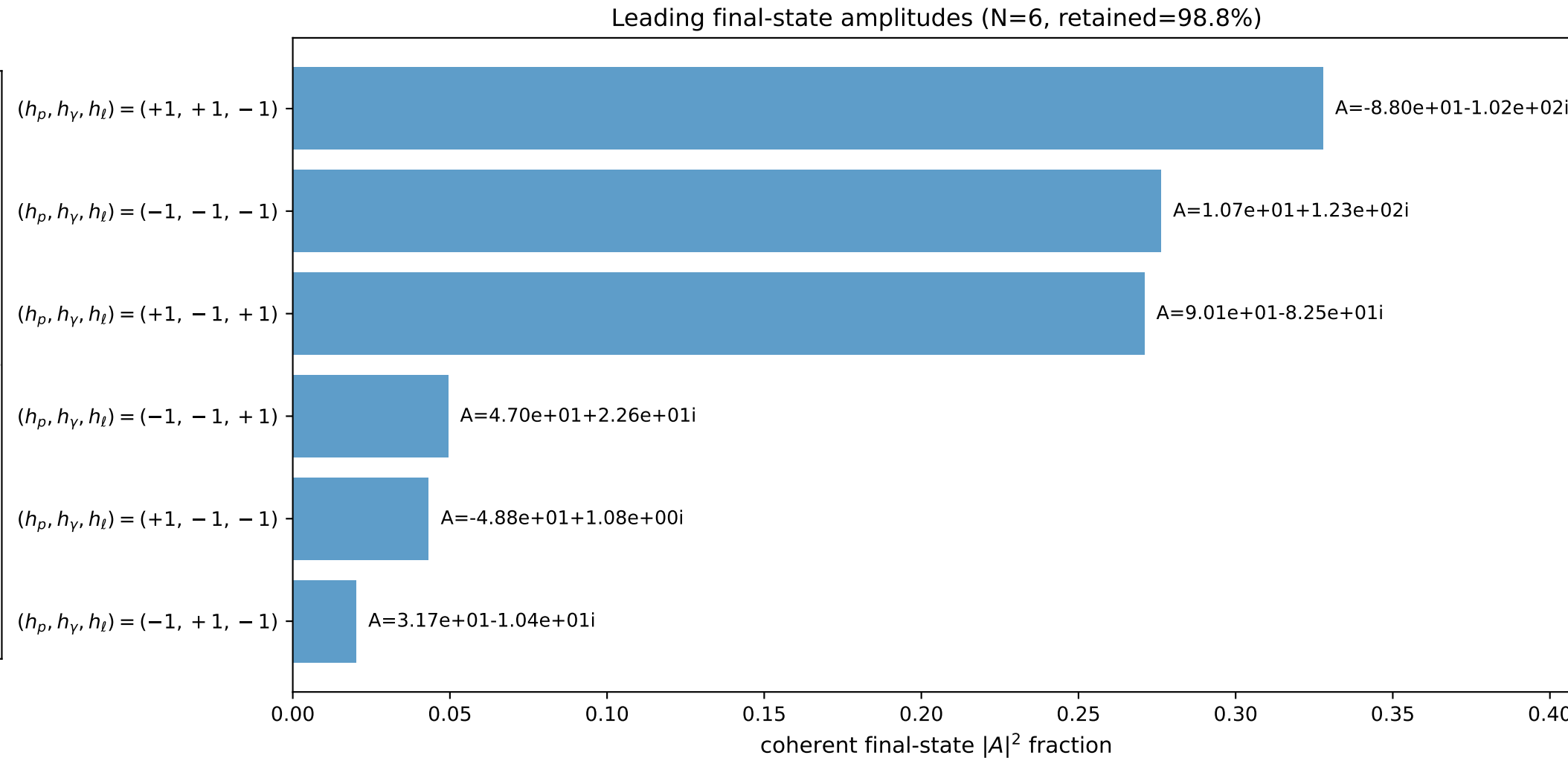
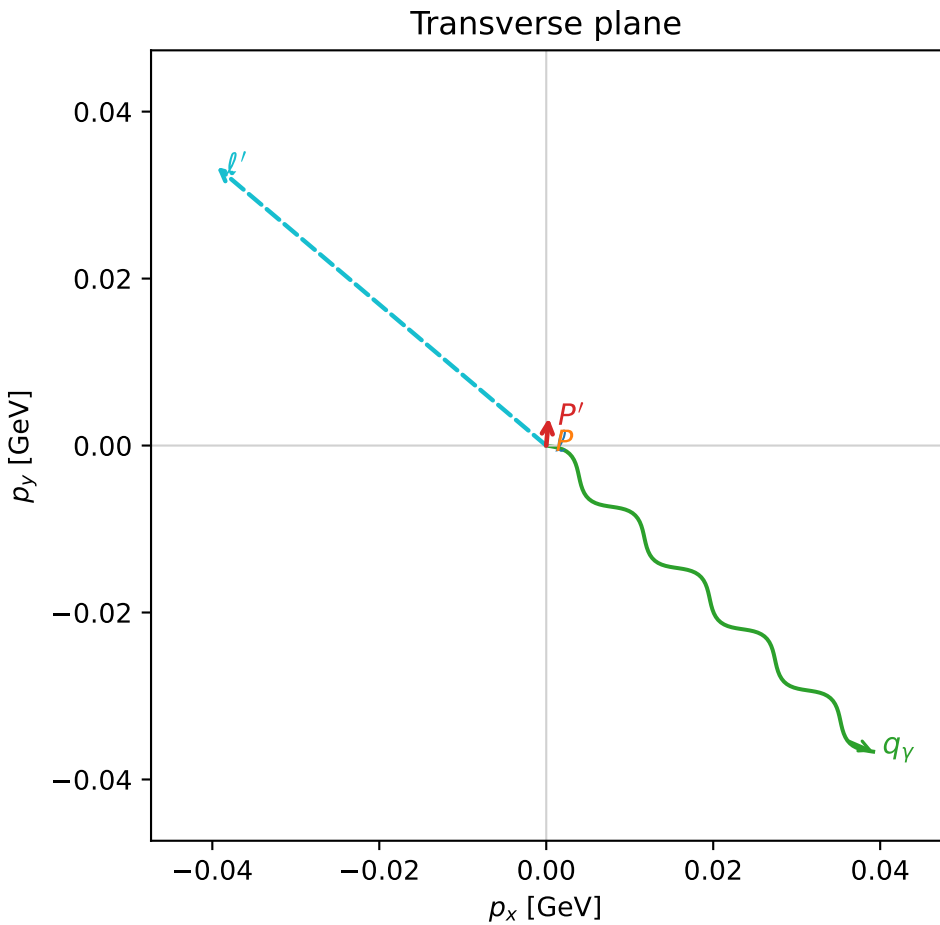
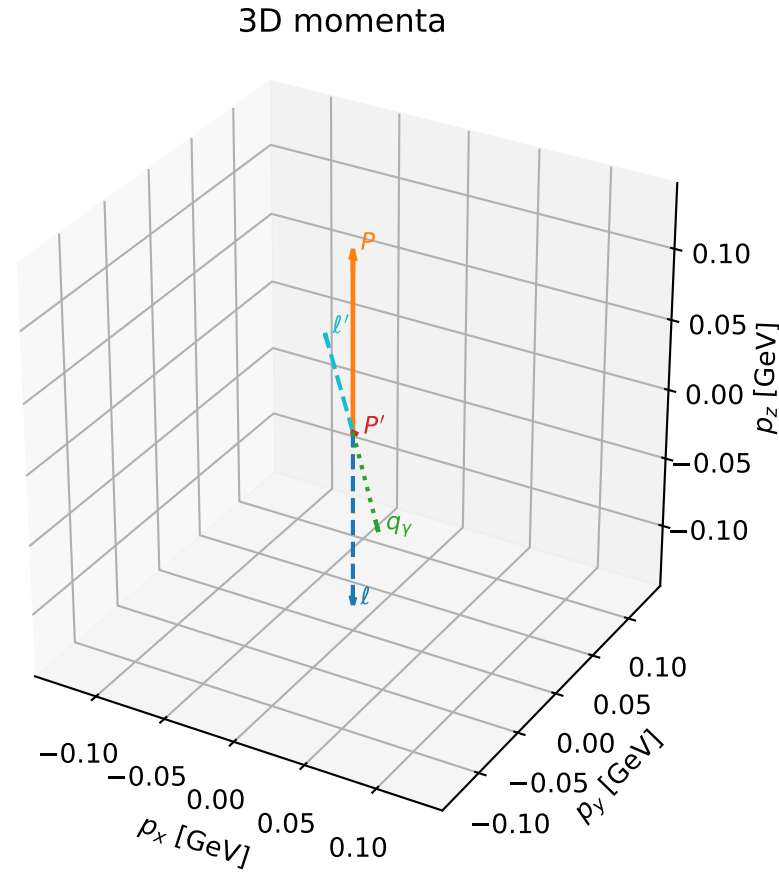


coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

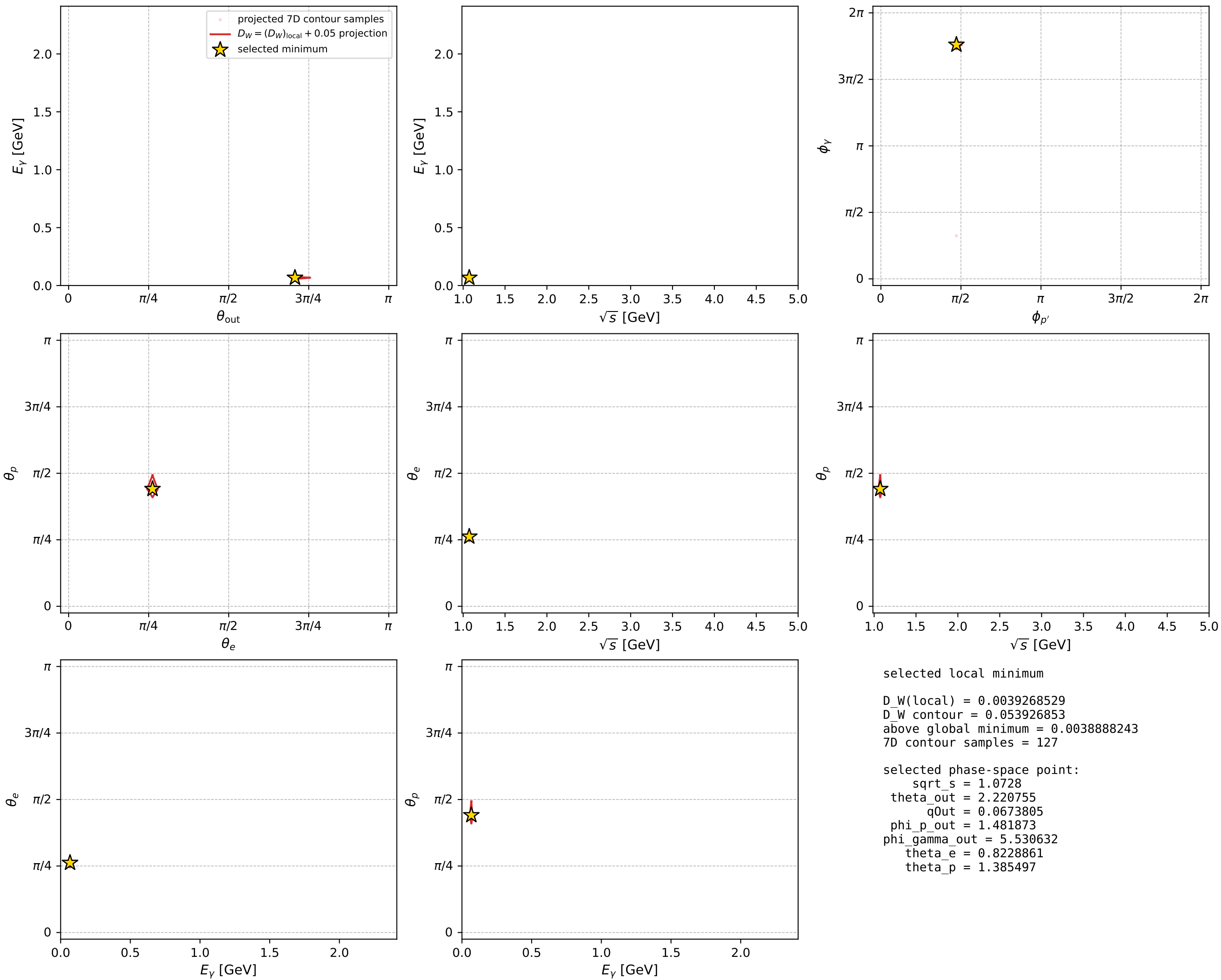
dw_mixing_angles_polarization_03_local_minimum_457 D_W=0.00392685
 region: polarization_cluster_03_local_minimum_457
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\theta_e=0.8228861$ rad
 incoming lepton: $0.680108|+\rangle + 0.733112|-\rangle$
 proton mixing angle: $\theta_p=1.3854972$ rad
 incoming proton: $0.184241|+\rangle + 0.982881|-\rangle$

$s=1.1509$, $\sqrt{s}=1.0728$, $|\vec{P}|=0.12633$, $|\vec{P}'|=0.0041635$
 $\theta_{\text{out}}=2.22076$, $\phi_{p'}=1.48187$, $\phi_\gamma=5.53063$
 $E_\gamma=0.0673805$, $\theta_\ell=0.873326$, $\phi_{\ell'}=2.4396$
 $Q^2=0.0279705$, $x_B=0.181166$, $t=-0.0165417$, $W^2=1.00626$, $y=0.569592$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1263$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1263$
 P $E= 0.9465$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1263$
 ℓ' $E= 0.0674$ $p_x= -0.0395$ $p_y= 0.0334$ $p_z= 0.0433$
 P' $E= 0.9380$ $p_x= 0.0003$ $p_y= 0.0033$ $p_z= -0.0025$
 q_γ $E= 0.0674$ $p_x= 0.0392$ $p_y= -0.0367$ $p_z= -0.0408$



electron: polarization cluster P3, local minimum 457; pairwise projections of the 7D $D_W = (D_W)_{\text{local}} + 0.05$ contour

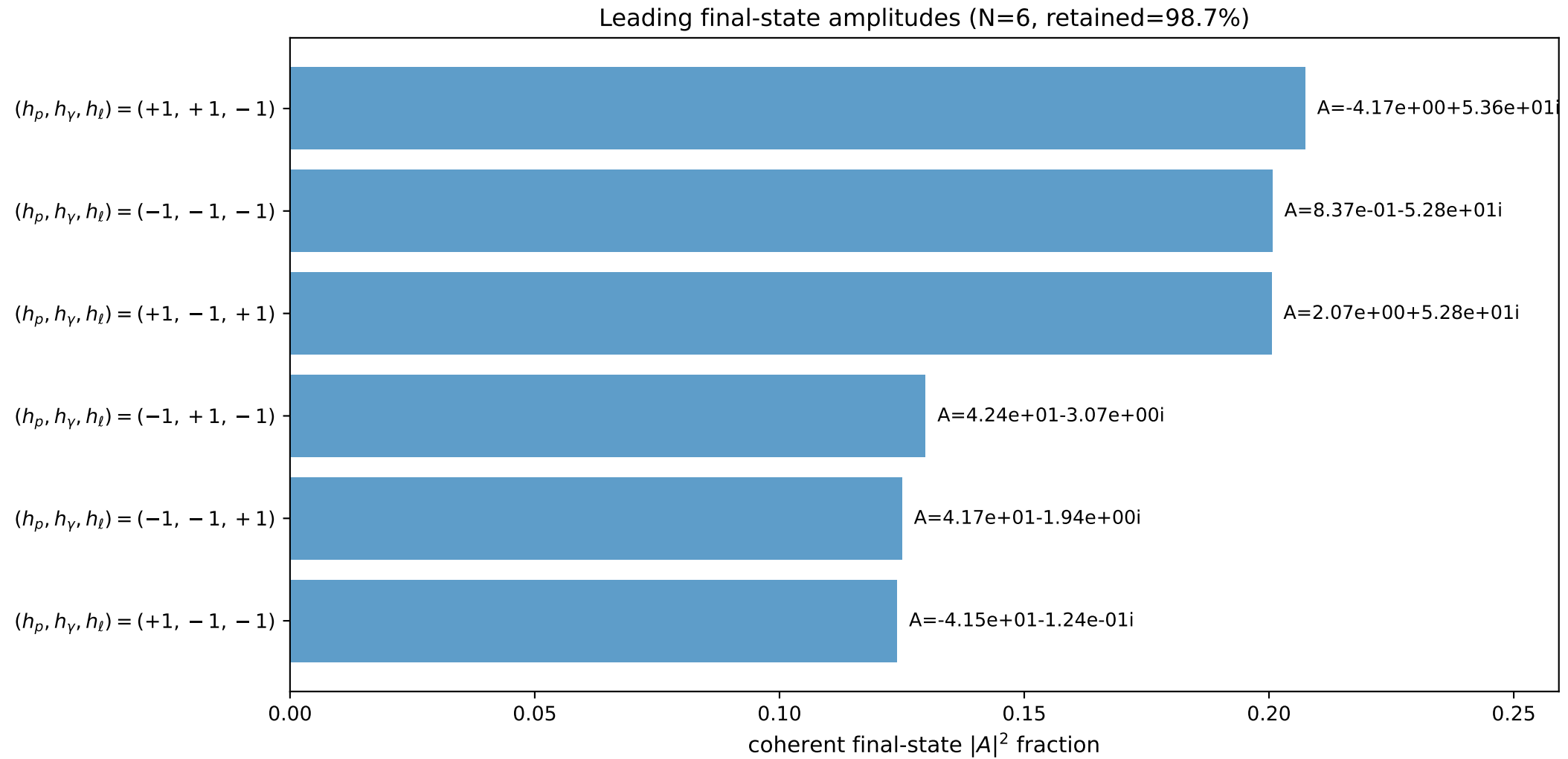
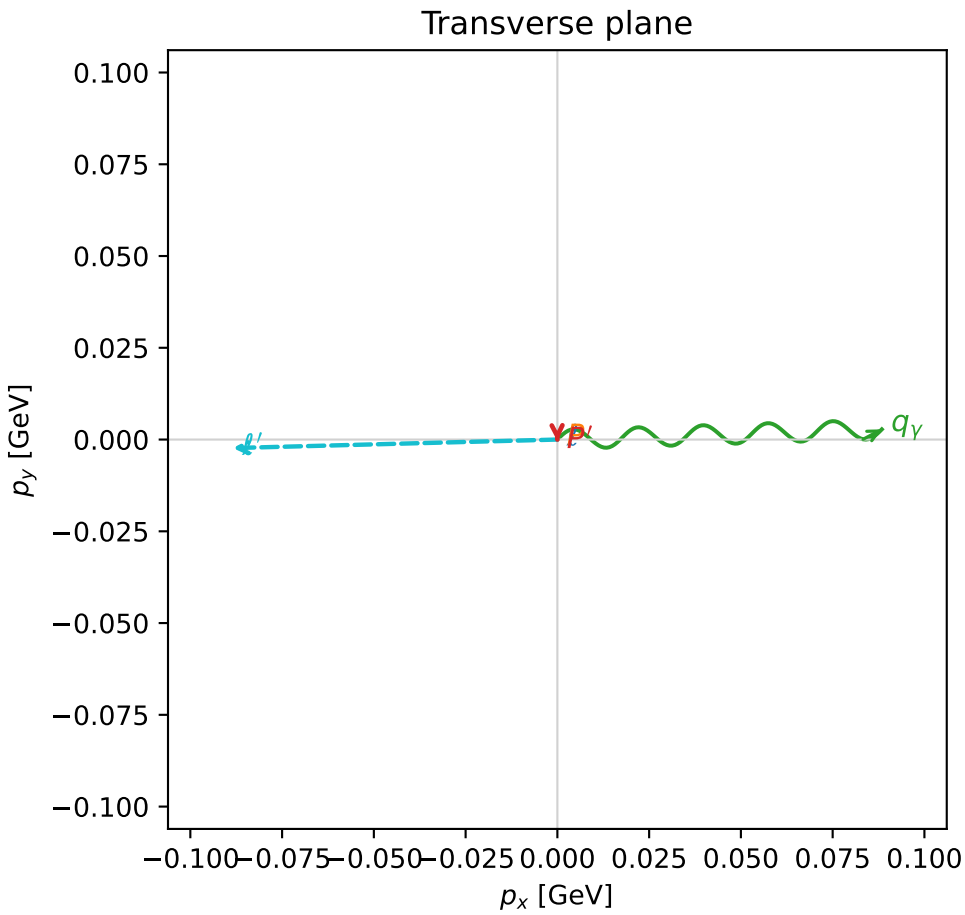
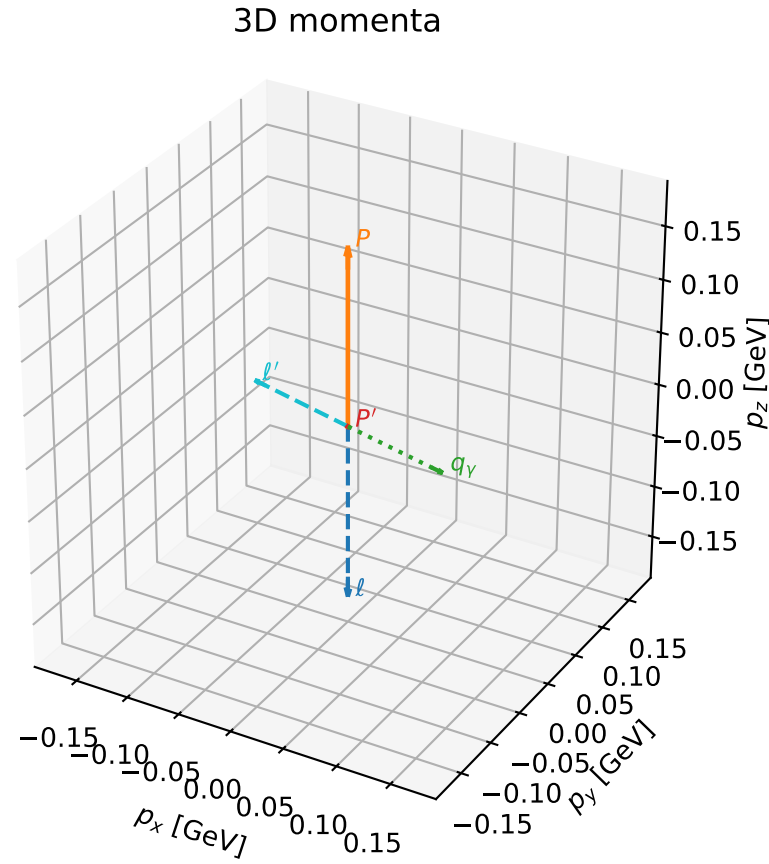


coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

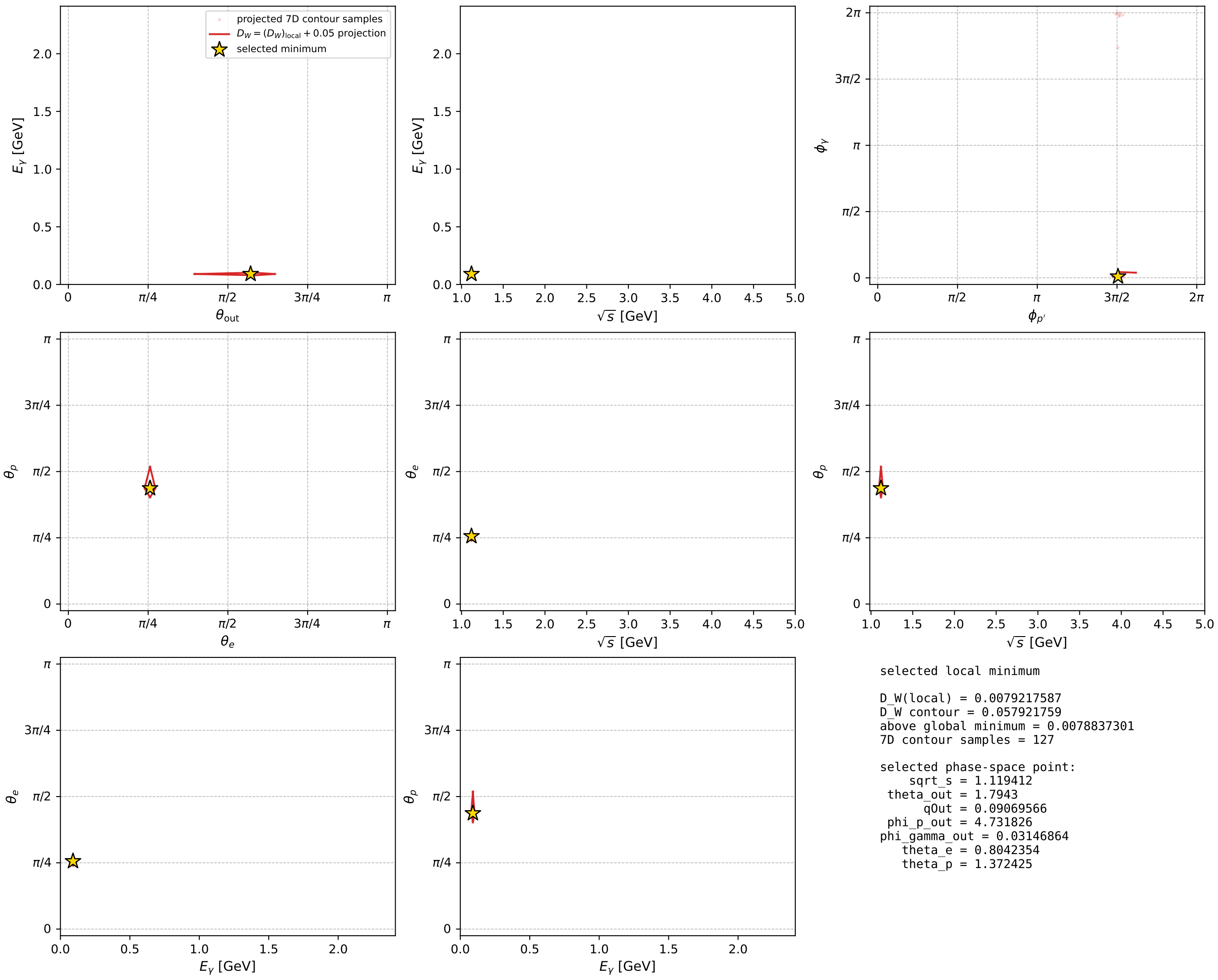
dw_mixing_angles_polarization_03_local_minimum_529 D_W=0.00792176
 region: polarization_cluster_03_local_minimum_529
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\theta_e=0.80423539$ rad
 incoming lepton: $0.693662|+\rangle +0.7203|-\rangle$
 proton mixing angle: $\theta_p=1.3724246$ rad
 incoming proton: $0.197073|+\rangle +0.980389|-\rangle$

$s=1.25308$, $\sqrt{s}=1.11941$, $|\vec{P}|=0.166712$, $|\vec{P}'|=0.000475325$
 $\theta_{\text{out}}=1.7943$, $\phi_{p'}=4.73183$, $\phi_Y=0.0314686$
 $E_Y=0.0906957$, $\theta_{\ell'}=1.34615$, $\phi_{\ell'}=3.16782$
 $Q^2=0.0369843$, $x_B=0.178559$, $t=-0.027612$, $W^2=1.04999$, $y=0.554942$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1667$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1667$
 P $E= 0.9527$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1667$
 ℓ' $E= 0.0907$ $p_x= -0.0884$ $p_y= -0.0023$ $p_z= 0.0202$
 P' $E= 0.9380$ $p_x= 0.0000$ $p_y= -0.0005$ $p_z= -0.0001$
 q_Y $E= 0.0907$ $p_x= 0.0884$ $p_y= 0.0028$ $p_z= -0.0201$



electron: polarization cluster P3, local minimum 529; pairwise projections of the 7D $D_W = (D_W)_{\text{local}} + 0.05$ contour

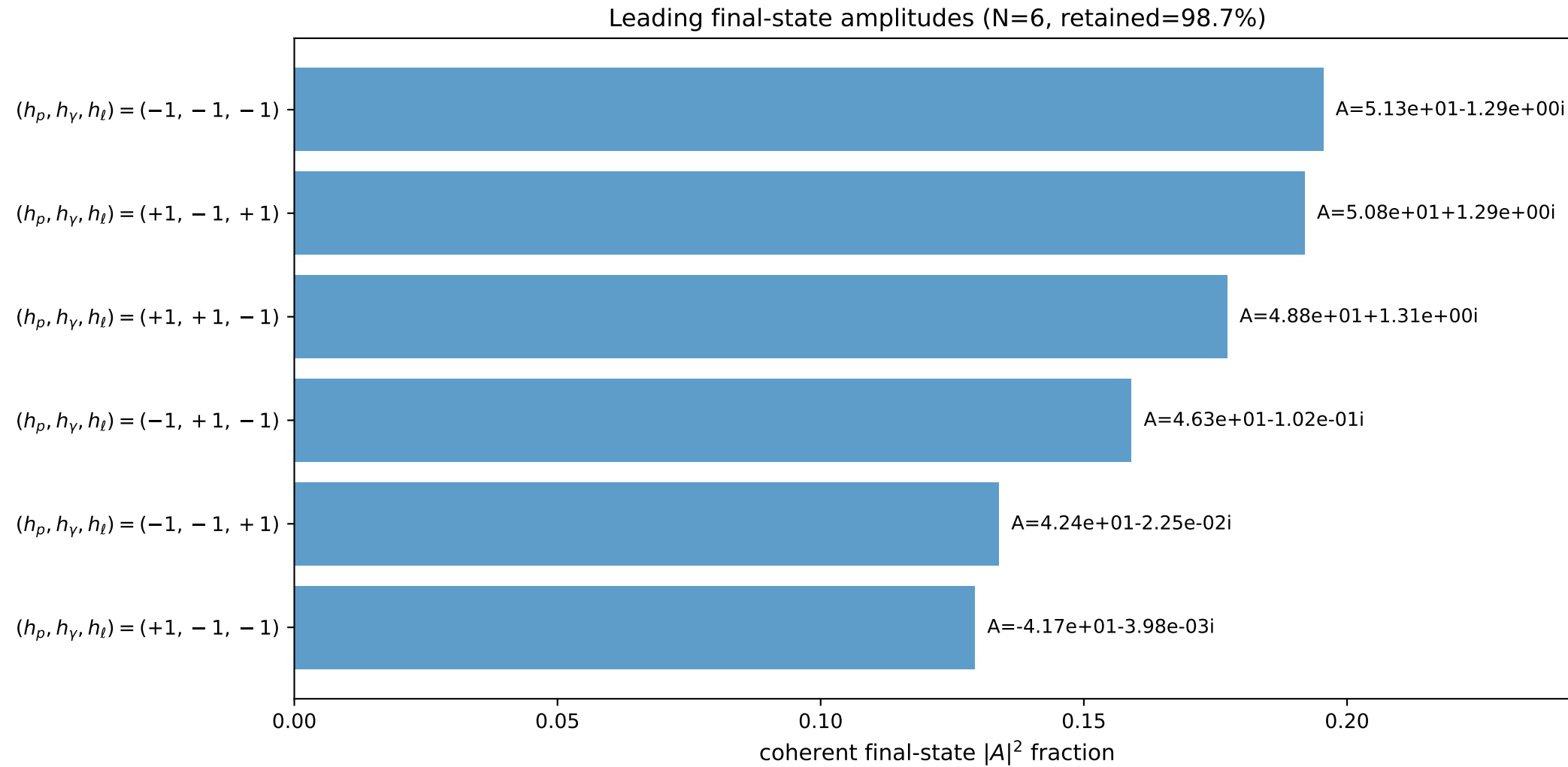
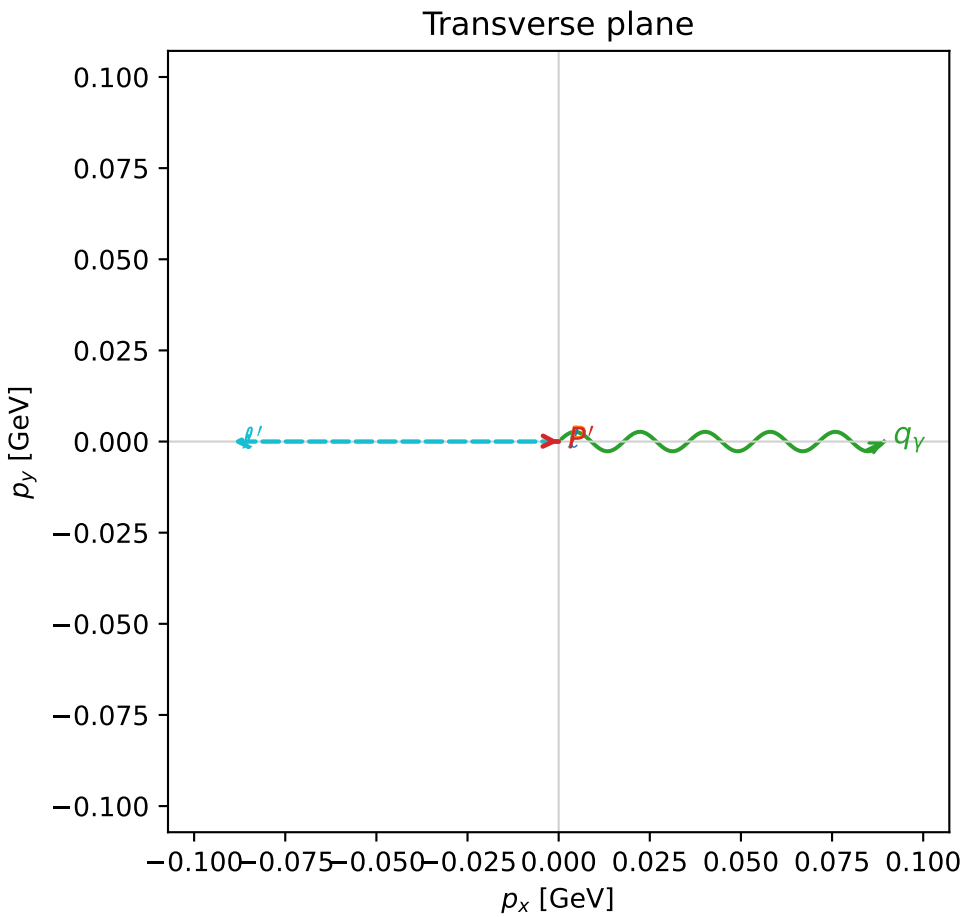
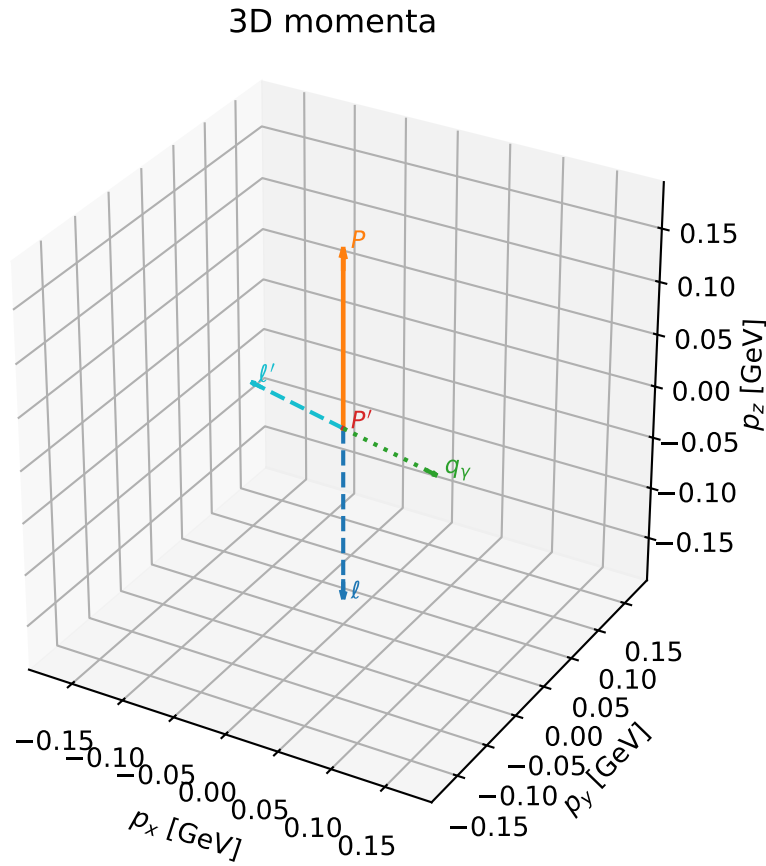


coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

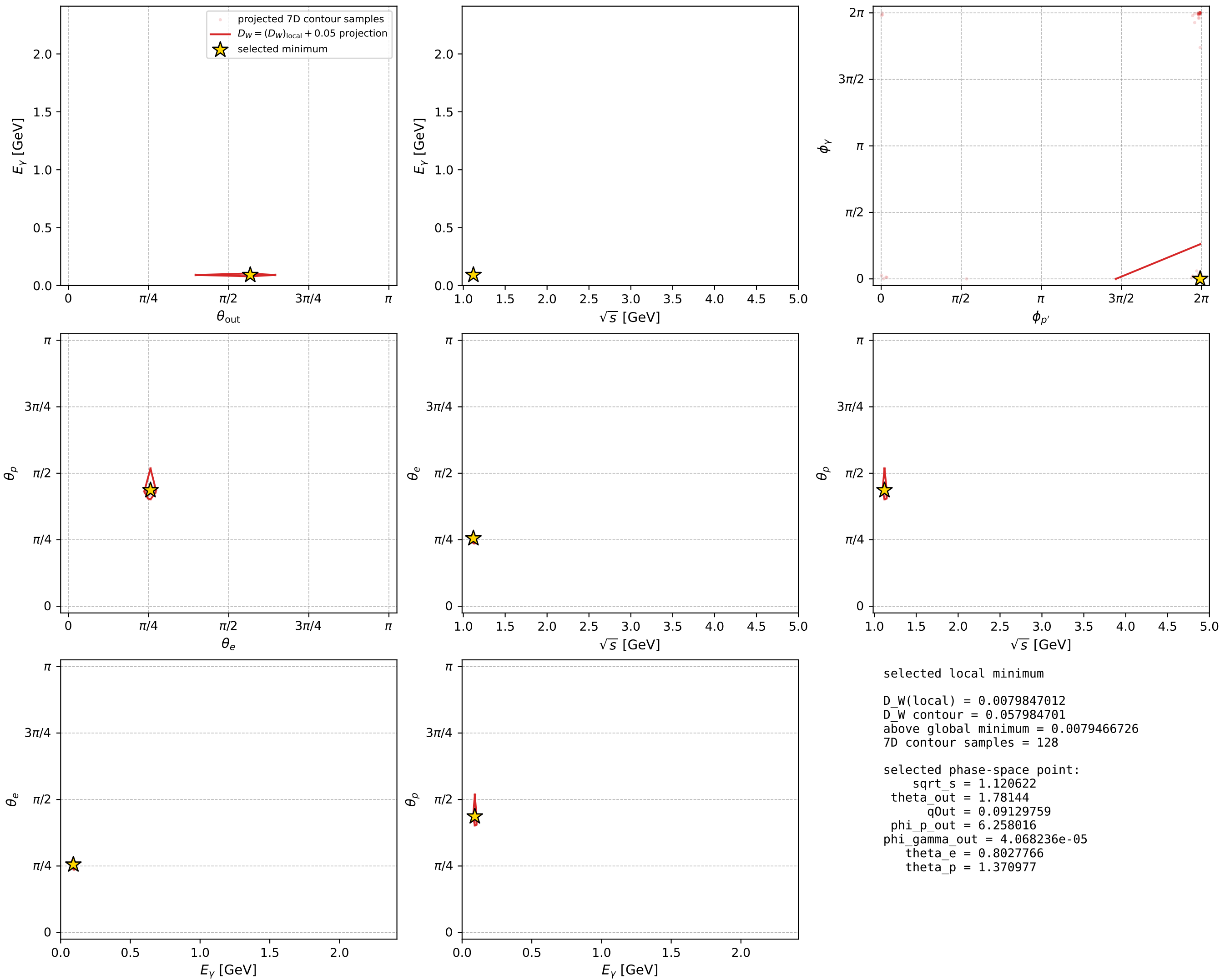
dw_mixing_angles_polarization_03_local_minimum_531 D_W=0.0079847
 region: polarization_cluster_03_local_minimum_531
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: theta_e=0.80277656 rad
 incoming lepton: 0.694712|+> +0.719288|->
 proton mixing angle: theta_p=1.3709775 rad
 incoming proton: 0.198492|+> +0.980103|->

$s=1.25579$, $\sqrt{s}=1.12062$, $|\vec{P}|=0.167741$, $|\vec{P}'|=2.5843e-05$
 $\theta_{\text{out}}=1.78144$, $\phi_{p'}=6.25802$, $\phi_\gamma=4.06824e-05$
 $E_\gamma=0.0912976$, $\theta_{\ell'}=1.36015$, $\phi_{\ell'}=3.14163$
 $Q^2=0.0370435$, $x_B=0.177826$, $t=-0.0279175$, $W^2=1.05111$, $y=0.554097$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1677$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1677$
 P $E= 0.9529$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1677$
 ℓ' $E= 0.0913$ $p_x= -0.0893$ $p_y= -0.0000$ $p_z= 0.0191$
 P' $E= 0.9380$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -0.0000$
 q_γ $E= 0.0913$ $p_x= 0.0893$ $p_y= 0.0000$ $p_z= -0.0191$



electron: polarization cluster P3, local minimum 531; pairwise projections of the 7D $D_W = (D_W)_{\text{local}} + 0.05$ contour

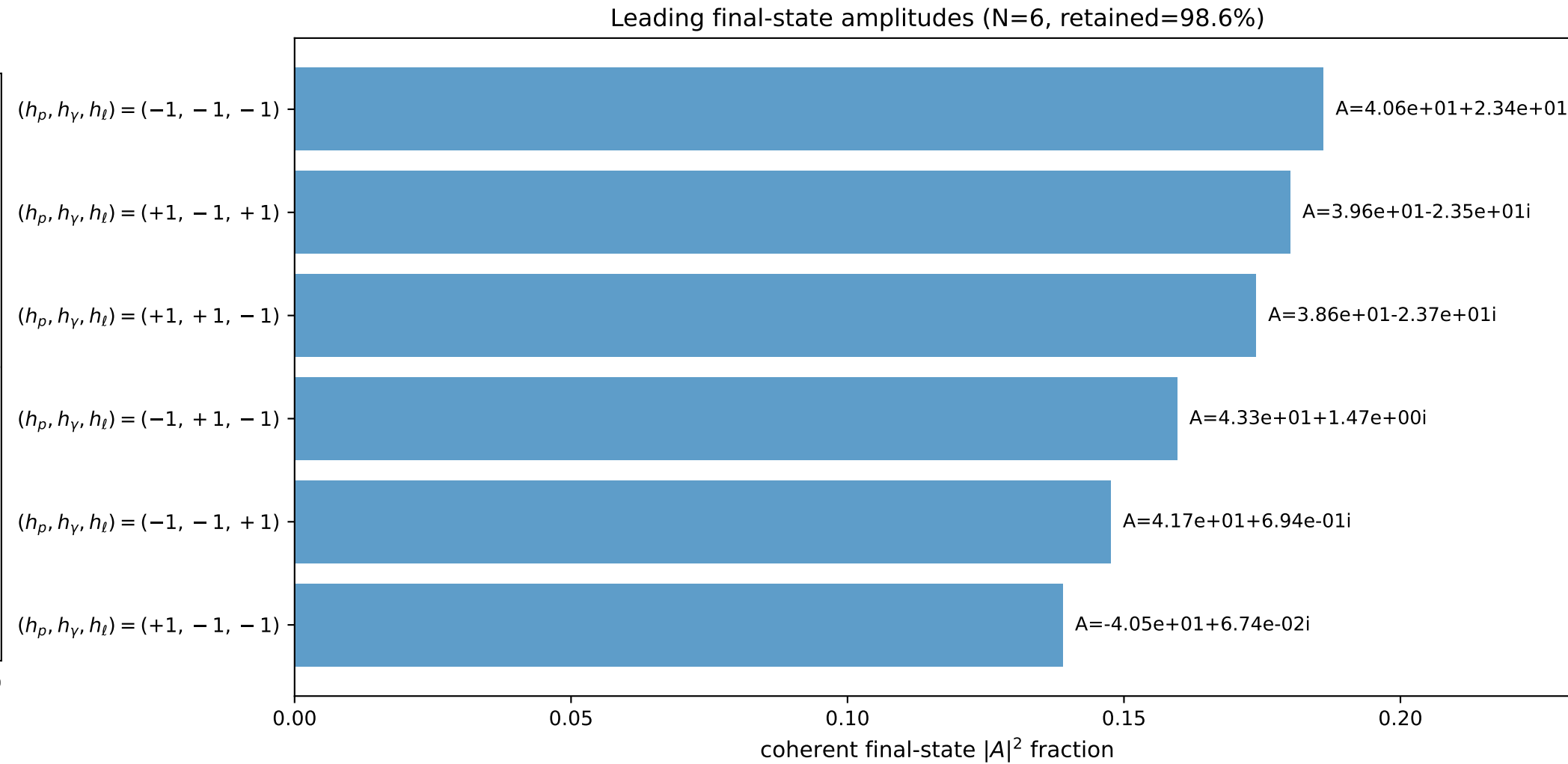
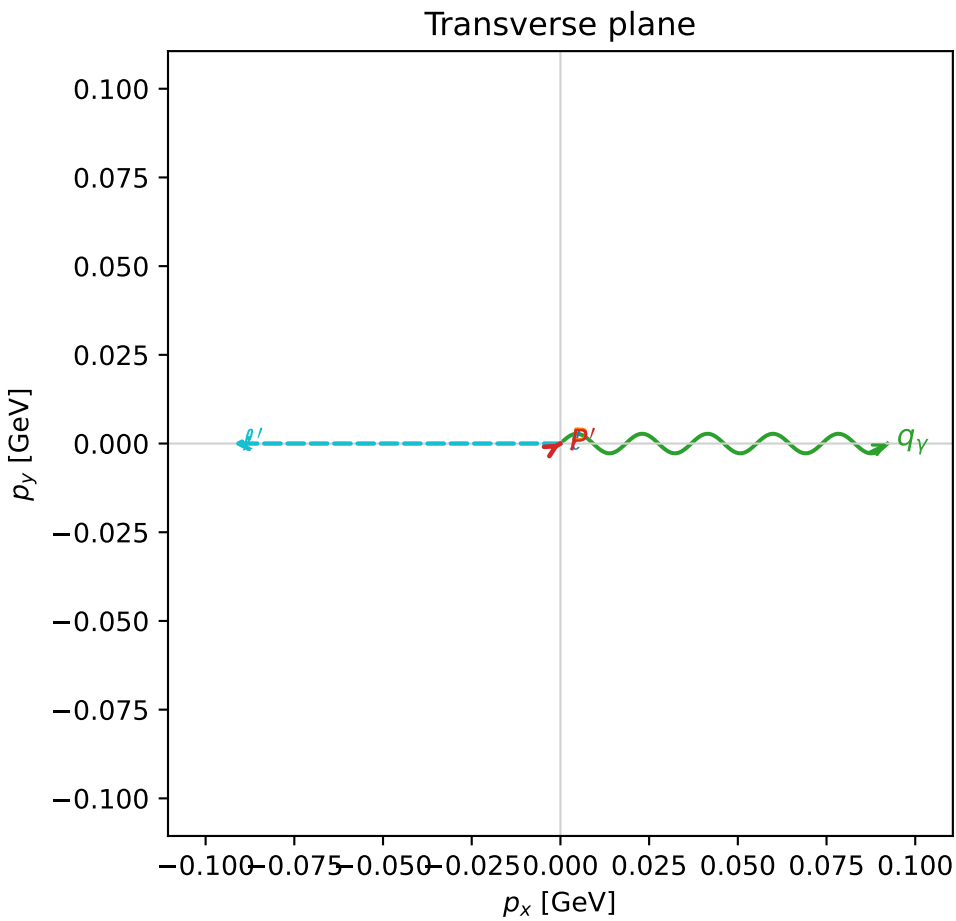
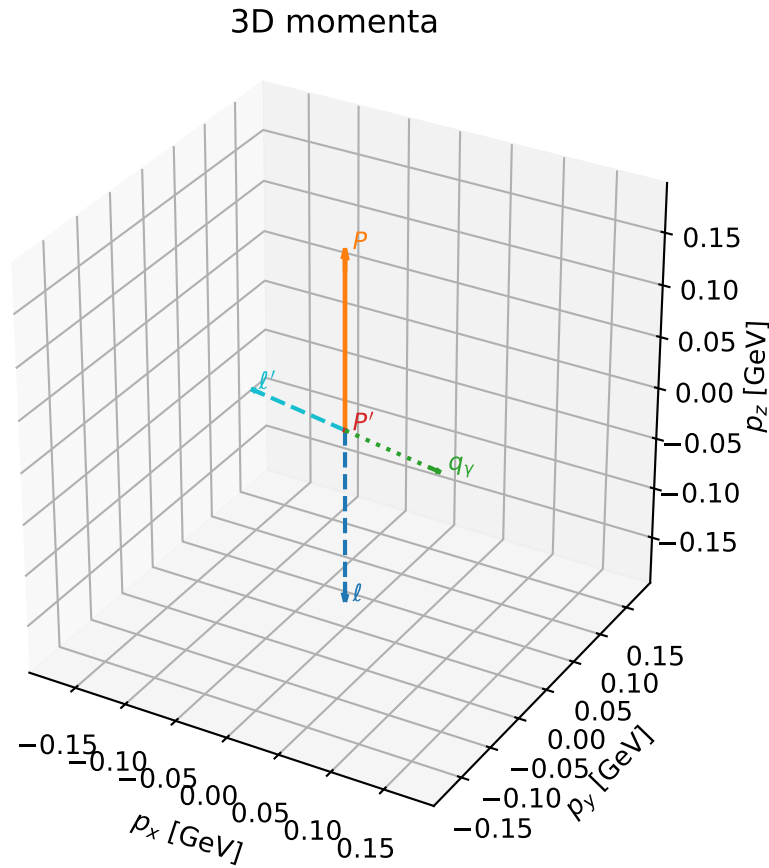


coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

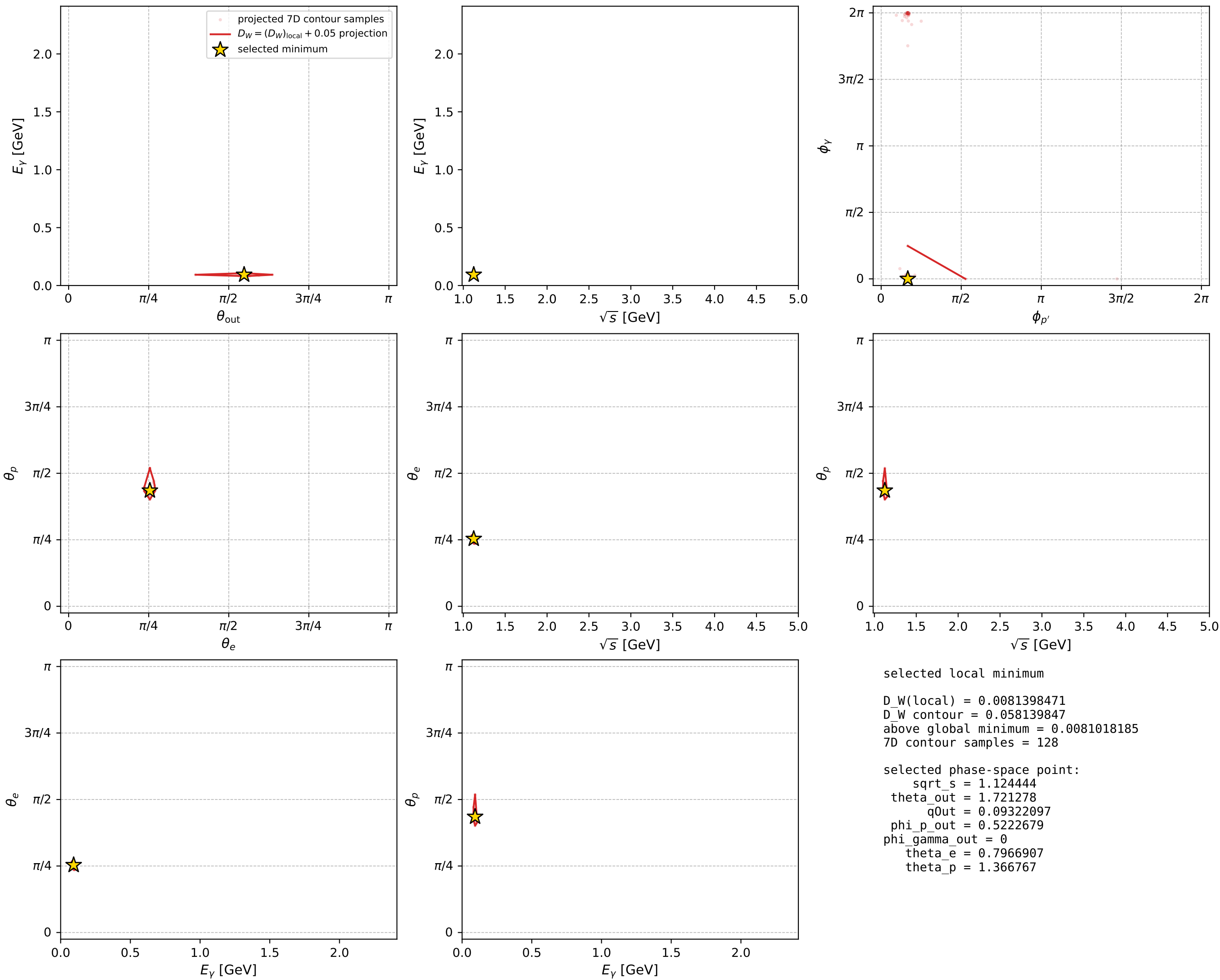
dw_mixing_angles_polarization_03_local_minimum_537 D_W=0.00813985
 region: polarization_cluster_03_local_minimum_537
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\theta_e=0.79669068$ rad
 incoming lepton: $0.699077|+\rangle +0.715047|-\rangle$
 proton mixing angle: $\theta_p=1.3667674$ rad
 incoming proton: $0.202616|+\rangle +0.979258|-\rangle$

$s=1.26437$, $\sqrt{s}=1.12444$, $|\vec{P}|=0.170986$, $|\vec{P}'|=7.07051e-07$
 $\theta_{\text{out}}=1.72128$, $\phi_{P'}=0.522268$, $\phi_Y=0$
 $E_Y=0.093221$, $\theta_{\ell'}=1.42031$, $\phi_{\ell'}=3.1416$
 $Q^2=0.0366585$, $x_B=0.173293$, $t=-0.0289974$, $W^2=1.05473$, $y=0.550128$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E=0.1710$ $p_x=0.0000$ $p_y=0.0000$ $p_z=-0.1710$
 P $E=0.9535$ $p_x=0.0000$ $p_y=0.0000$ $p_z=0.1710$
 ℓ' $E=0.0932$ $p_x=-0.0922$ $p_y=-0.0000$ $p_z=0.0140$
 P' $E=0.9380$ $p_x=0.0000$ $p_y=0.0000$ $p_z=-0.0000$
 q_Y $E=0.0932$ $p_x=0.0922$ $p_y=0.0000$ $p_z=-0.0140$



electron: polarization cluster P3, local minimum 537; pairwise projections of the 7D $D_W = (D_W)_{\text{local}} + 0.05$ contour

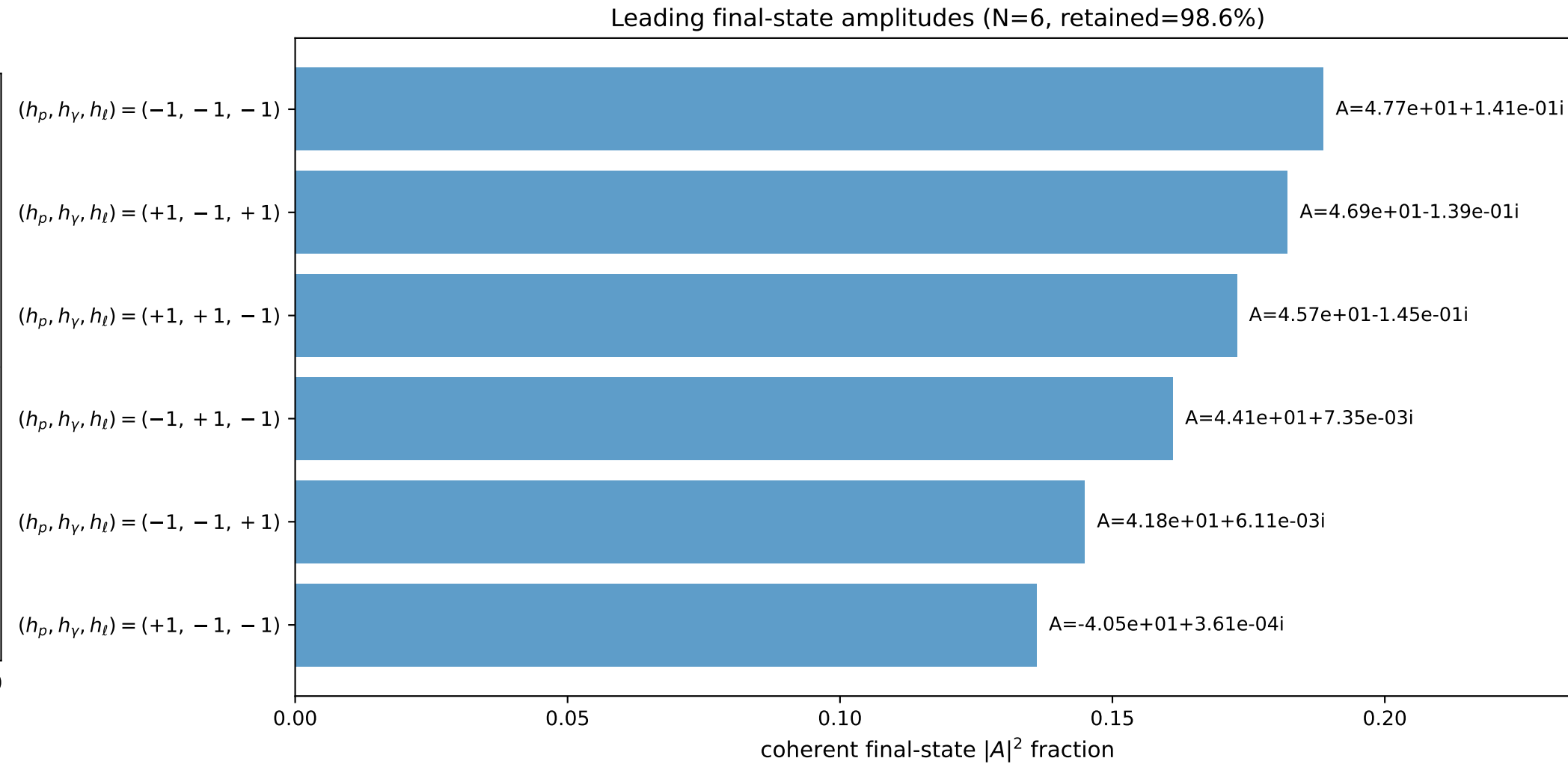
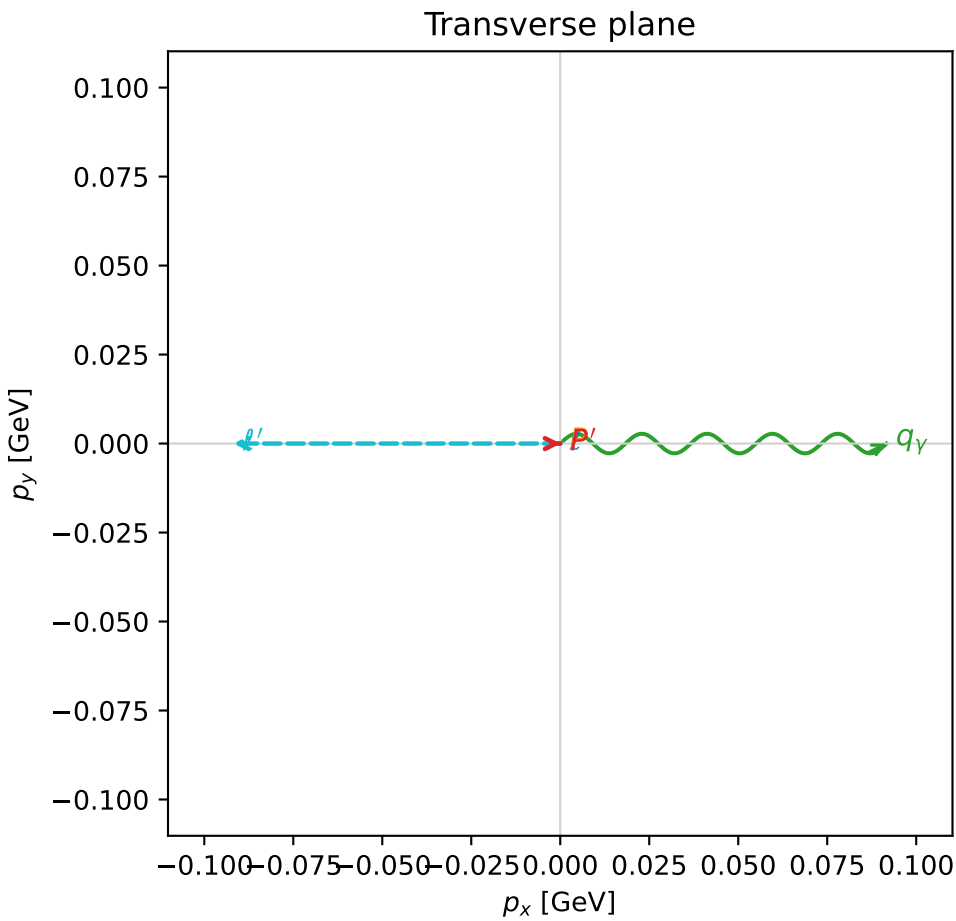
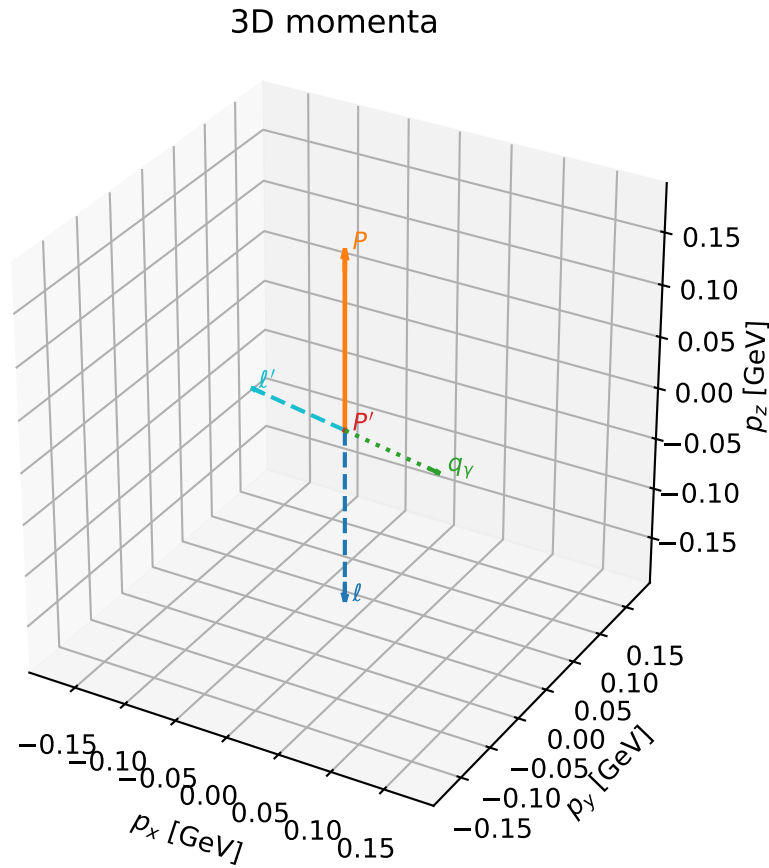


coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

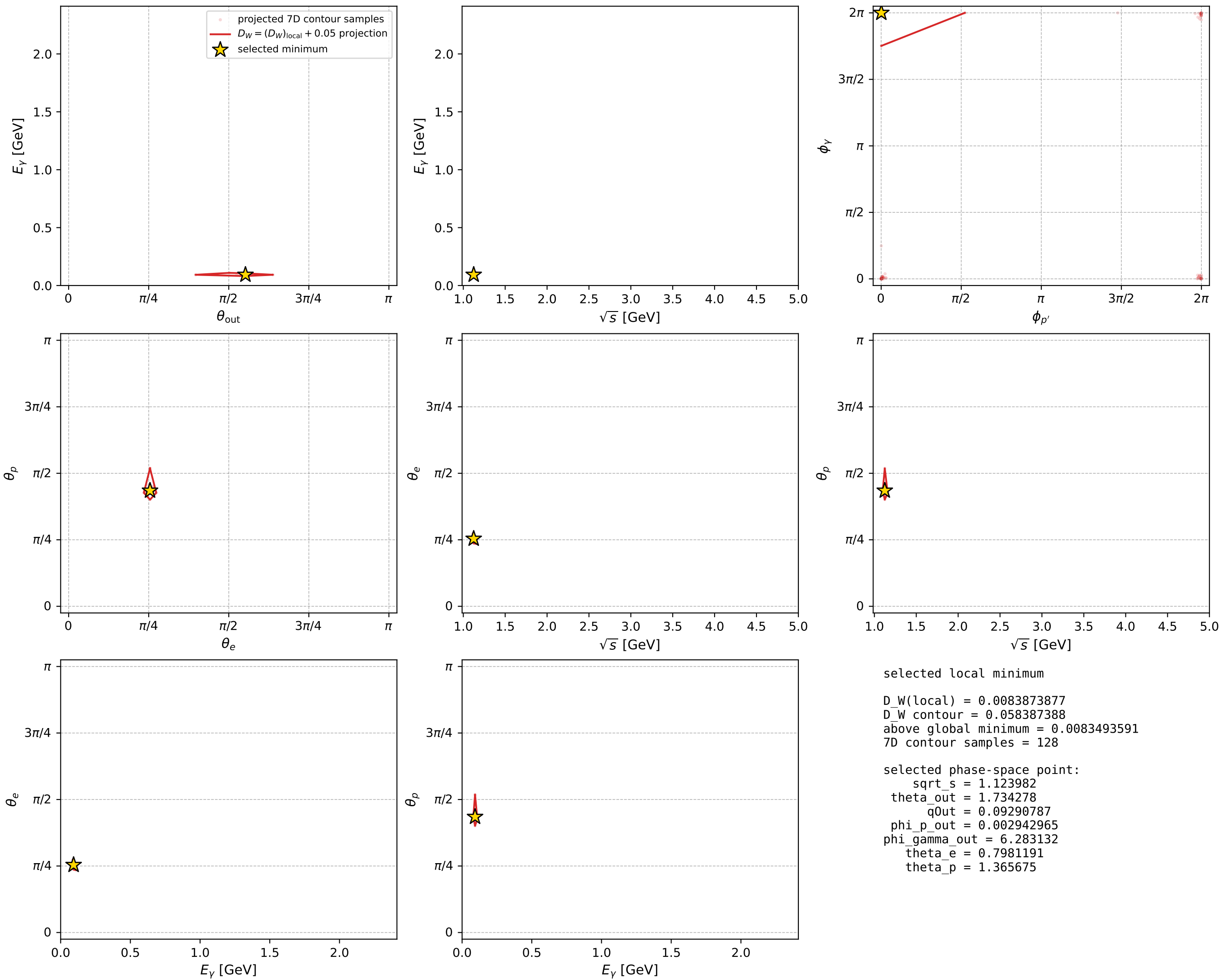
dw_mixing_angles_polarization_03_local_minimum_538 D_W=0.00838739
 region: polarization_cluster_03_local_minimum_538
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\theta_{\bar{e}}=0.79811909$ rad
 incoming lepton: $0.698055|+\rangle +0.716044|-\rangle$
 proton mixing angle: $\theta_{\bar{p}}=1.3656753$ rad
 incoming proton: $0.203686|+\rangle +0.979036|-\rangle$

$s=1.26334$, $\sqrt{s}=1.12398$, $|\vec{P}|=0.170594$, $|\vec{P}'|=0.000164858$
 $\theta_{\text{out}}=1.73428$, $\phi_{P'}=0.00294296$, $\phi_Y=6.28313$
 $E_Y=0.0929079$, $\theta_{\ell'}=1.40731$, $\phi_{\ell'}=3.14155$
 $Q^2=0.0369238$, $x_B=0.174838$, $t=-0.0288749$, $W^2=1.05411$, $y=0.550699$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E=0.1706$ $p_x=0.0000$ $p_y=0.0000$ $p_z=-0.1706$
 P $E=0.9534$ $p_x=0.0000$ $p_y=0.0000$ $p_z=0.1706$
 ℓ' $E=0.0931$ $p_x=-0.0918$ $p_y=0.0000$ $p_z=0.0151$
 P' $E=0.9380$ $p_x=0.0002$ $p_y=0.0000$ $p_z=-0.0000$
 q_Y $E=0.0929$ $p_x=0.0917$ $p_y=-0.0000$ $p_z=-0.0151$



electron: polarization cluster P3, local minimum 538; pairwise projections of the 7D $D_W = (D_W)_{\text{local}} + 0.05$ contour

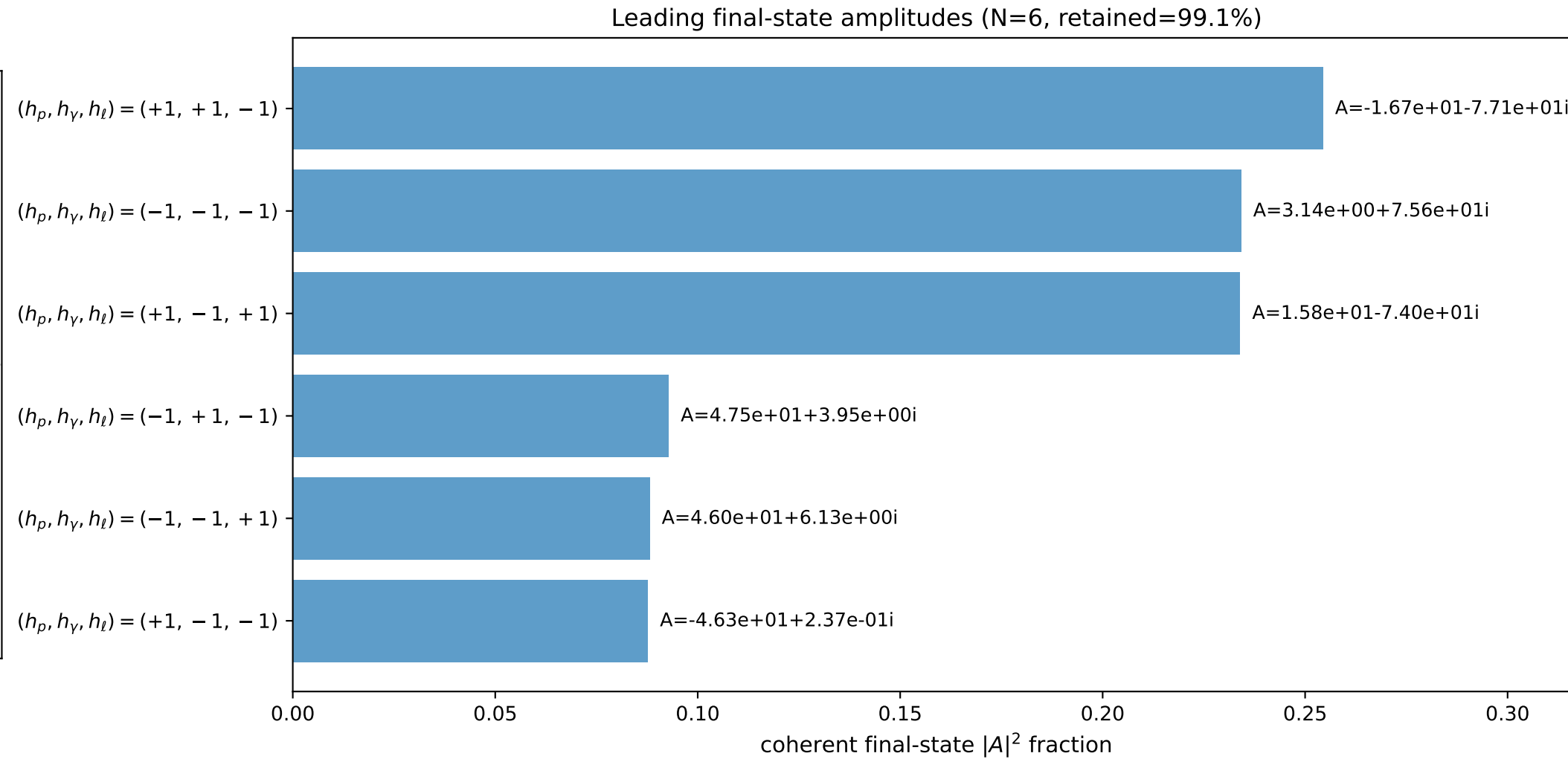
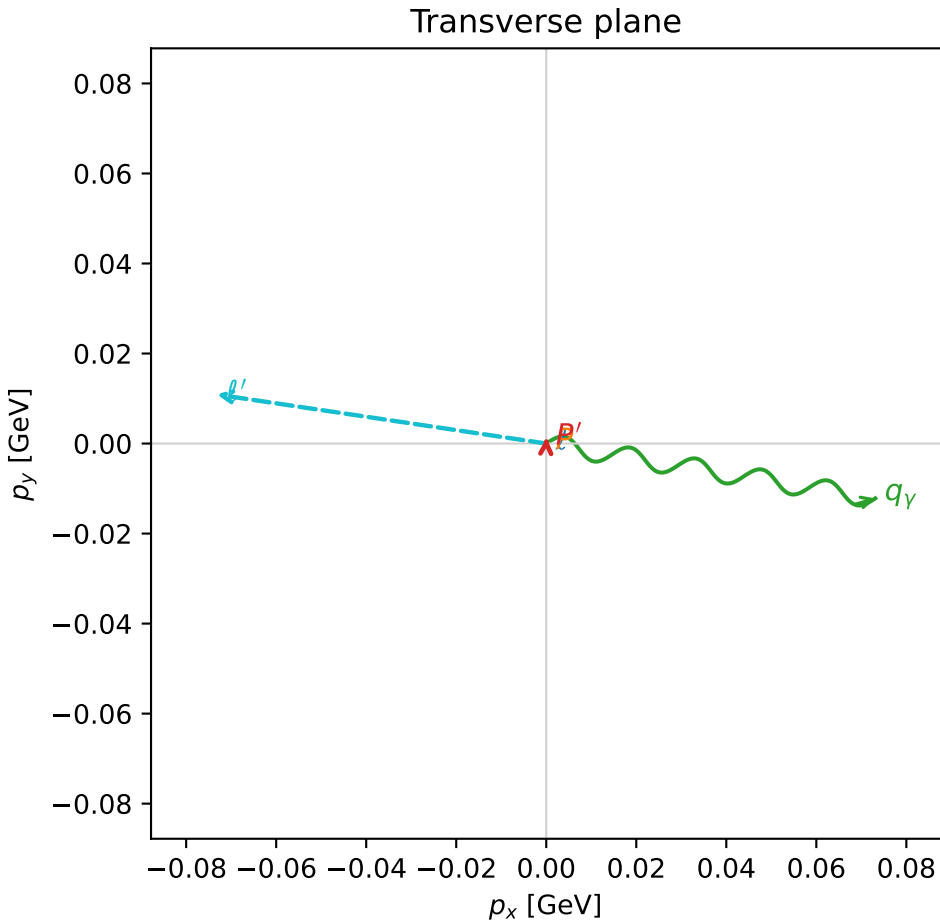
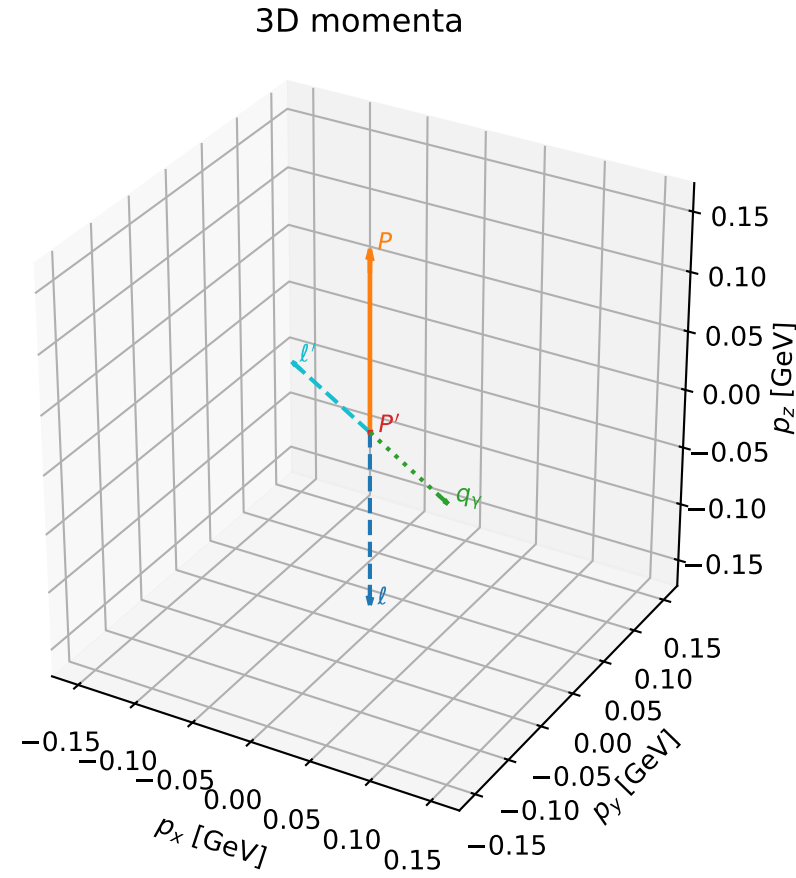


coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

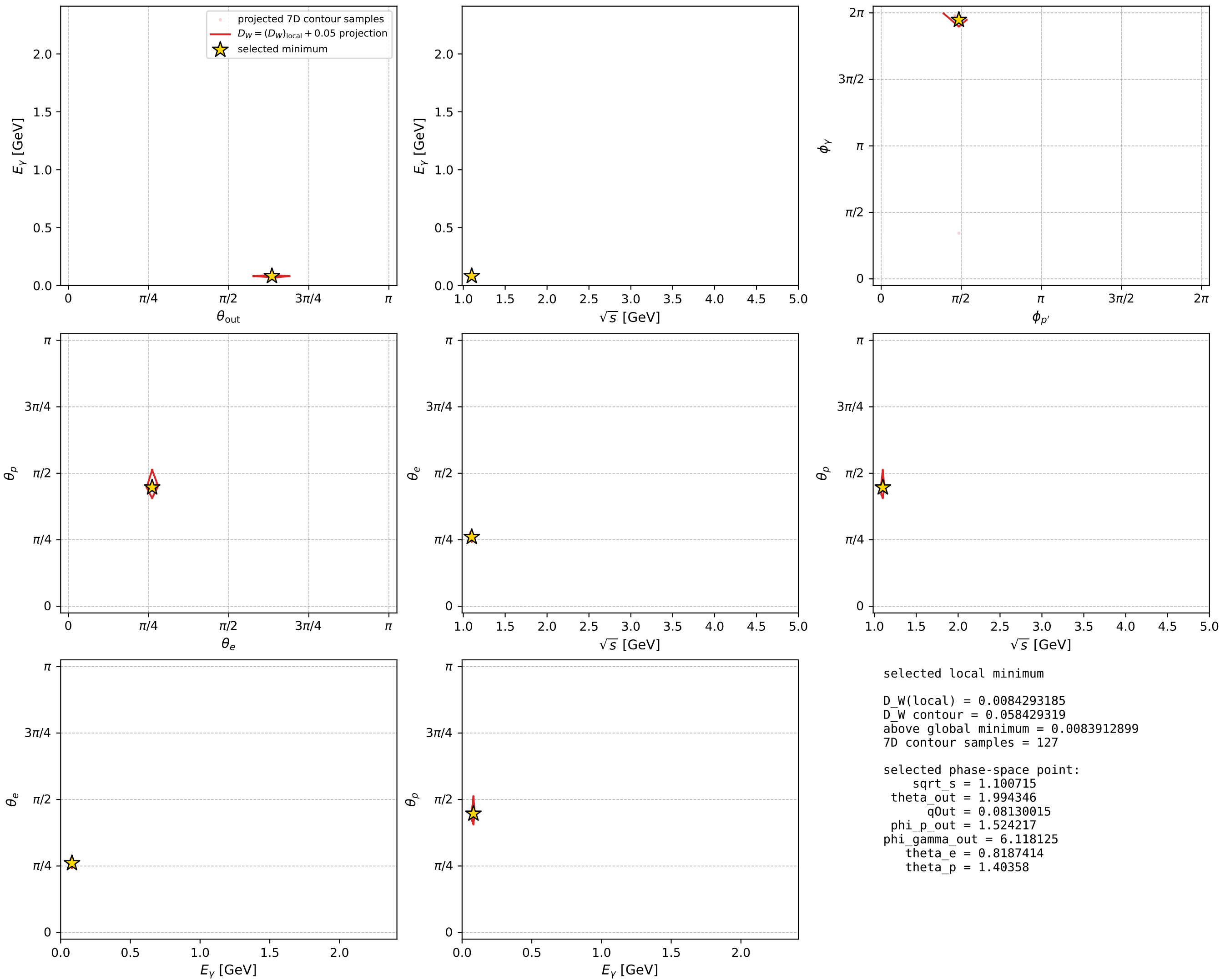
dw_mixing_angles_polarization_03_local_minimum_540 D_W=0.00842932
 region: polarization_cluster_03_local_minimum_540
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\theta_e=0.8187414$ rad
 incoming lepton: $0.683141|+\rangle + 0.730287|-\rangle$
 proton mixing angle: $\theta_p=1.4035797$ rad
 incoming proton: $0.166438|+\rangle + 0.986052|-\rangle$

$s=1.21157$, $\sqrt{s}=1.10071$, $|\vec{P}|=0.150687$, $|\vec{P}'|=0.00140952$
 $\theta_{\text{out}}=1.99435$, $\phi_{P'}=1.52422$, $\phi_Y=6.11812$
 $E_Y=0.0813002$, $\theta_{\ell'}=1.14005$, $\phi_{\ell'}=2.99378$
 $Q^2=0.0347804$, $x_B=0.18571$, $t=-0.0227386$, $W^2=1.03235$, $y=0.564568$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1507$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1507$
 P $E= 0.9500$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1507$
 ℓ' $E= 0.0814$ $p_x= -0.0732$ $p_y= 0.0109$ $p_z= 0.0340$
 P' $E= 0.9380$ $p_x= 0.0001$ $p_y= 0.0013$ $p_z= -0.0006$
 q_Y $E= 0.0813$ $p_x= 0.0731$ $p_y= -0.0122$ $p_z= -0.0334$



electron: polarization cluster P3, local minimum 540; pairwise projections of the 7D $D_W = (D_W)_{\text{local}} + 0.05$ contour



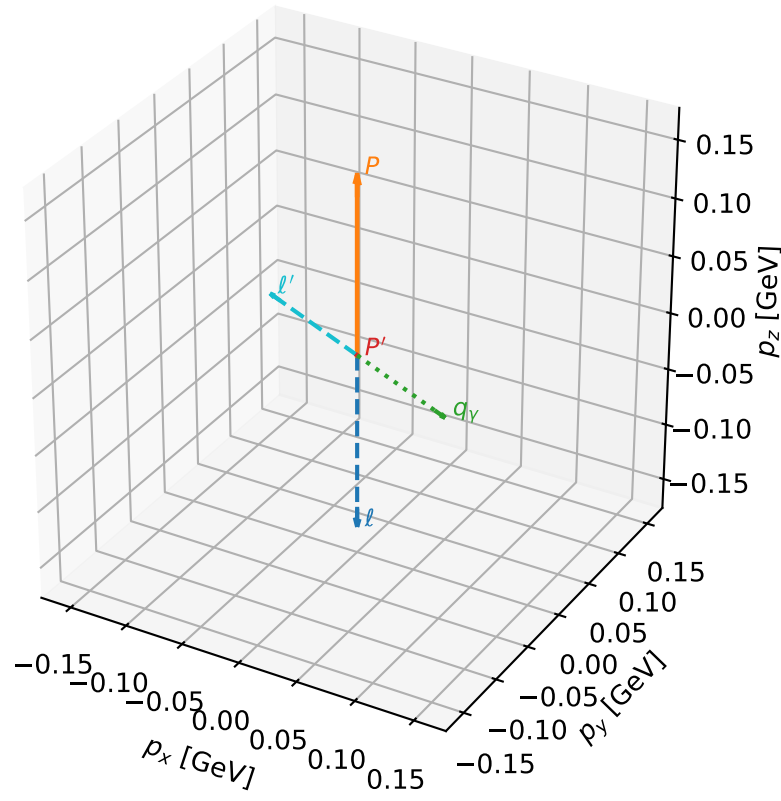
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

dw_mixing_angles_polarization_03_local_minimum_542 D_W=0.00846388
 region: polarization_cluster_03_local_minimum_542
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\theta_e=0.81725862$ rad
 incoming lepton: $0.684223|+\rangle +0.729273|-\rangle$
 proton mixing angle: $\theta_p=1.4000691$ rad
 incoming proton: $0.169899|+\rangle +0.985461|-\rangle$

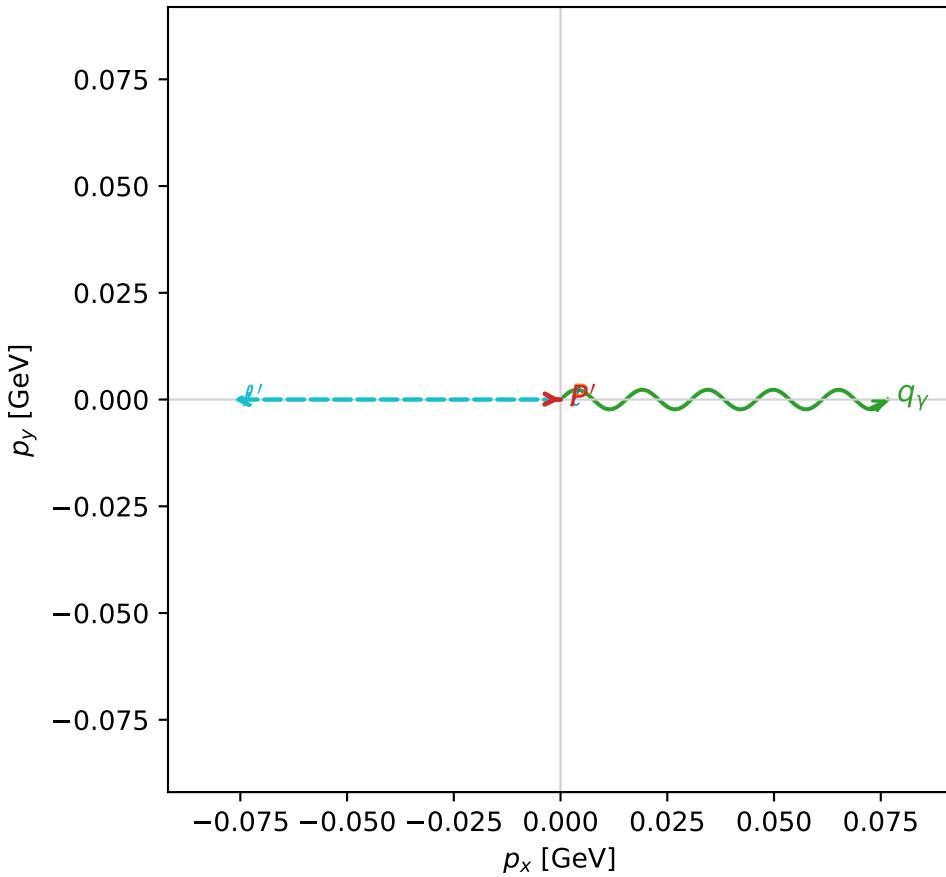
$s=1.21945$, $\sqrt{s}=1.10429$, $|\vec{P}|=0.153766$, $|\vec{P}'|=1.20042e-06$
 $\theta_{out}=1.9697$, $\phi_{p'}=6.25578$, $\phi_Y=6.28318$
 $E_Y=0.0831418$, $\theta_{\ell'}=1.17189$, $\phi_{\ell'}=3.14158$
 $Q^2=0.0355004$, $x_B=0.185406$, $t=-0.0234873$, $W^2=1.03582$, $y=0.563816$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1538$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1538$
 P $E= 0.9505$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1538$
 ℓ' $E= 0.0831$ $p_x= -0.0766$ $p_y= 0.0000$ $p_z= 0.0323$
 P' $E= 0.9380$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -0.0000$
 q_Y $E= 0.0831$ $p_x= 0.0766$ $p_y= -0.0000$ $p_z= -0.0323$

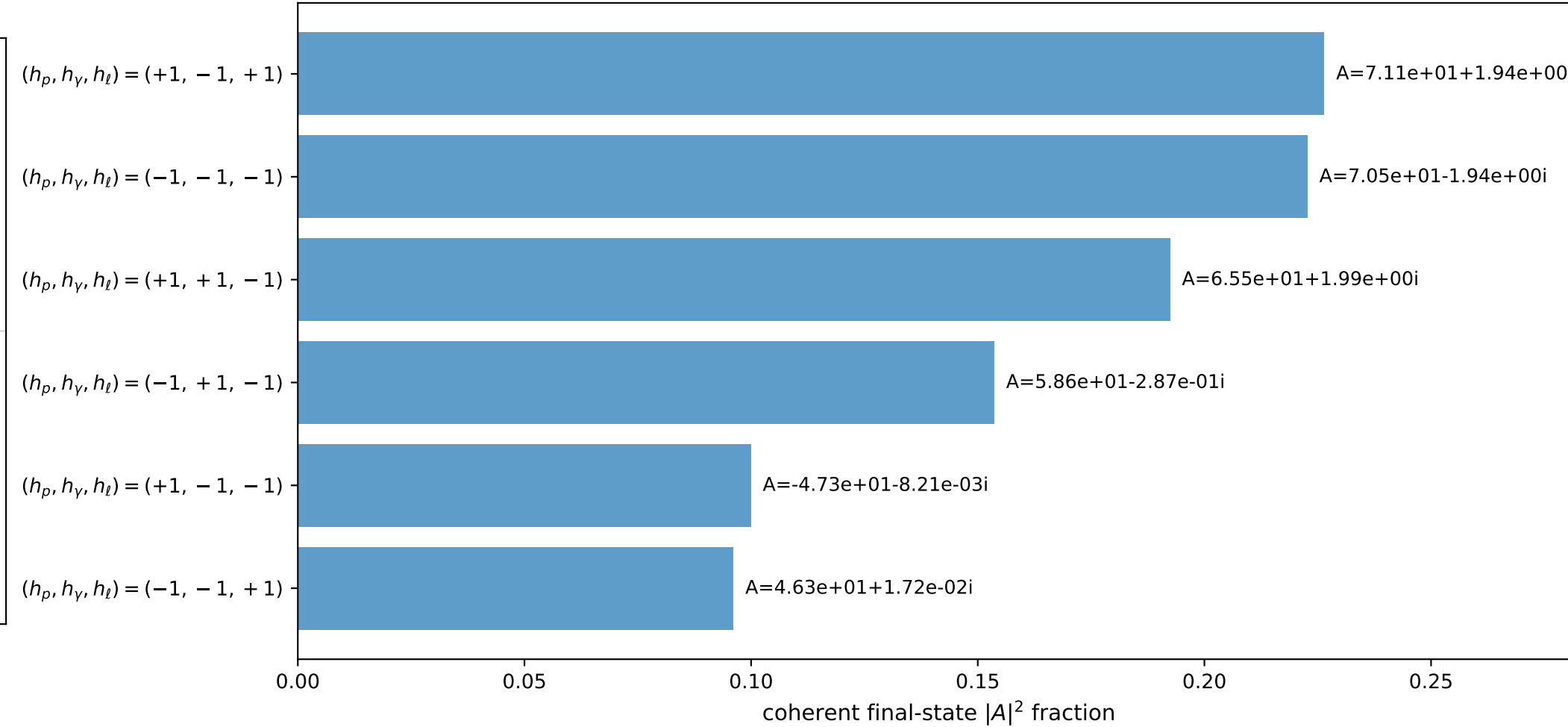
3D momenta



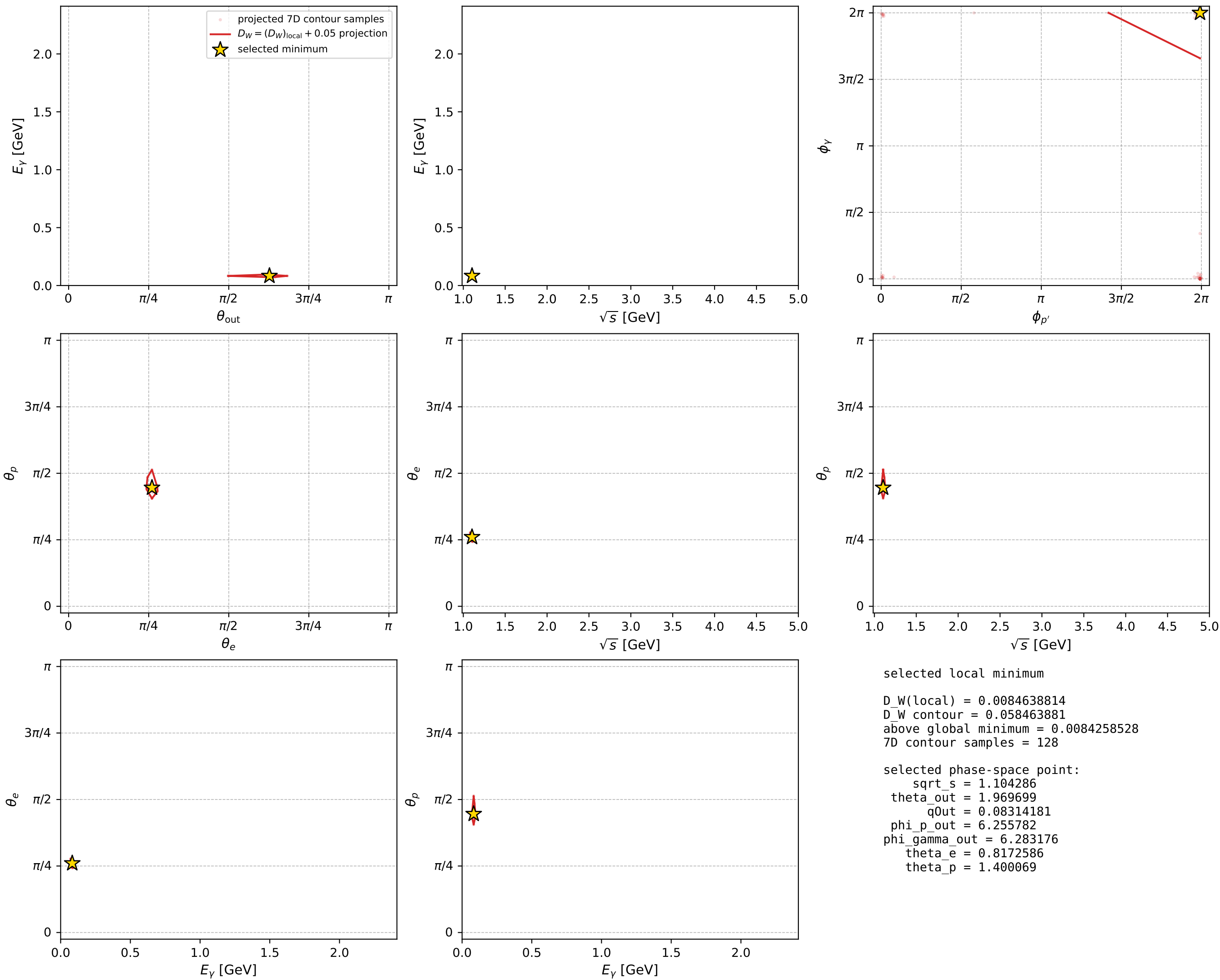
Transverse plane



Leading final-state amplitudes (N=6, retained=99.1%)



electron: polarization cluster P3, local minimum 542; pairwise projections of the 7D $D_W = (D_W)_{\text{local}} + 0.05$ contour

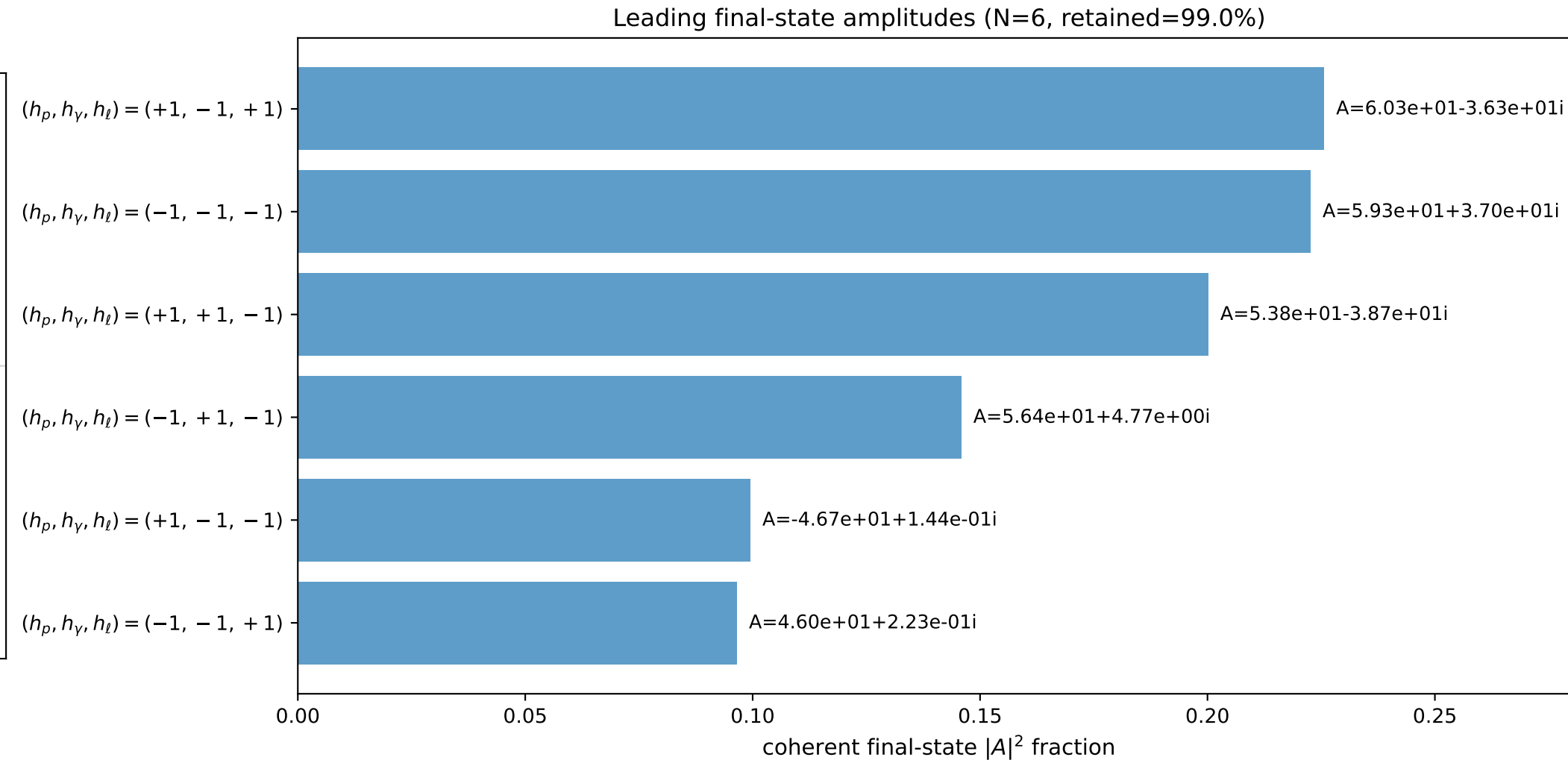
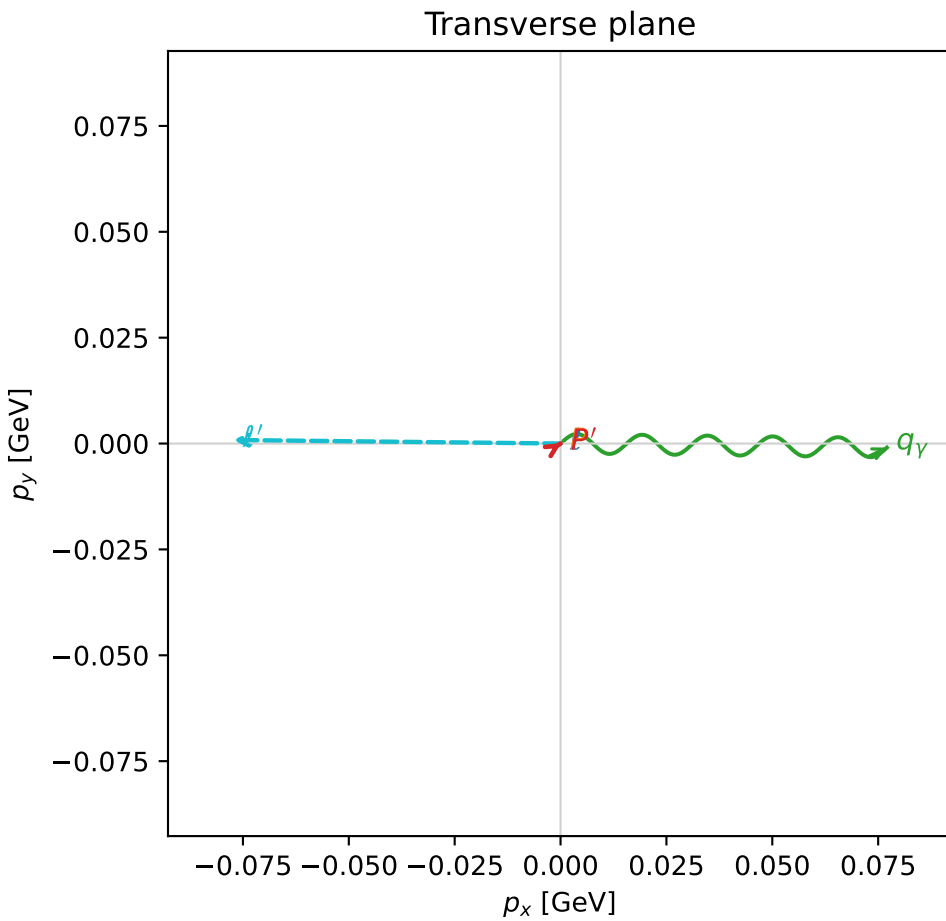
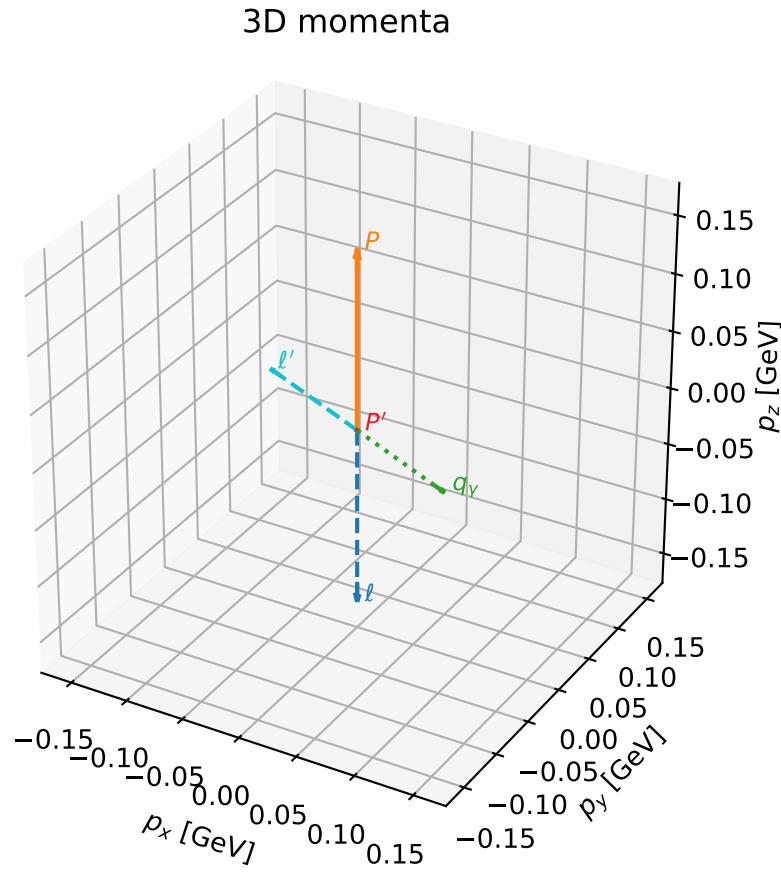


coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

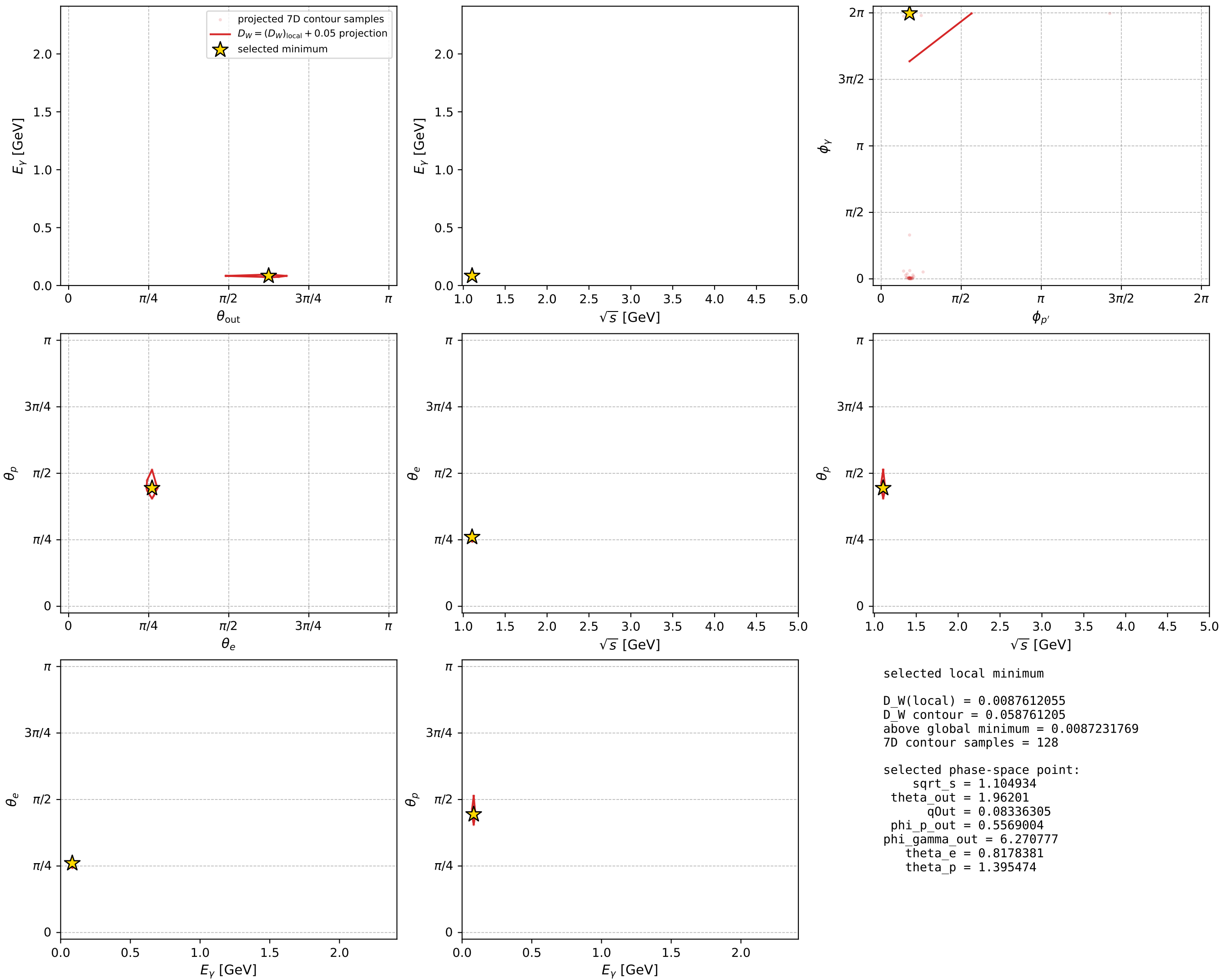
dw_mixing_angles_polarization_03_local_minimum_545 D_W=0.00876121
 region: polarization_cluster_03_local_minimum_545
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\theta_e=0.81783806$ rad
 incoming lepton: $0.6838|+\rangle +0.729669|-\rangle$
 proton mixing angle: $\theta_p=1.3954735$ rad
 incoming proton: $0.174426|+\rangle +0.98467|-\rangle$

$s=1.22088$, $\sqrt{s}=1.10493$, $|\vec{P}|=0.154323$, $|\vec{P}'|=0.000238861$
 $\theta_{\text{out}}=1.96201$, $\phi_{P'}=0.5569$, $\phi_Y=6.27078$
 $E_Y=0.083363$, $\theta_{\ell'}=1.17942$, $\phi_{\ell'}=3.13073$
 $Q^2=0.0356328$, $x_B=0.1856$, $t=-0.0236849$, $W^2=1.0362$, $y=0.562954$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E=0.1543$ $p_x=0.0000$ $p_y=0.0000$ $p_z=-0.1543$
 P $E=0.9506$ $p_x=0.0000$ $p_y=0.0000$ $p_z=0.1543$
 ℓ' $E=0.0836$ $p_x=-0.0772$ $p_y=0.0008$ $p_z=0.0319$
 P' $E=0.9380$ $p_x=0.0002$ $p_y=0.0001$ $p_z=-0.0001$
 q_Y $E=0.0834$ $p_x=0.0771$ $p_y=-0.0010$ $p_z=-0.0318$



electron: polarization cluster P3, local minimum 545; pairwise projections of the 7D $D_W = (D_W)_{\text{local}} + 0.05$ contour

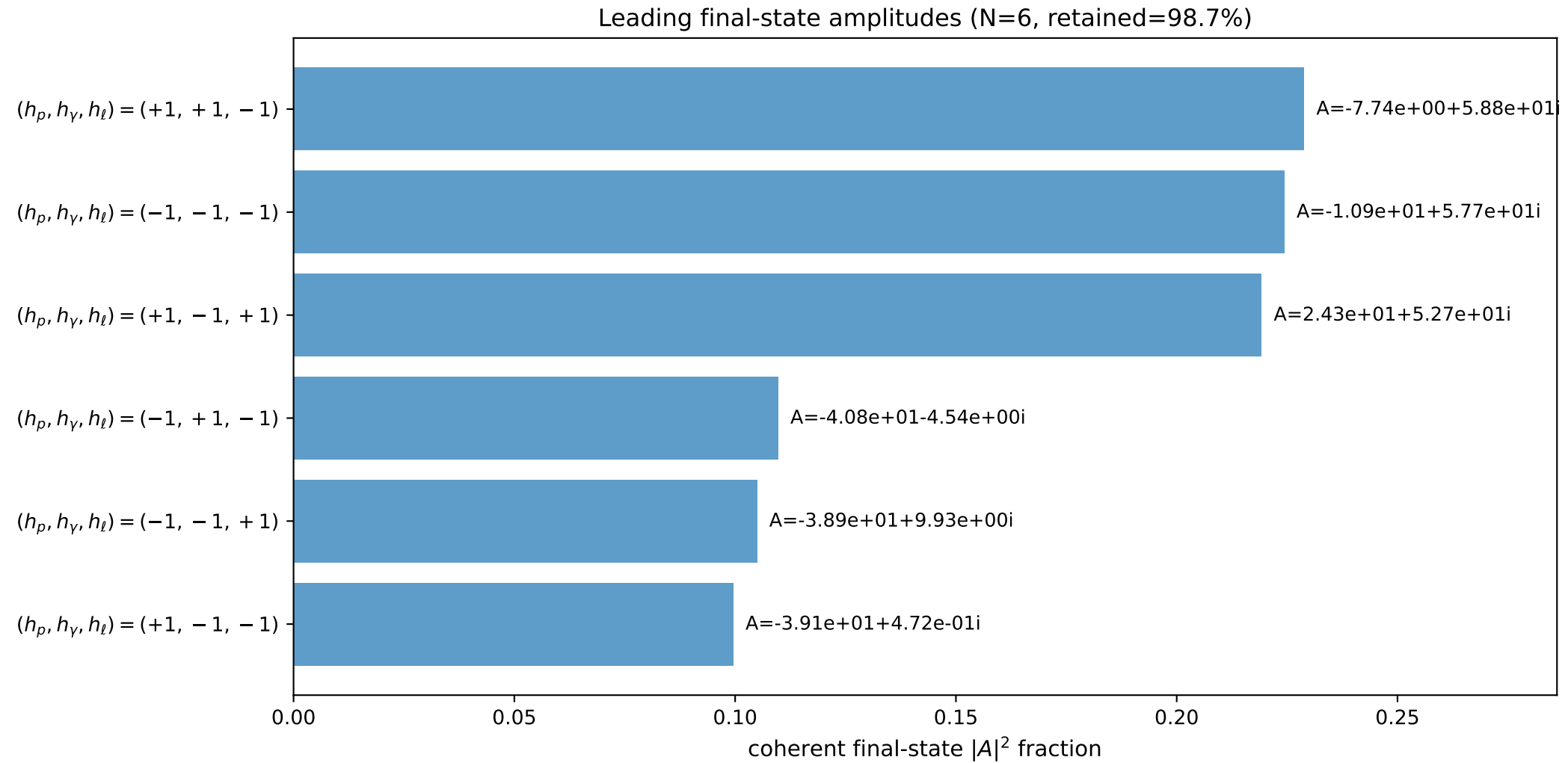
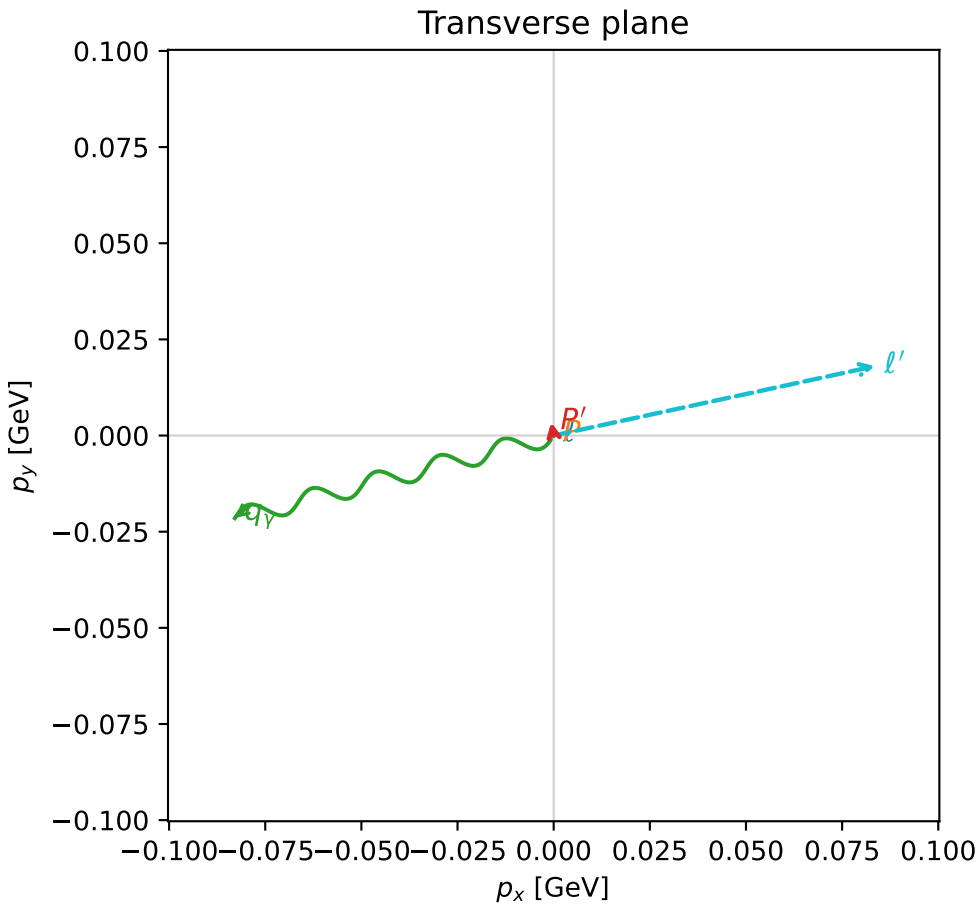
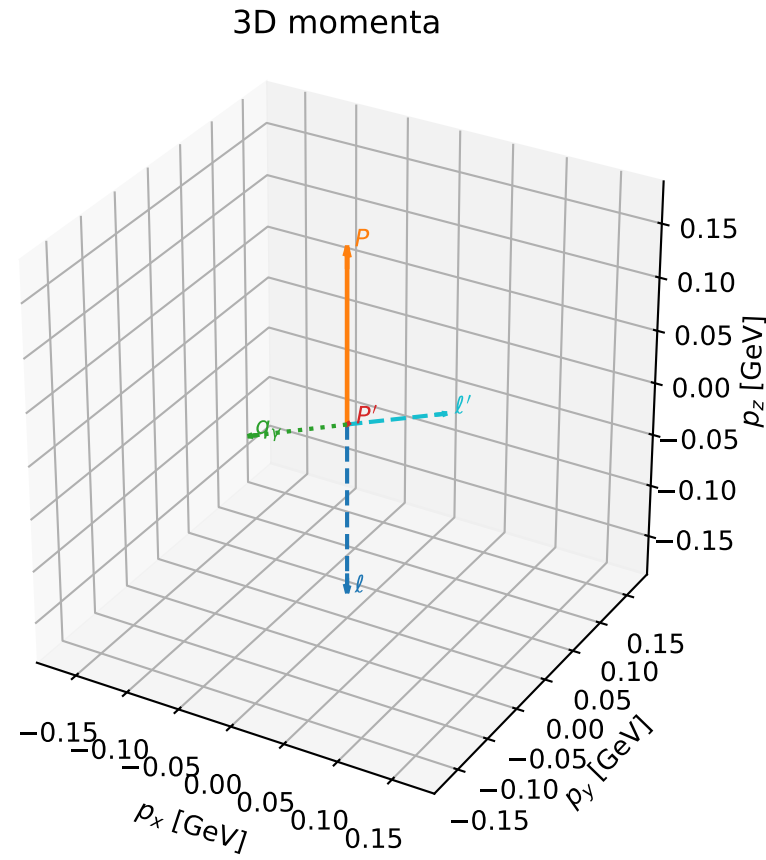


coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

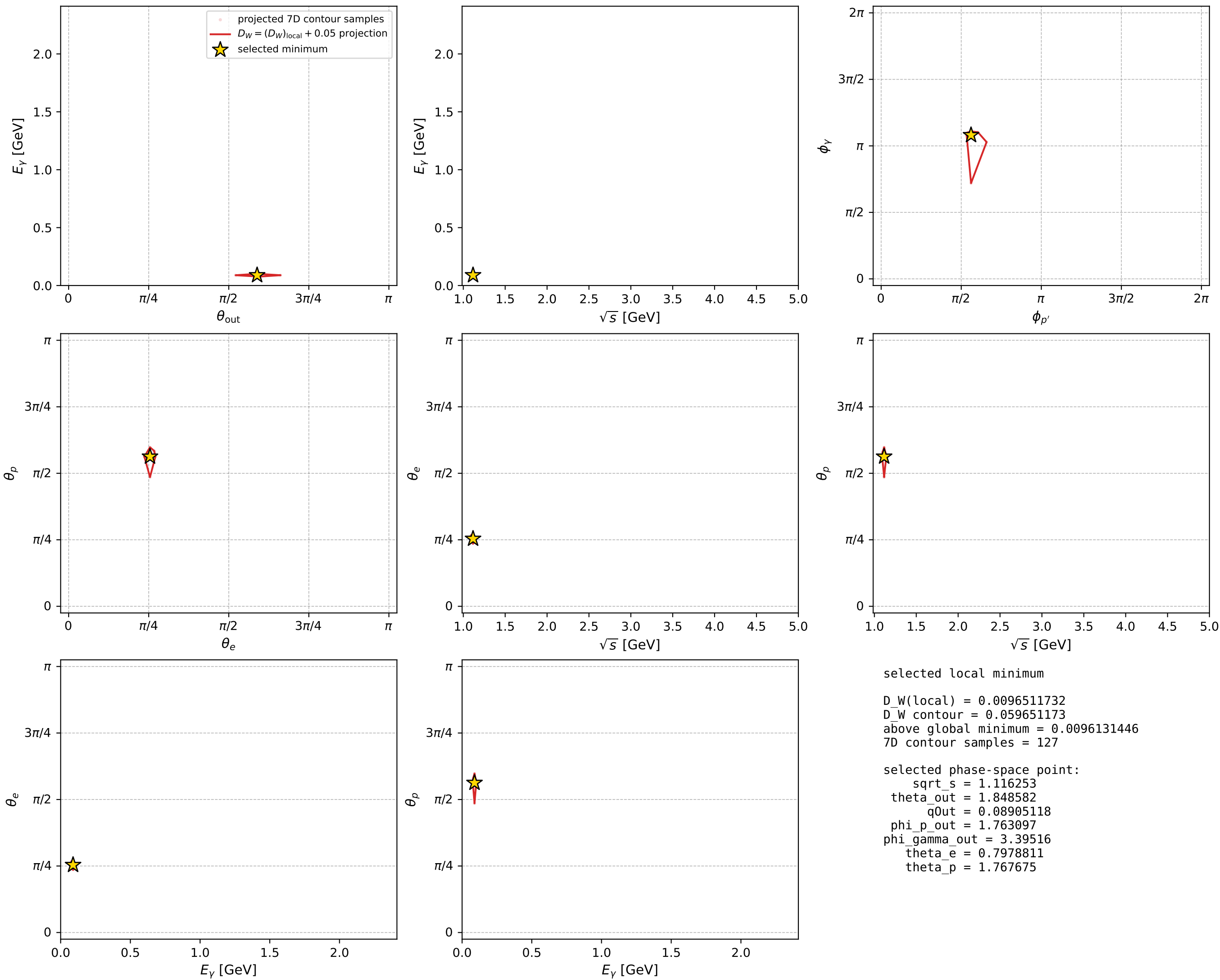
dw_mixing_angles_polarization_03_local_minimum_565 D_W=0.00965117
region: polarization_cluster_03_local_minimum_565
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\theta_e=0.79788109$ rad
incoming lepton: $0.698225|+\rangle +0.715878|-\rangle$
proton mixing angle: $\theta_p=1.7676754$ rad
incoming proton: $-0.19561|+\rangle +0.980682|-\rangle$

$s=1.24602$, $\sqrt{s}=1.11625$, $|\vec{P}|=0.16402$, $|\vec{P}'|=0.00364865$
 $\theta_{\text{out}}=1.84858$, $\phi_{P'}=1.7631$, $\phi_\gamma=3.39516$
 $E_\gamma=0.0890512$, $\theta_{l'}=1.28178$, $\phi_{l'}=0.212592$
 $Q^2=0.0375982$, $x_B=0.183722$, $t=-0.0270418$, $W^2=1.04689$, $y=0.558875$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 0.1640$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1640$
 P $E= 0.9522$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1640$
 l' $E= 0.0892$ $p_x= 0.0836$ $p_y= 0.0180$ $p_z= 0.0254$
 P' $E= 0.9380$ $p_x= -0.0007$ $p_y= 0.0034$ $p_z= -0.0010$
 q_γ $E= 0.0891$ $p_x= -0.0829$ $p_y= -0.0215$ $p_z= -0.0244$



electron: polarization cluster P3, local minimum 565; pairwise projections of the 7D $D_W = (D_W)_{\text{local}} + 0.05$ contour



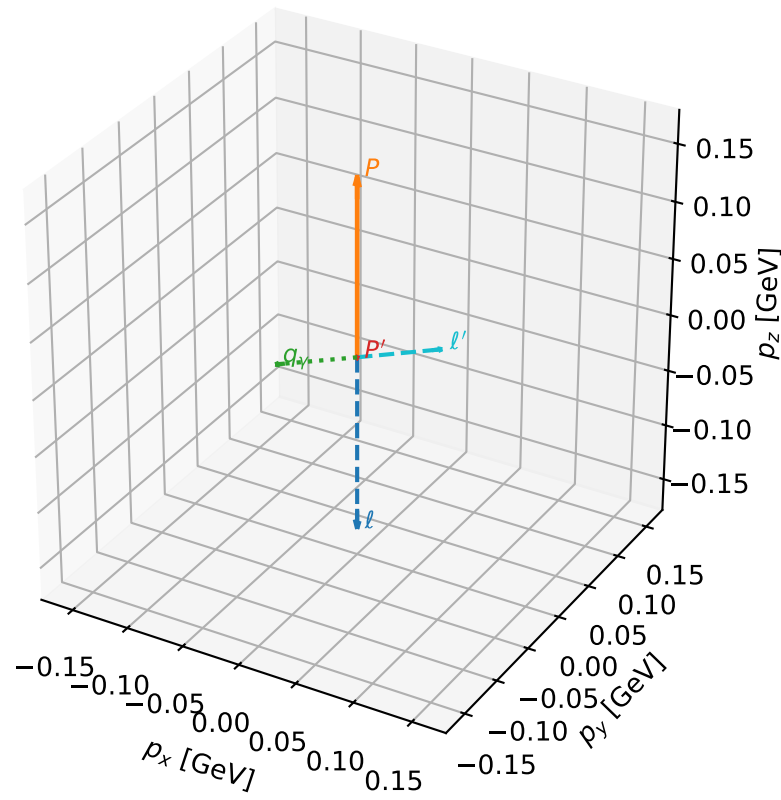
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

dw_mixing_angles_polarization_03_local_minimum_569 D_W=0.00995012
region: polarization_cluster_03_local_minimum_569
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\theta_e=0.8021483$ rad
incoming lepton: $0.695164|+\rangle +0.718851|-\rangle$
proton mixing angle: $\theta_p=1.7291778$ rad
incoming proton: $-0.15772|+\rangle +0.987484|-\rangle$

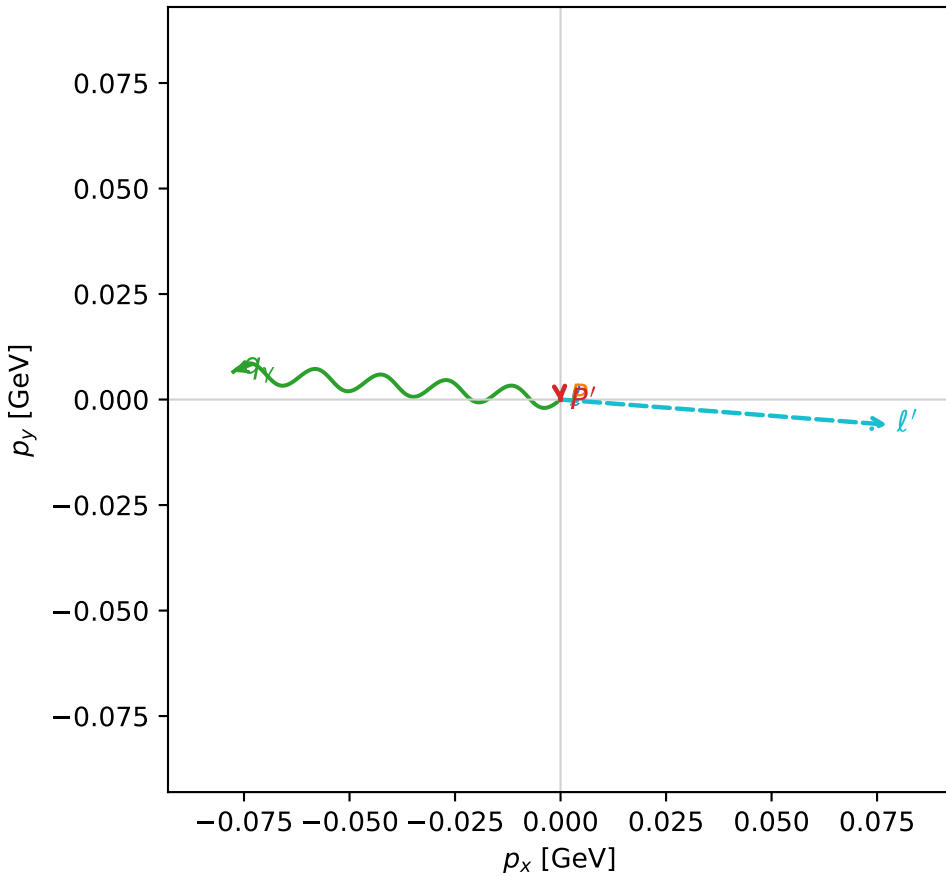
$s=1.22393$, $\sqrt{s}=1.10632$, $|\vec{P}|=0.155511$, $|\vec{P}'|=0.000683255$
 $\theta_{\text{out}}=1.96212$, $\phi_{P'}=4.78692$, $\phi_Y=3.05688$
 $E_Y=0.0841518$, $\theta_{\ell'}=1.17617$, $\phi_{\ell'}=6.2065$
 $Q^2=0.0362397$, $x_B=0.186699$, $t=-0.0241011$, $W^2=1.03771$, $y=0.56412$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1555$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1555$
 P $E= 0.9508$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1555$
 ℓ' $E= 0.0842$ $p_x= 0.0775$ $p_y= -0.0060$ $p_z= 0.0324$
 P' $E= 0.9380$ $p_x= 0.0000$ $p_y= -0.0006$ $p_z= -0.0003$
 q_Y $E= 0.0842$ $p_x= -0.0775$ $p_y= 0.0066$ $p_z= -0.0321$

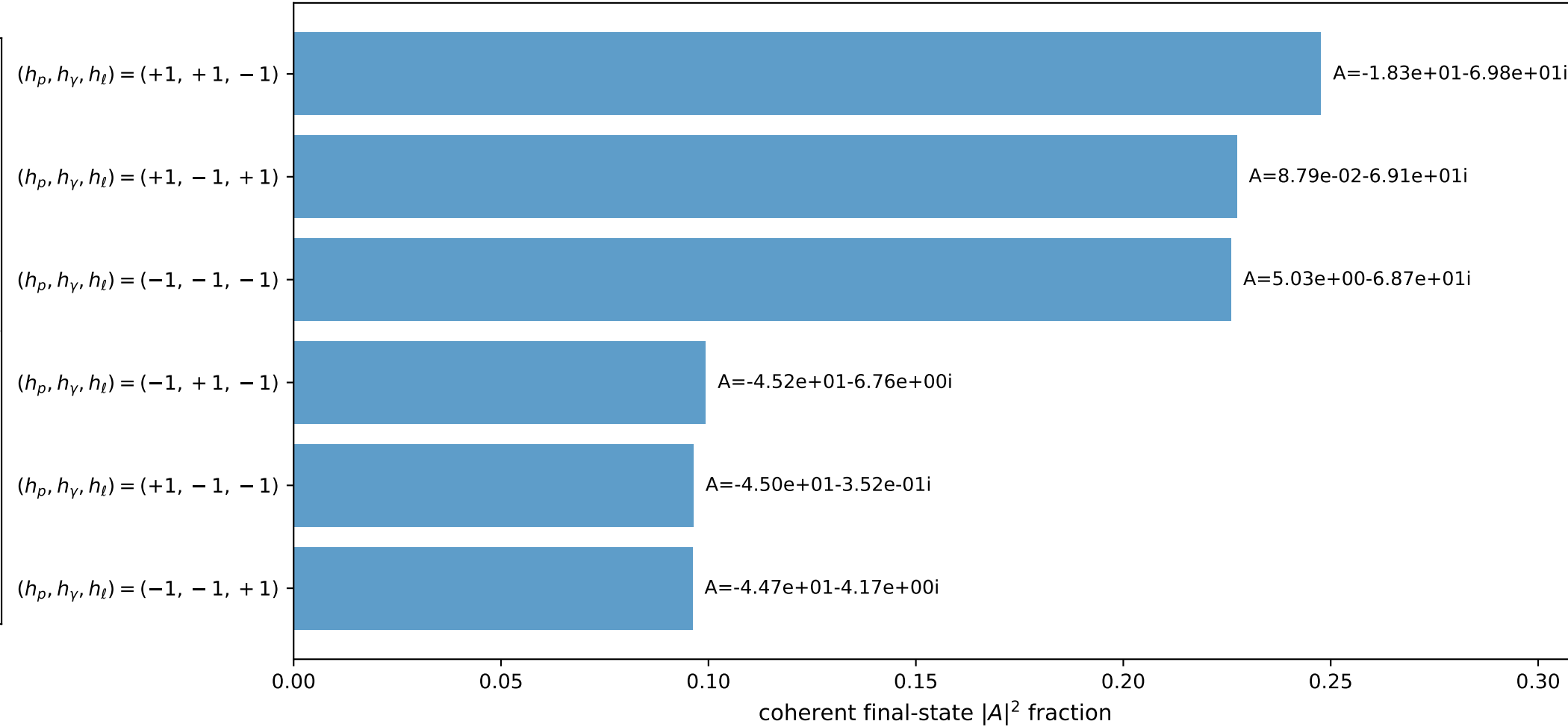
3D momenta



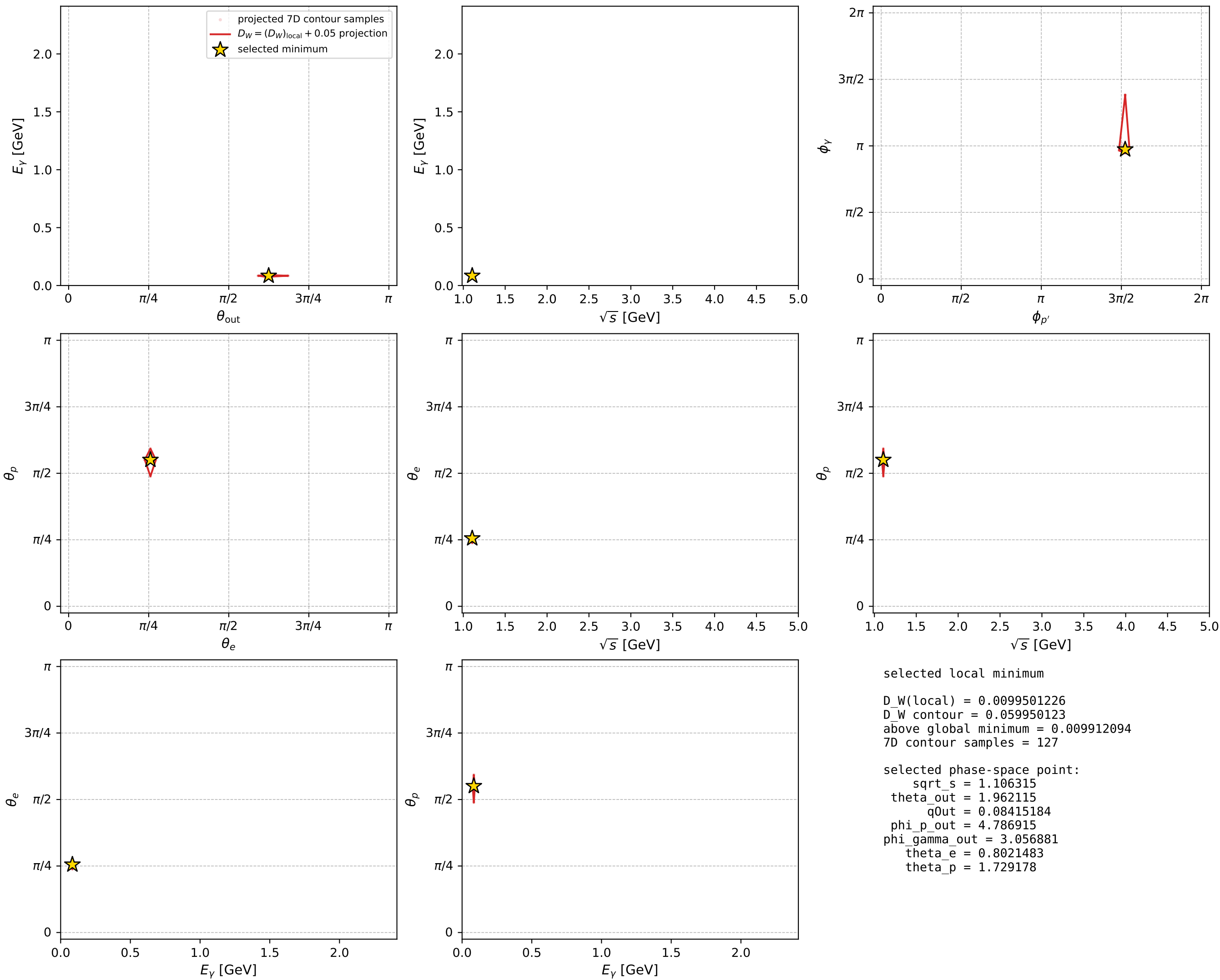
Transverse plane



Leading final-state amplitudes (N=6, retained=99.3%)



electron: polarization cluster P3, local minimum 569; pairwise projections of the 7D $D_W = (D_W)_{\text{local}} + 0.05$ contour

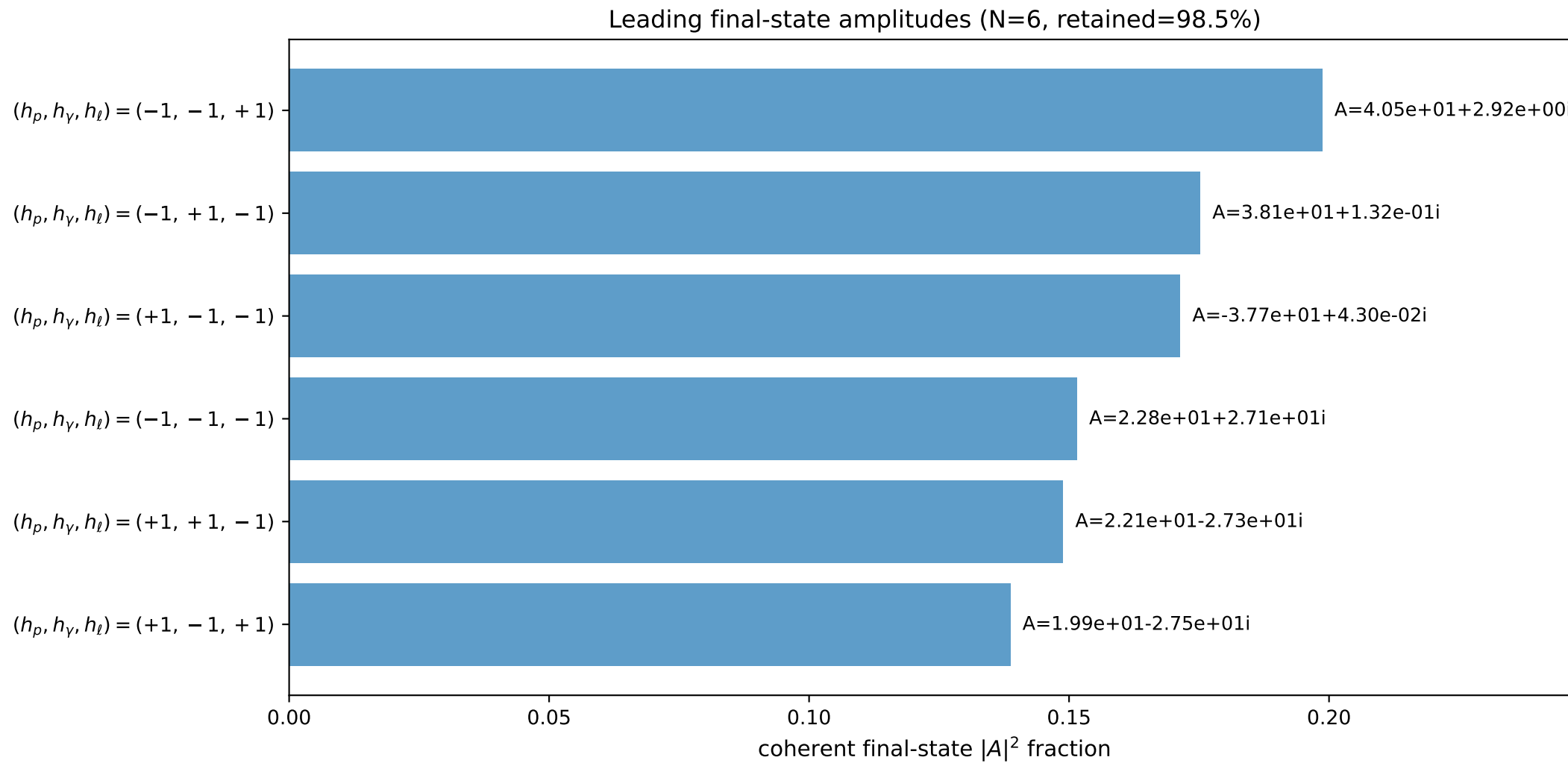
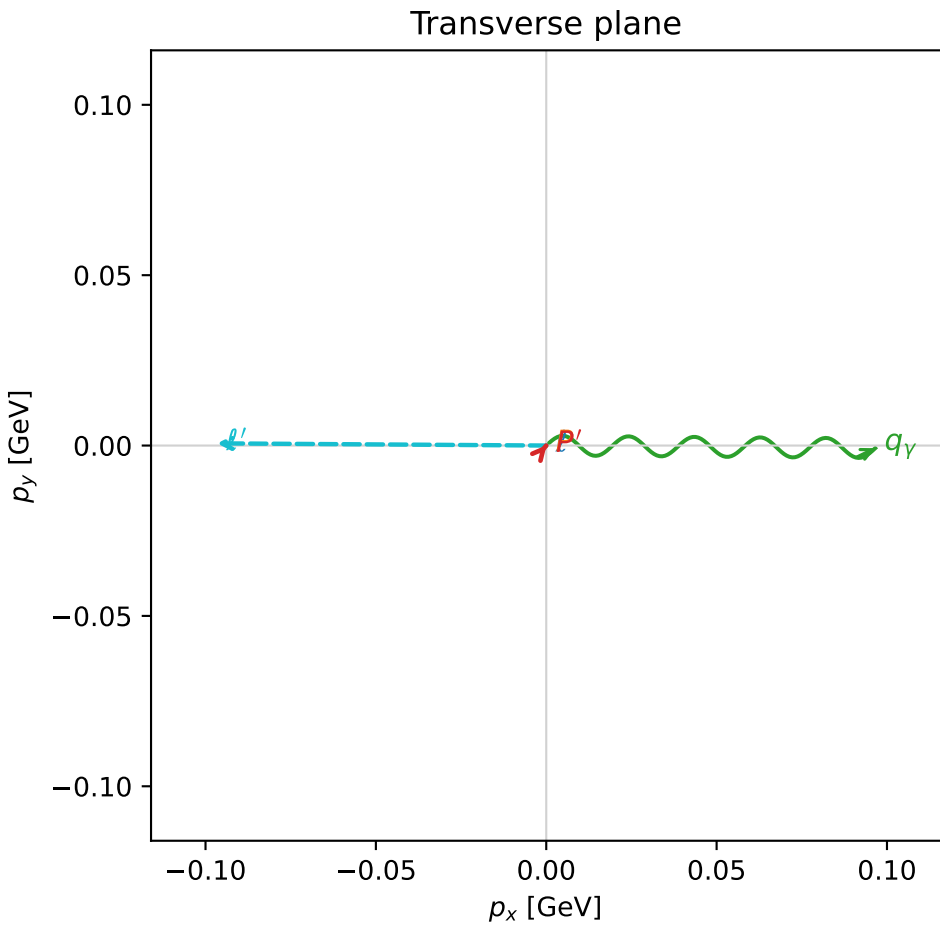
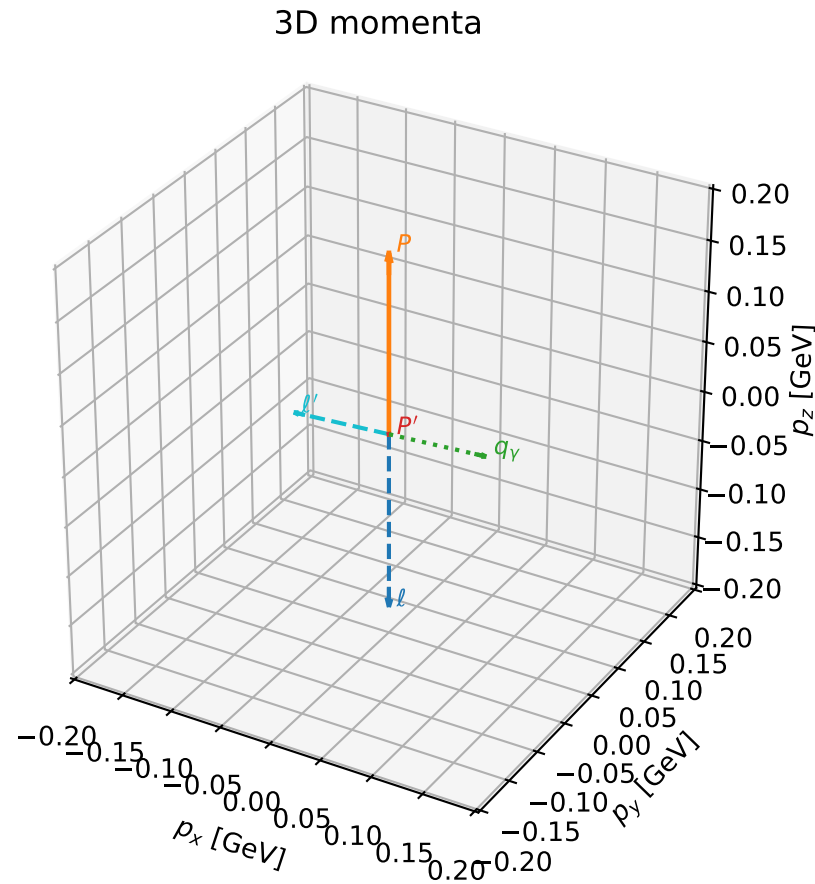


coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

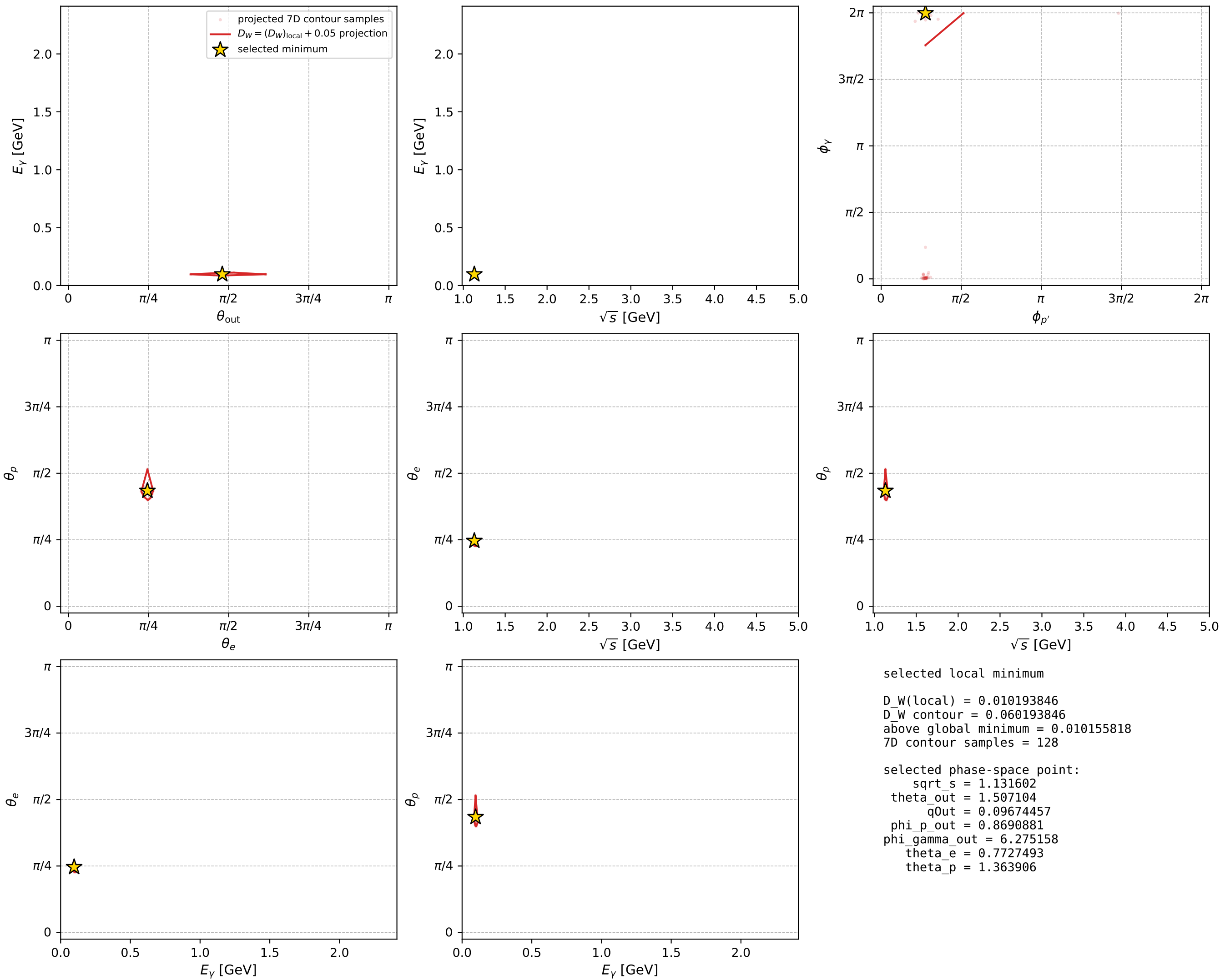
dw_mixing_angles_polarization_03_local_minimum_579 D_W=0.0101938
 region: polarization_cluster_03_local_minimum_579
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\theta_e=0.77274927$ rad
 incoming lepton: $0.715994|+\rangle + 0.698106|-\rangle$
 proton mixing angle: $\theta_p=1.3639061$ rad
 incoming proton: $0.205417|+\rangle + 0.978674|-\rangle$

$s=1.28052$, $\sqrt{s}=1.1316$, $|\vec{P}|=0.17704$, $|\vec{P}'|=0.00017393$
 $\theta_{\text{out}}=1.5071$, $\phi_{P'}=0.869088$, $\phi_Y=6.27516$
 $E_Y=0.0967446$, $\theta_{\ell'}=1.63453$, $\phi_{\ell'}=3.13495$
 $Q^2=0.0321107$, $x_B=0.150344$, $t=-0.031065$, $W^2=1.06132$, $y=0.53305$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1770$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1770$
 P $E= 0.9546$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1770$
 ℓ' $E= 0.0969$ $p_x= -0.0967$ $p_y= 0.0006$ $p_z= -0.0062$
 P' $E= 0.9380$ $p_x= 0.0001$ $p_y= 0.0001$ $p_z= 0.0000$
 q_Y $E= 0.0967$ $p_x= 0.0965$ $p_y= -0.0008$ $p_z= 0.0062$



electron: polarization cluster P3, local minimum 579; pairwise projections of the 7D $D_W = (D_W)_{\text{local}} + 0.05$ contour



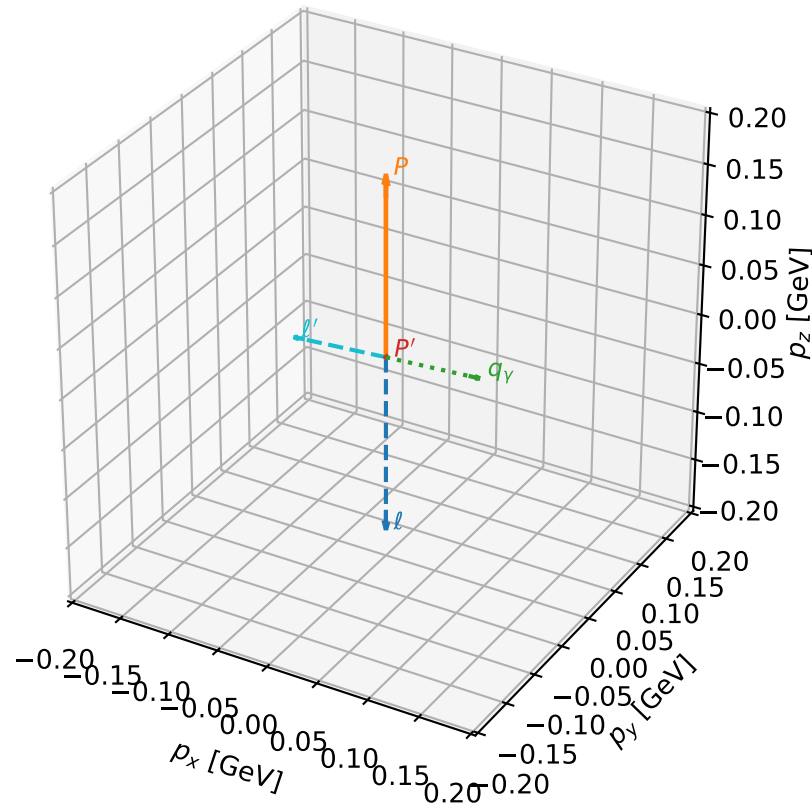
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

dw_mixing_angles_polarization_03_local_minimum_595 D_W=0.0109373
 region: polarization_cluster_03_local_minimum_595
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\theta_e=0.76747828$ rad
 incoming lepton: $0.719664|+\rangle + 0.694323|-\rangle$
 proton mixing angle: $\theta_p=1.3653668$ rad
 incoming proton: $0.203988|+\rangle + 0.978973|-\rangle$

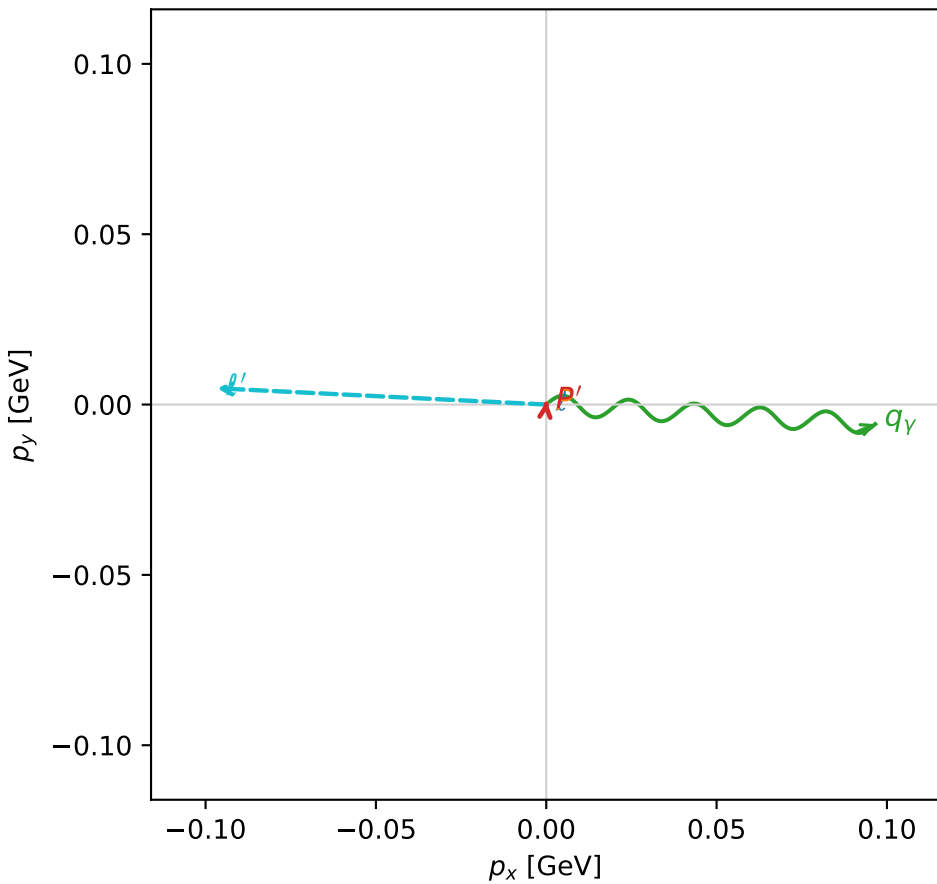
$s=1.28257$, $\sqrt{s}=1.13251$, $|\vec{P}|=0.177802$, $|\vec{P}'|=0.000964474$
 $\theta_{\text{out}}=1.47132$, $\phi_{P'}=1.46555$, $\phi_\gamma=6.22383$
 $E_\gamma=0.0972227$, $\theta_{\ell'}=1.67121$, $\phi_{\ell'}=3.09214$
 $Q^2=0.0311258$, $x_B=0.145785$, $t=-0.0313013$, $W^2=1.06222$, $y=0.530153$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1778$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1778$
 P $E= 0.9547$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1778$
 ℓ' $E= 0.0973$ $p_x= -0.0967$ $p_y= 0.0048$ $p_z= -0.0098$
 P' $E= 0.9380$ $p_x= 0.0001$ $p_y= 0.0010$ $p_z= 0.0001$
 q_γ $E= 0.0972$ $p_x= 0.0966$ $p_y= -0.0057$ $p_z= 0.0097$

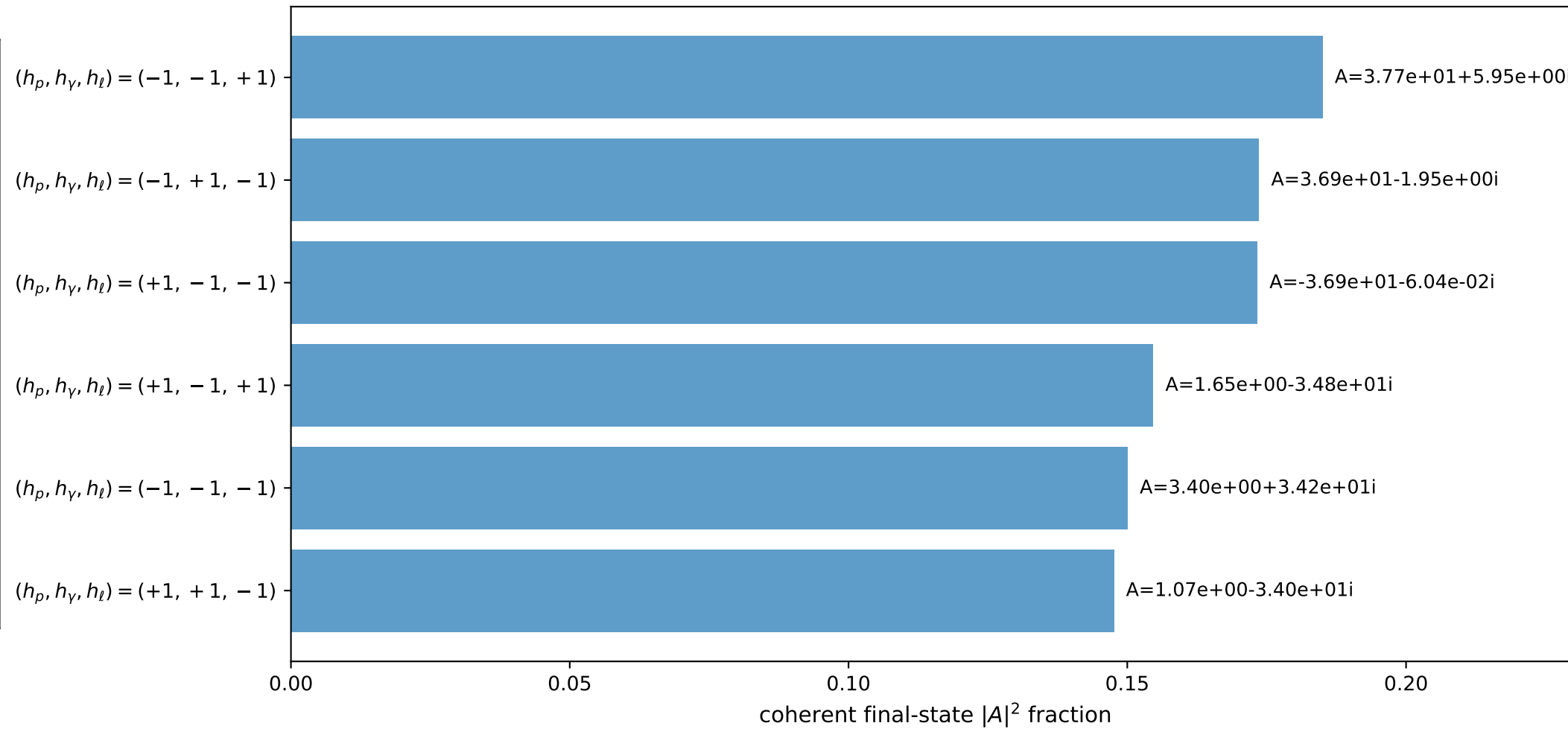
3D momenta



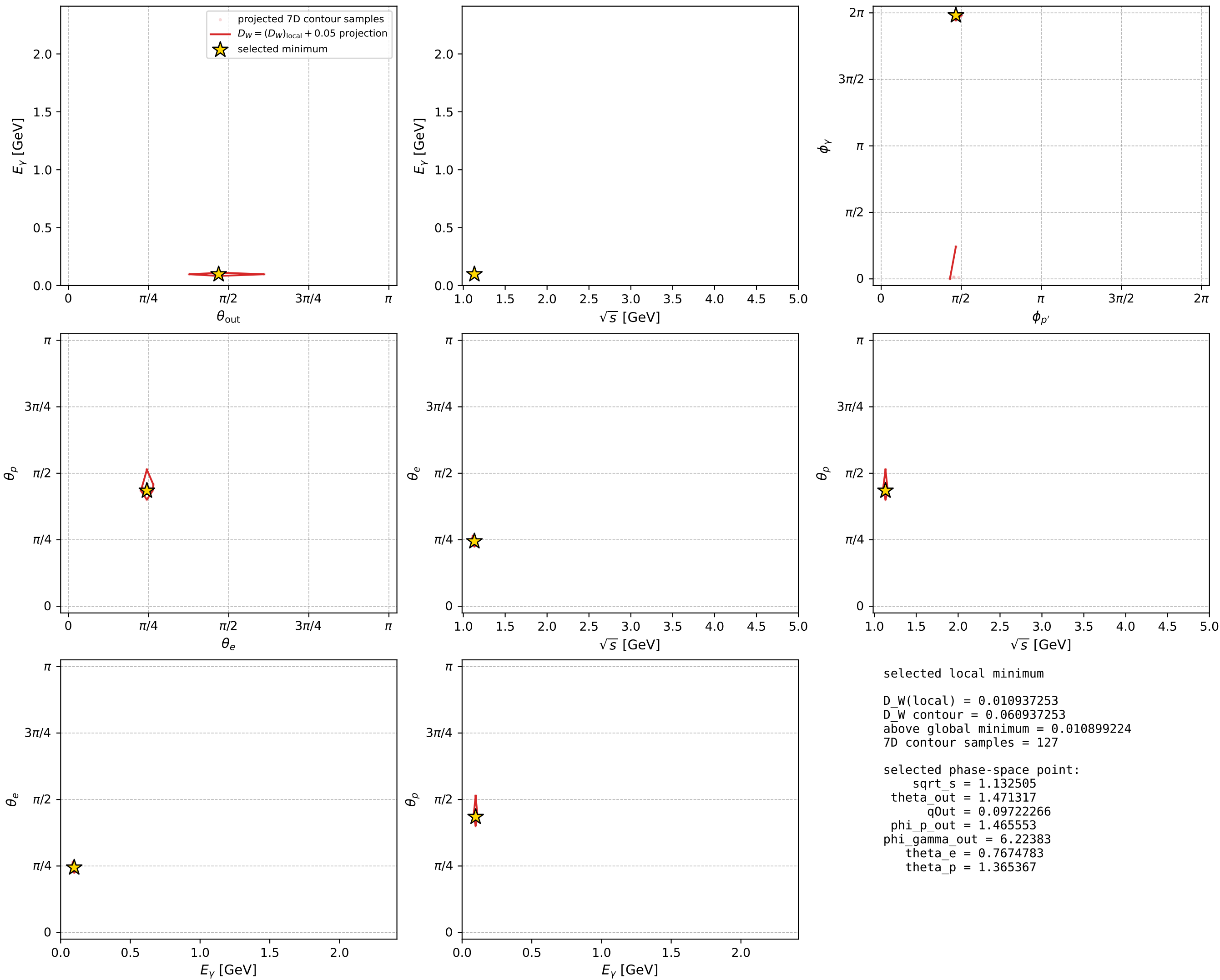
Transverse plane



Leading final-state amplitudes (N=6, retained=98.4%)



electron: polarization cluster P3, local minimum 595; pairwise projections of the 7D $D_W = (D_W)_{\text{local}} + 0.05$ contour



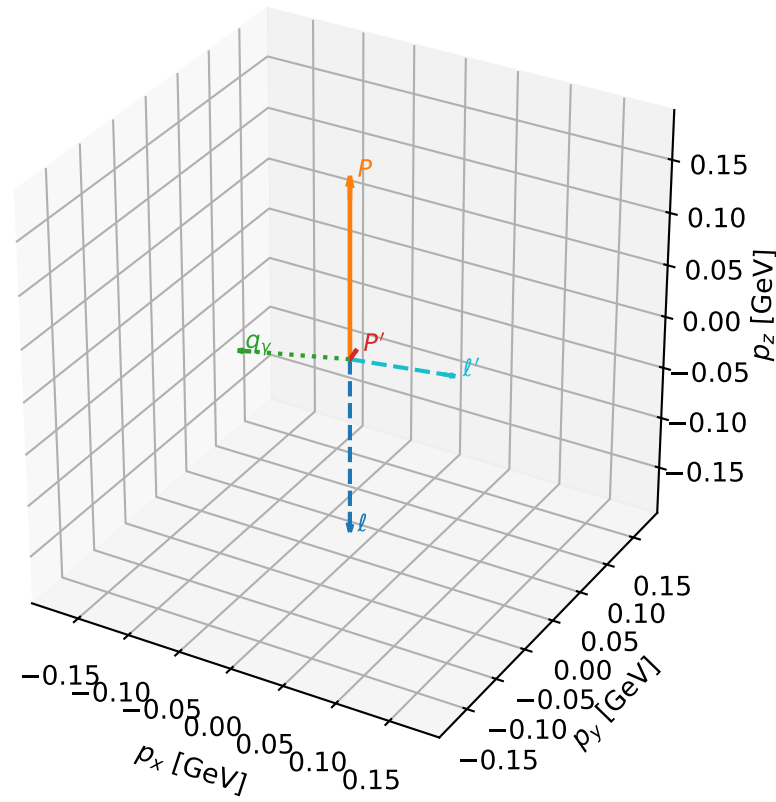
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

dw_mixing_angles_polarization_03_local_minimum_596 D_W=0.0109559
 region: polarization_cluster_03_local_minimum_596
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\theta_e=0.75613588$ rad
 incoming lepton: $0.727493|+\rangle + 0.686115|-\rangle$
 proton mixing angle: $\theta_p=1.808871$ rad
 incoming proton: $-0.235832|+\rangle + 0.971794|-\rangle$

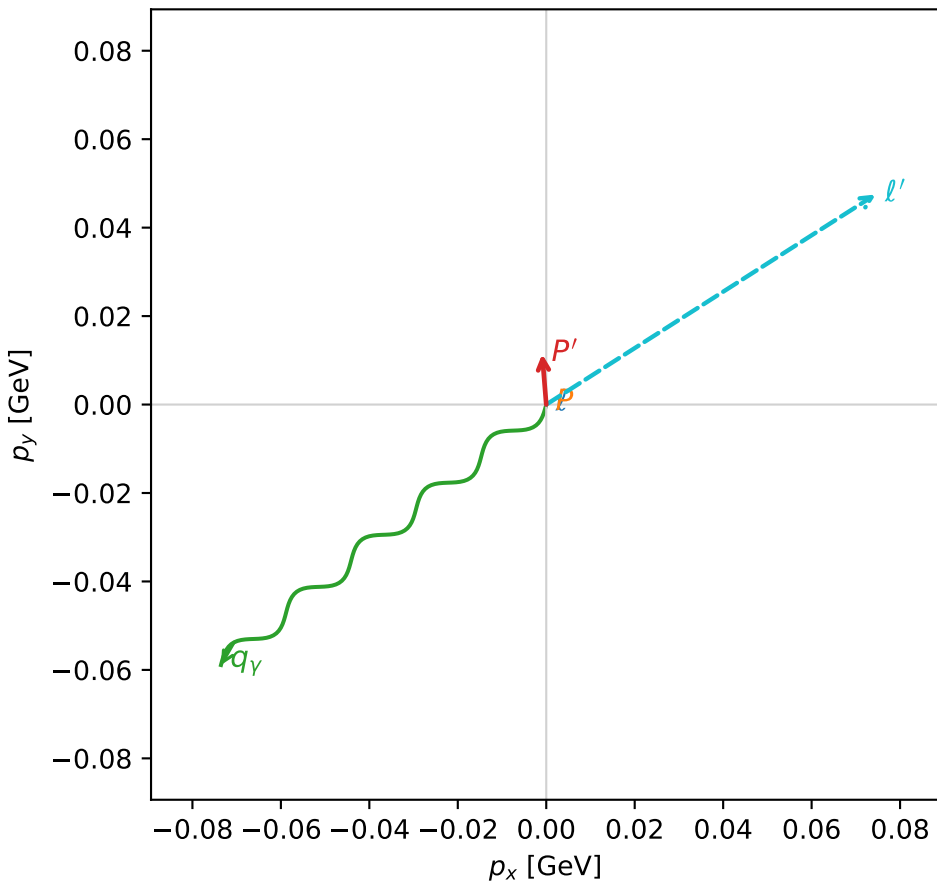
$s=1.26344$, $\sqrt{s}=1.12403$, $|\vec{P}|=0.170634$, $|\vec{P}'|=0.0116126$
 $\theta_{\text{out}}=1.3959$, $\phi_{P'}=1.65119$, $\phi_\gamma=3.8168$
 $E_\gamma=0.0956812$, $\theta_{\ell'}=1.77911$, $\phi_{\ell'}=0.567764$
 $Q^2=0.0244367$, $x_B=0.119152$, $t=-0.0283266$, $W^2=1.0605$, $y=0.534646$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1706$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1706$
 P $E= 0.9534$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1706$
 ℓ' $E= 0.0903$ $p_x= 0.0745$ $p_y= 0.0475$ $p_z= -0.0187$
 P' $E= 0.9381$ $p_x= -0.0009$ $p_y= 0.0114$ $p_z= 0.0020$
 q_γ $E= 0.0957$ $p_x= -0.0735$ $p_y= -0.0589$ $p_z= 0.0166$

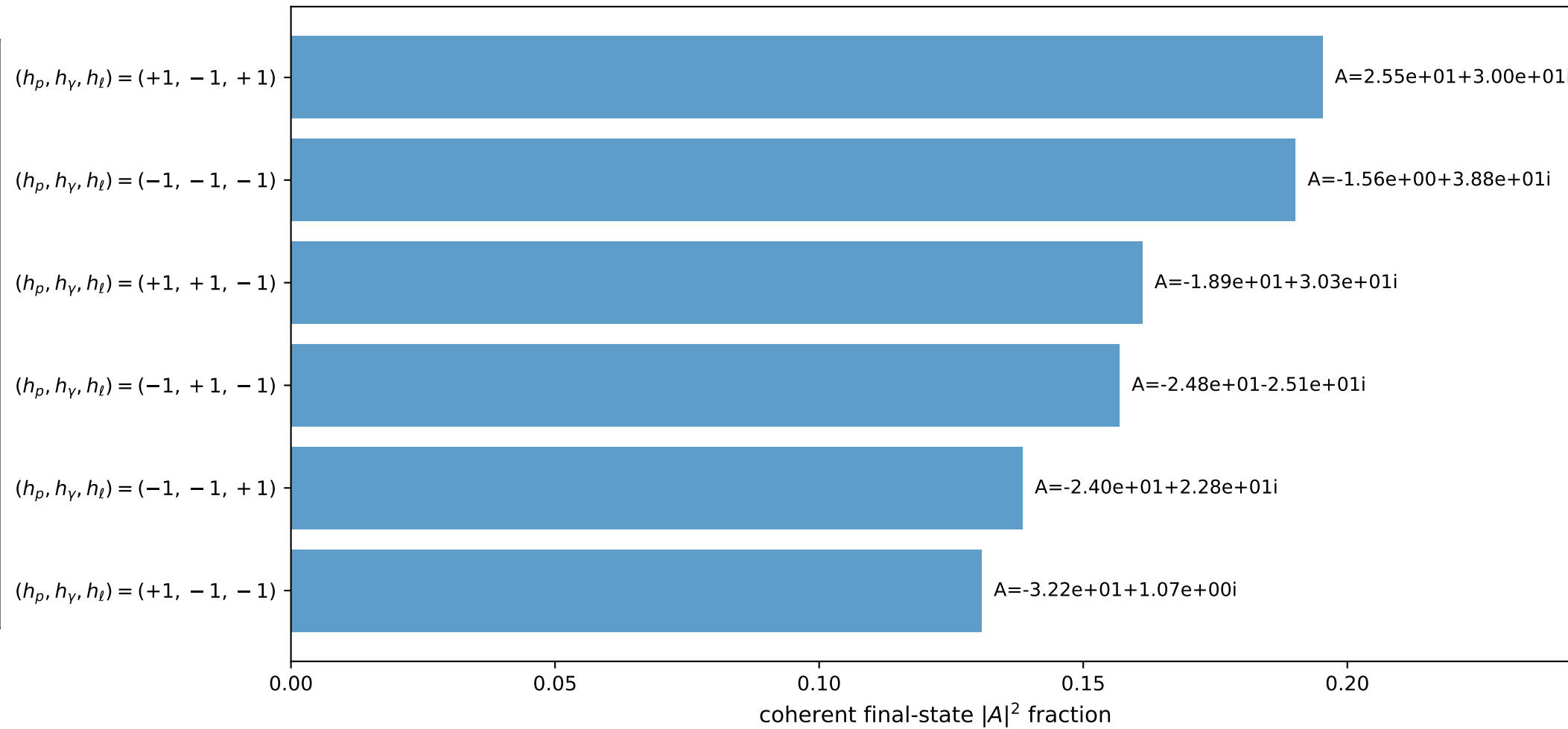
3D momenta



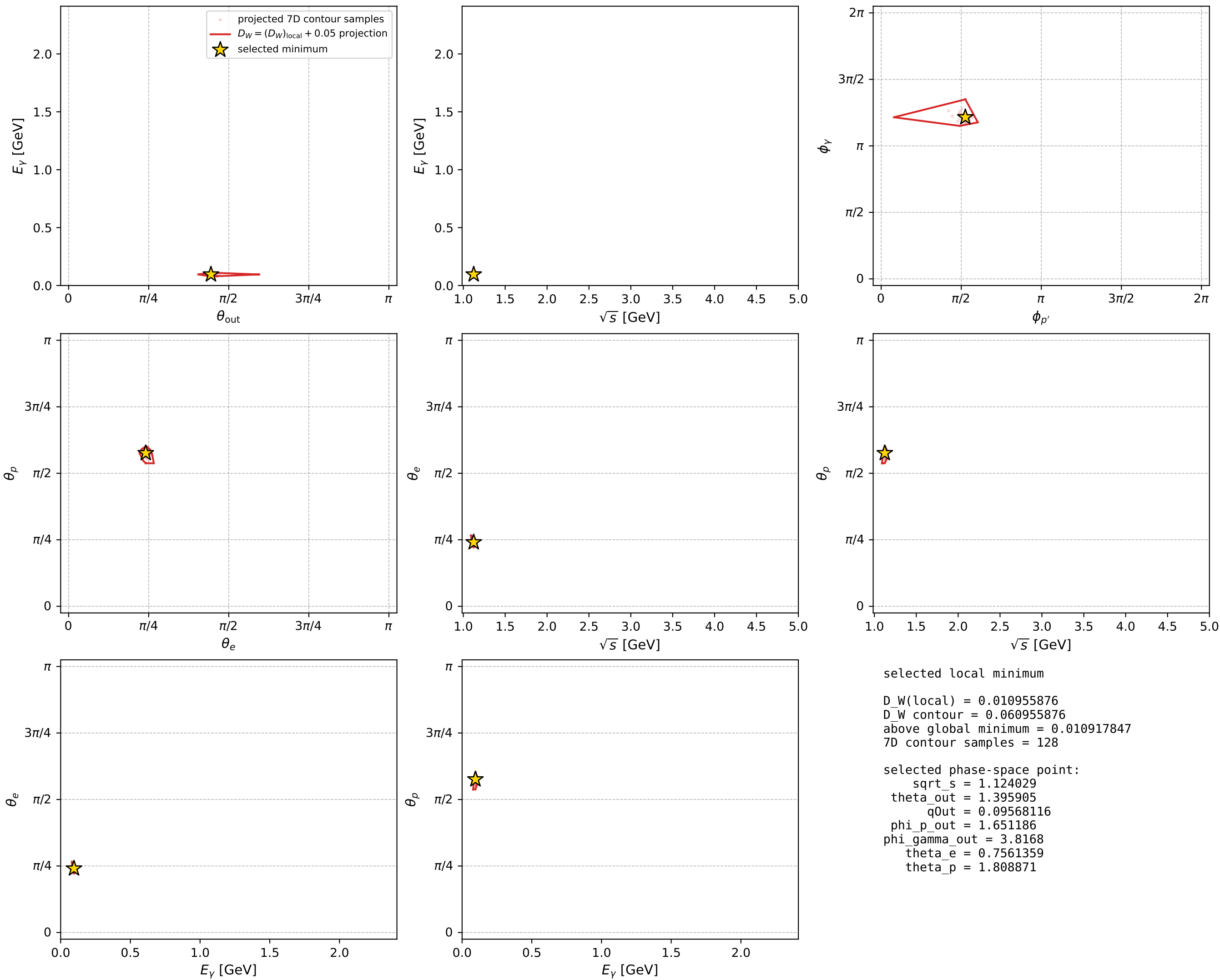
Transverse plane



Leading final-state amplitudes (N=6, retained=97.3%)



electron: polarization cluster P3, local minimum 596; pairwise projections of the 7D $D_W = (D_W)_{\text{local}} + 0.05$ contour

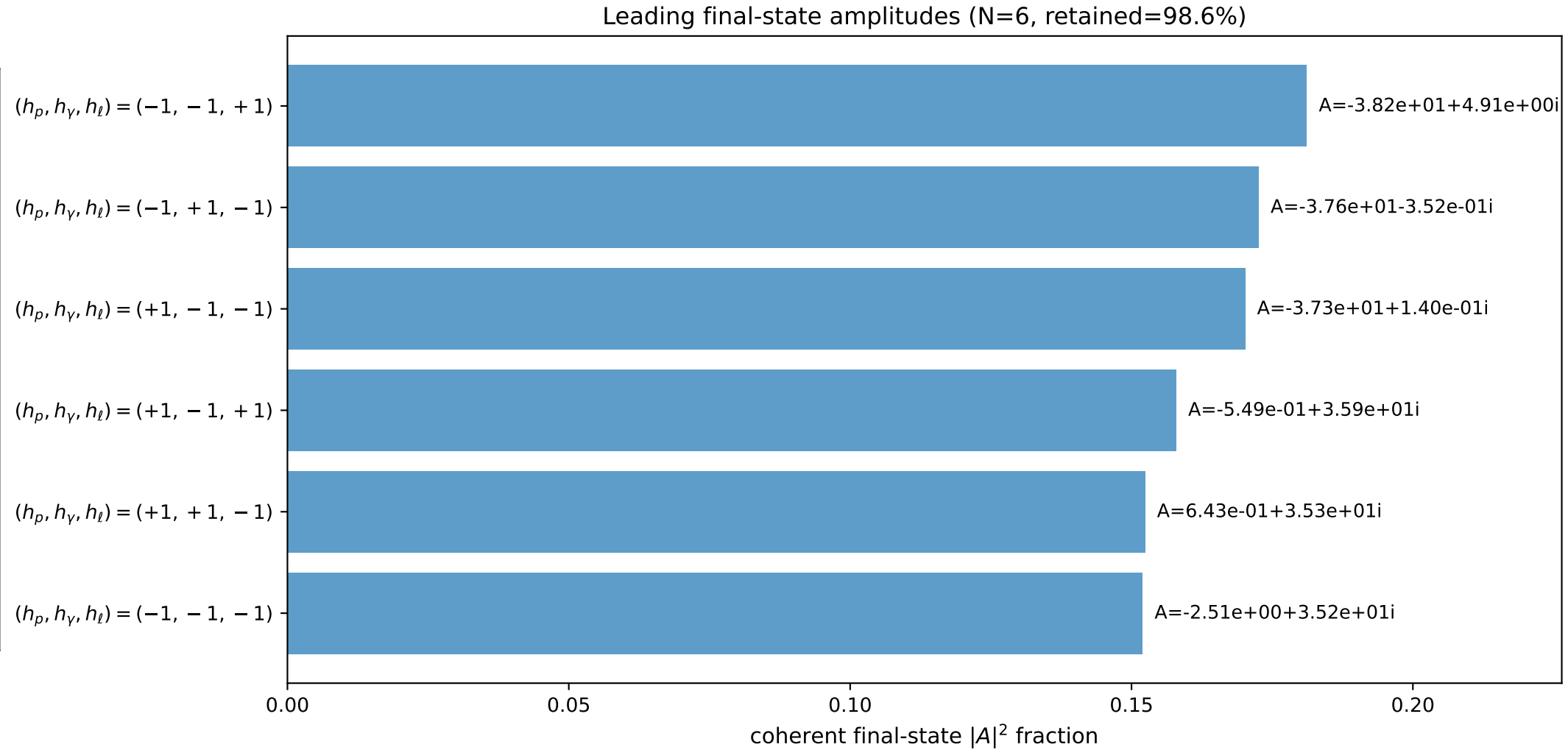
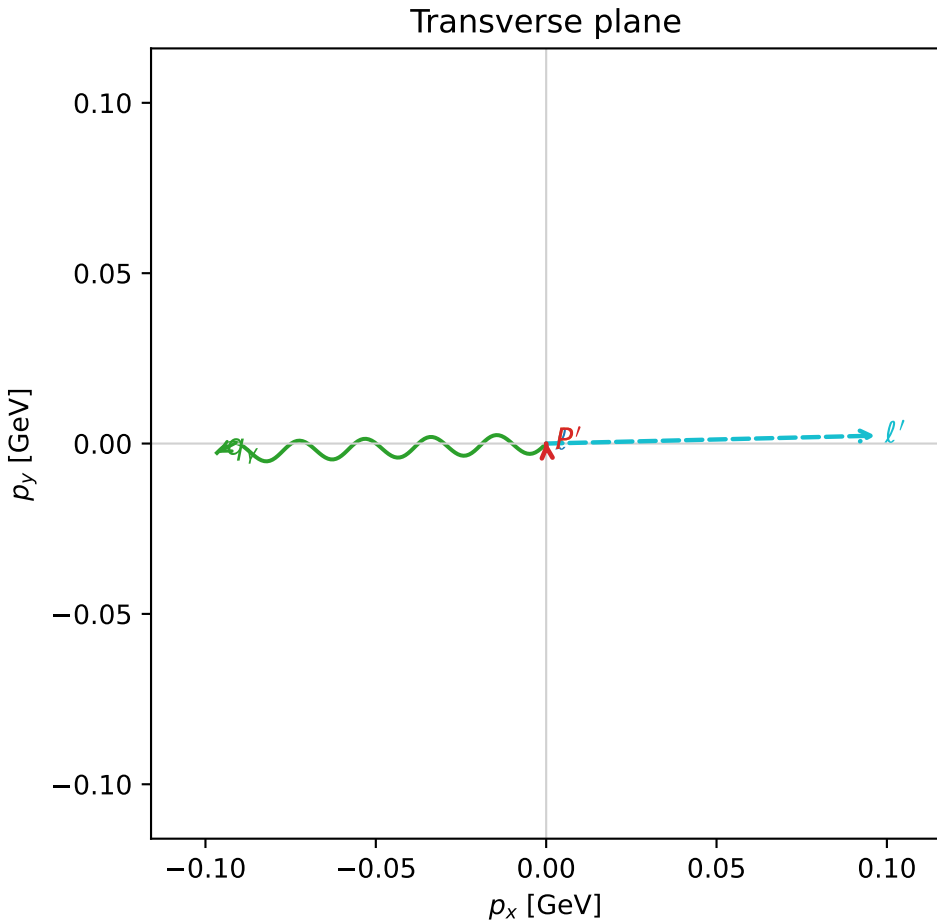
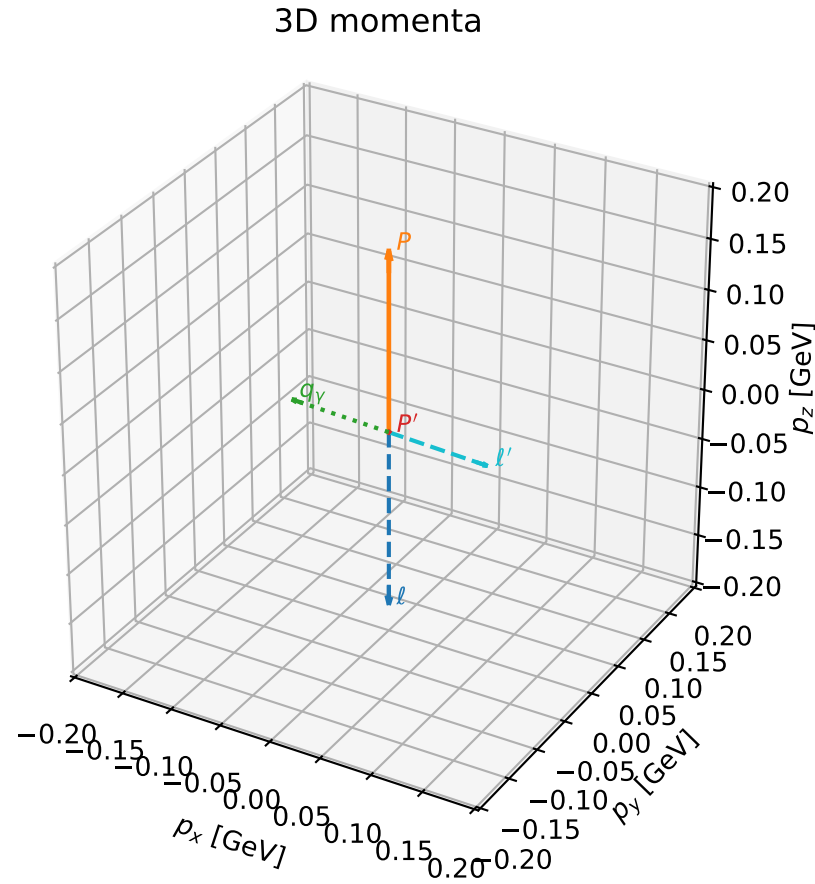


coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

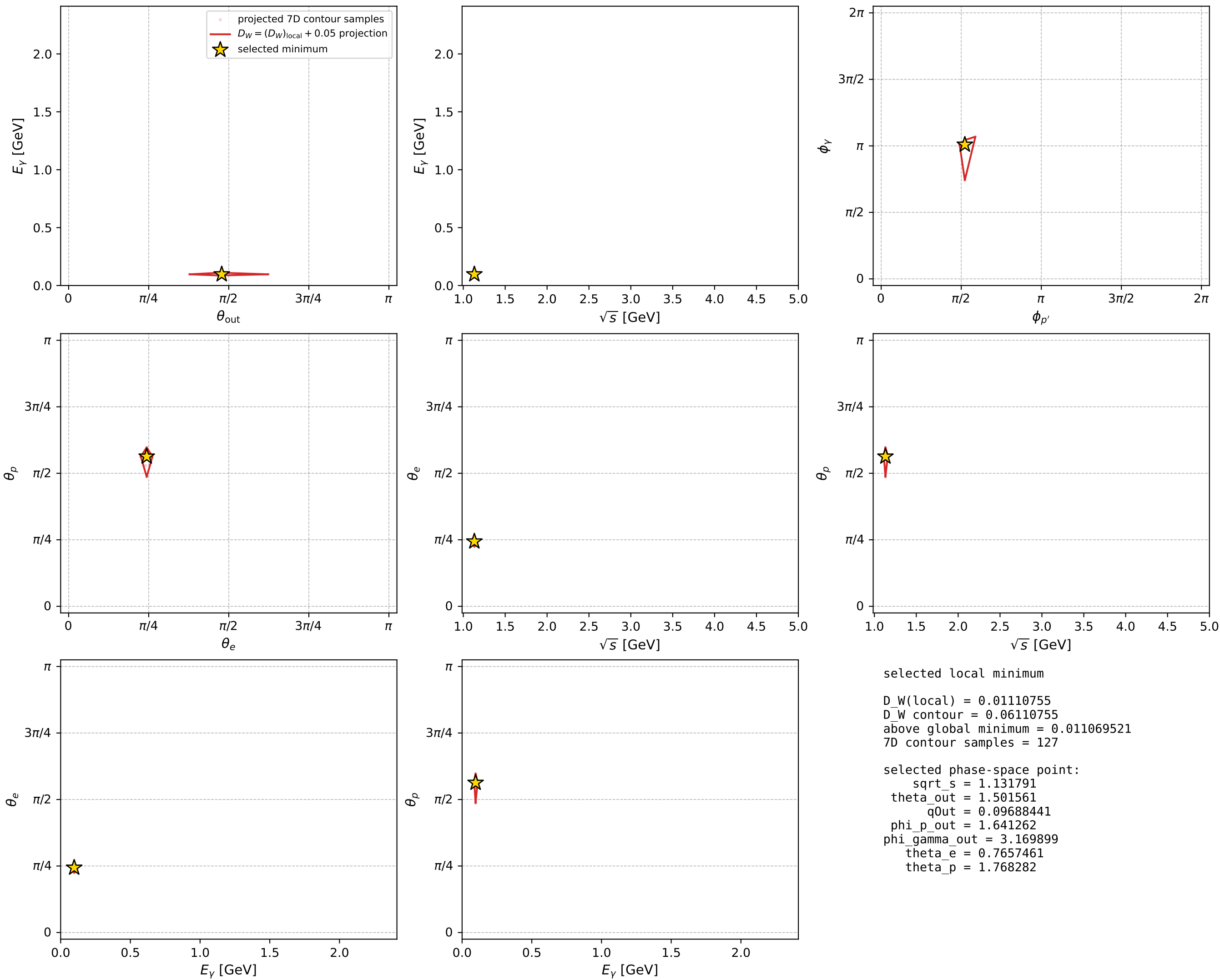
dw_mixing_angles_polarization_03_local_minimum_599 D_W=0.0111075
region: polarization_cluster_03_local_minimum_599
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\theta_e=0.76574606$ rad
incoming lepton: $0.720865|+\rangle + 0.693075|-\rangle$
proton mixing angle: $\theta_p=1.7682818$ rad
incoming proton: $-0.196204|+\rangle + 0.980563|-\rangle$

$s=1.28095$, $\sqrt{s}=1.13179$, $|\vec{P}|=0.177199$, $|\vec{P}'|=0.000422297$
 $\theta_{\text{out}}=1.50156$, $\phi_{P'}=1.64126$, $\phi_\gamma=3.1699$
 $E_\gamma=0.0968844$, $\theta_{\ell'}=1.64032$, $\phi_{\ell'}=0.0239524$
 $Q^2=0.0319574$, $x_B=0.149537$, $t=-0.0311142$, $W^2=1.0616$, $y=0.532798$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1772$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1772$
 P $E= 0.9546$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1772$
 ℓ' $E= 0.0969$ $p_x= 0.0966$ $p_y= 0.0023$ $p_z= -0.0067$
 P' $E= 0.9380$ $p_x= -0.0000$ $p_y= 0.0004$ $p_z= 0.0000$
 q_γ $E= 0.0969$ $p_x= -0.0966$ $p_y= -0.0027$ $p_z= 0.0067$



electron: polarization cluster P3, local minimum 599; pairwise projections of the 7D $D_W = (D_W)_{\text{local}} + 0.05$ contour

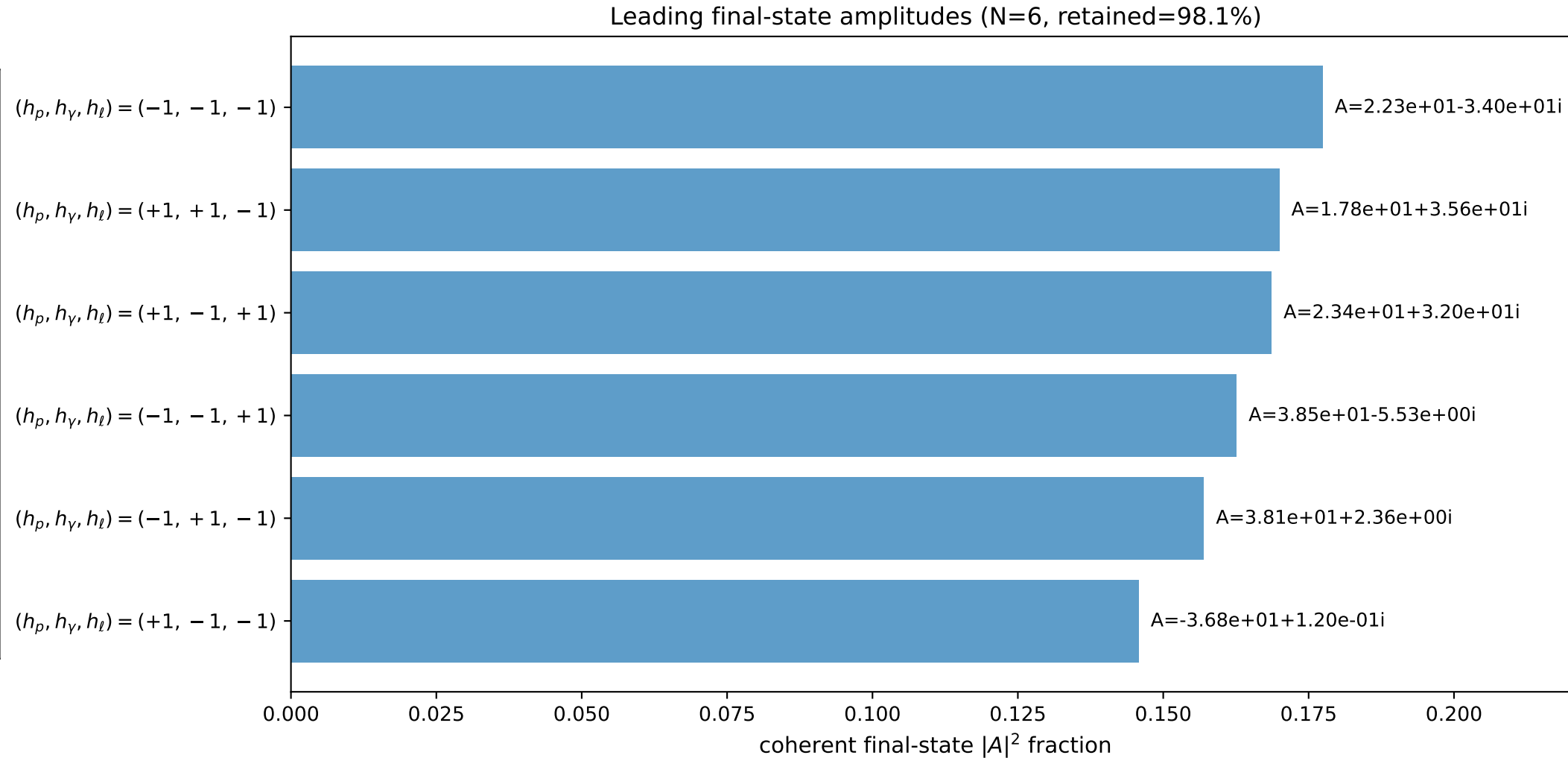
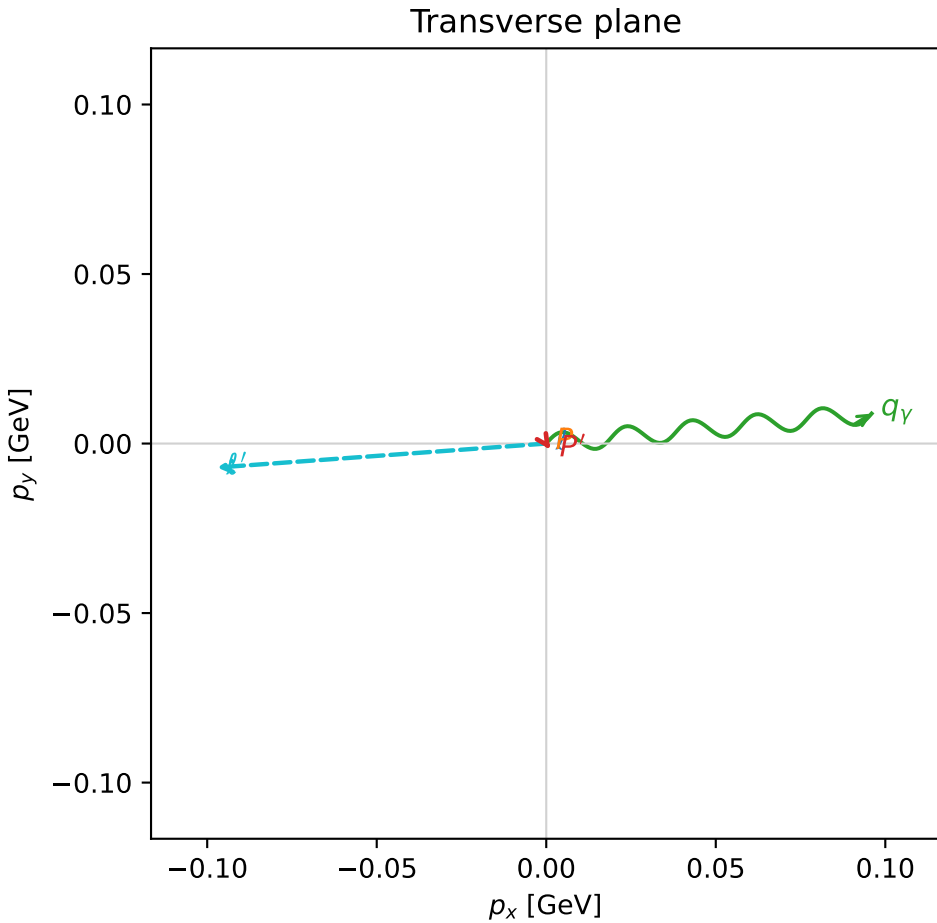
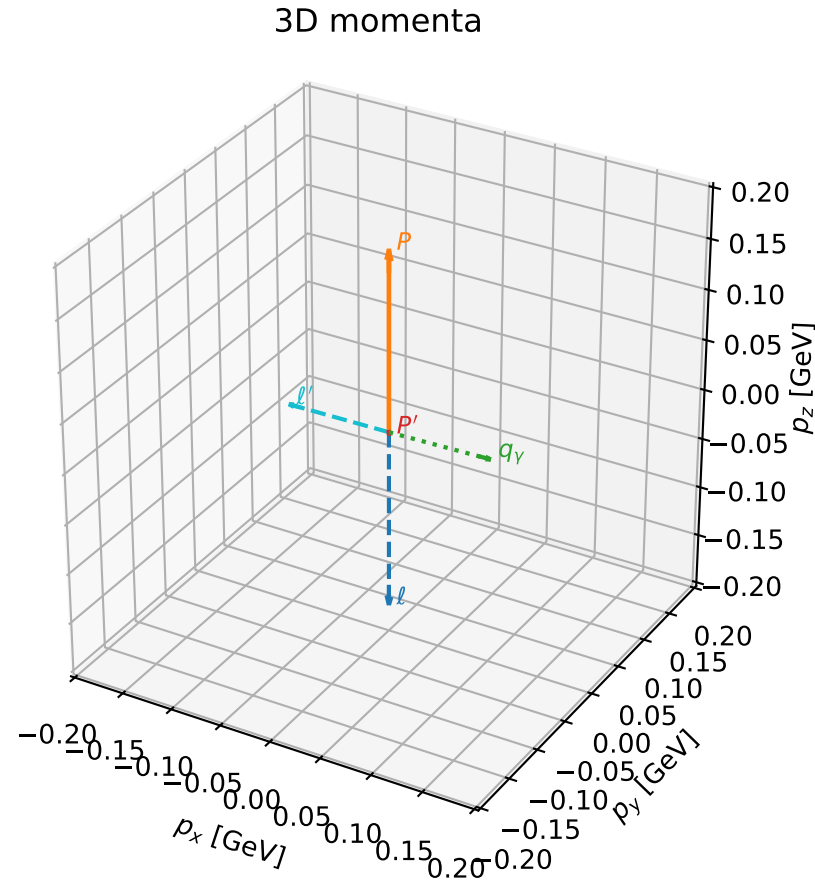


coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

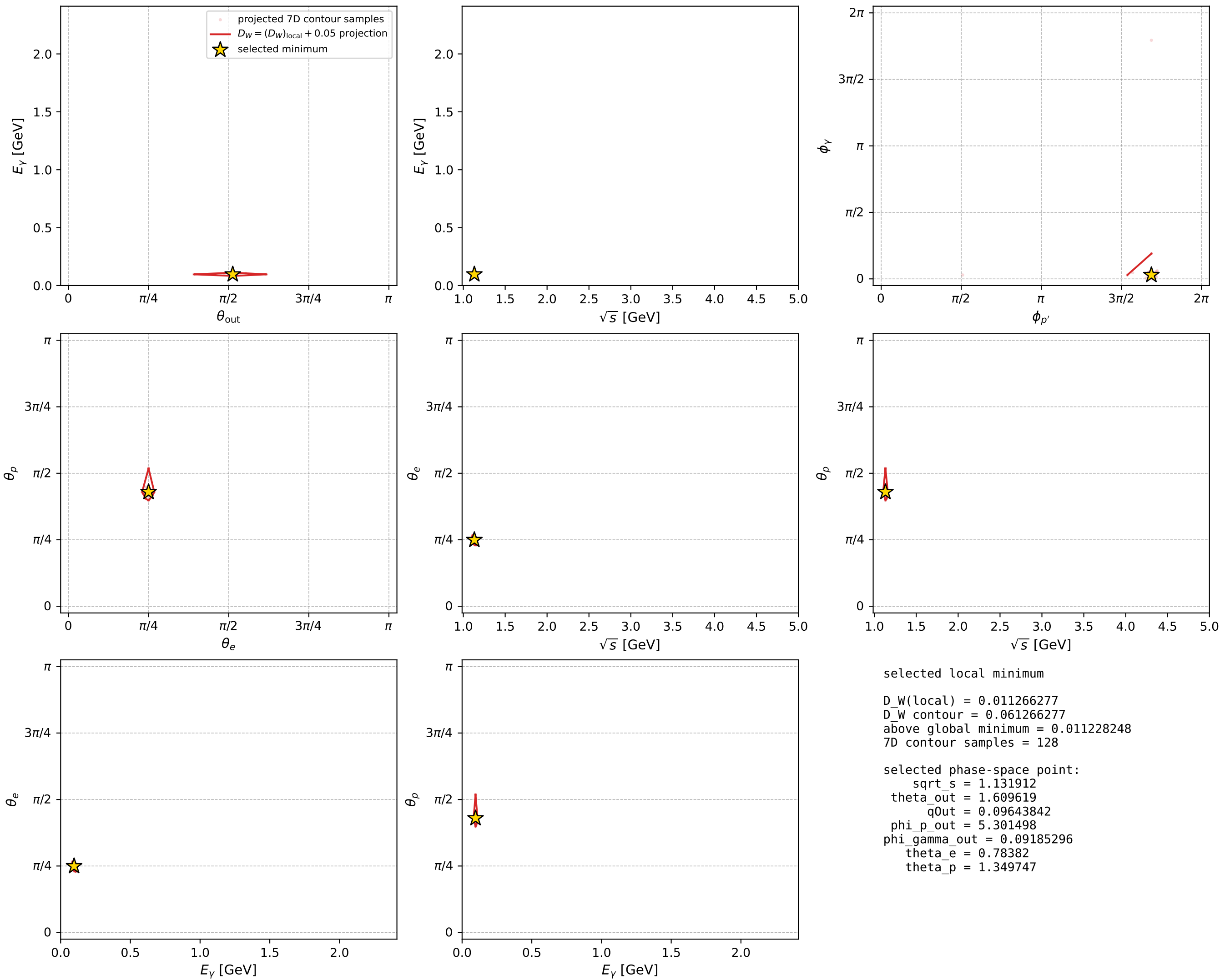
dw_mixing_angles_polarization_03_local_minimum_601 D_W=0.0112663
 region: polarization_cluster_03_local_minimum_601
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\theta_e=0.78382004$ rad
 incoming lepton: $0.708222|+\rangle + 0.70599|-\rangle$
 proton mixing angle: $\theta_p=1.3497474$ rad
 incoming proton: $0.219253|+\rangle + 0.975668|-\rangle$

$s=1.28122$, $\sqrt{s}=1.13191$, $|\vec{P}|=0.177301$, $|\vec{P}'|=0.00212054$
 $\theta_{\text{out}}=1.60962$, $\phi_{P'}=5.3015$, $\phi_\gamma=0.091853$
 $E_\gamma=0.0964384$, $\theta_{\ell'}=1.53154$, $\phi_{\ell'}=3.21432$
 $Q^2=0.0359195$, $x_B=0.1658$, $t=-0.0311936$, $W^2=1.06057$, $y=0.539746$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1773$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1773$
 P $E= 0.9546$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1773$
 ℓ' $E= 0.0975$ $p_x= -0.0971$ $p_y= -0.0071$ $p_z= 0.0038$
 P' $E= 0.9380$ $p_x= 0.0012$ $p_y= -0.0018$ $p_z= -0.0001$
 q_γ $E= 0.0964$ $p_x= 0.0960$ $p_y= 0.0088$ $p_z= -0.0037$



electron: polarization cluster P3, local minimum 601; pairwise projections of the 7D $D_W = (D_W)_{\text{local}} + 0.05$ contour

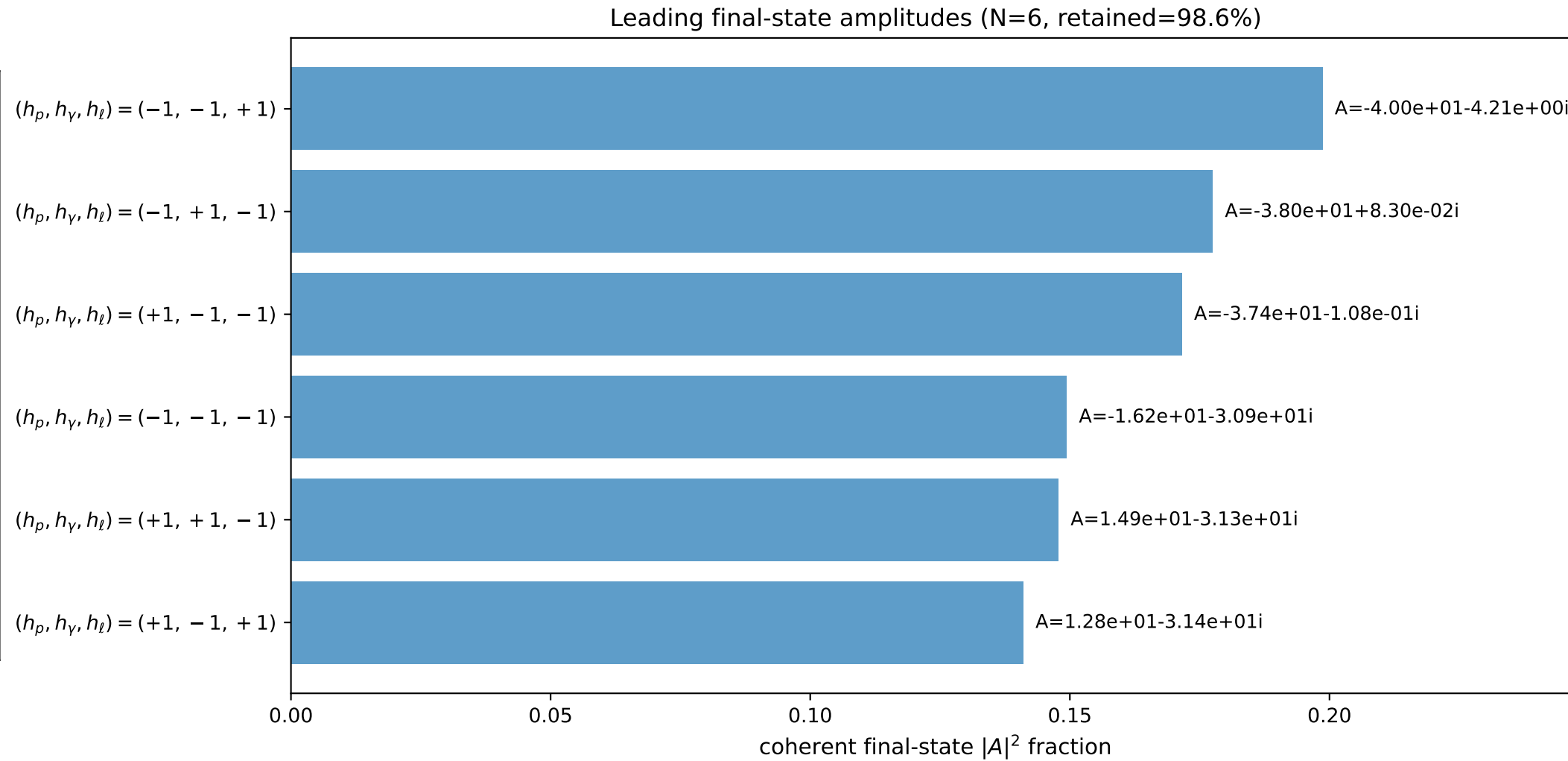
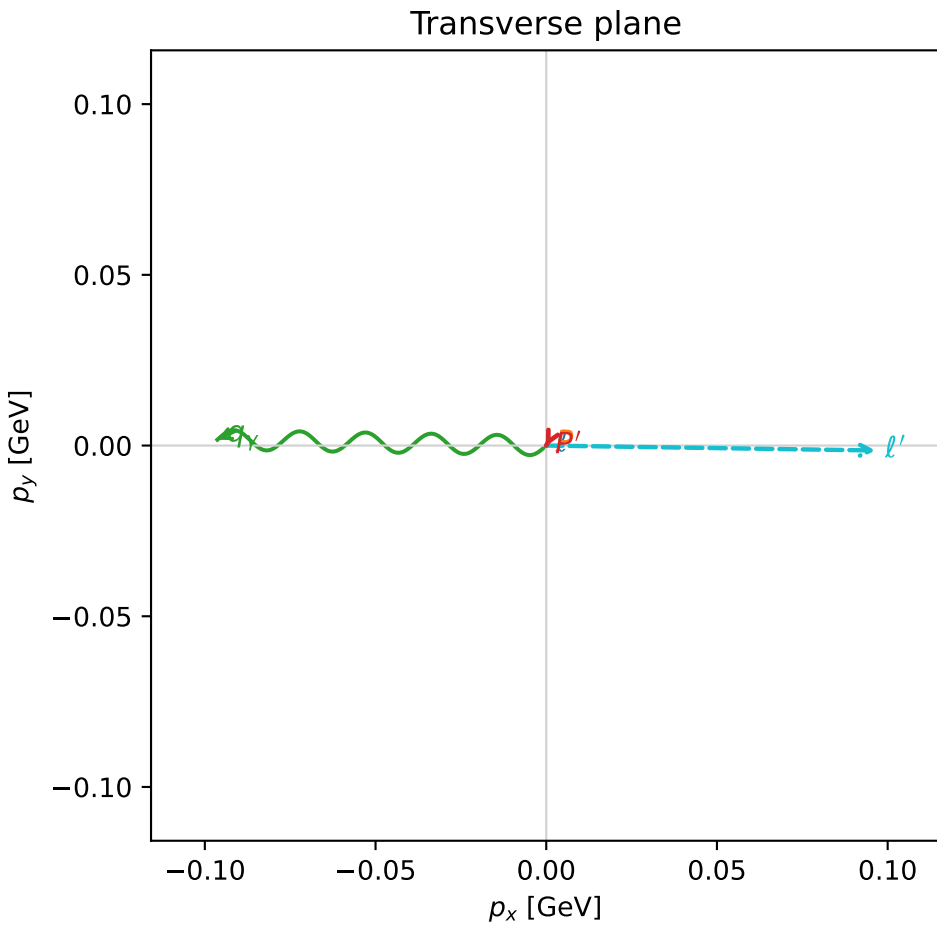
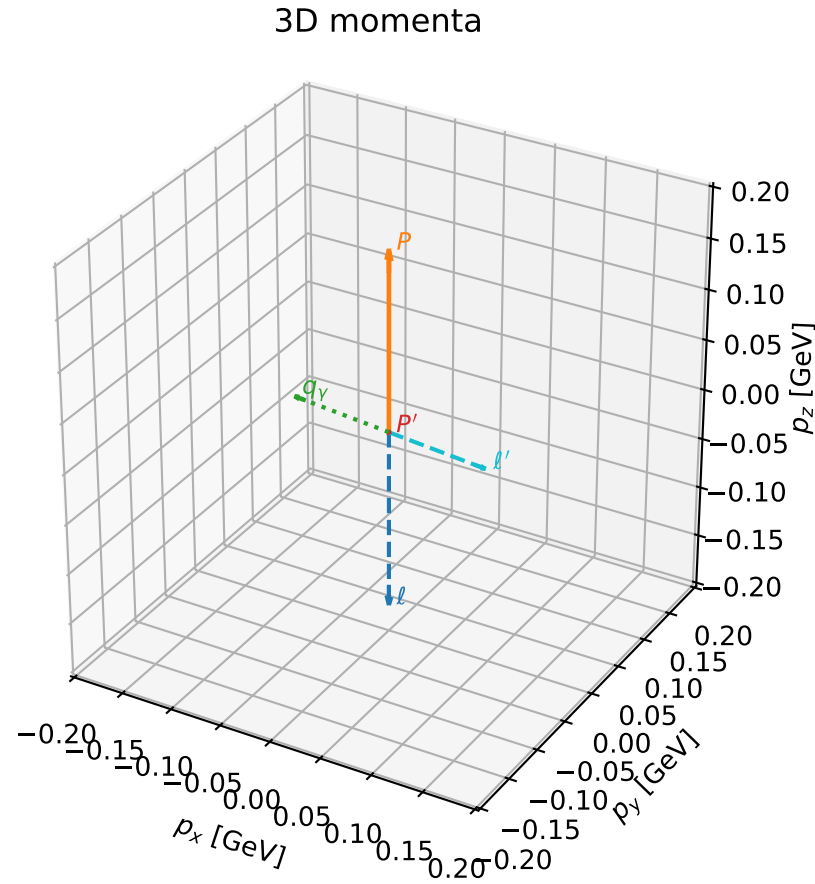


coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

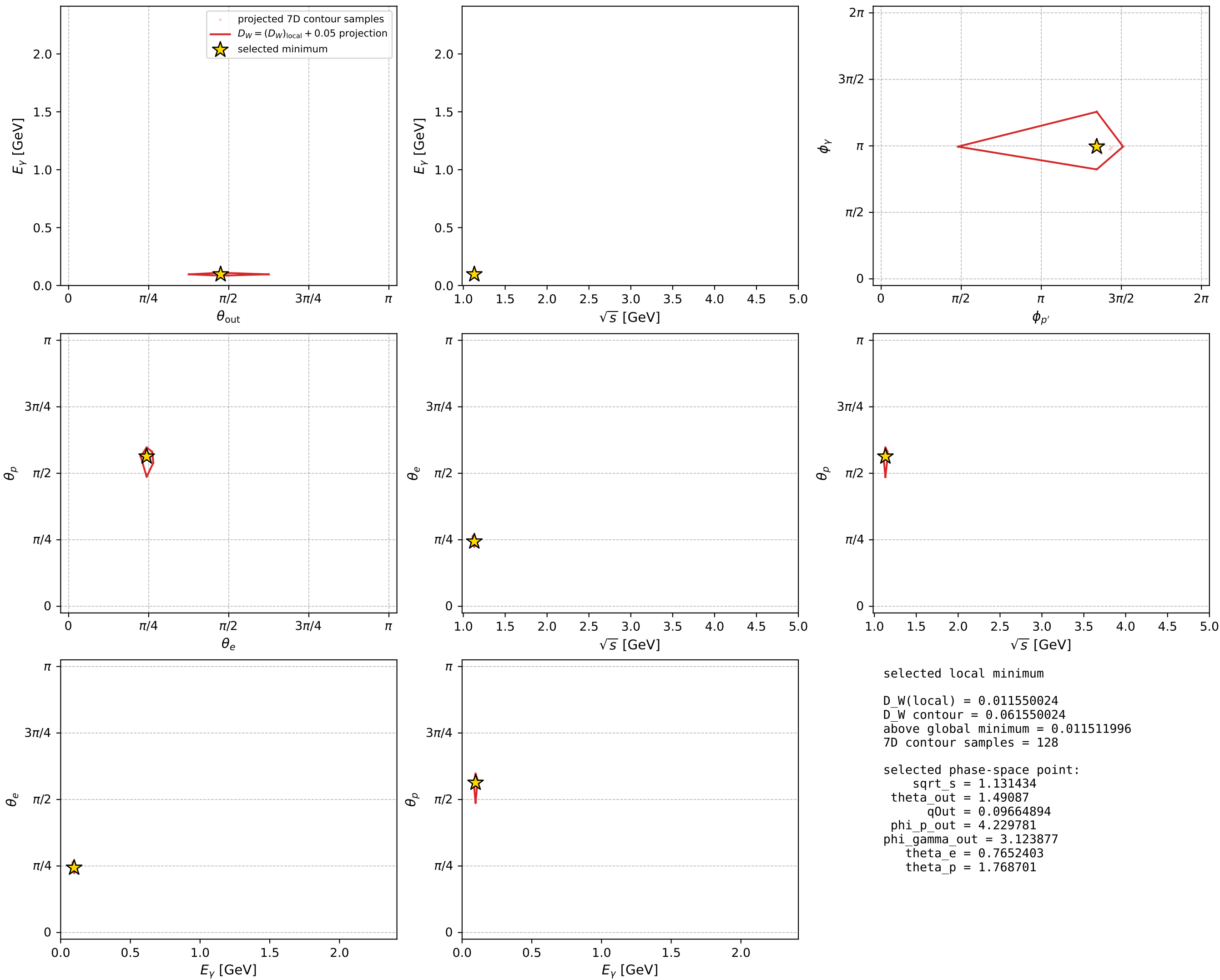
dw_mixing_angles_polarization_03_local_minimum_609 D_W=0.01155
 region: polarization_cluster_03_local_minimum_609
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\theta_e=0.76524026$ rad
 incoming lepton: $0.721216|+\rangle + 0.69271|-\rangle$
 proton mixing angle: $\theta_p=1.7687007$ rad
 incoming proton: $-0.196615|+\rangle + 0.980481|-\rangle$

$s=1.28014$, $\sqrt{s}=1.13143$, $|\vec{P}|=0.176898$, $|\vec{P}'|=0.000296751$
 $\theta_{\text{out}}=1.49087$, $\phi_{p'}=4.22978$, $\phi_\gamma=3.12388$
 $E_\gamma=0.0966489$, $\theta_\ell=1.65086$, $\phi_{\ell'}=6.26821$
 $Q^2=0.0315032$, $x_B=0.148048$, $t=-0.0310112$, $W^2=1.06113$, $y=0.531581$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1769$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1769$
 P $E= 0.9545$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1769$
 ℓ' $E= 0.0968$ $p_x= 0.0965$ $p_y= -0.0014$ $p_z= -0.0077$
 P' $E= 0.9380$ $p_x= -0.0001$ $p_y= -0.0003$ $p_z= 0.0000$
 q_γ $E= 0.0966$ $p_x= -0.0963$ $p_y= 0.0017$ $p_z= 0.0077$



electron: polarization cluster P3, local minimum 609; pairwise projections of the 7D $D_W = (D_W)_{\text{local}} + 0.05$ contour



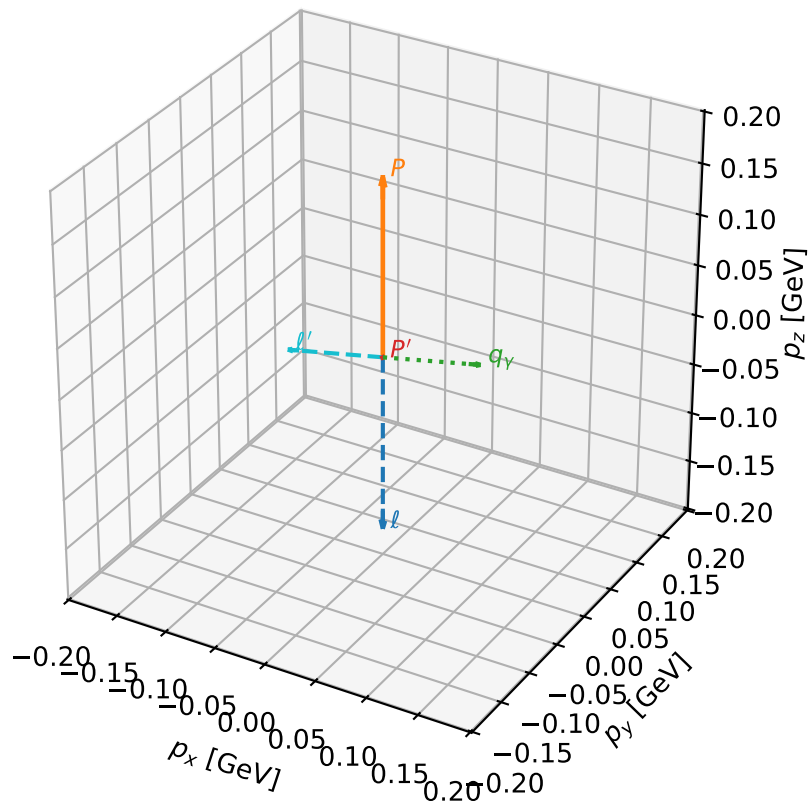
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

dw_mixing_angles_polarization_03_local_minimum_613 D_W=0.0117038
 region: polarization_cluster_03_local_minimum_613
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\theta_e=0.76087558$ rad
 incoming lepton: $0.724233|+\rangle +0.689556|-\rangle$
 proton mixing angle: $\theta_p=1.3753714$ rad
 incoming proton: $0.194183|+\rangle +0.980965|-\rangle$

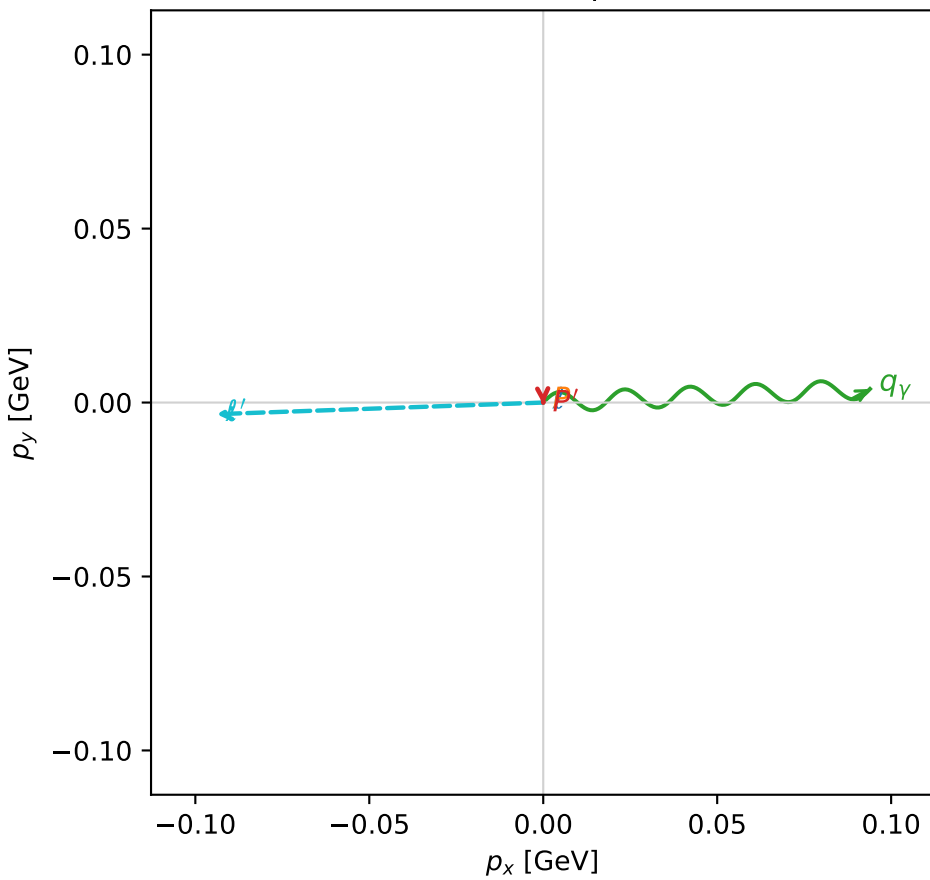
$s=1.27534$, $\sqrt{s}=1.12931$, $|\vec{P}|=0.175106$, $|\vec{P}'|=0.000571782$
 $\theta_{\text{out}}=1.38521$, $\phi_{P'}=4.75122$, $\phi_Y=0.0415765$
 $E_Y=0.0956451$, $\theta_{\ell'}=1.75747$, $\phi_{\ell'}=3.17719$
 $Q^2=0.0272854$, $x_B=0.131997$, $t=-0.030363$, $W^2=1.05927$, $y=0.52266$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E=0.1751$ $p_x=0.0000$ $p_y=0.0000$ $p_z=-0.1751$
 P $E=0.9542$ $p_x=0.0000$ $p_y=0.0000$ $p_z=0.1751$
 ℓ' $E=0.0957$ $p_x=-0.0939$ $p_y=-0.0033$ $p_z=-0.0178$
 P' $E=0.9380$ $p_x=0.0000$ $p_y=-0.0006$ $p_z=0.0001$
 q_Y $E=0.0956$ $p_x=0.0939$ $p_y=0.0039$ $p_z=0.0176$

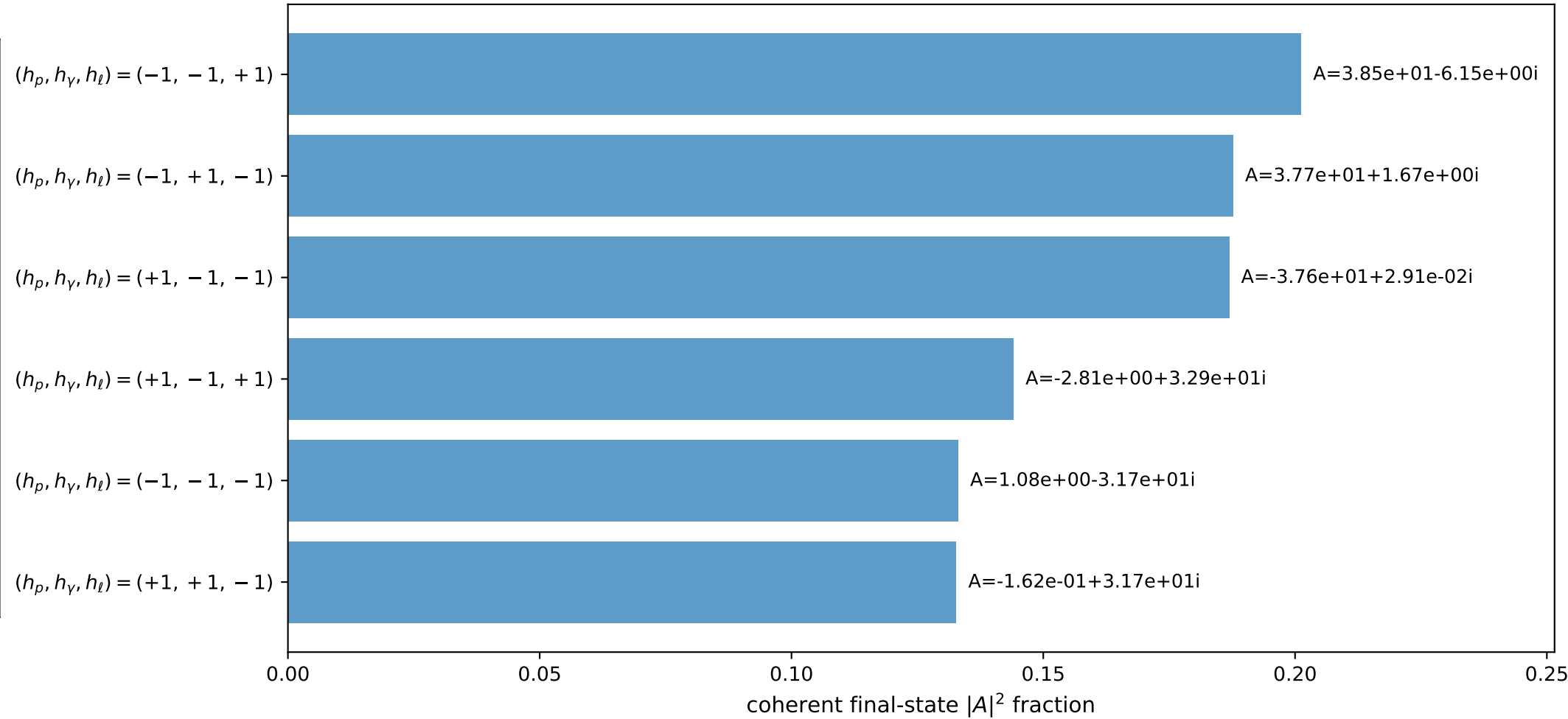
3D momenta



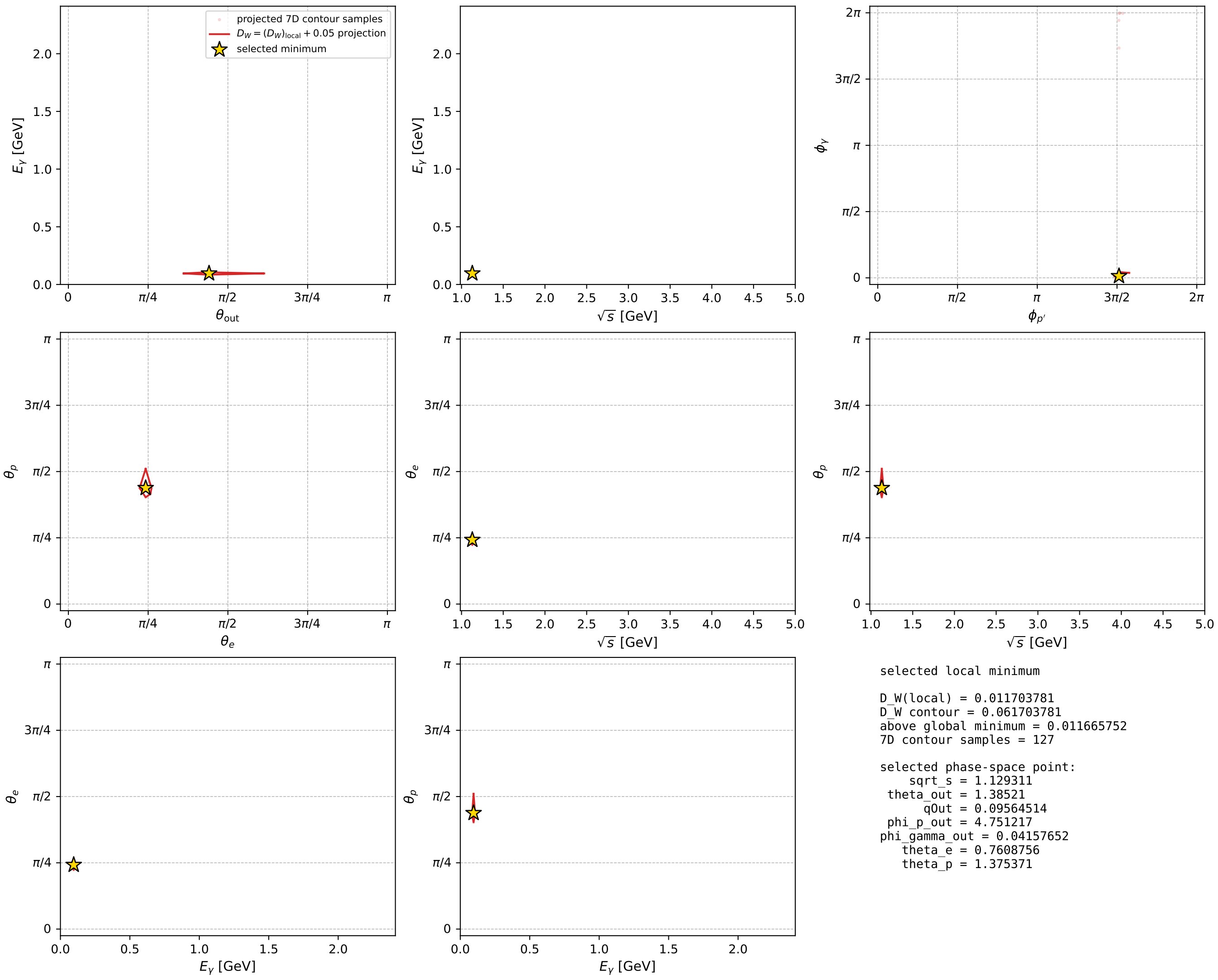
Transverse plane



Leading final-state amplitudes (N=6, retained=98.6%)



electron: polarization cluster P3, local minimum 613; pairwise projections of the 7D $D_W = (D_W)_{\text{local}} + 0.05$ contour

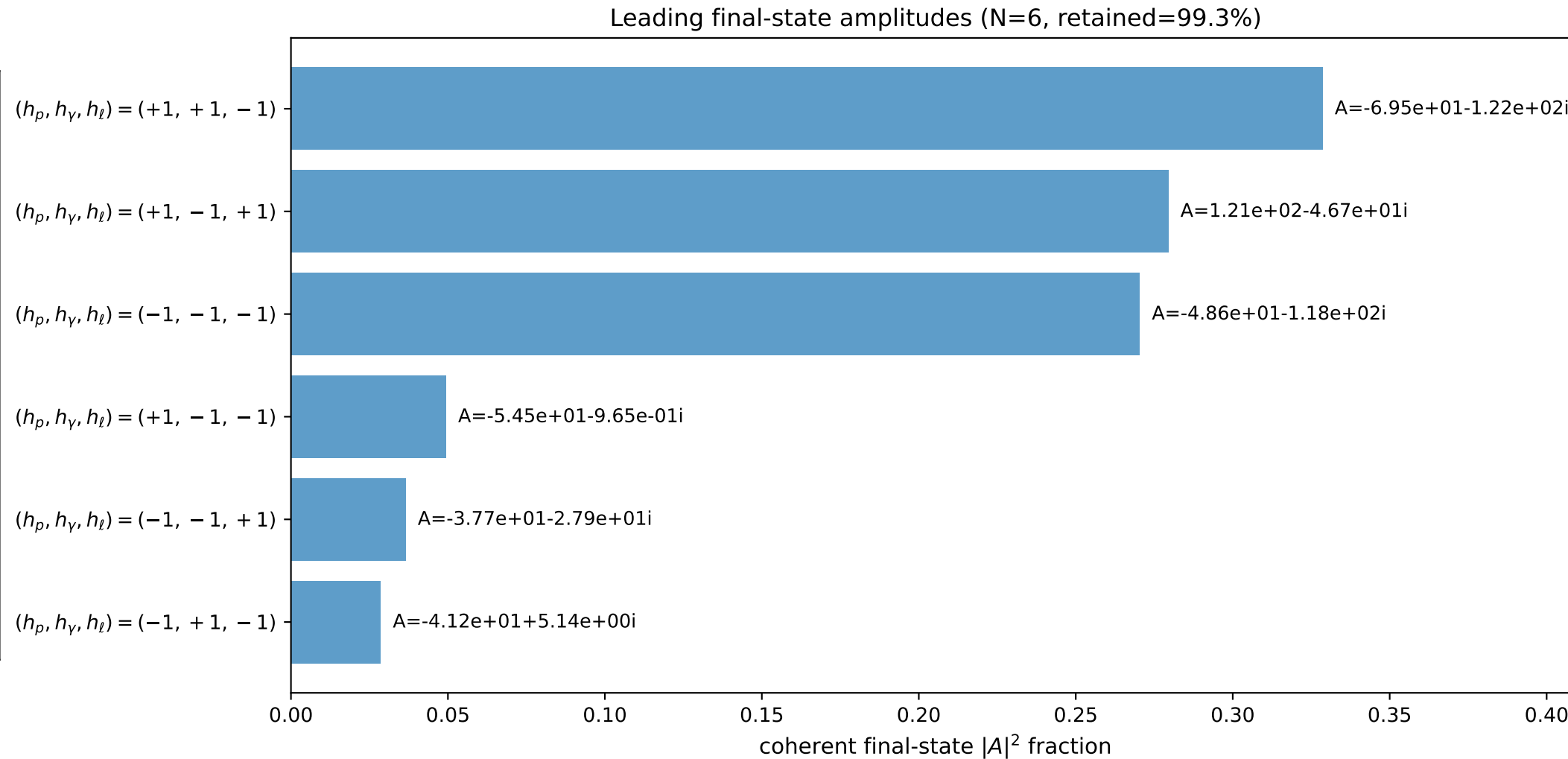
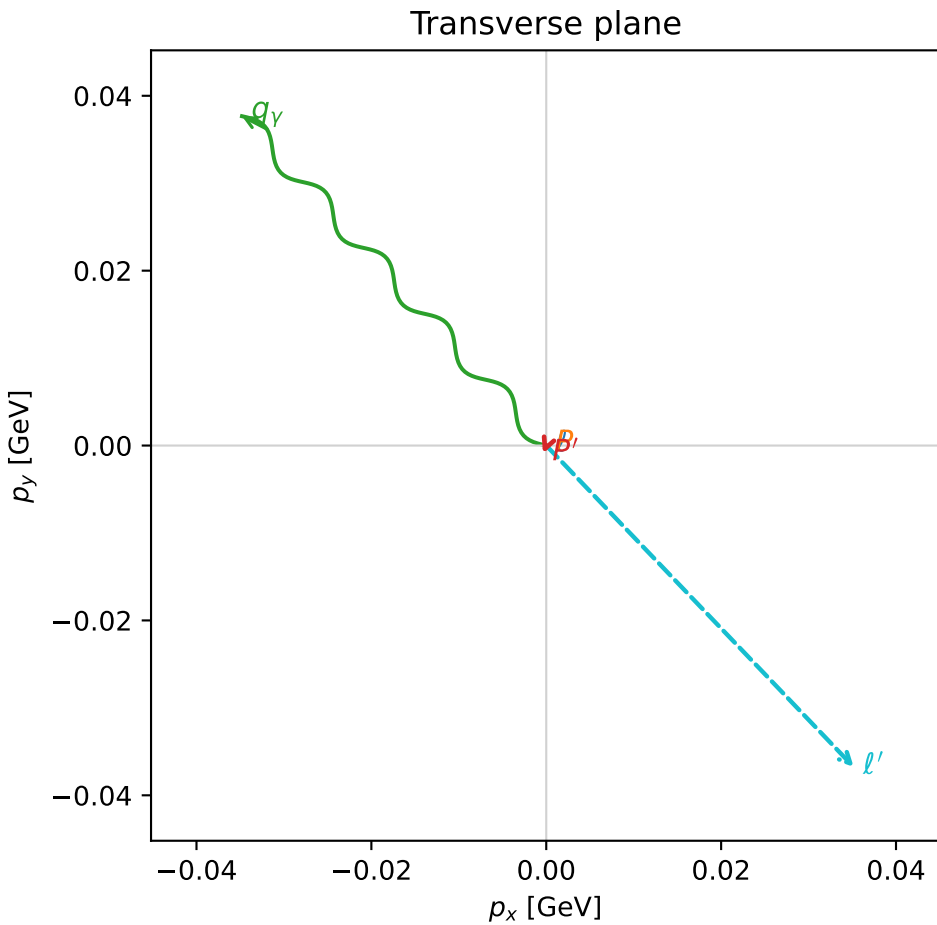
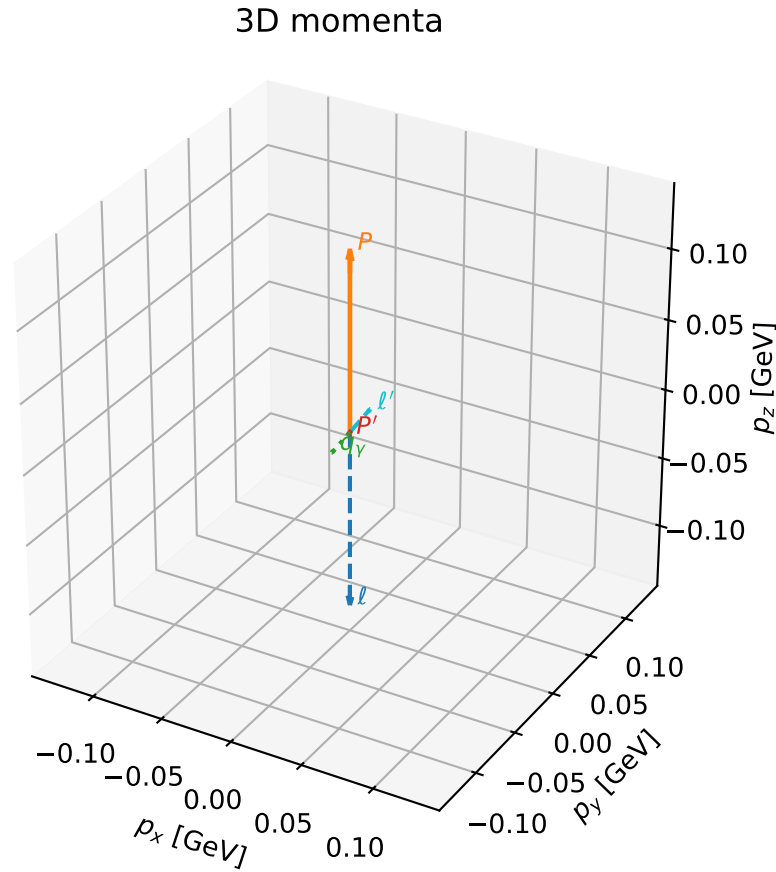


coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

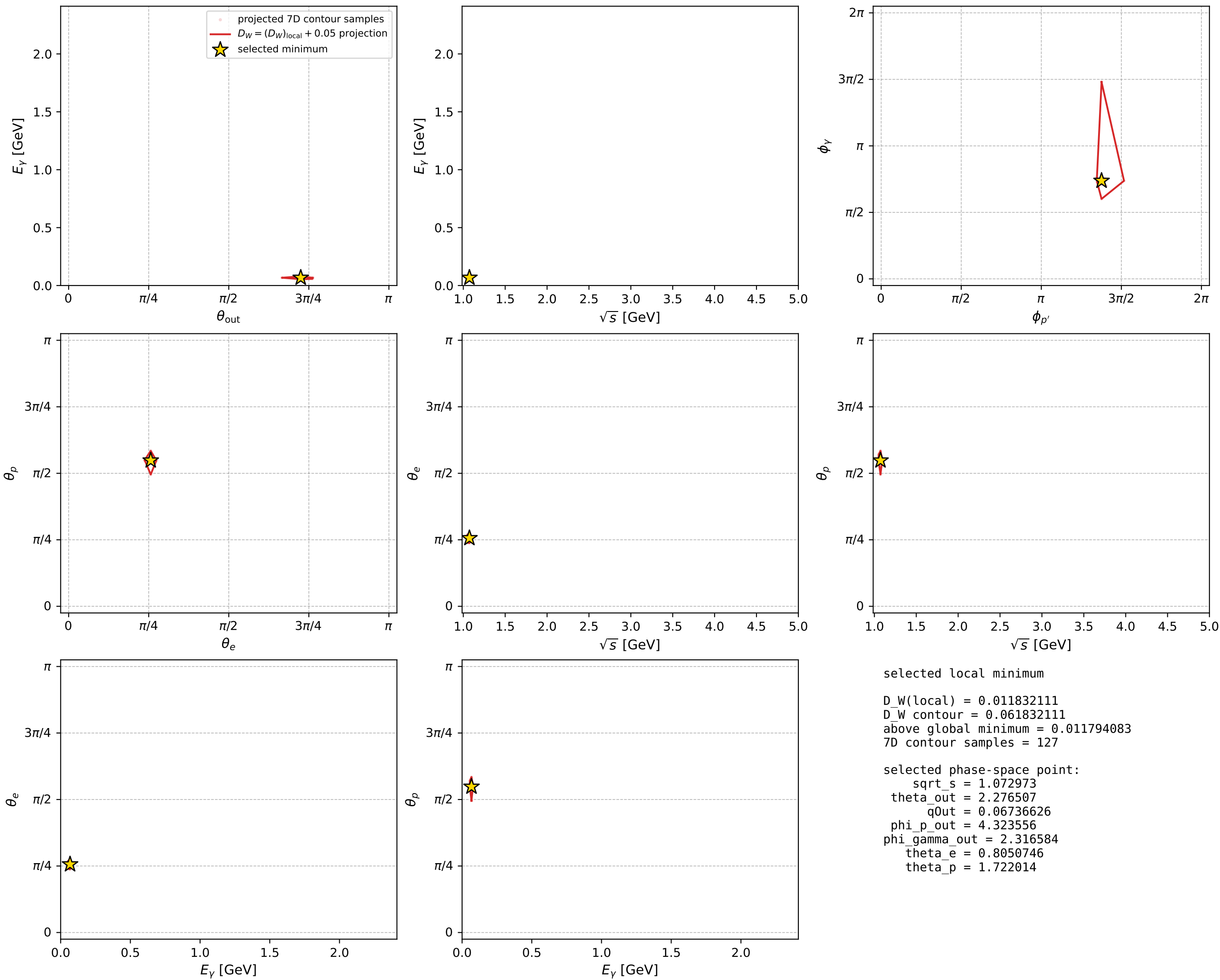
dw_mixing_angles_polarization_03_local_minimum_617 D_W=0.0118321
 region: polarization_cluster_03_local_minimum_617
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\theta_e=0.80507464$ rad
 incoming lepton: $0.693057|+\rangle + 0.720882|-\rangle$
 proton mixing angle: $\theta_p=1.7220145$ rad
 incoming proton: $-0.150643|+\rangle + 0.988588|-\rangle$

$s=1.15127$, $\sqrt{s}=1.07297$, $|\vec{P}|=0.126482$, $|\vec{P}'|=0.00128174$
 $\theta_{\text{out}}=2.27651$, $\phi_{P'}=4.32356$, $\phi_Y=2.31658$
 $E_Y=0.0673663$, $\theta_{\ell'}=0.851845$, $\phi_{\ell'}=5.47556$
 $Q^2=0.0283642$, $x_B=0.183334$, $t=-0.0161377$, $W^2=1.00619$, $y=0.570001$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1265$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1265$
 P $E= 0.9465$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1265$
 ℓ' $E= 0.0676$ $p_x= 0.0352$ $p_y= -0.0368$ $p_z= 0.0445$
 P' $E= 0.9380$ $p_x= -0.0004$ $p_y= -0.0009$ $p_z= -0.0008$
 q_Y $E= 0.0674$ $p_x= -0.0348$ $p_y= 0.0377$ $p_z= -0.0437$



electron: polarization cluster P3, local minimum 617; pairwise projections of the 7D $D_W = (D_W)_{\text{local}} + 0.05$ contour



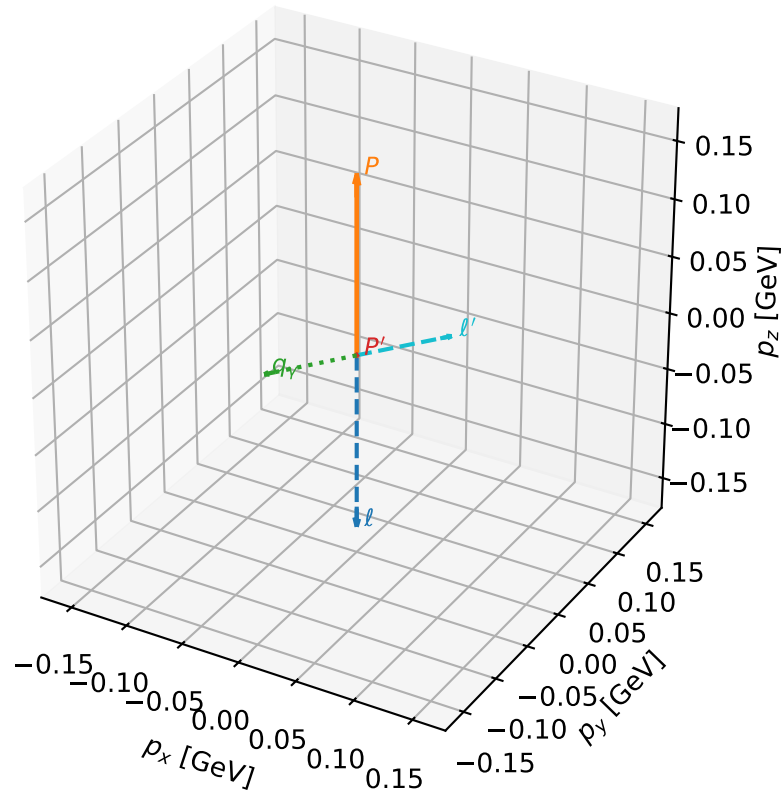
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

dw_mixing_angles_polarization_03_local_minimum_618 D_W=0.0118518
 region: polarization_cluster_03_local_minimum_618
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\theta_e=0.80704159$ rad
 incoming lepton: $0.691638|+\rangle +0.722244|-\rangle$
 proton mixing angle: $\theta_p=1.7463698$ rad
 incoming proton: $-0.174673|+\rangle +0.984627|-\rangle$

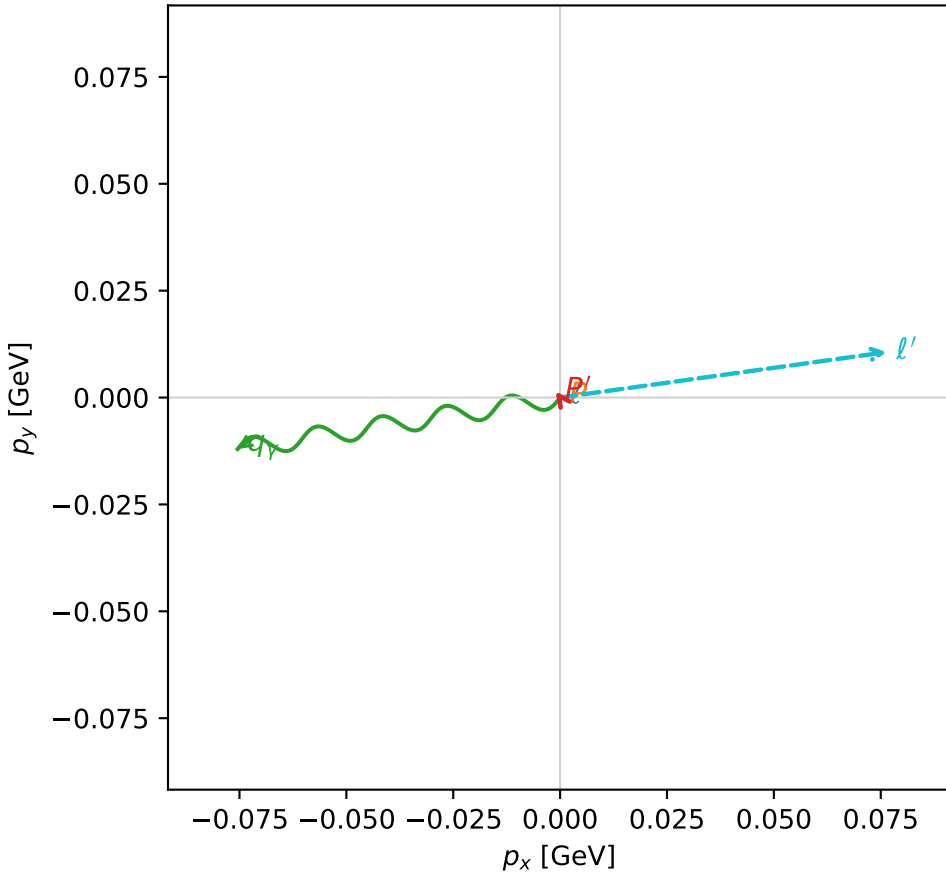
$s=1.22149$, $\sqrt{s}=1.10521$, $|\vec{P}|=0.154562$, $|\vec{P}'|=0.00190208$
 $\theta_{\text{out}}=1.97546$, $\phi_{p'}=2.1842$, $\phi_\gamma=3.30007$
 $E_\gamma=0.0830959$, $\theta_{\ell'}=1.16162$, $\phi_{\ell'}=0.138124$
 $Q^2=0.036346$, $x_B=0.189235$, $t=-0.0239646$, $W^2=1.03557$, $y=0.562179$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E=0.1546$ $p_x=0.0000$ $p_y=0.0000$ $p_z=-0.1546$
 P $E=0.9506$ $p_x=0.0000$ $p_y=0.0000$ $p_z=0.1546$
 ℓ' $E=0.0841$ $p_x=0.0764$ $p_y=0.0106$ $p_z=0.0335$
 P' $E=0.9380$ $p_x=-0.0010$ $p_y=0.0014$ $p_z=-0.0007$
 q_γ $E=0.0831$ $p_x=-0.0754$ $p_y=-0.0121$ $p_z=-0.0327$

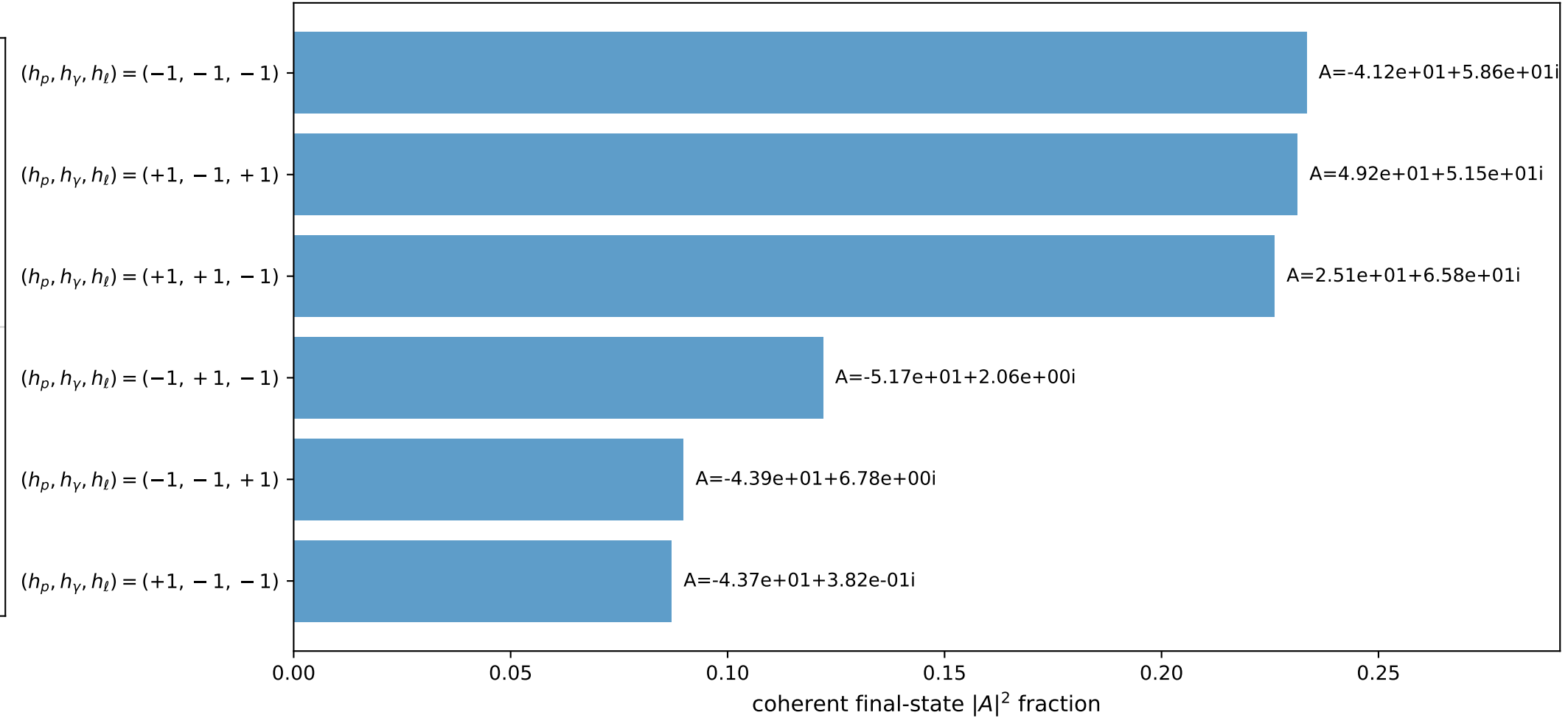
3D momenta



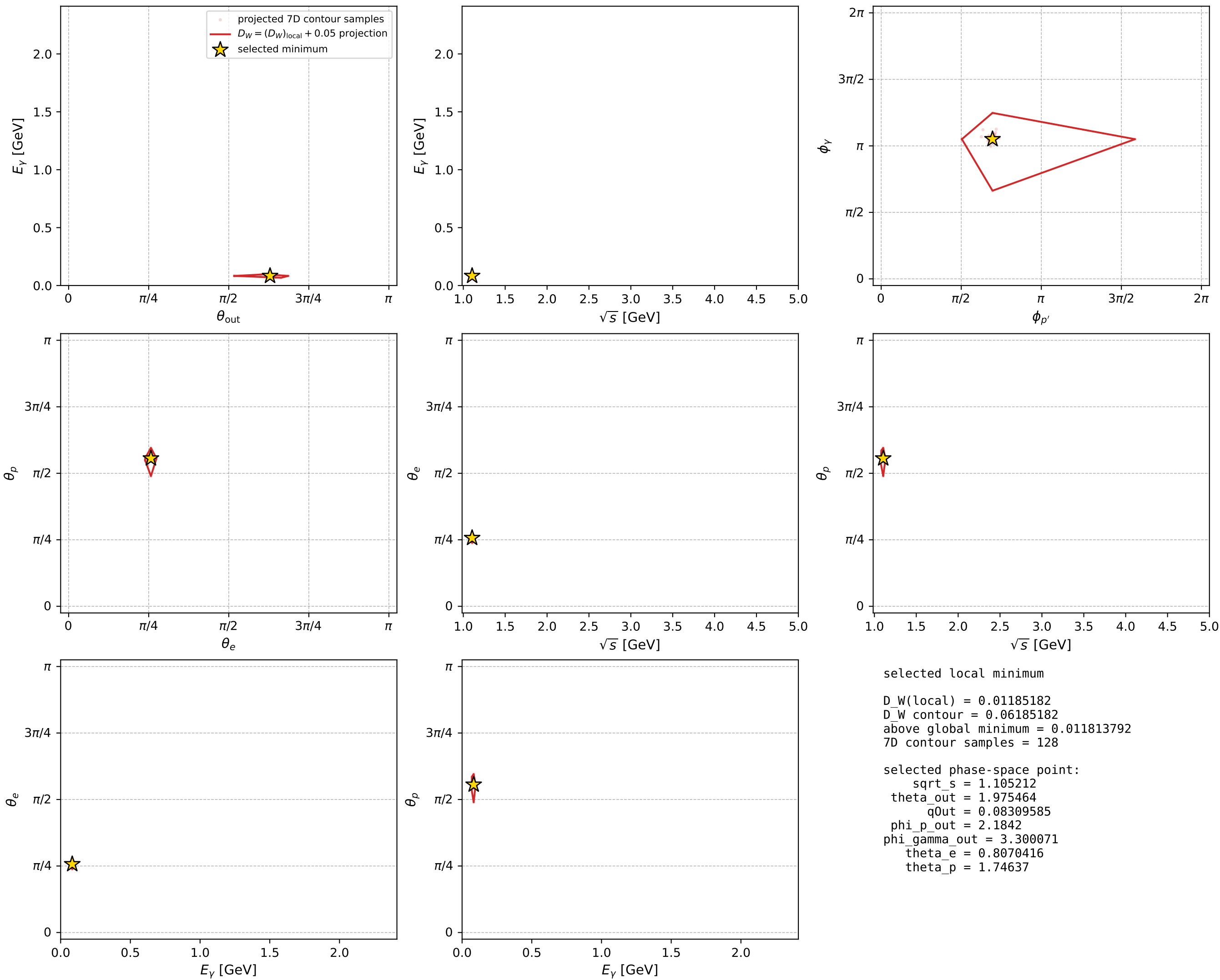
Transverse plane



Leading final-state amplitudes (N=6, retained=99.0%)



electron: polarization cluster P3, local minimum 618; pairwise projections of the 7D $D_W = (D_W)_{\text{local}} + 0.05$ contour

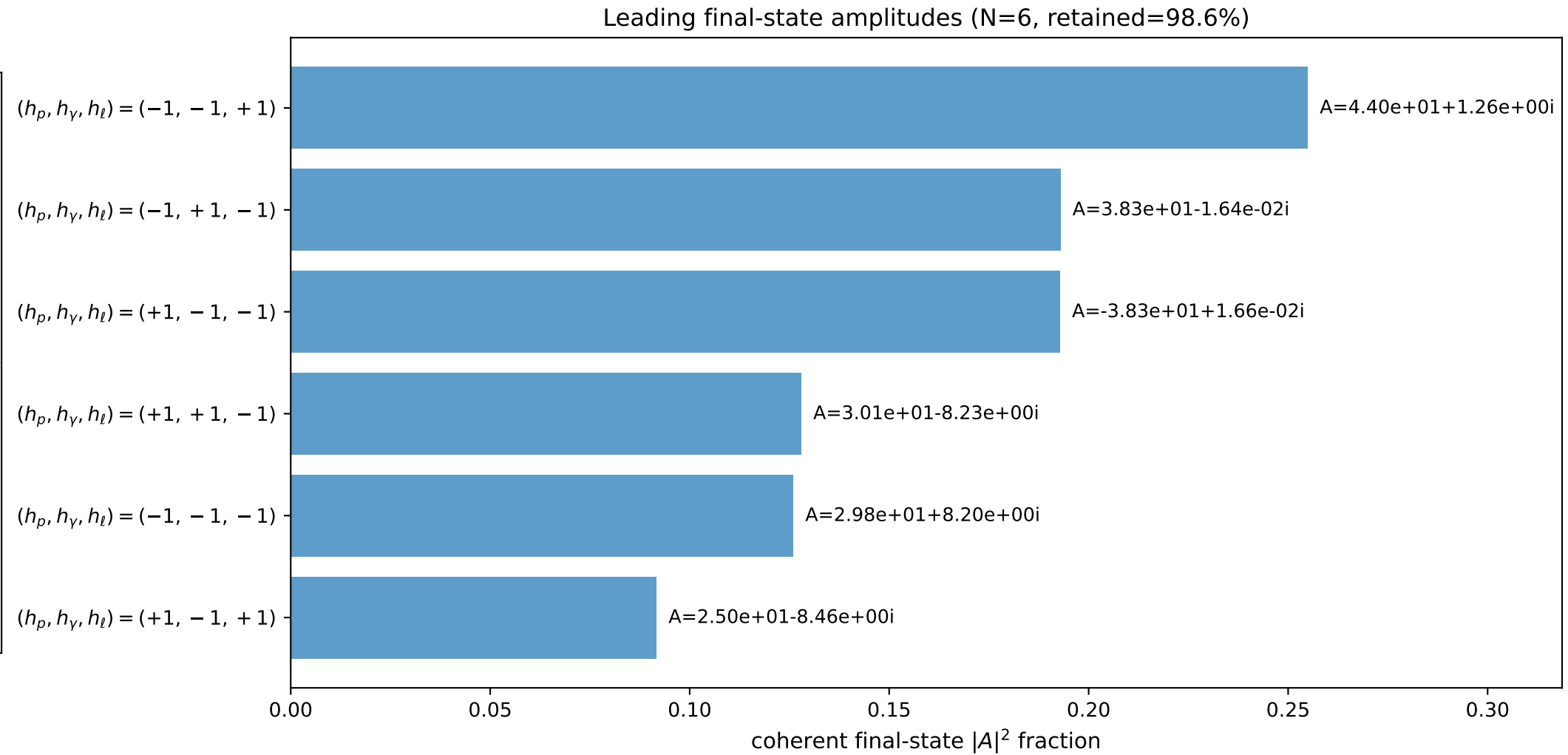
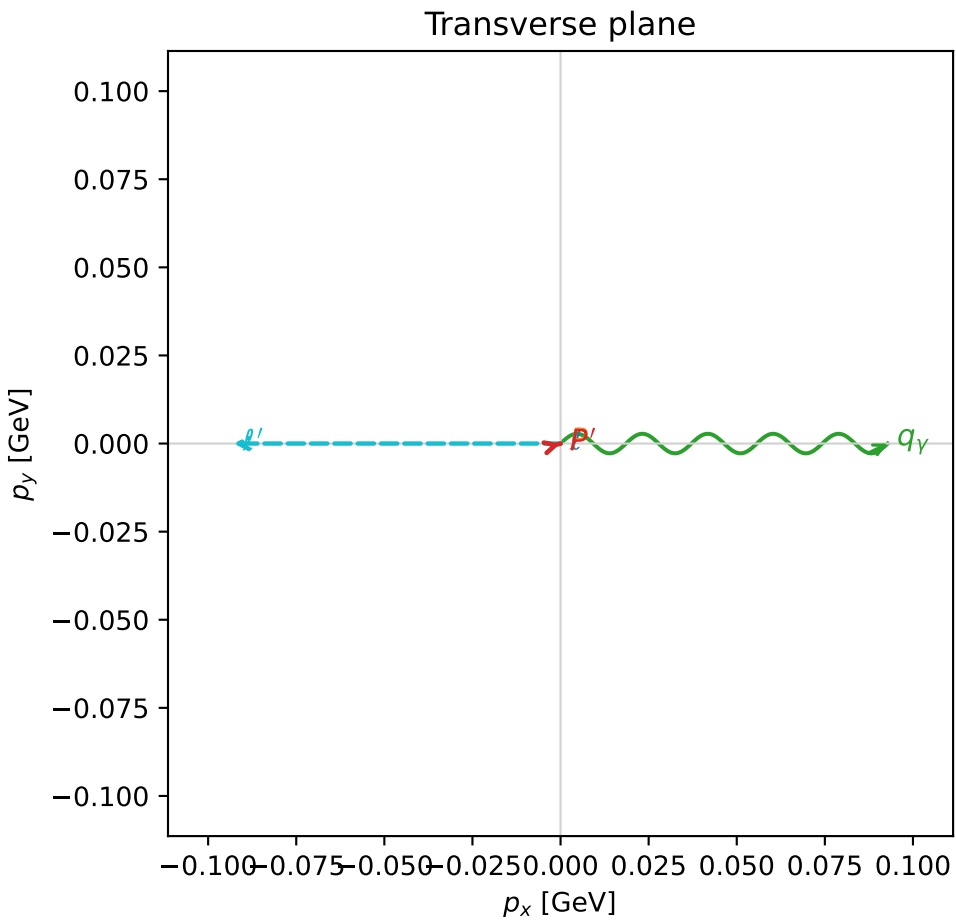
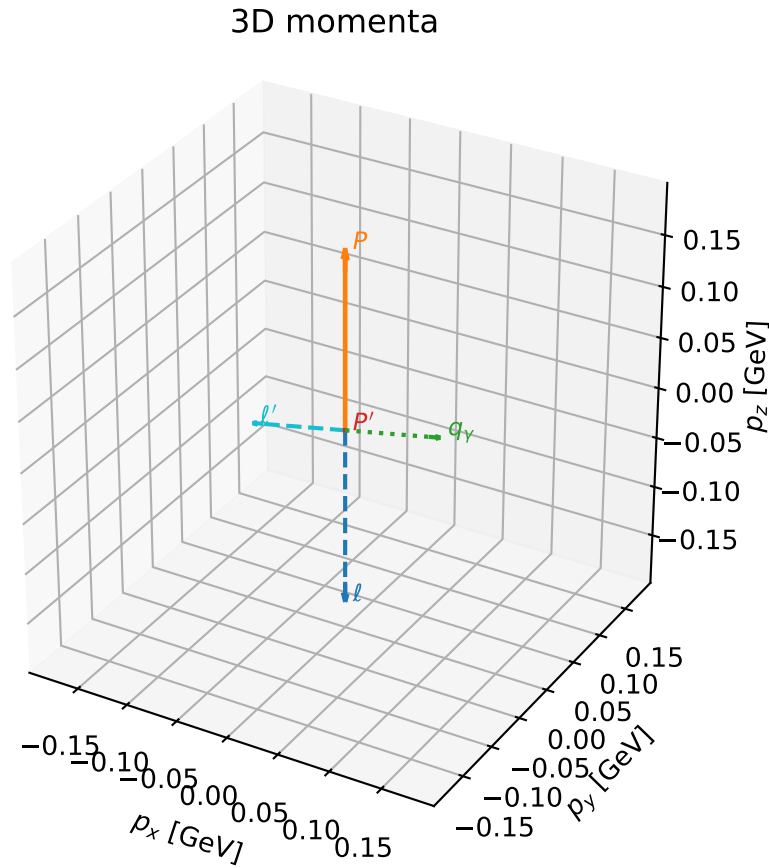


coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

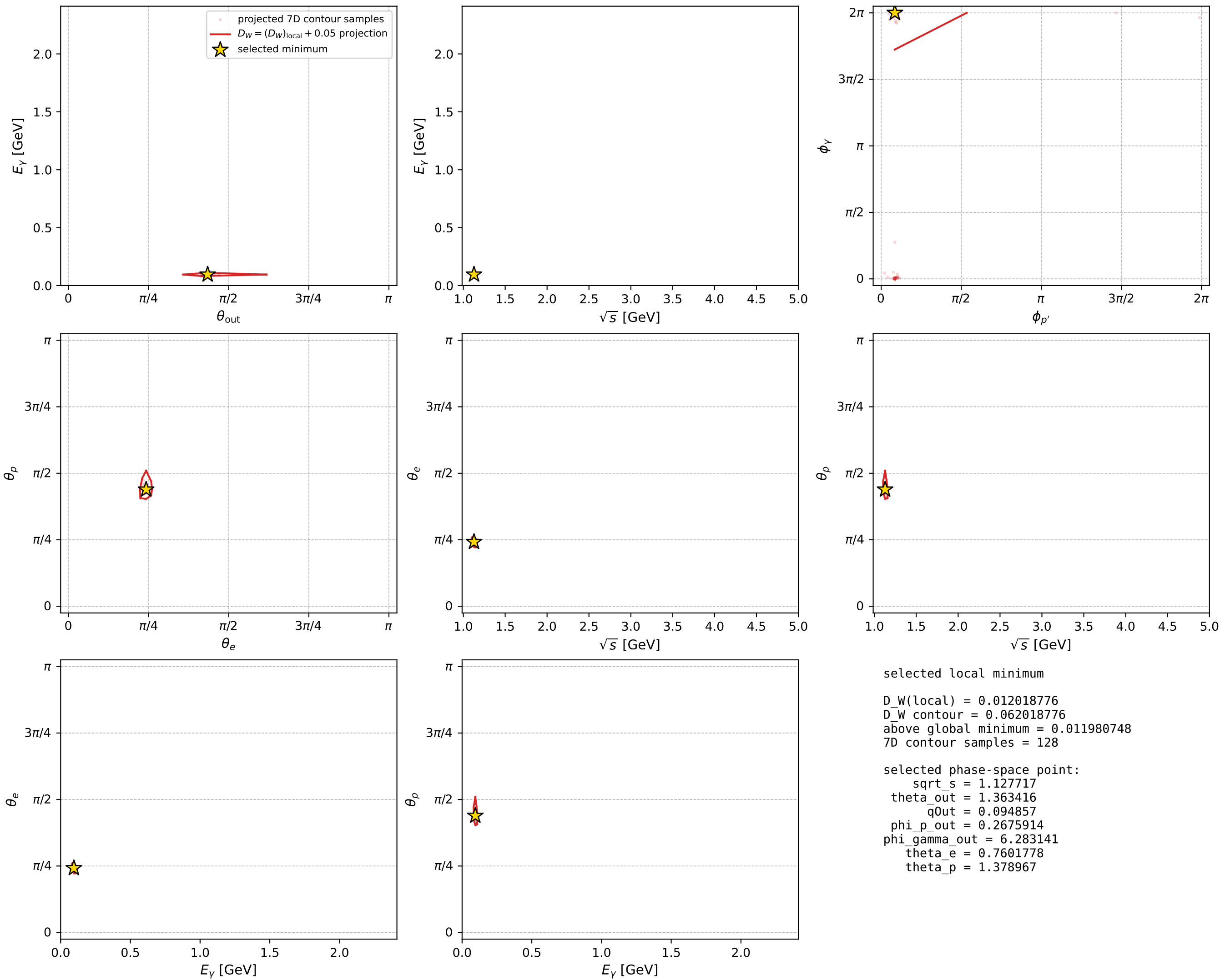
dw_mixing_angles_polarization_03_local_minimum_620 D_W=0.0120188
 region: polarization_cluster_03_local_minimum_620
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\theta_e=0.76017785$ rad
 incoming lepton: $0.724713|+\rangle +0.68905|-\rangle$
 proton mixing angle: $\theta_p=1.3789672$ rad
 incoming proton: $0.190655|+\rangle +0.981657|-\rangle$

$s=1.27175$, $\sqrt{s}=1.12772$, $|\vec{P}|=0.173758$, $|\vec{P}'|=1.52878\text{e-}06$
 $\theta_{\text{out}}=1.36342$, $\phi_{P'}=0.267591$, $\phi_Y=6.28314$
 $E_Y=0.094857$, $\theta_{\ell'}=1.77818$, $\phi_{\ell'}=3.14155$
 $Q^2=0.0261776$, $x_B=0.12824$, $t=-0.0299371$, $W^2=1.0578$, $y=0.520869$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1738$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1738$
 P $E= 0.9540$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1738$
 ℓ' $E= 0.0949$ $p_x= -0.0928$ $p_y= 0.0000$ $p_z= -0.0195$
 P' $E= 0.9380$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.0000$
 q_Y $E= 0.0949$ $p_x= 0.0928$ $p_y= -0.0000$ $p_z= 0.0195$



electron: polarization cluster P3, local minimum 620; pairwise projections of the 7D $D_W = (D_W)_{\text{local}} + 0.05$ contour

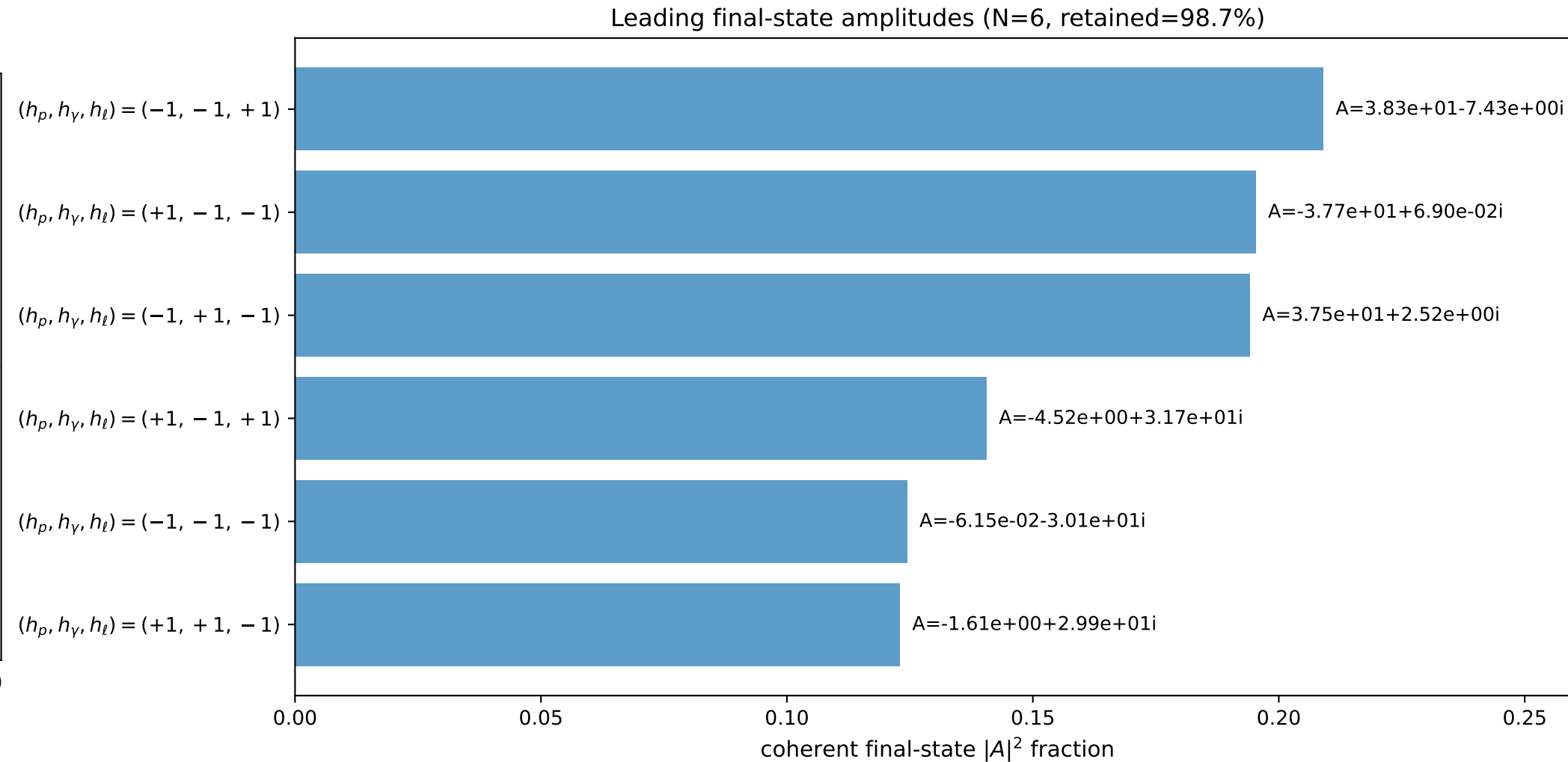
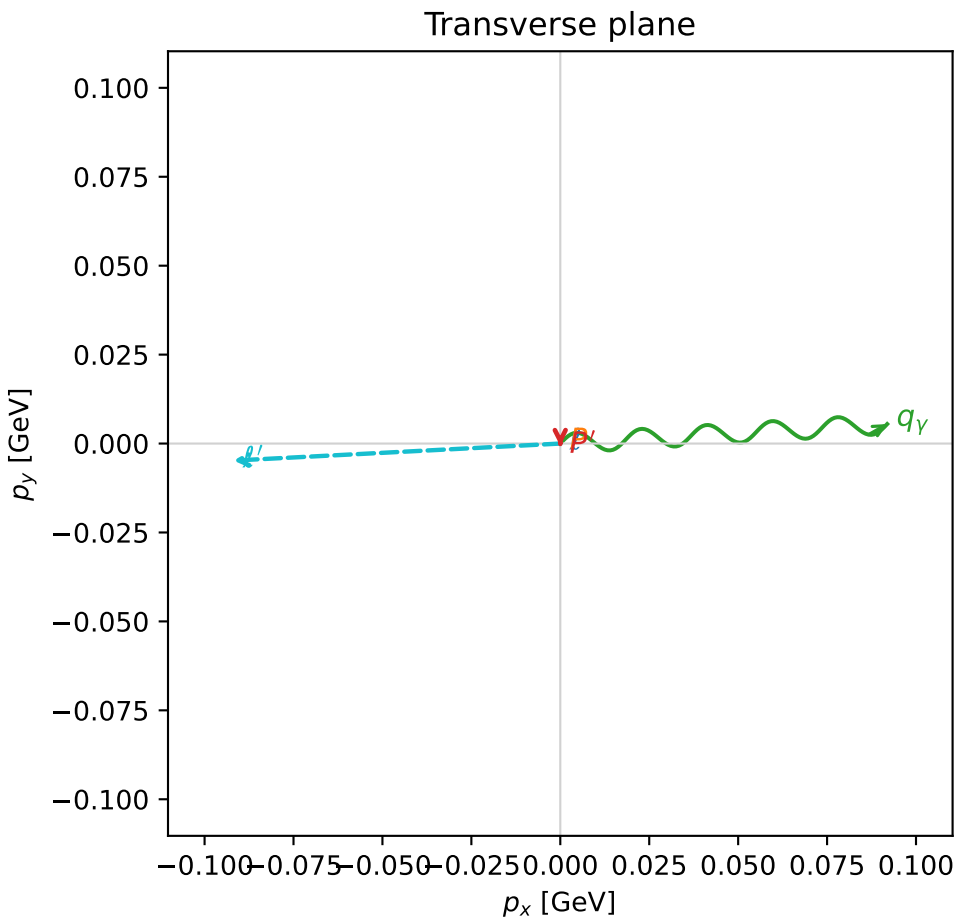
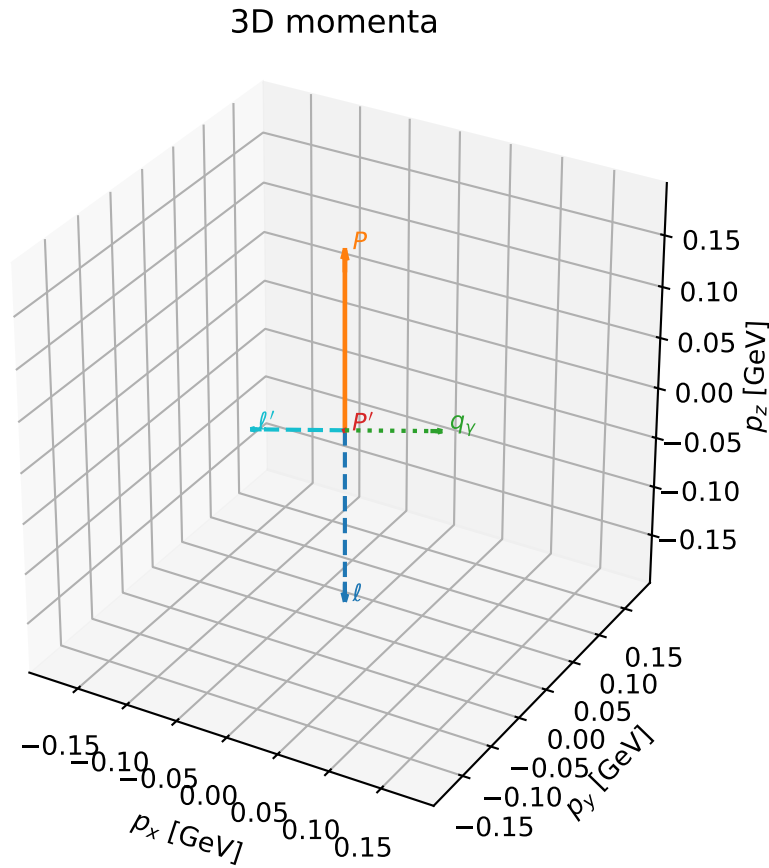


coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

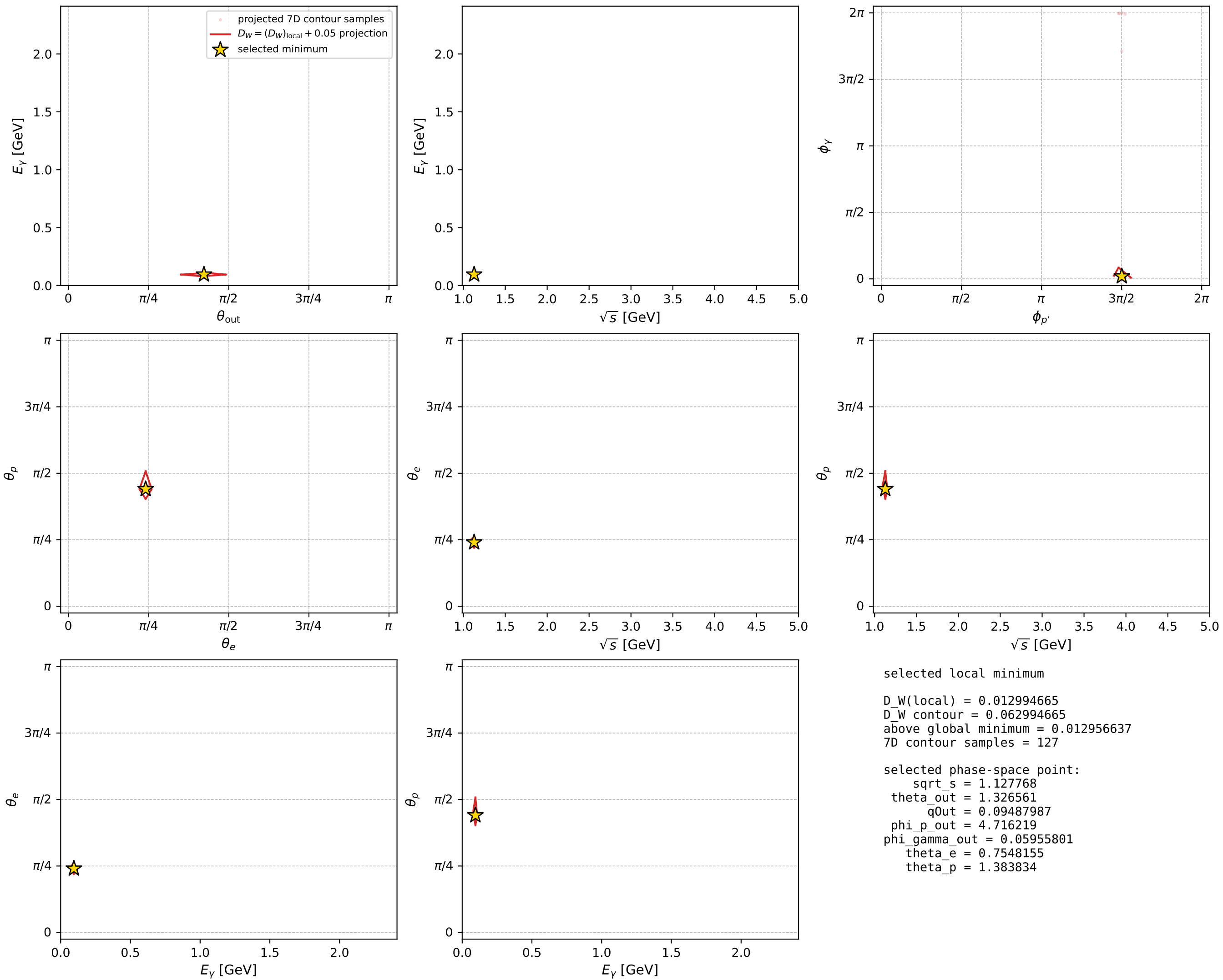
dw_mixing_angles_polarization_03_local_minimum_634 D_W=0.0129947
 region: polarization_cluster_03_local_minimum_634
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\theta_e=0.75481549$ rad
 incoming lepton: $0.728398|+> +0.685154|->$
 proton mixing angle: $\theta_p=1.383834$ rad
 incoming proton: $0.185875|+> +0.982573|->$

$s=1.27186$, $\sqrt{s}=1.12777$, $|\vec{P}|=0.173802$, $|\vec{P}'|=0.000709199$
 $\theta_{\text{out}}=1.32656$, $\phi_{P'}=4.71622$, $\phi_Y=0.059558$
 $E_Y=0.0948799$, $\theta_{\ell'}=1.81688$, $\phi_{\ell'}=3.19368$
 $Q^2=0.0249483$, $x_B=0.122933$, $t=-0.029893$, $W^2=1.05784$, $y=0.517687$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1738$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1738$
 P $E= 0.9540$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1738$
 ℓ' $E= 0.0949$ $p_x= -0.0919$ $p_y= -0.0048$ $p_z= -0.0231$
 P' $E= 0.9380$ $p_x= 0.0000$ $p_y= -0.0007$ $p_z= 0.0002$
 q_Y $E= 0.0949$ $p_x= 0.0919$ $p_y= 0.0055$ $p_z= 0.0229$



electron: polarization cluster P3, local minimum 634; pairwise projections of the 7D $D_W = (D_W)_{\text{local}} + 0.05$ contour



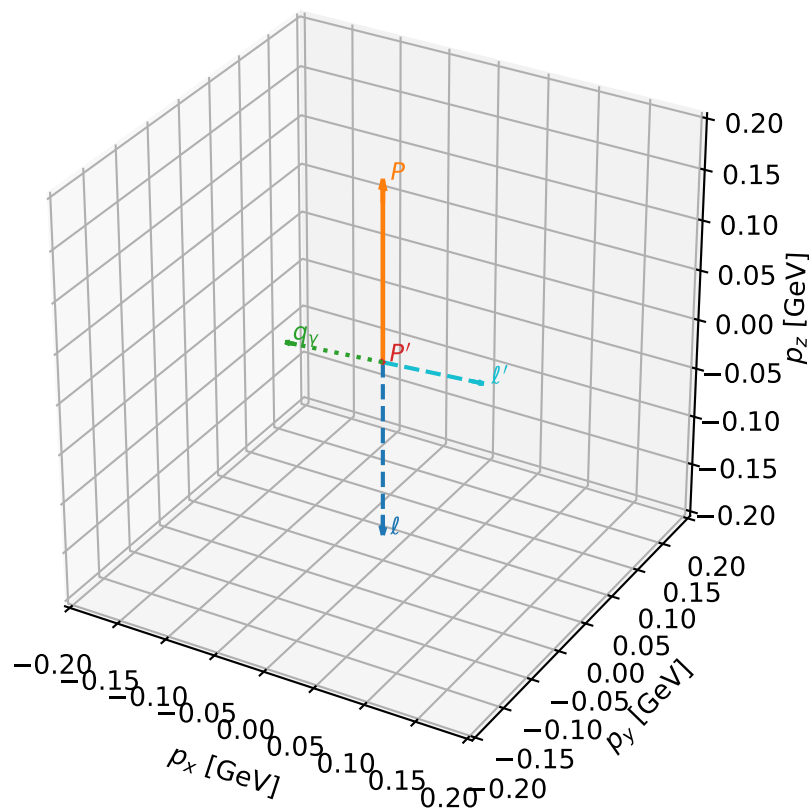
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

dw_mixing_angles_polarization_03_local_minimum_635 D_W=0.0130119
 region: polarization_cluster_03_local_minimum_635
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\theta_e=0.77721244$ rad
 incoming lepton: $0.712871|+\rangle + 0.701295|-\rangle$
 proton mixing angle: $\theta_p=1.7883127$ rad
 incoming proton: $-0.215805|+\rangle + 0.976436|-\rangle$

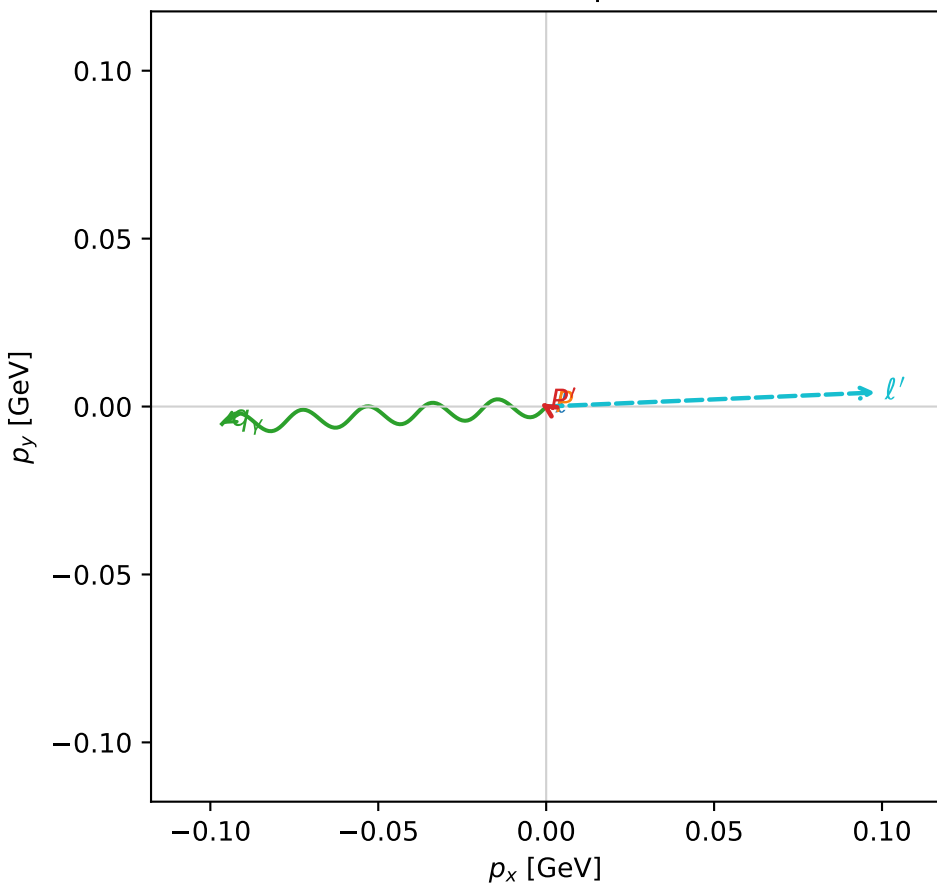
$s=1.28374$, $\sqrt{s}=1.13302$, $|\vec{P}|=0.178239$, $|\vec{P}'|=0.00183303$
 $\theta_{\text{out}}=1.62024$, $\phi_{P'}=2.56942$, $\phi_\gamma=3.19521$
 $E_\gamma=0.0967643$, $\theta_{\ell'}=1.52118$, $\phi_{\ell'}=0.0426938$
 $Q^2=0.0367634$, $x_B=0.168635$, $t=-0.0315231$, $W^2=1.06109$, $y=0.539753$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1782$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1782$
 P $E= 0.9548$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1782$
 ℓ' $E= 0.0983$ $p_x= 0.0980$ $p_y= 0.0042$ $p_z= 0.0049$
 P' $E= 0.9380$ $p_x= -0.0015$ $p_y= 0.0010$ $p_z= -0.0001$
 q_γ $E= 0.0968$ $p_x= -0.0965$ $p_y= -0.0052$ $p_z= -0.0048$

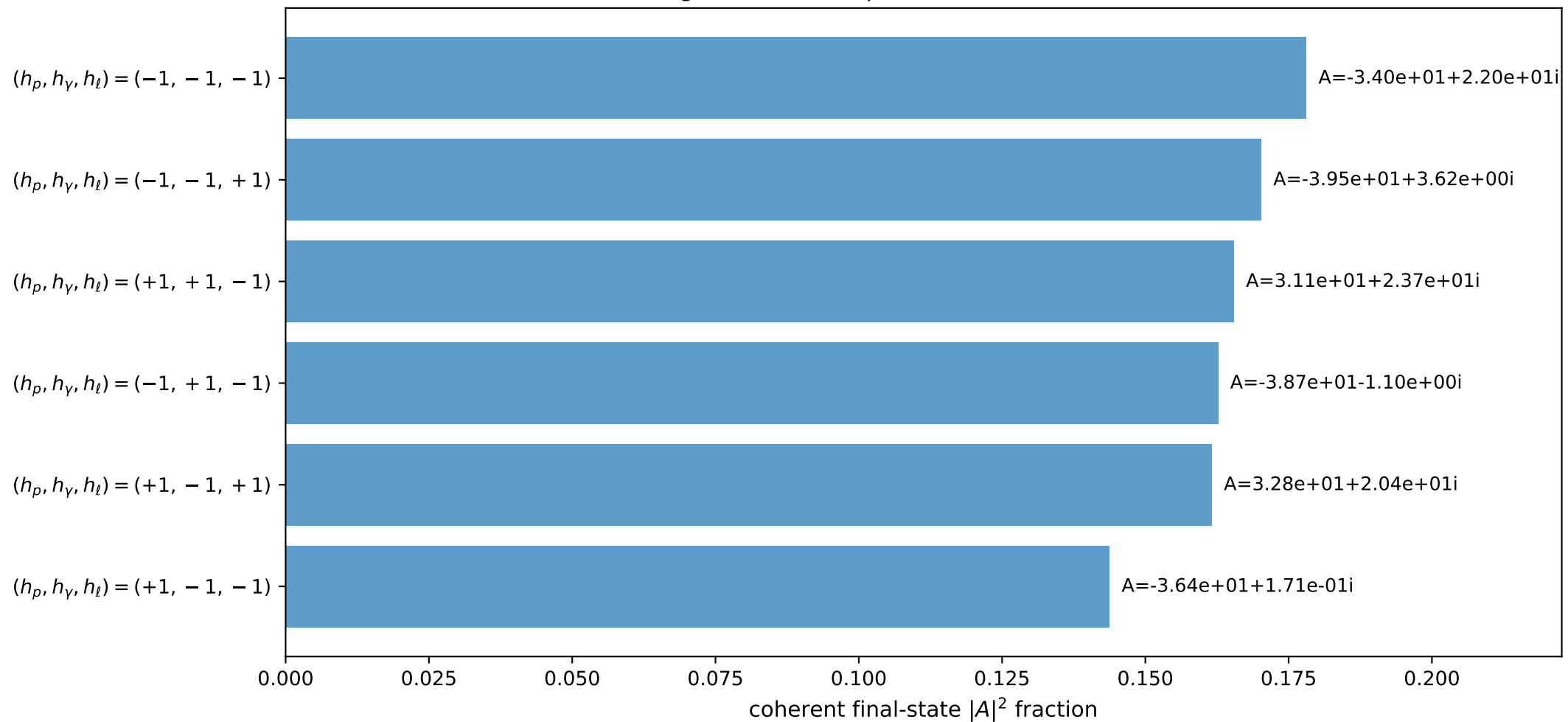
3D momenta



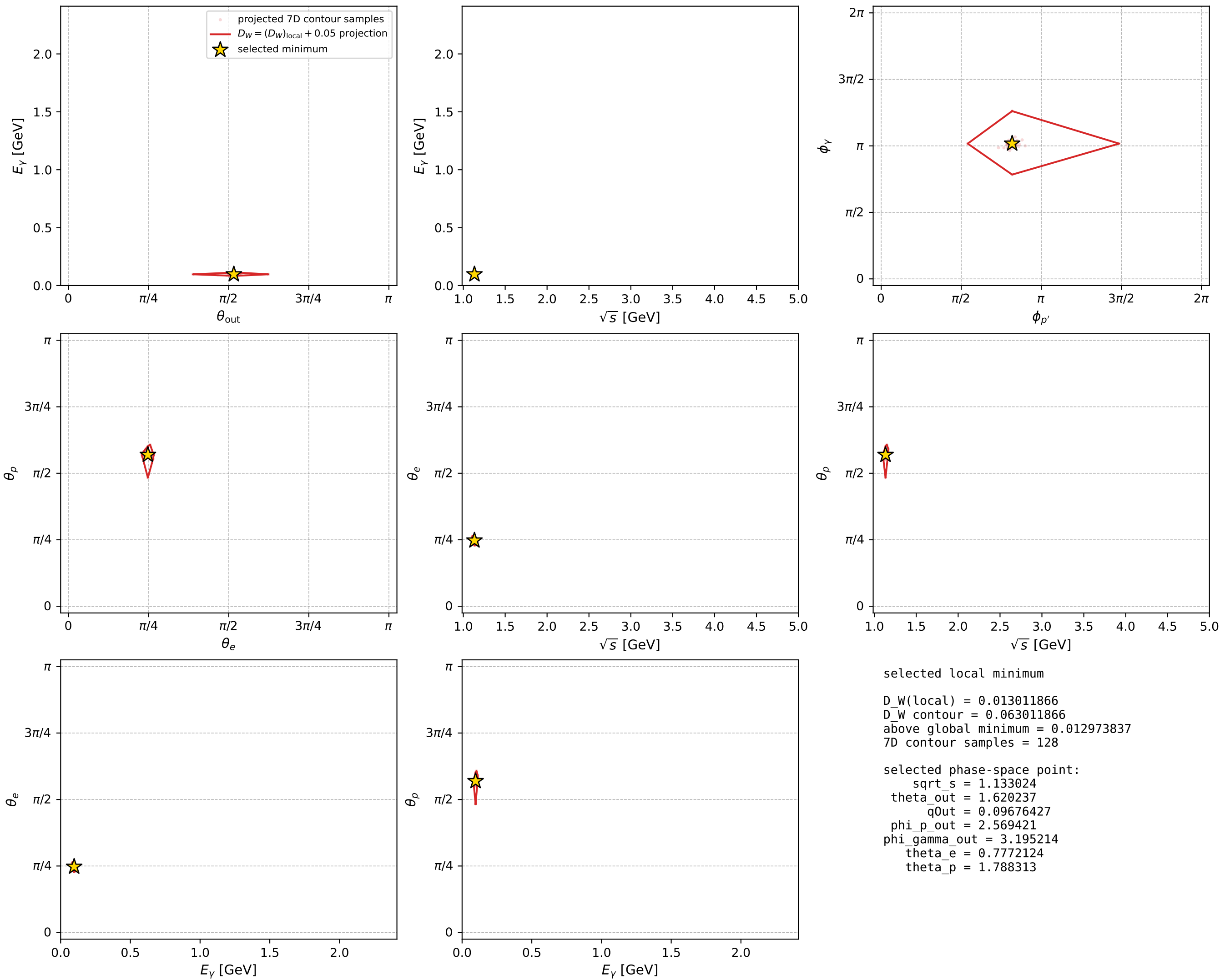
Transverse plane



Leading final-state amplitudes (N=6, retained=98.2%)



electron: polarization cluster P3, local minimum 635; pairwise projections of the 7D $D_W = (D_W)_{\text{local}} + 0.05$ contour

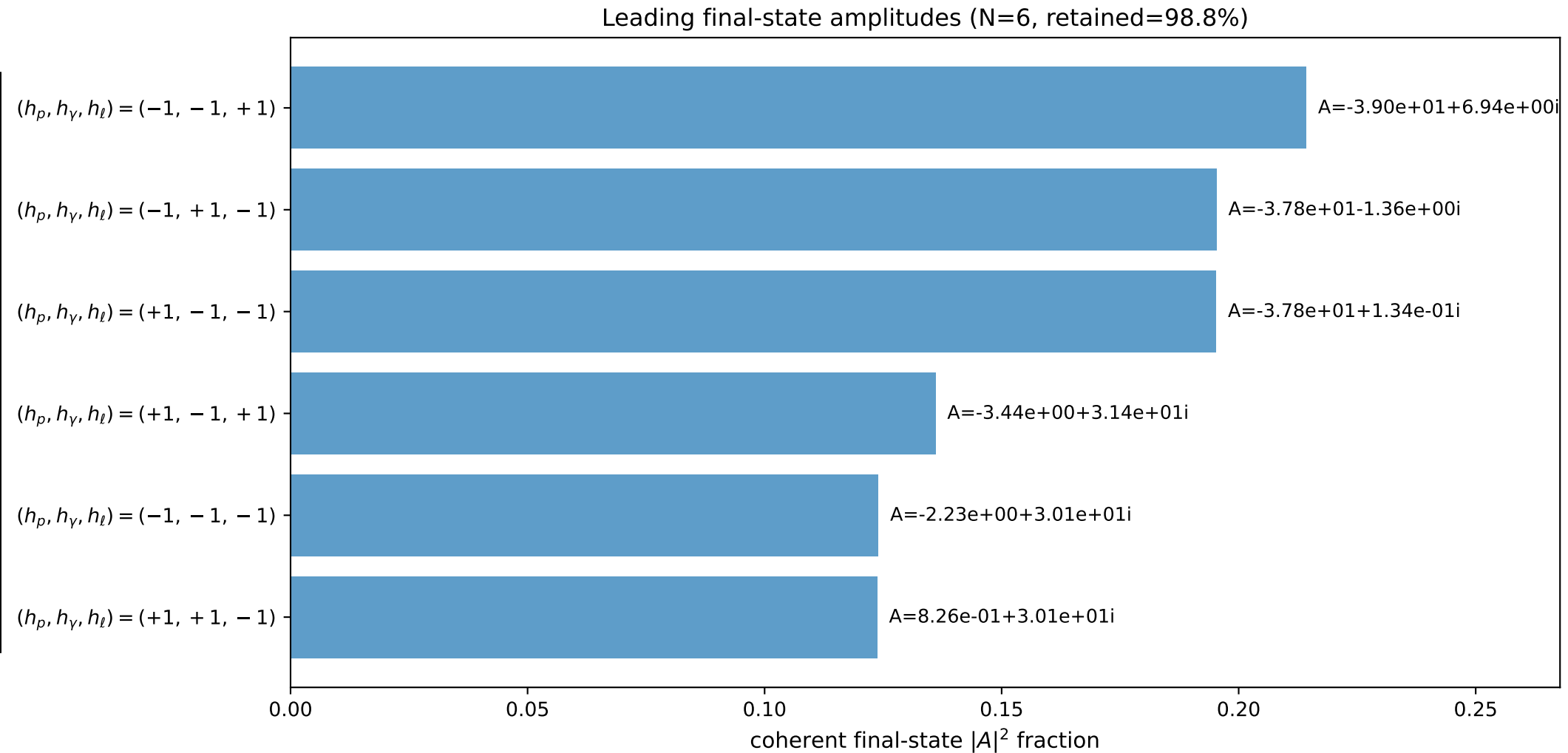
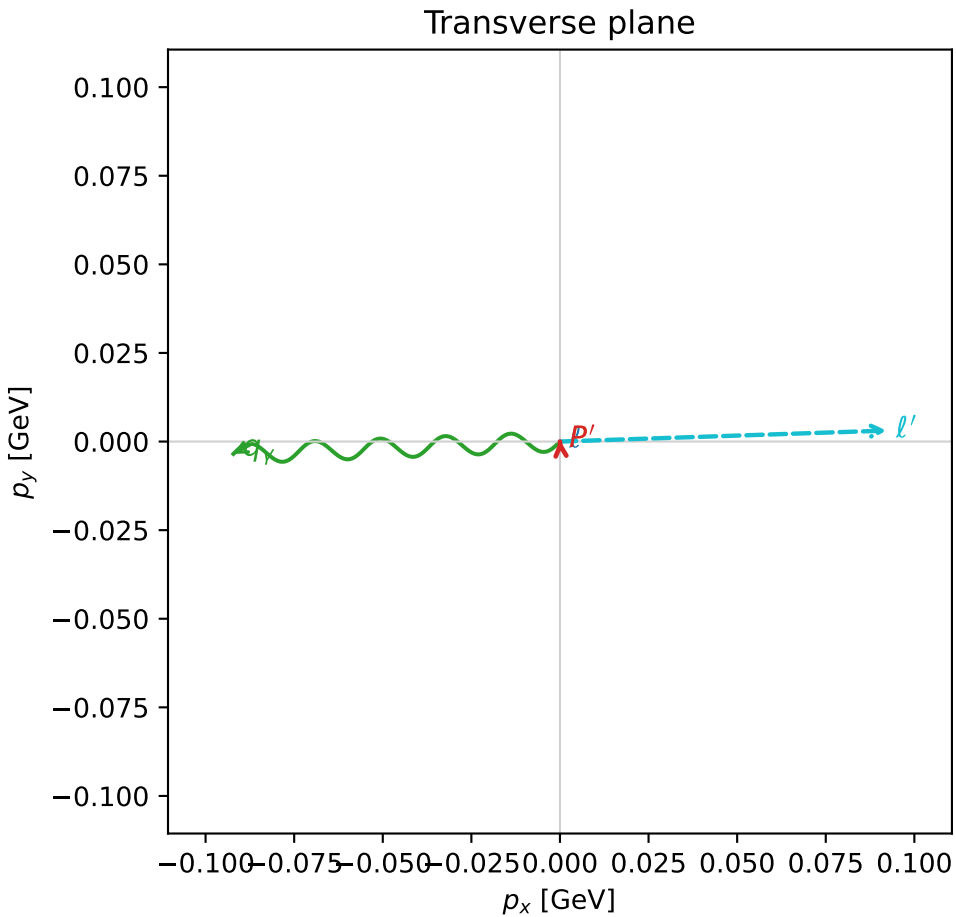
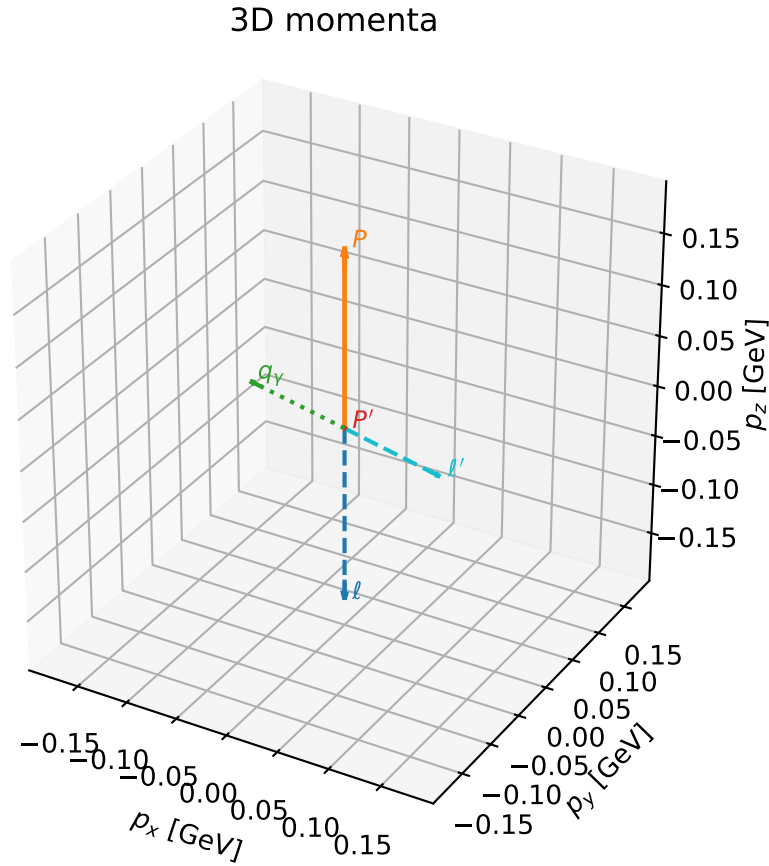


coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

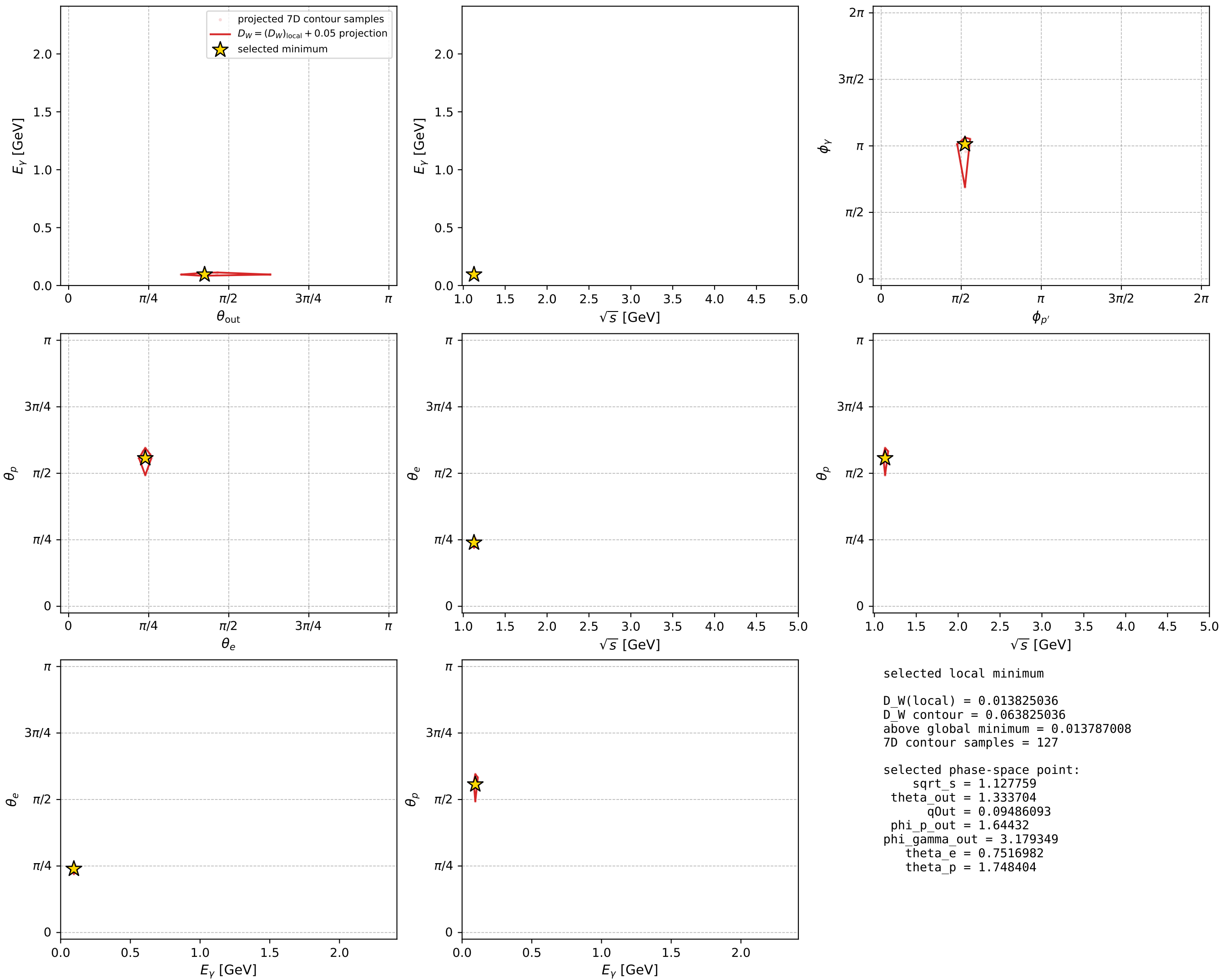
dw_mixing_angles_polarization_03_local_minimum_642 D_W=0.013825
 region: polarization_cluster_03_local_minimum_642
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\theta_e=0.75169822$ rad
 incoming lepton: $0.73053|+\rangle + 0.68288|-\rangle$
 proton mixing angle: $\theta_p=1.7484043$ rad
 incoming proton: $-0.176676|+\rangle + 0.984269|-\rangle$

$s=1.27184$, $\sqrt{s}=1.12776$, $|\vec{P}|=0.173794$, $|\vec{P}'|=0.000396429$
 $\theta_{\text{out}}=1.3337$, $\phi_{P'}=1.64432$, $\phi_Y=3.17935$
 $E_Y=0.0948609$, $\theta_{\ell'}=1.80881$, $\phi_{\ell'}=0.033581$
 $Q^2=0.0252083$, $x_B=0.124081$, $t=-0.0299173$, $W^2=1.0578$, $y=0.518271$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1738$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1738$
 P $E= 0.9540$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1738$
 ℓ' $E= 0.0949$ $p_x= 0.0922$ $p_y= 0.0031$ $p_z= -0.0224$
 P' $E= 0.9380$ $p_x= -0.0000$ $p_y= 0.0004$ $p_z= 0.0001$
 q_Y $E= 0.0949$ $p_x= -0.0921$ $p_y= -0.0035$ $p_z= 0.0223$



electron: polarization cluster P3, local minimum 642; pairwise projections of the 7D $D_W = (D_W)_{\text{local}} + 0.05$ contour



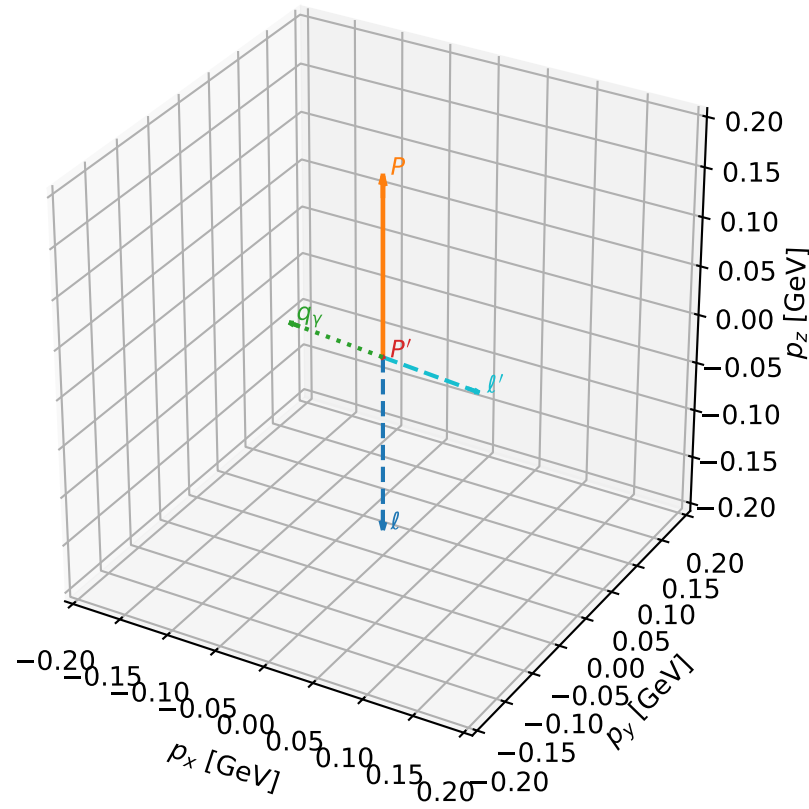
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

dw_mixing_angles_polarization_03_local_minimum_649 D_W=0.0145831
 region: polarization_cluster_03_local_minimum_649
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\theta_e=0.76068516$ rad
 incoming lepton: $0.724364|+\rangle + 0.689418|-\rangle$
 proton mixing angle: $\theta_p=1.7722708$ rad
 incoming proton: $-0.200114|+\rangle + 0.979773|-\rangle$

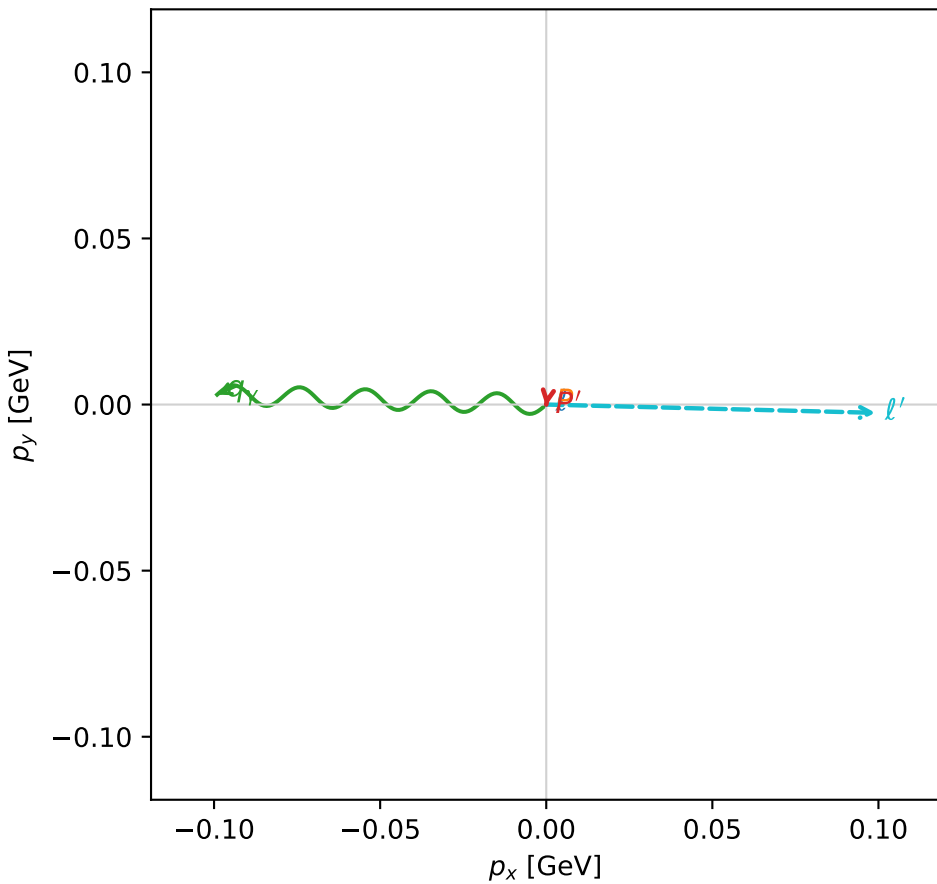
$s=1.29195$, $\sqrt{s}=1.13664$, $|\vec{P}|=0.181282$, $|\vec{P}'|=0.000488167$
 $\theta_{\text{out}}=1.5108$, $\phi_{P'}=4.56151$, $\phi_\gamma=3.11154$
 $E_\gamma=0.0992883$, $\theta_\ell=1.63105$, $\phi_{\ell'}=6.25801$
 $Q^2=0.0338517$, $x_B=0.153798$, $t=-0.0325515$, $W^2=1.0661$, $y=0.534098$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E=0.1813$ $p_x=0.0000$ $p_y=0.0000$ $p_z=-0.1813$
 P $E=0.9554$ $p_x=0.0000$ $p_y=0.0000$ $p_z=0.1813$
 ℓ' $E=0.0994$ $p_x=0.0991$ $p_y=-0.0025$ $p_z=-0.0060$
 P' $E=0.9380$ $p_x=-0.0001$ $p_y=-0.0005$ $p_z=0.0000$
 q_γ $E=0.0993$ $p_x=-0.0991$ $p_y=0.0030$ $p_z=0.0060$

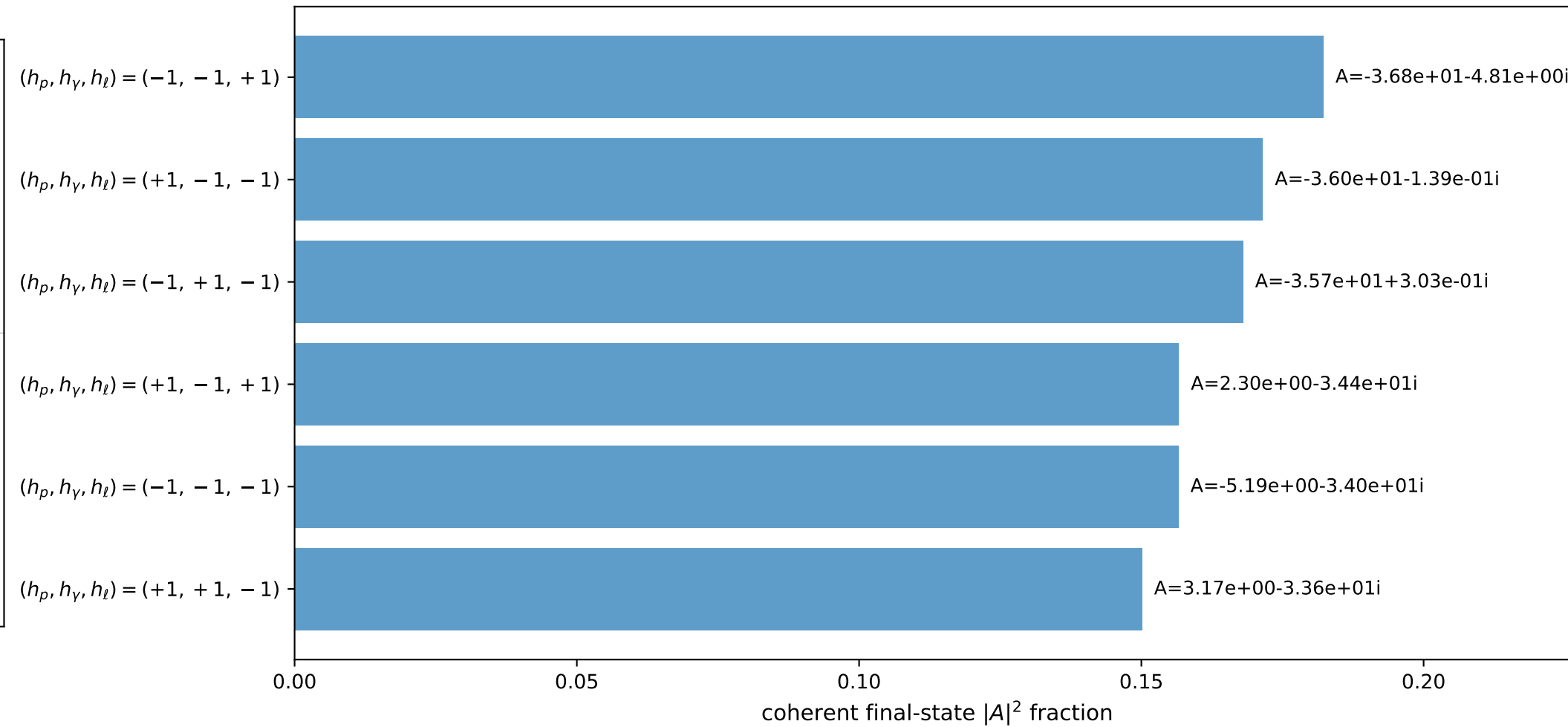
3D momenta



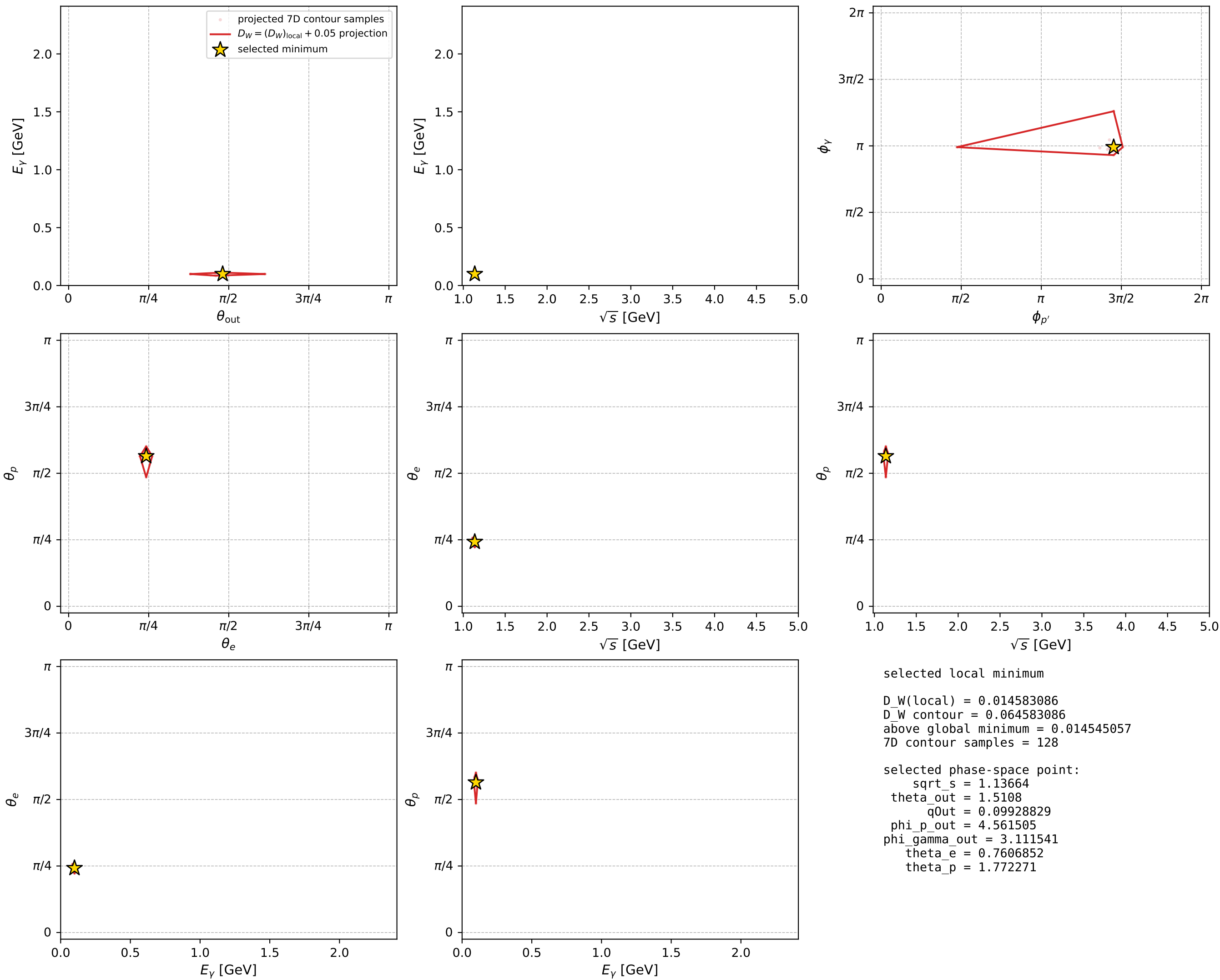
Transverse plane



Leading final-state amplitudes (N=6, retained=98.5%)



electron: polarization cluster P3, local minimum 649; pairwise projections of the 7D $D_W = (D_W)_{\text{local}} + 0.05$ contour

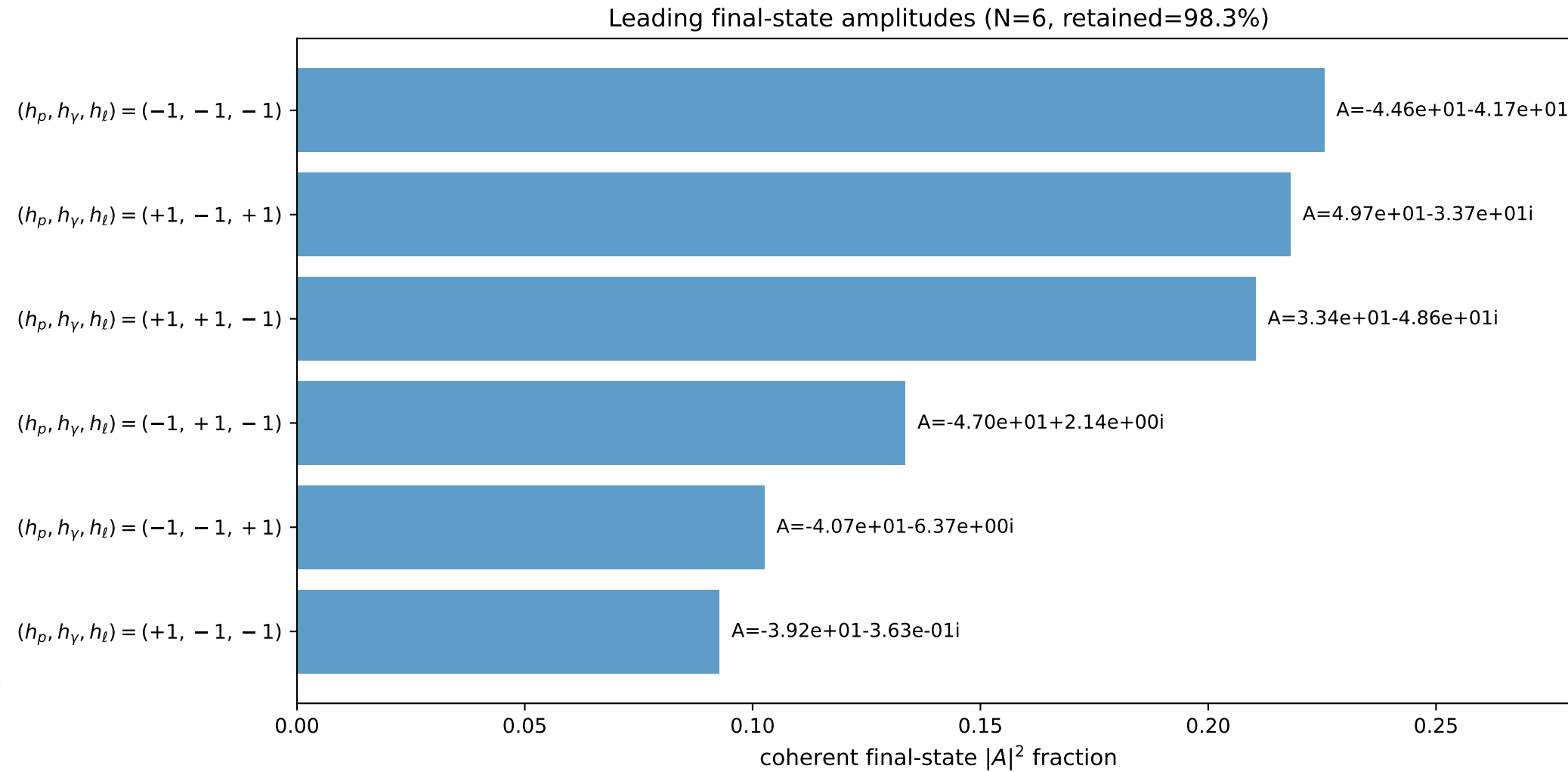
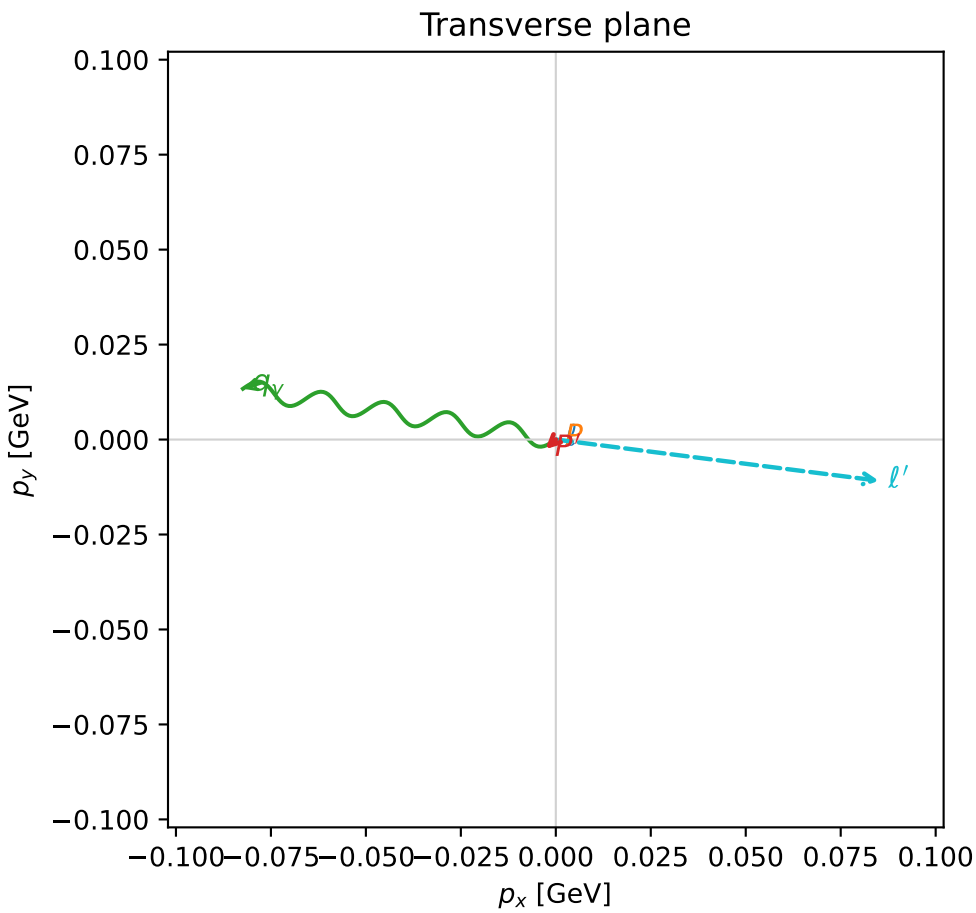
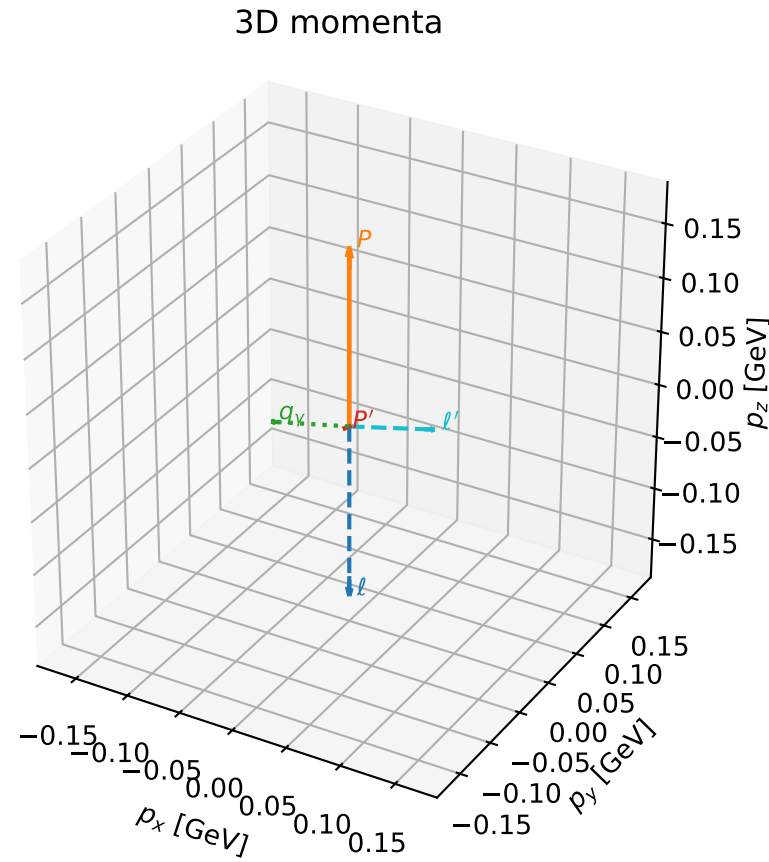


coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

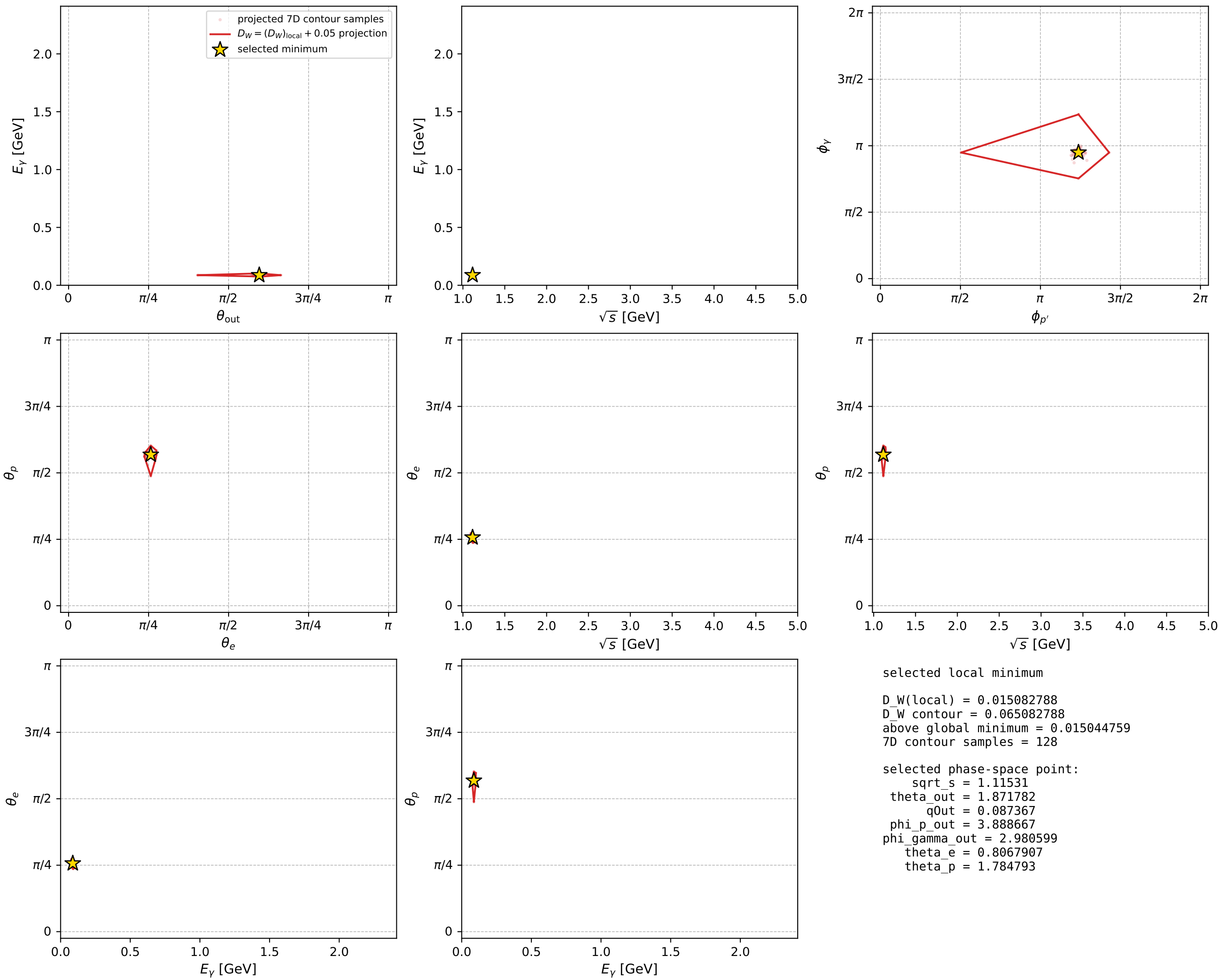
dw_mixing_angles_polarization_03_local_minimum_655 D_W=0.0150828
region: polarization_cluster_03_local_minimum_655
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\theta_e=0.80679075$ rad
incoming lepton: $0.691819|+\rangle + 0.722071|-\rangle$
proton mixing angle: $\theta_p=1.7847932$ rad
incoming proton: $-0.212367|+\rangle + 0.97719|-\rangle$

$s=1.24392$, $\sqrt{s}=1.11531$, $|\vec{P}|=0.163215$, $|\vec{P}'|=0.00387916$
 $\theta_{\text{out}}=1.87178$, $\phi_{P'}=3.88867$, $\phi_Y=2.9806$
 $E_Y=0.087367$, $\theta_{\ell'}=1.26528$, $\phi_{\ell'}=6.15625$
 $Q^2=0.0381873$, $x_B=0.189375$, $t=-0.0268311$, $W^2=1.04331$, $y=0.553872$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1632$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1632$
 P $E= 0.9521$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1632$
 ℓ' $E= 0.0899$ $p_x= 0.0851$ $p_y= -0.0109$ $p_z= 0.0271$
 P' $E= 0.9380$ $p_x= -0.0027$ $p_y= -0.0025$ $p_z= -0.0012$
 q_Y $E= 0.0874$ $p_x= -0.0824$ $p_y= 0.0134$ $p_z= -0.0259$



electron: polarization cluster P3, local minimum 655; pairwise projections of the 7D $D_W = (D_W)_{\text{local}} + 0.05$ contour



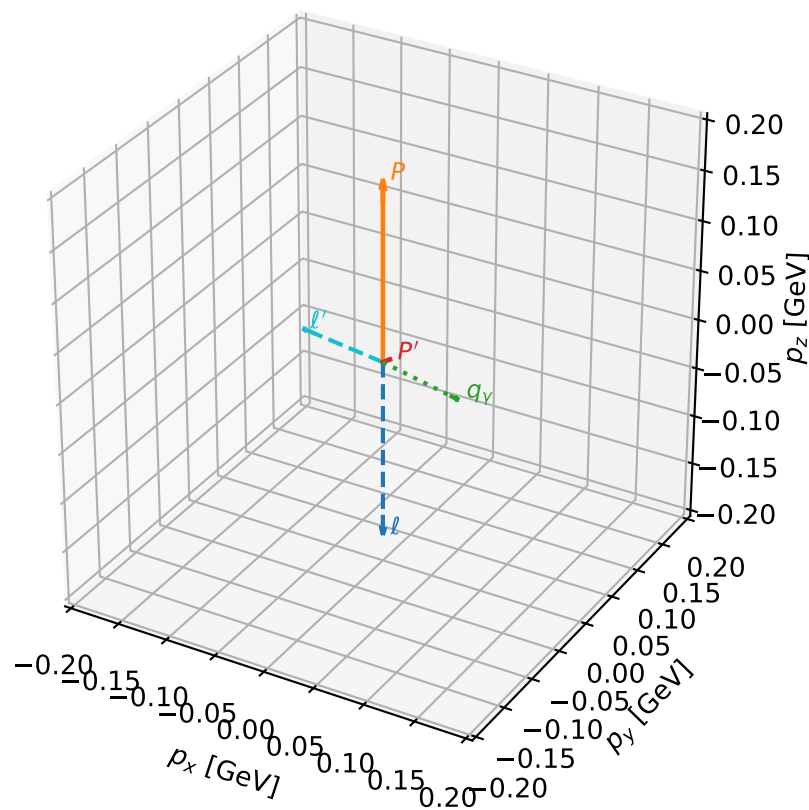
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

dw_mixing_angles_polarization_03_local_minimum_662 D_W=0.0166499
 region: polarization_cluster_03_local_minimum_662
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\theta_e=0.77371069$ rad
 incoming lepton: $0.715323|+\rangle + 0.698794|-\rangle$
 proton mixing angle: $\theta_p=1.3257751$ rad
 incoming proton: $0.242577|+\rangle + 0.970132|-\rangle$

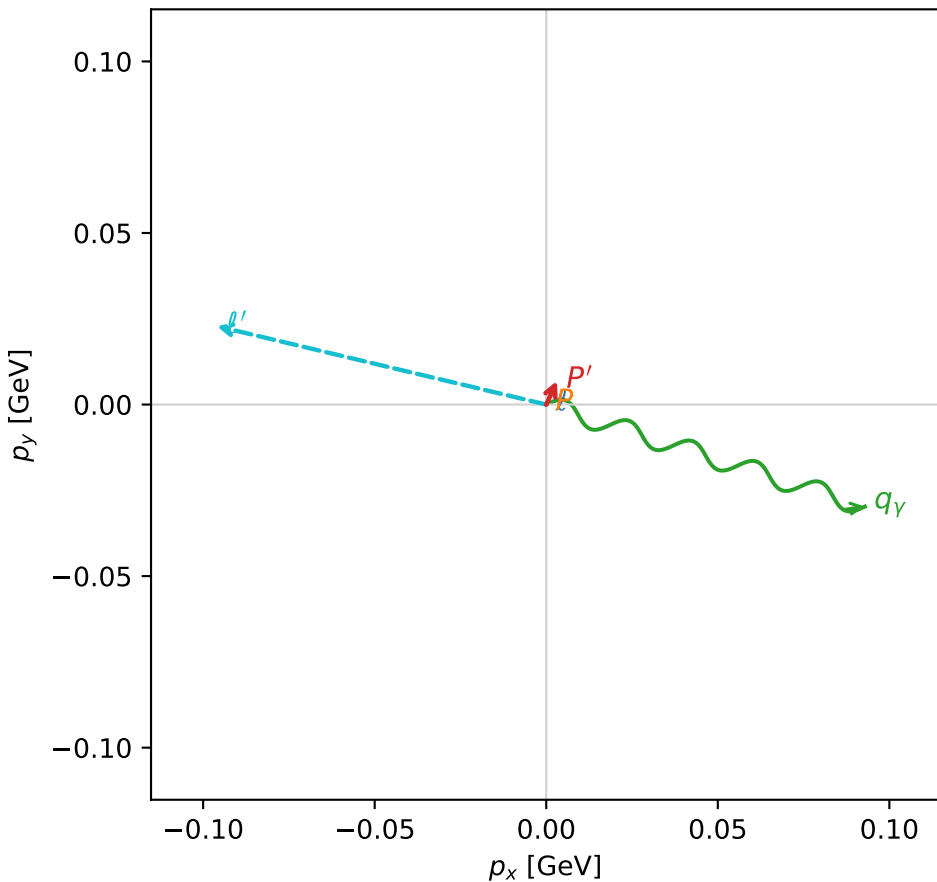
$s=1.28678$, $\sqrt{s}=1.13436$, $|\vec{P}|=0.179368$, $|\vec{P}'|=0.00758198$
 $\theta_{\text{out}}=1.52994$, $\phi_{P'}=1.14412$, $\phi_Y=5.97331$
 $E_Y=0.0975753$, $\theta_{\ell'}=1.6143$, $\phi_{\ell'}=2.90805$
 $Q^2=0.0338871$, $x_B=0.156328$, $t=-0.0318315$, $W^2=1.06273$, $y=0.532684$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1794$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1794$
 P $E= 0.9550$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1794$
 ℓ' $E= 0.0988$ $p_x= -0.0960$ $p_y= 0.0228$ $p_z= -0.0043$
 P' $E= 0.9380$ $p_x= 0.0031$ $p_y= 0.0069$ $p_z= 0.0003$
 q_Y $E= 0.0976$ $p_x= 0.0929$ $p_y= -0.0297$ $p_z= 0.0040$

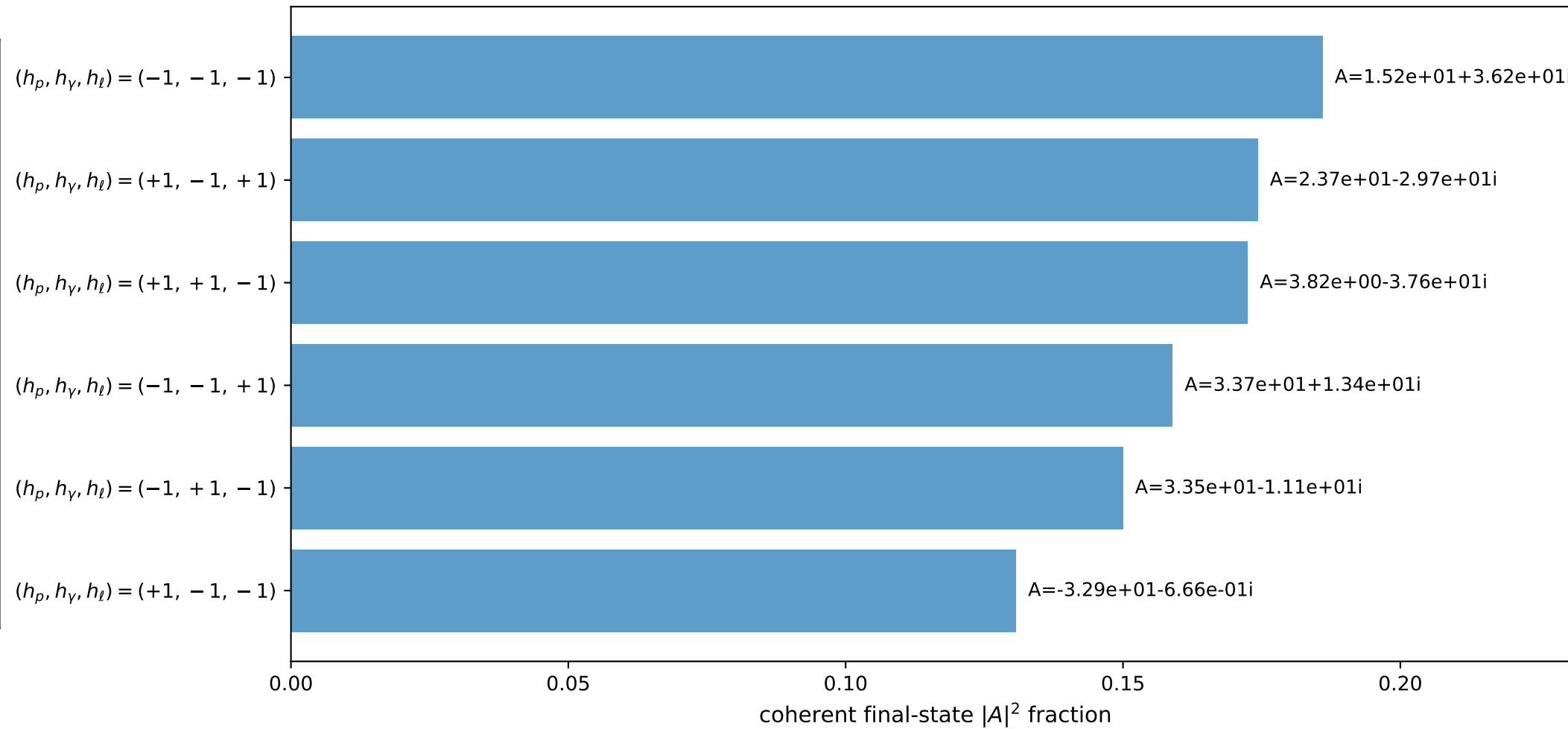
3D momenta



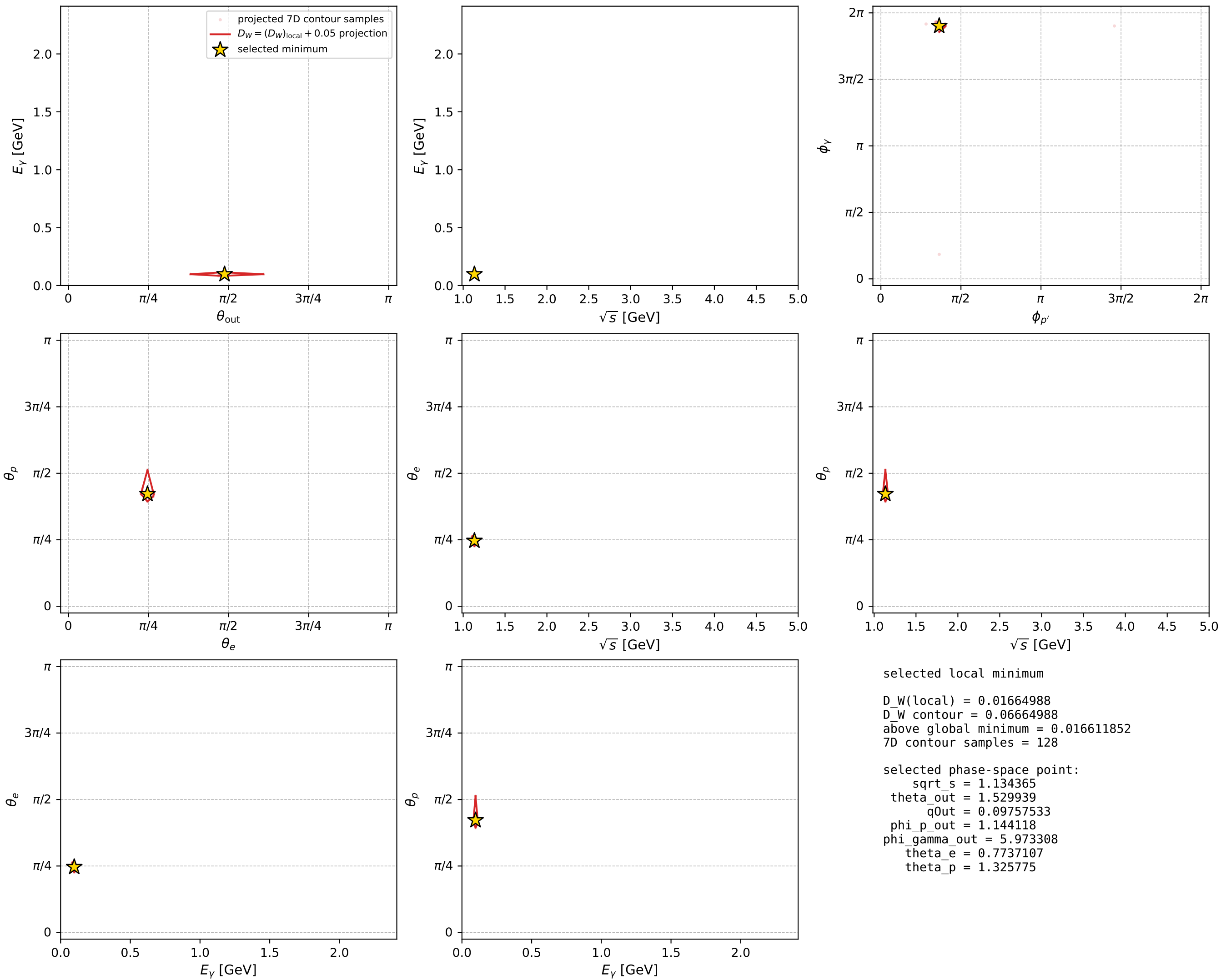
Transverse plane



Leading final-state amplitudes (N=6, retained=97.2%)



electron: polarization cluster P3, local minimum 662; pairwise projections of the 7D $D_W = (D_W)_{\text{local}} + 0.05$ contour

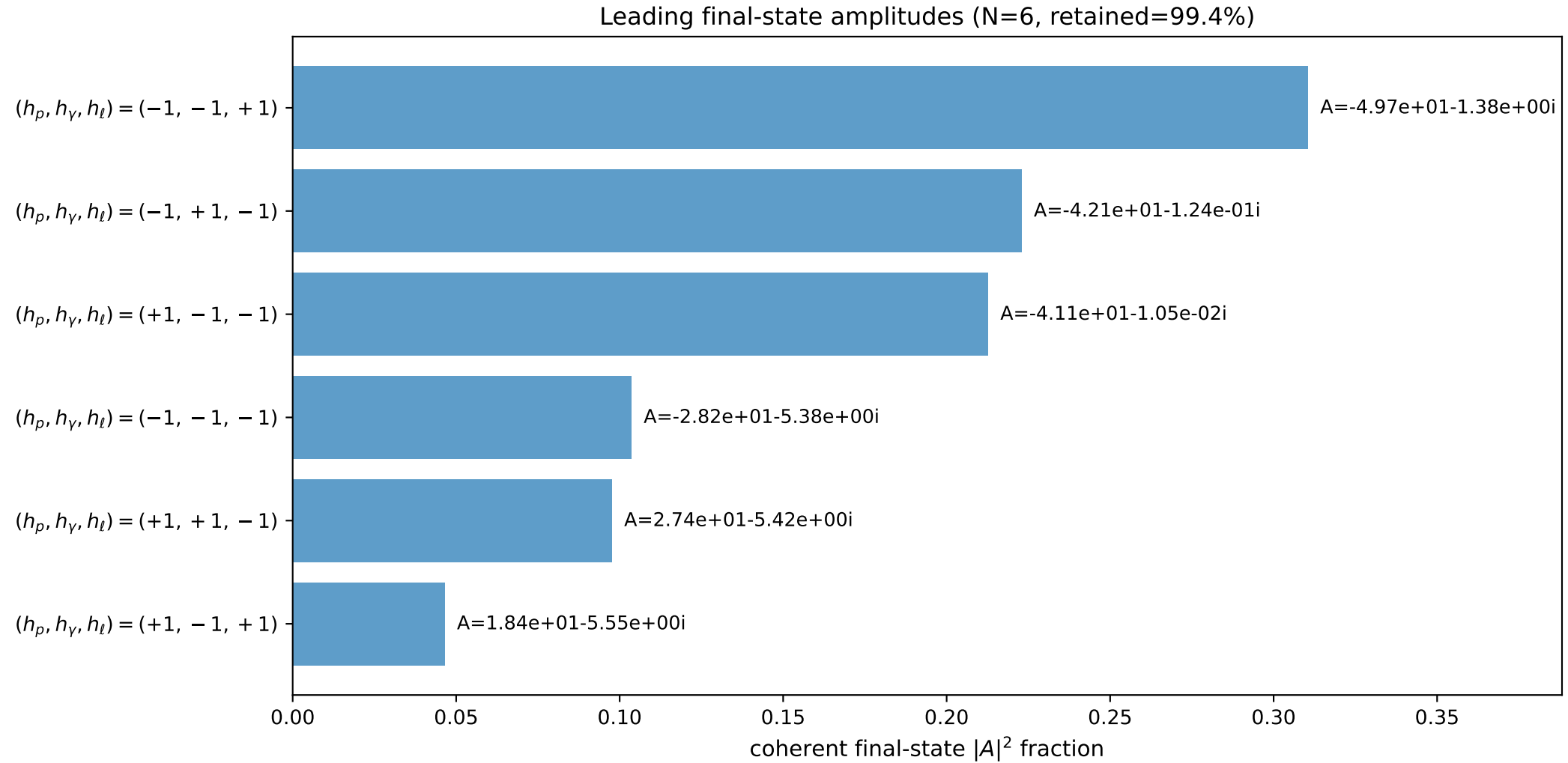
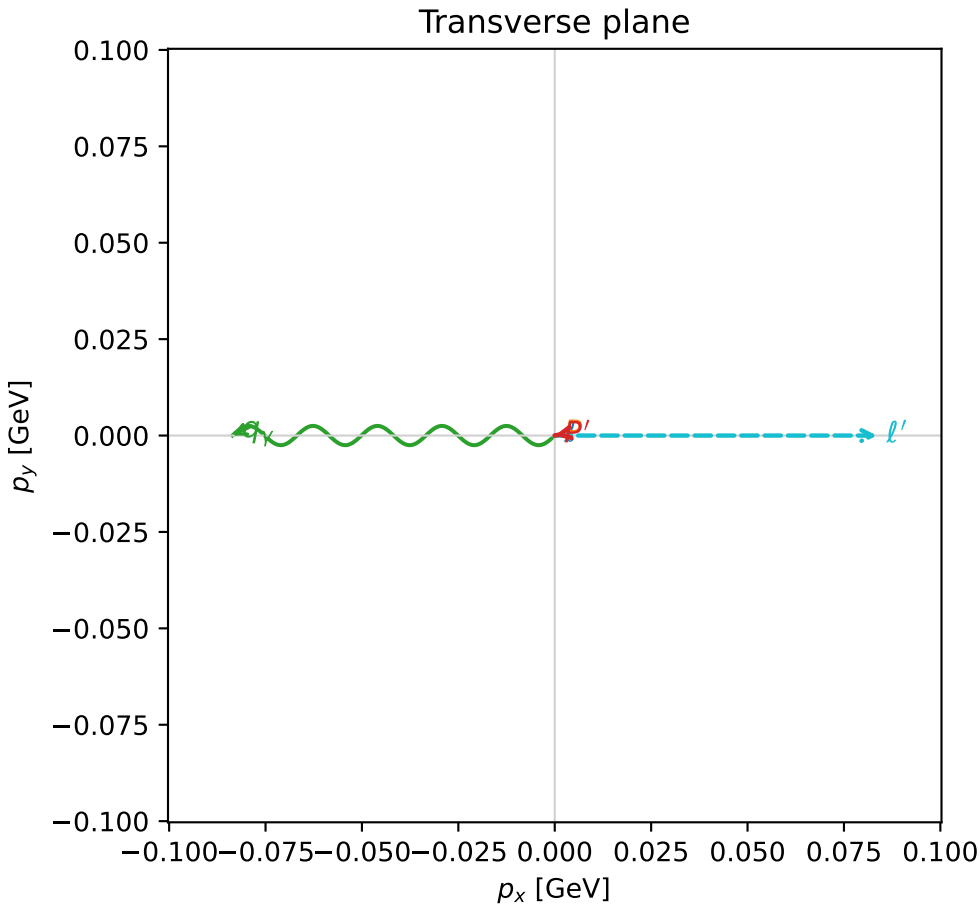
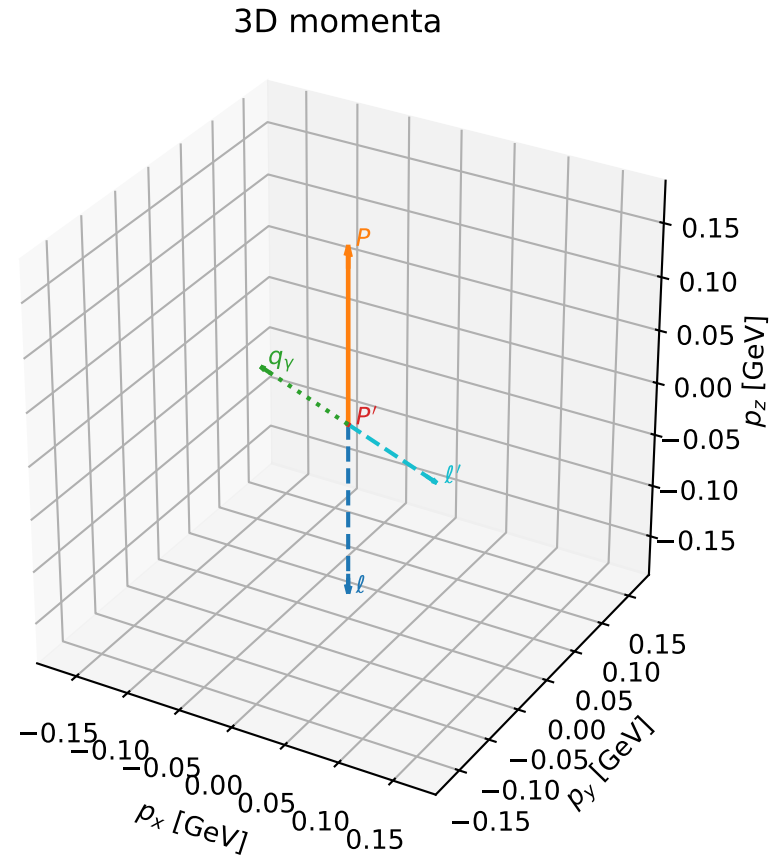


coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

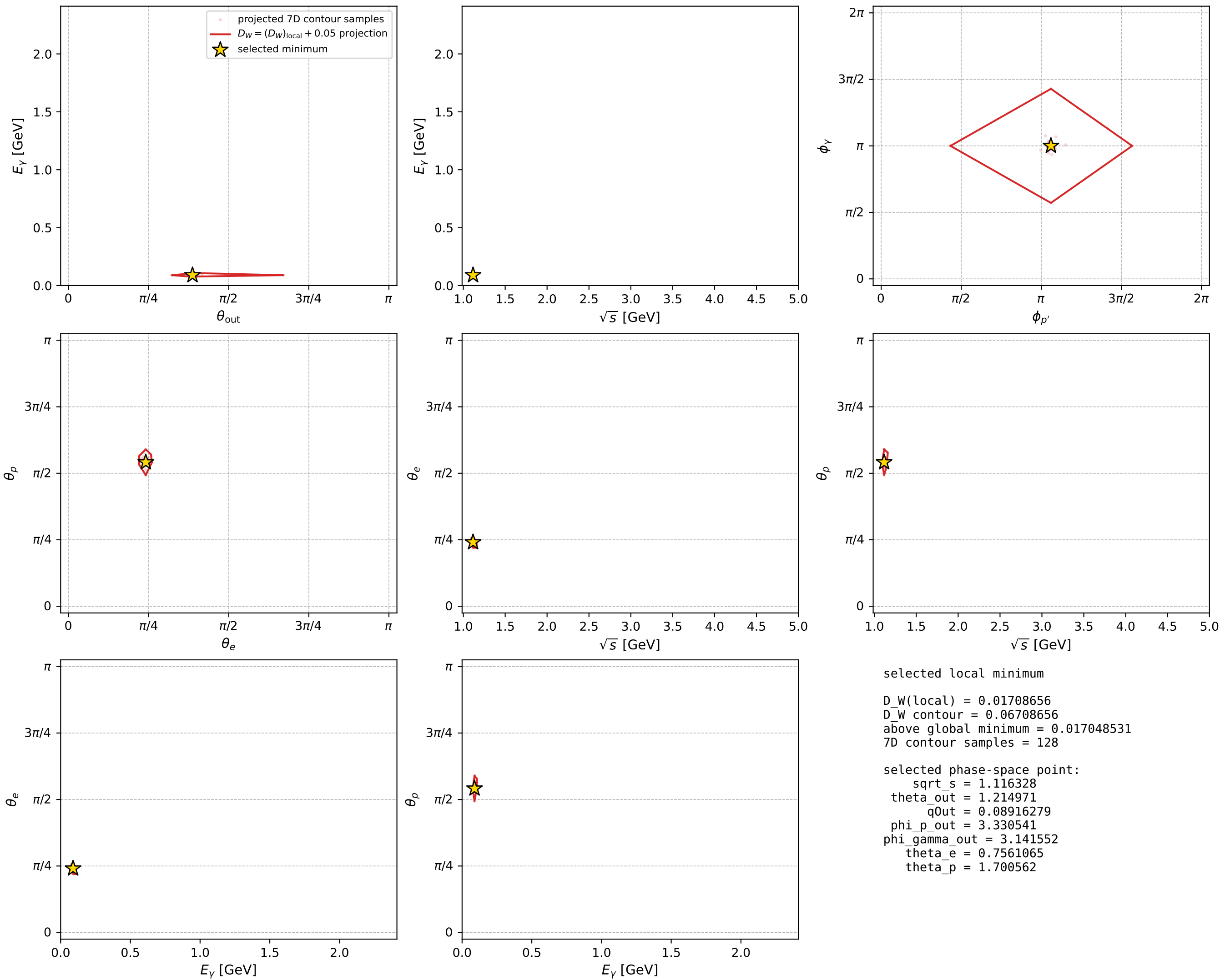
dw_mixing_angles_polarization_03_local_minimum_668 D_W=0.0170866
 region: polarization_cluster_03_local_minimum_668
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\theta_e=0.75610652$ rad
 incoming lepton: $0.727513|+\rangle + 0.686094|-\rangle$
 proton mixing angle: $\theta_p=1.7005618$ rad
 incoming proton: $-0.129402|+\rangle + 0.991592|-\rangle$

$s=1.24619$, $\sqrt{s}=1.11633$, $|\vec{P}|=0.164083$, $|\vec{P}'|=5.49179\text{e-}07$
 $\theta_{\text{out}}=1.21497$, $\phi_{p'}=3.33054$, $\phi_Y=3.14155$
 $E_Y=0.0891628$, $\theta_{\ell'}=1.92662$, $\phi_{\ell'}=6.28315$
 $Q^2=0.0190673$, $x_B=0.102327$, $t=-0.0267204$, $W^2=1.04711$, $y=0.50864$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1641$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1641$
 P $E= 0.9522$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1641$
 ℓ' $E= 0.0892$ $p_x= 0.0836$ $p_y= -0.0000$ $p_z= -0.0311$
 P' $E= 0.9380$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 0.0000$
 q_Y $E= 0.0892$ $p_x= -0.0836$ $p_y= 0.0000$ $p_z= 0.0311$



electron: polarization cluster P3, local minimum 668; pairwise projections of the 7D $D_W = (D_W)_{\text{local}} + 0.05$ contour

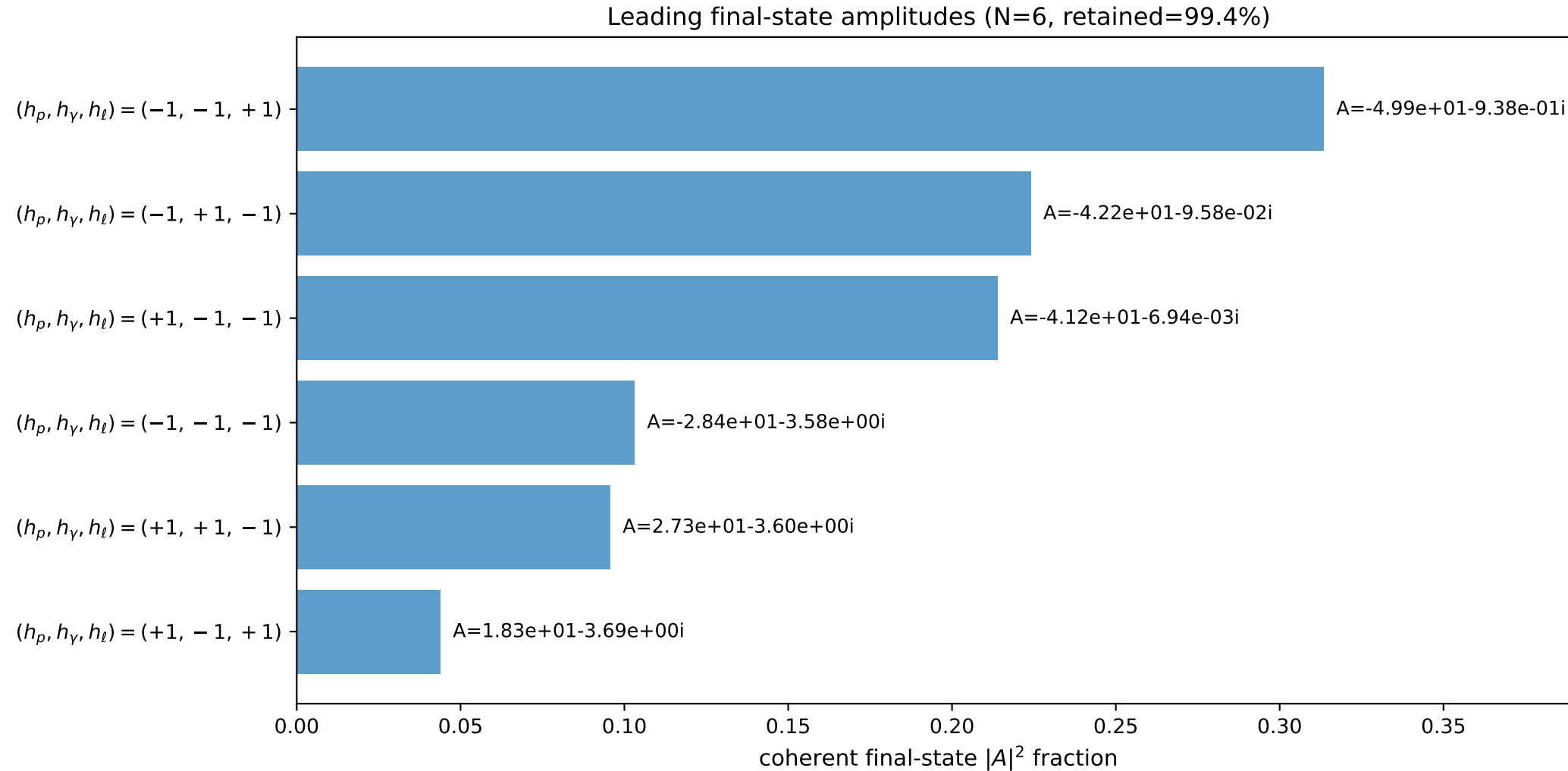
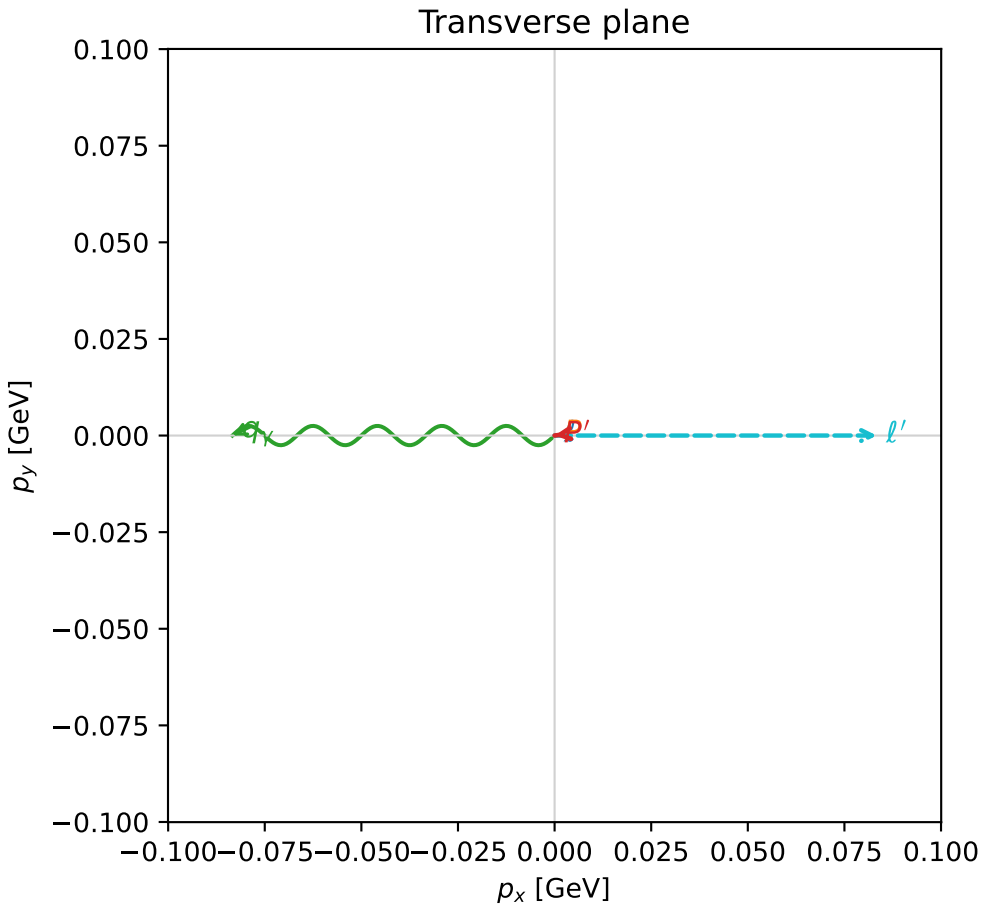
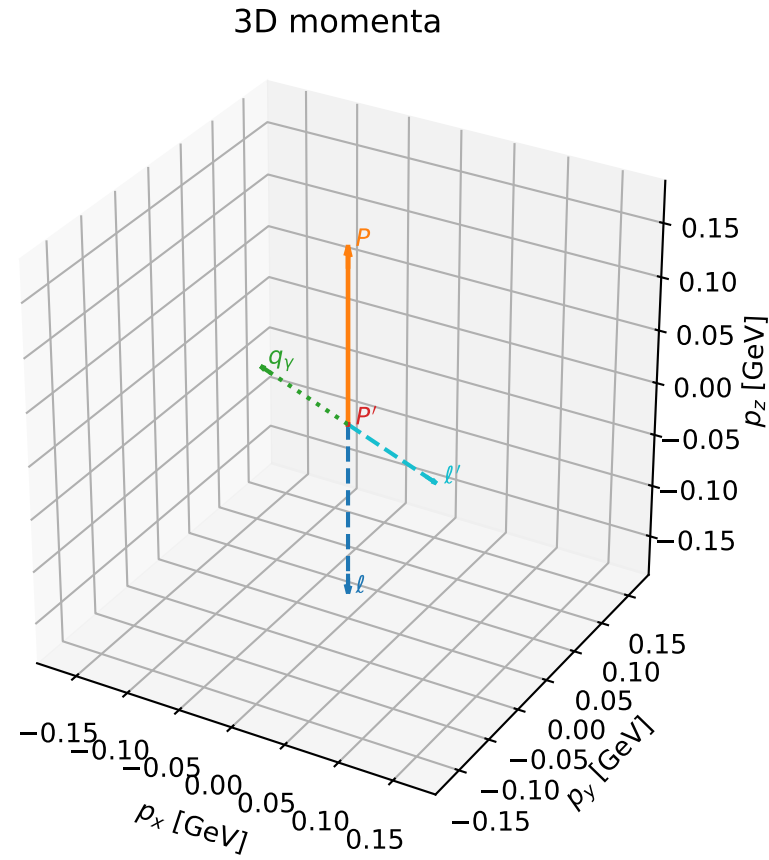


coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

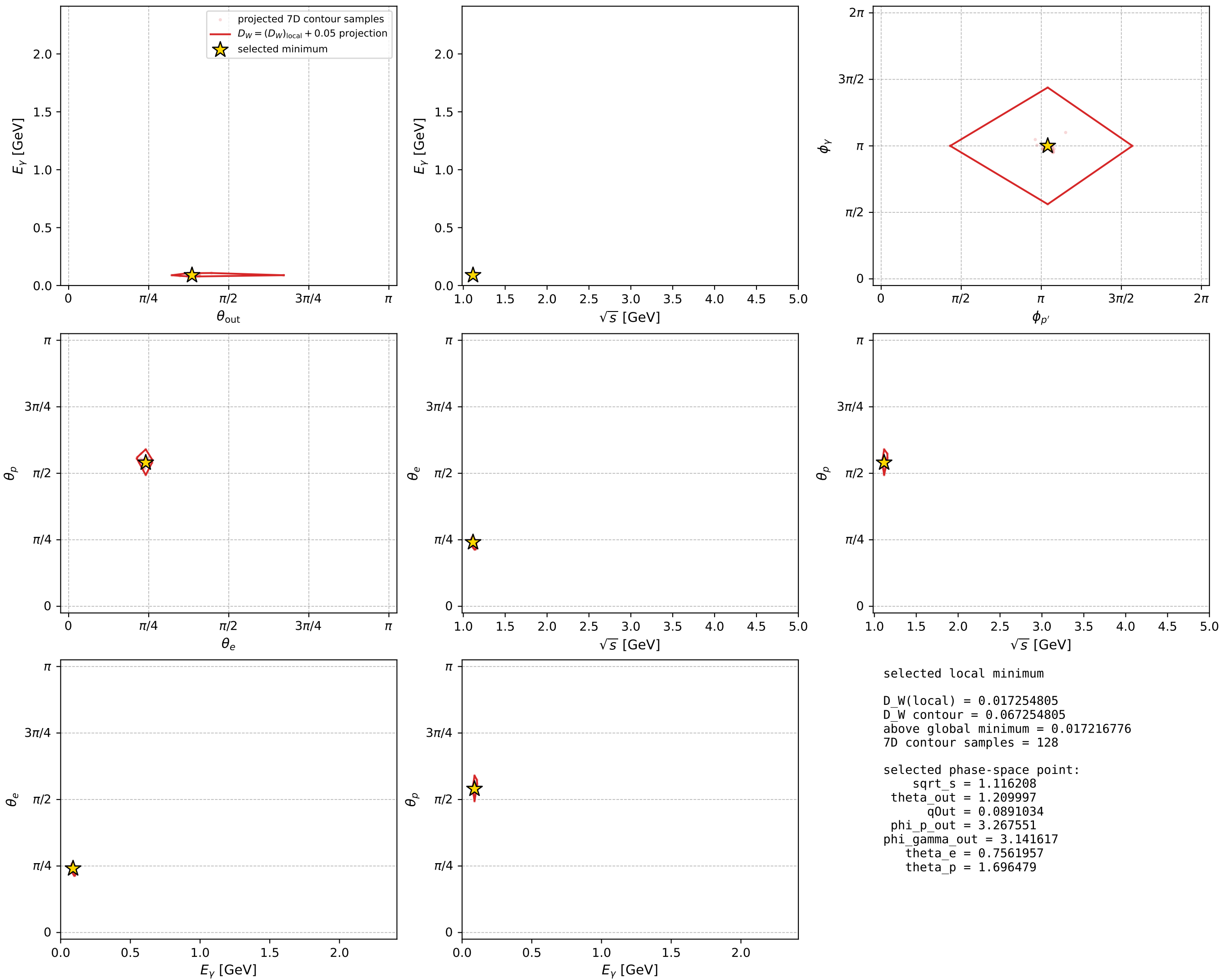
dw_mixing_angles_polarization_03_local_minimum_669 D_W=0.0172548
 region: polarization_cluster_03_local_minimum_669
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\theta_e=0.75619571$ rad
 incoming lepton: $0.727452|+\rangle + 0.686159|-\rangle$
 proton mixing angle: $\theta_p=1.6964792$ rad
 incoming proton: $-0.125352|+\rangle + 0.992112|-\rangle$

$s=1.24592$, $\sqrt{s}=1.11621$, $|\vec{P}|=0.163982$, $|\vec{P}'|=2.43178e-08$
 $\theta_{\text{out}}=1.21$, $\phi_{p'}=3.26755$, $\phi_\gamma=3.14162$
 $E_\gamma=0.0891034$, $\theta_{\ell'}=1.9316$, $\phi_{\ell'}=2.47966e-05$
 $Q^2=0.0189065$, $x_B=0.101613$, $t=-0.0266876$, $W^2=1.047$, $y=0.508266$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1640$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1640$
 P $E= 0.9522$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1640$
 ℓ' $E= 0.0891$ $p_x= 0.0834$ $p_y= 0.0000$ $p_z= -0.0315$
 P' $E= 0.9380$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 0.0000$
 q_γ $E= 0.0891$ $p_x= -0.0834$ $p_y= -0.0000$ $p_z= 0.0315$



electron: polarization cluster P3, local minimum 669; pairwise projections of the 7D $D_W = (D_W)_{\text{local}} + 0.05$ contour

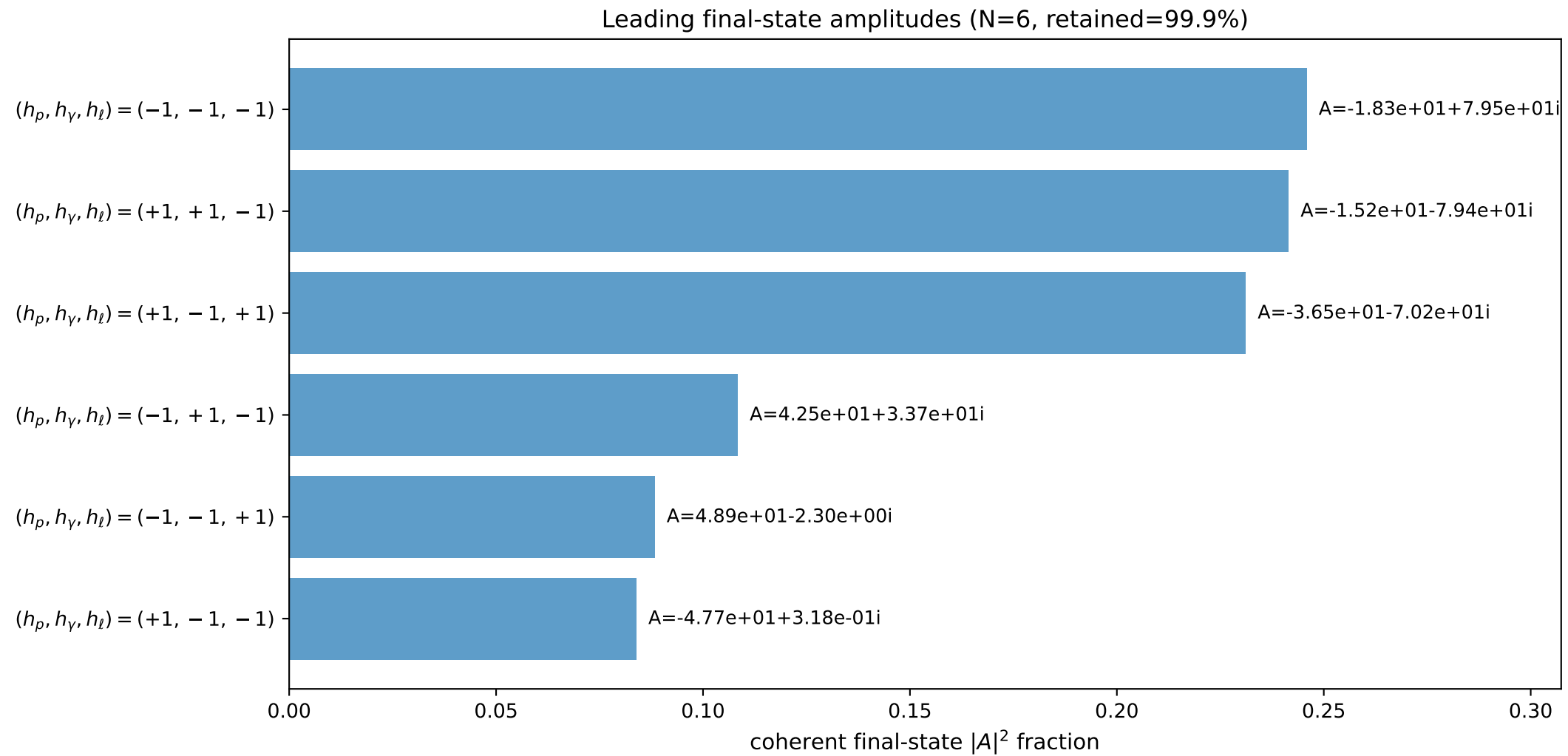
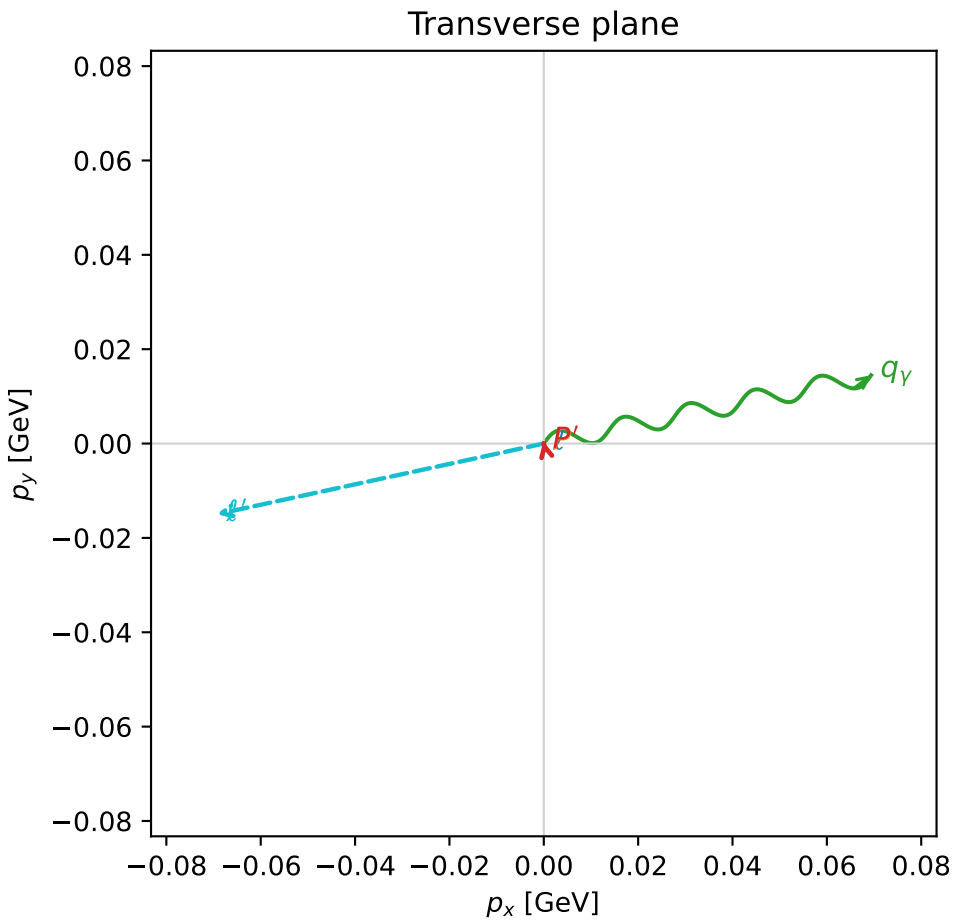
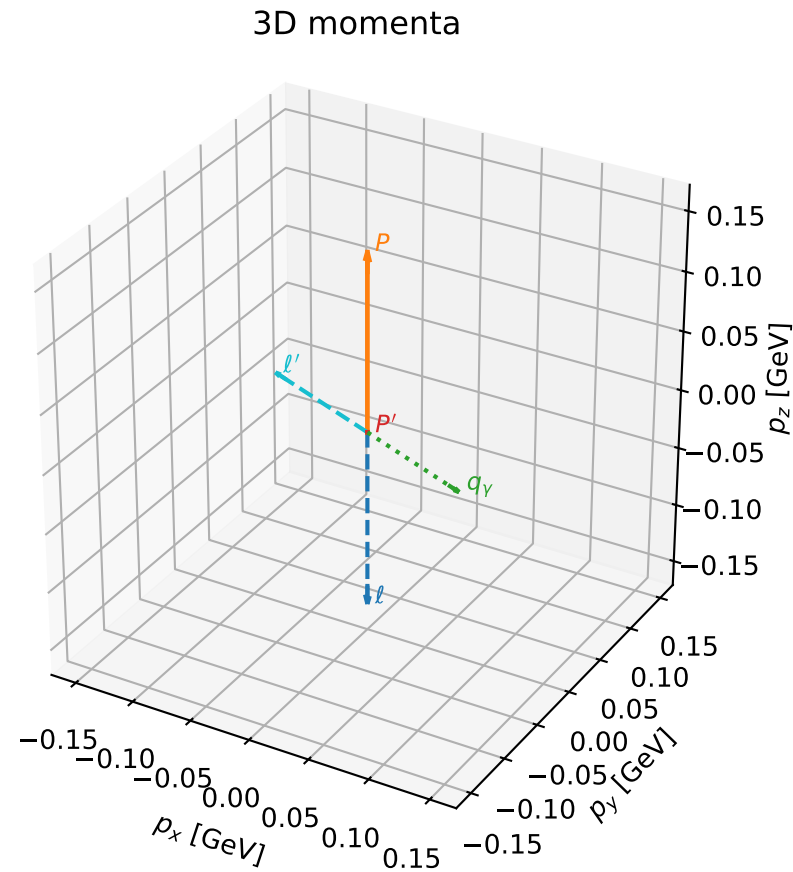


coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

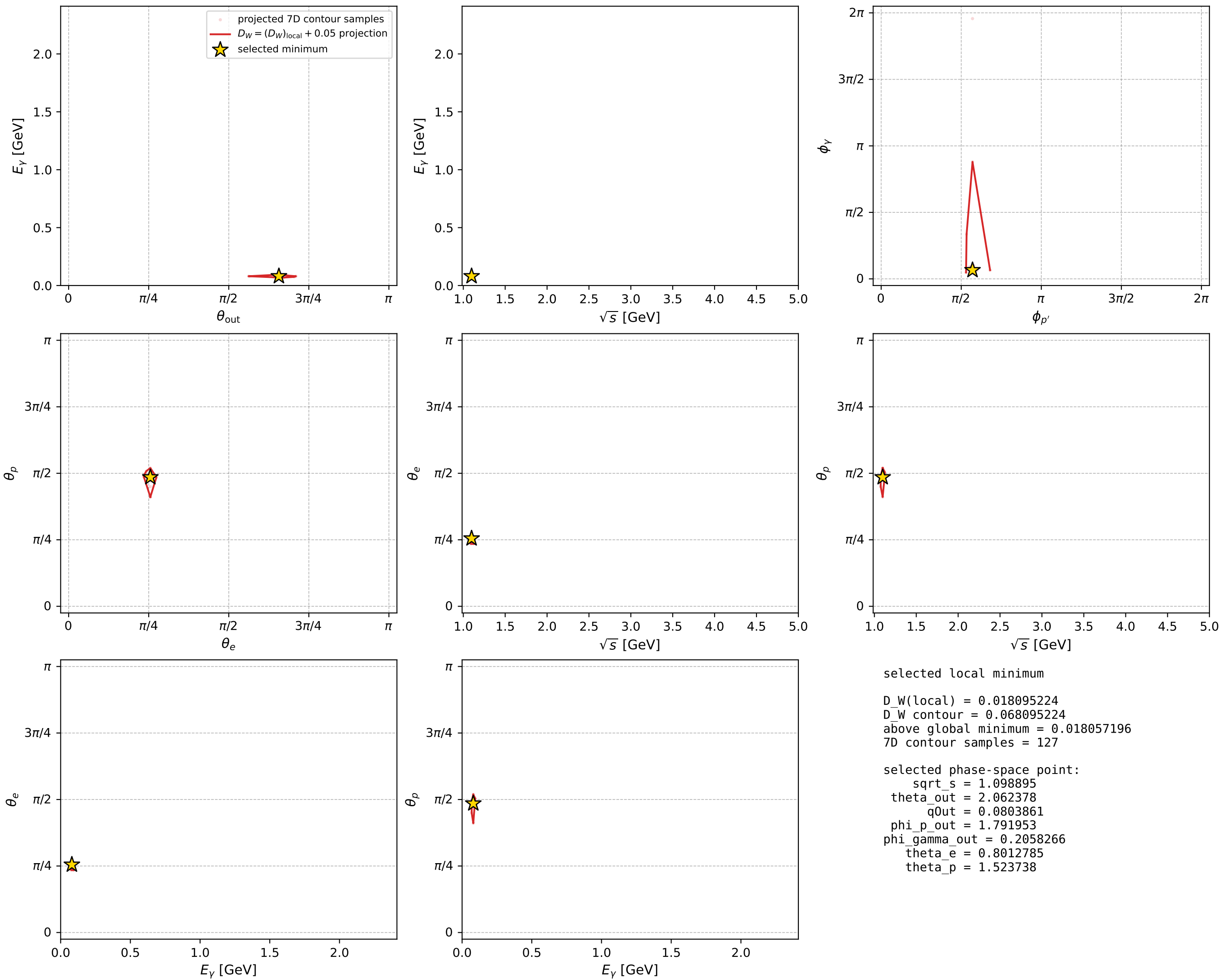
dw_mixing_angles_polarization_03_local_minimum_680 D_W=0.0180952
region: polarization_cluster_03_local_minimum_680
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\theta_e=0.80127846$ rad
incoming lepton: $0.695789|+\rangle + 0.718246|-\rangle$
proton mixing angle: $\theta_p=1.5237376$ rad
incoming proton: $0.0470414|+\rangle + 0.998893|-\rangle$

$s=1.20757$, $\sqrt{s}=1.0989$, $|\vec{P}|=0.149116$, $|\vec{P}'|=0.000566423$
 $\theta_{\text{out}}=2.06238$, $\phi_{P'}=1.79195$, $\phi_Y=0.205827$
 $E_Y=0.0803861$, $\theta_{l'}=1.07625$, $\phi_{l'}=3.35447$
 $Q^2=0.0354058$, $x_B=0.190158$, $t=-0.0221768$, $W^2=1.03063$, $y=0.568129$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 0.1491$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1491$
 P $E= 0.9498$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1491$
 l' $E= 0.0805$ $p_x= -0.0693$ $p_y= -0.0150$ $p_z= 0.0382$
 P' $E= 0.9380$ $p_x= -0.0001$ $p_y= 0.0005$ $p_z= -0.0003$
 q_Y $E= 0.0804$ $p_x= 0.0694$ $p_y= 0.0145$ $p_z= -0.0379$



electron: polarization cluster P3, local minimum 680; pairwise projections of the 7D $D_W = (D_W)_{\text{local}} + 0.05$ contour

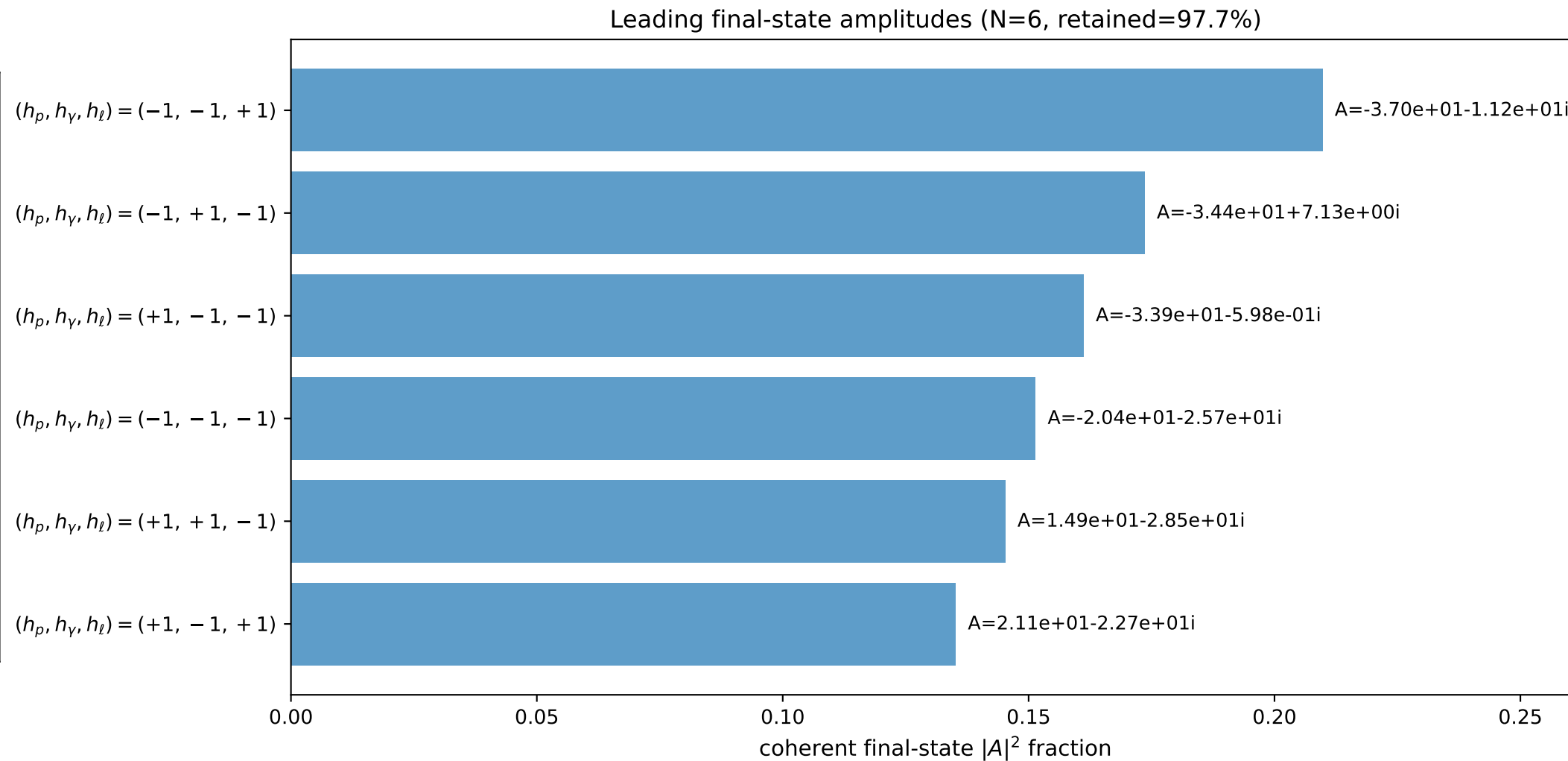
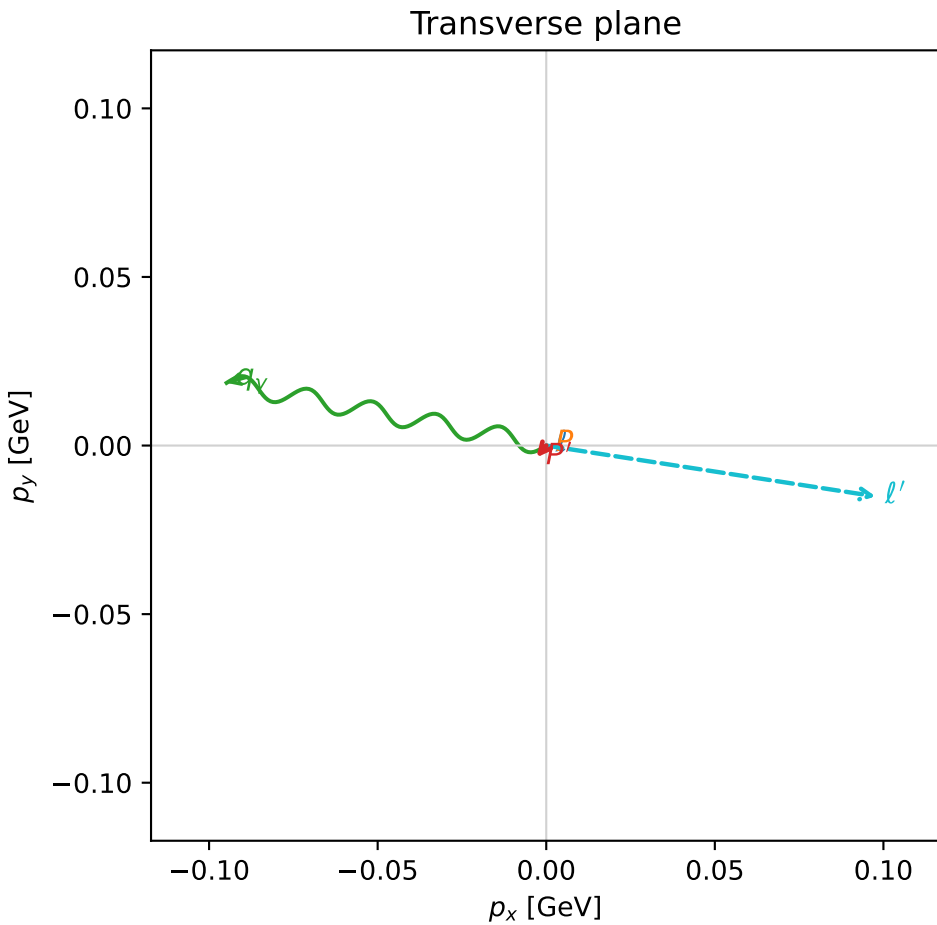
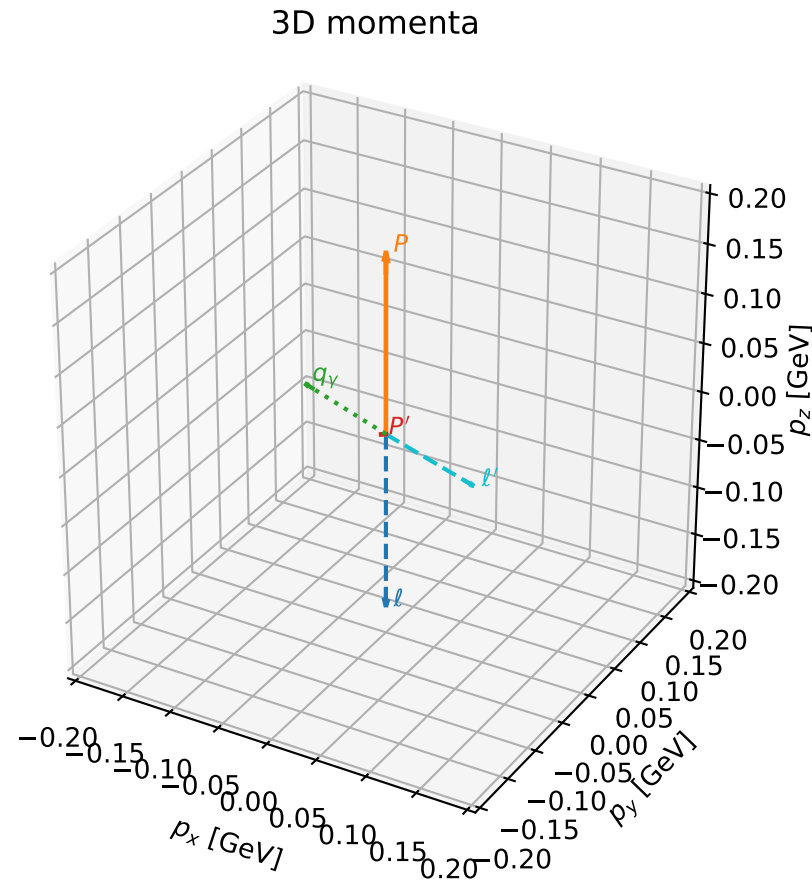


coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

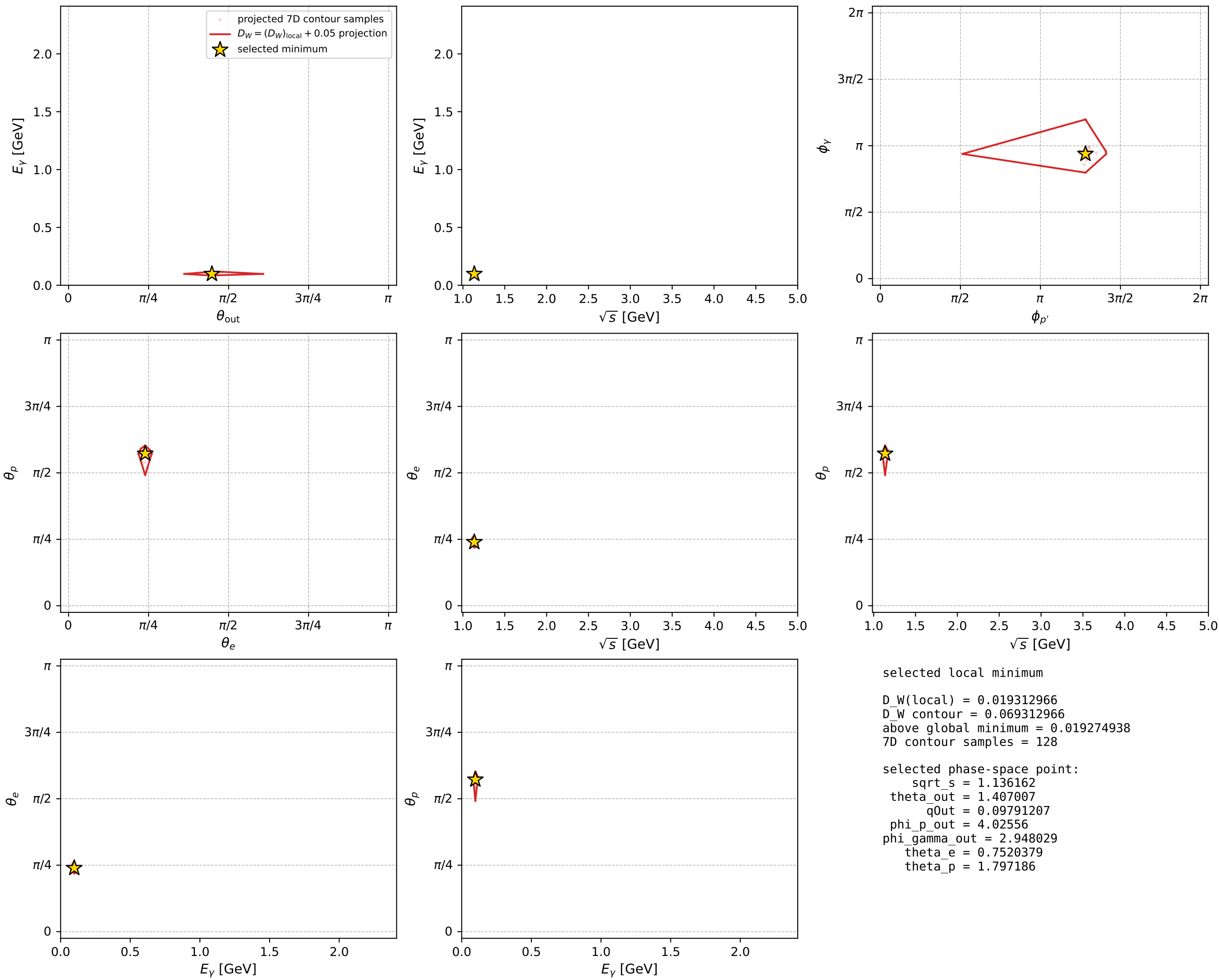
dw_mixing_angles_polarization_03_local_minimum_686 D_W=0.019313
 region: polarization_cluster_03_local_minimum_686
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\theta_e=0.75203785$ rad
 incoming lepton: $0.730298|+\rangle + 0.683128|-\rangle$
 proton mixing angle: $\theta_p=1.7971858$ rad
 incoming proton: $-0.224461|+\rangle + 0.974483|-\rangle$

$s=1.29086$, $\sqrt{s}=1.13616$, $|\vec{P}|=0.18088$, $|\vec{P}'|=0.00460362$
 $\theta_{\text{out}}=1.40701$, $\phi_{P'}=4.02556$, $\phi_Y=2.94803$
 $E_Y=0.0979121$, $\theta_{\ell'}=1.73834$, $\phi_{\ell'}=6.13011$
 $Q^2=0.0302146$, $x_B=0.141547$, $t=-0.032169$, $W^2=1.06309$, $y=0.519344$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1809$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1809$
 P $E= 0.9553$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1809$
 ℓ' $E= 0.1002$ $p_x= 0.0977$ $p_y= -0.0151$ $p_z= -0.0167$
 P' $E= 0.9380$ $p_x= -0.0029$ $p_y= -0.0035$ $p_z= 0.0008$
 q_Y $E= 0.0979$ $p_x= -0.0948$ $p_y= 0.0186$ $p_z= 0.0160$



electron: polarization cluster P3, local minimum 686; pairwise projections of the 7D $D_W = (D_W)_{\text{local}} + 0.05$ contour

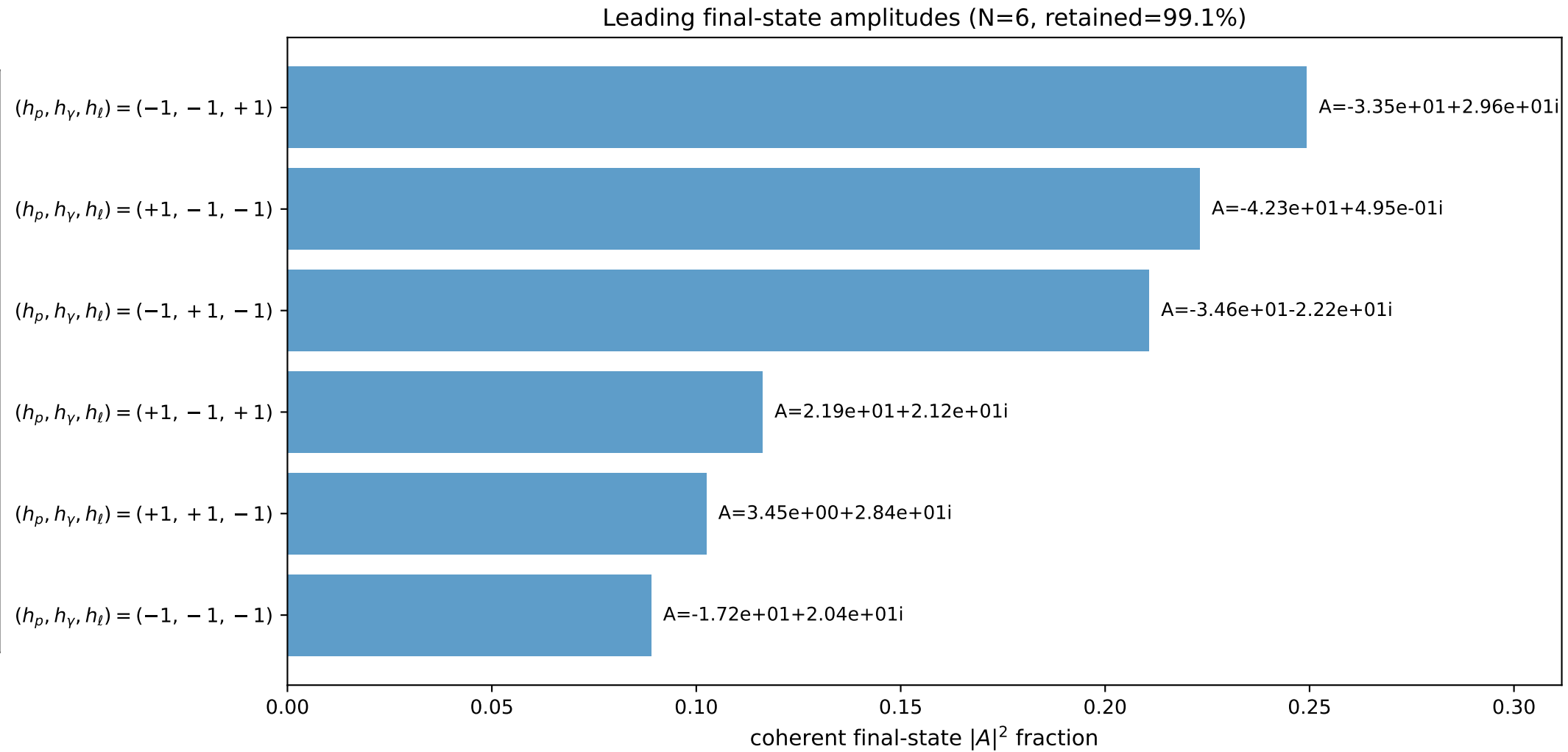
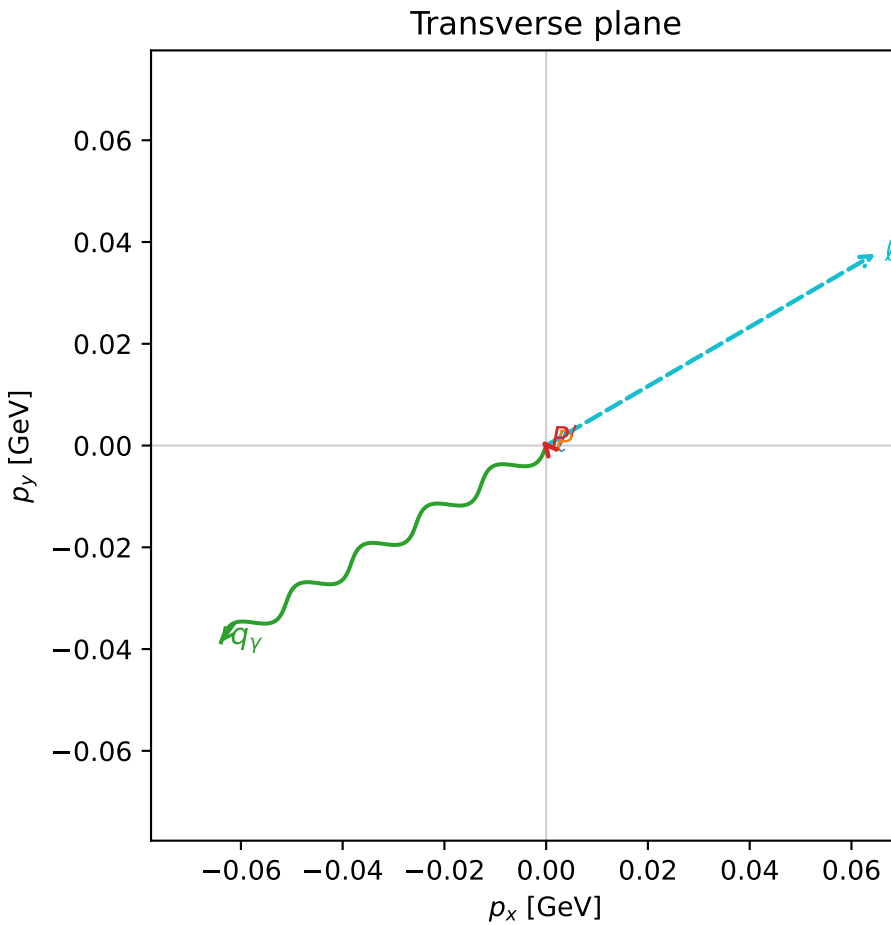
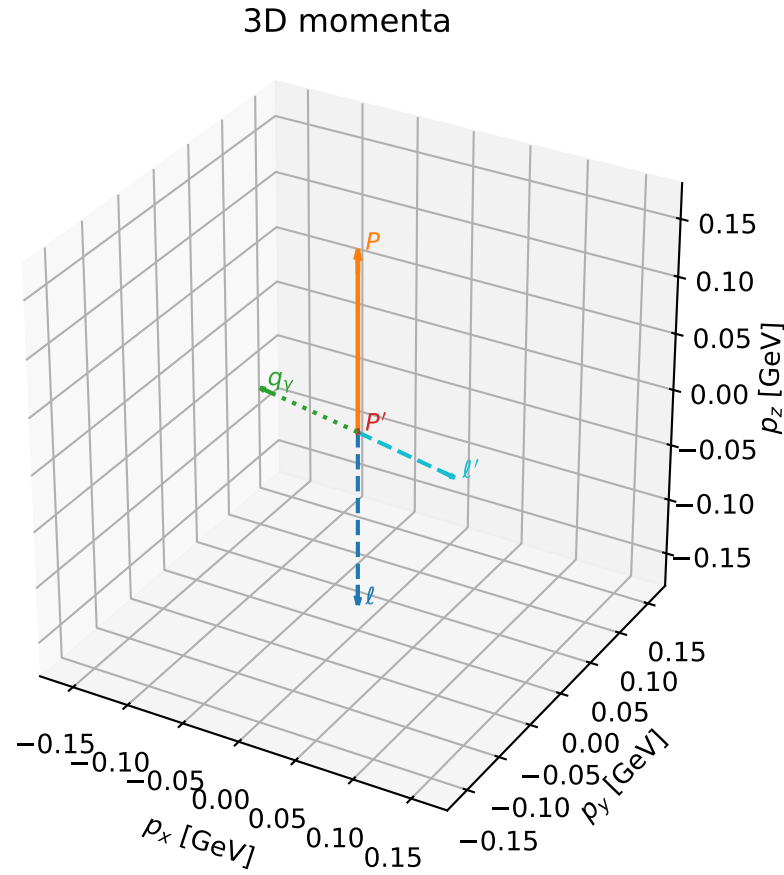


coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

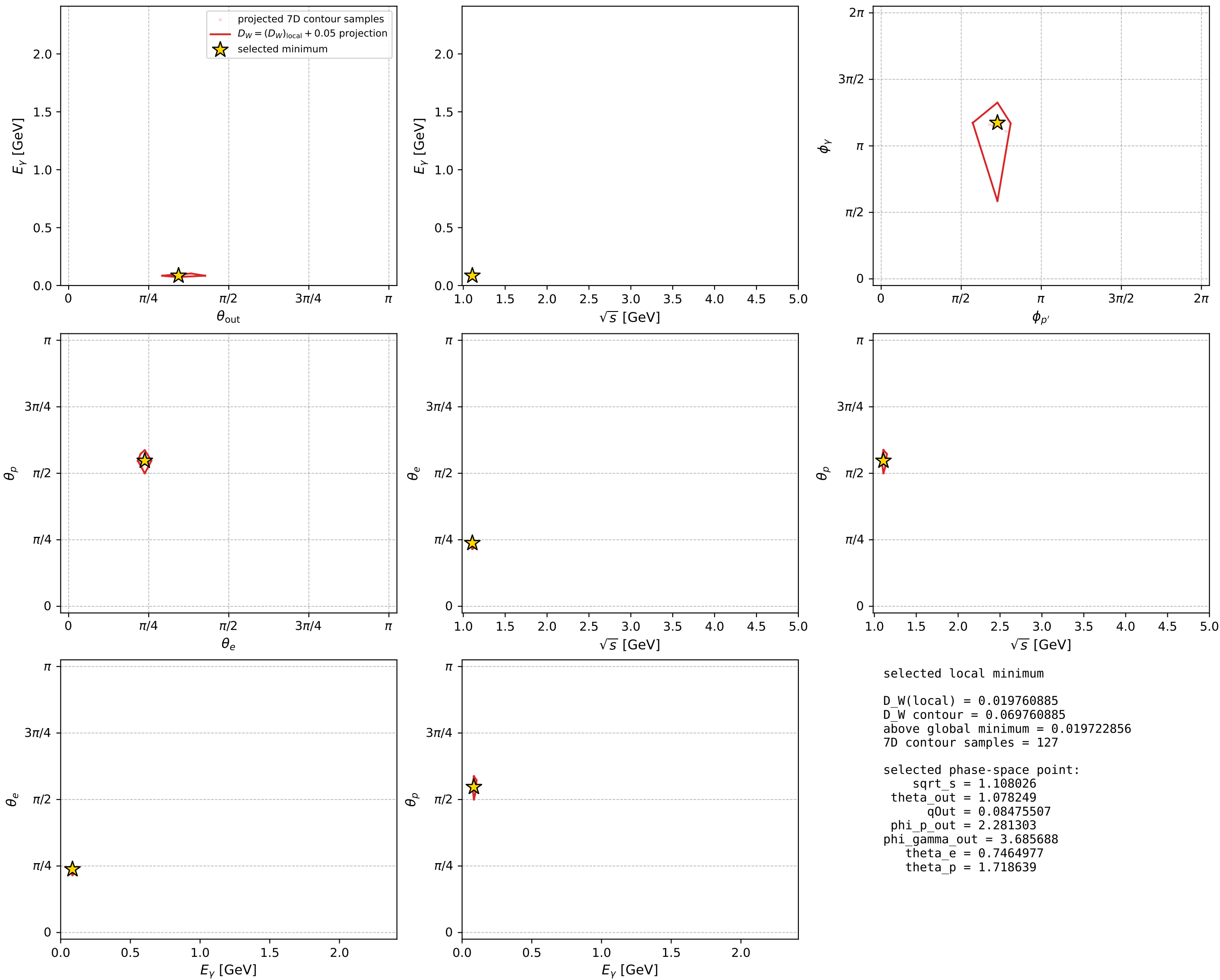
dw_mixing_angles_polarization_03_local_minimum_688 D_W=0.0197609
 region: polarization_cluster_03_local_minimum_688
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\theta_e=0.74649772$ rad
 incoming lepton: $0.734072|+\rangle + 0.679072|-\rangle$
 proton mixing angle: $\theta_p=1.718639$ rad
 incoming proton: $-0.147305|+\rangle + 0.989091|-\rangle$

$s=1.22772$, $\sqrt{s}=1.10803$, $|\vec{P}|=0.15698$, $|\vec{P}'|=0.0014283$
 $\theta_{\text{out}}=1.07825$, $\phi_{P'}=2.2813$, $\phi_Y=3.68569$
 $E_Y=0.0847551$, $\theta_{\ell'}=2.06911$, $\phi_{\ell'}=0.527524$
 $Q^2=0.013976$, $x_B=0.080837$, $t=-0.0242627$, $W^2=1.03876$, $y=0.496989$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1570$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1570$
 P $E= 0.9510$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1570$
 ℓ' $E= 0.0853$ $p_x= 0.0647$ $p_y= 0.0377$ $p_z= -0.0408$
 P' $E= 0.9380$ $p_x= -0.0008$ $p_y= 0.0010$ $p_z= 0.0007$
 q_Y $E= 0.0848$ $p_x= -0.0639$ $p_y= -0.0387$ $p_z= 0.0401$



electron: polarization cluster P3, local minimum 688; pairwise projections of the 7D $D_W = (D_W)_{\text{local}} + 0.05$ contour



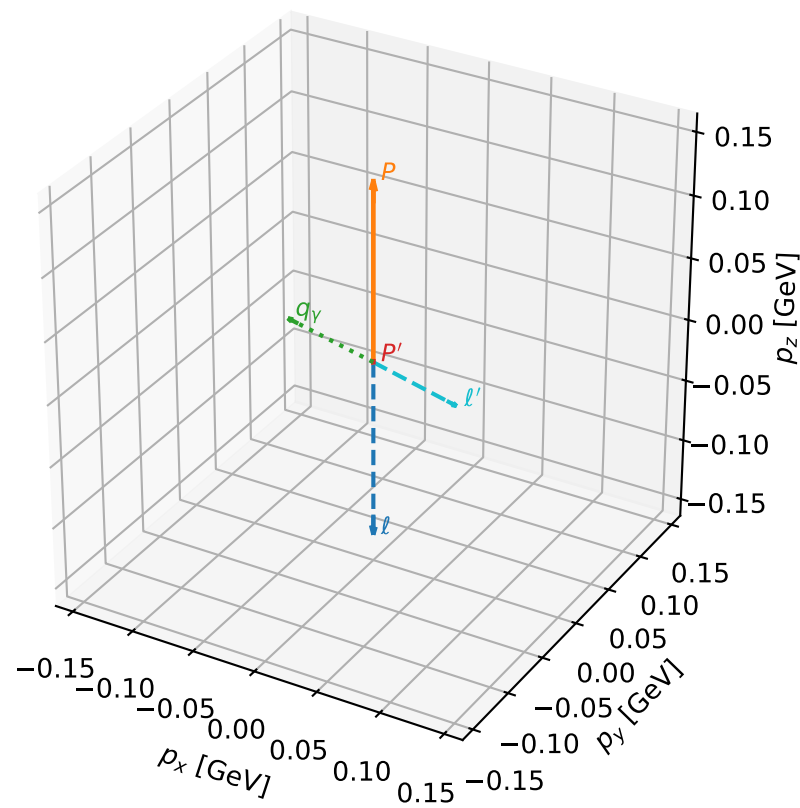
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

dw_mixing_angles_polarization_03_local_minimum_692 D_W=0.0206179
 region: polarization_cluster_03_local_minimum_692
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\theta_e=0.74693041$ rad
 incoming lepton: $0.733778|+\rangle + 0.67939|-\rangle$
 proton mixing angle: $\theta_p=1.7336761$ rad
 incoming proton: $-0.162161|+\rangle + 0.986764|-\rangle$

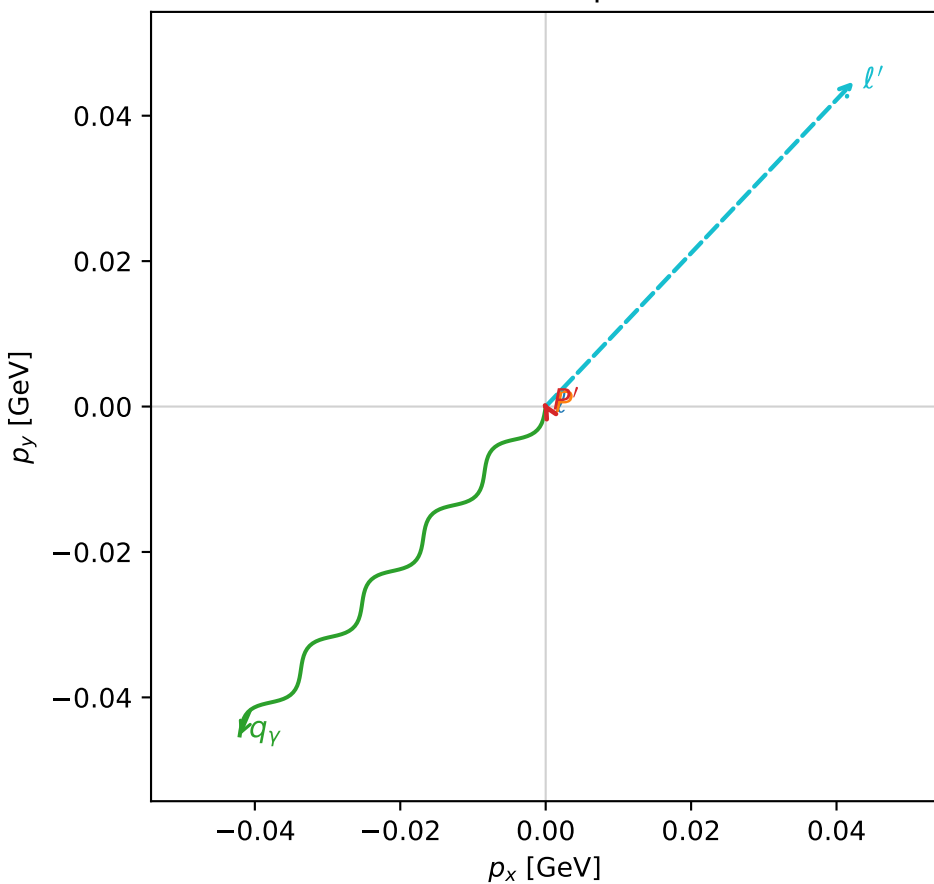
$s=1.19173$, $\sqrt{s}=1.09166$, $|\vec{P}|=0.142849$, $|\vec{P}'|=0.000745213$
 $\theta_{\text{out}}=0.935223$, $\phi_{P'}=2.08509$, $\phi_Y=3.96304$
 $E_Y=0.0767711$, $\theta_{\ell'}=2.21238$, $\phi_{\ell'}=0.812164$
 $Q^2=0.00882091$, $x_B=0.0577188$, $t=-0.020163$, $W^2=1.02385$, $y=0.490003$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E=0.1428$ $p_x=0.0000$ $p_y=0.0000$ $p_z=-0.1428$
 P $E=0.9488$ $p_x=0.0000$ $p_y=0.0000$ $p_z=0.1428$
 ℓ' $E=0.0769$ $p_x=0.0424$ $p_y=0.0447$ $p_z=-0.0460$
 P' $E=0.9380$ $p_x=-0.0003$ $p_y=0.0005$ $p_z=0.0004$
 q_Y $E=0.0768$ $p_x=-0.0421$ $p_y=-0.0452$ $p_z=0.0456$

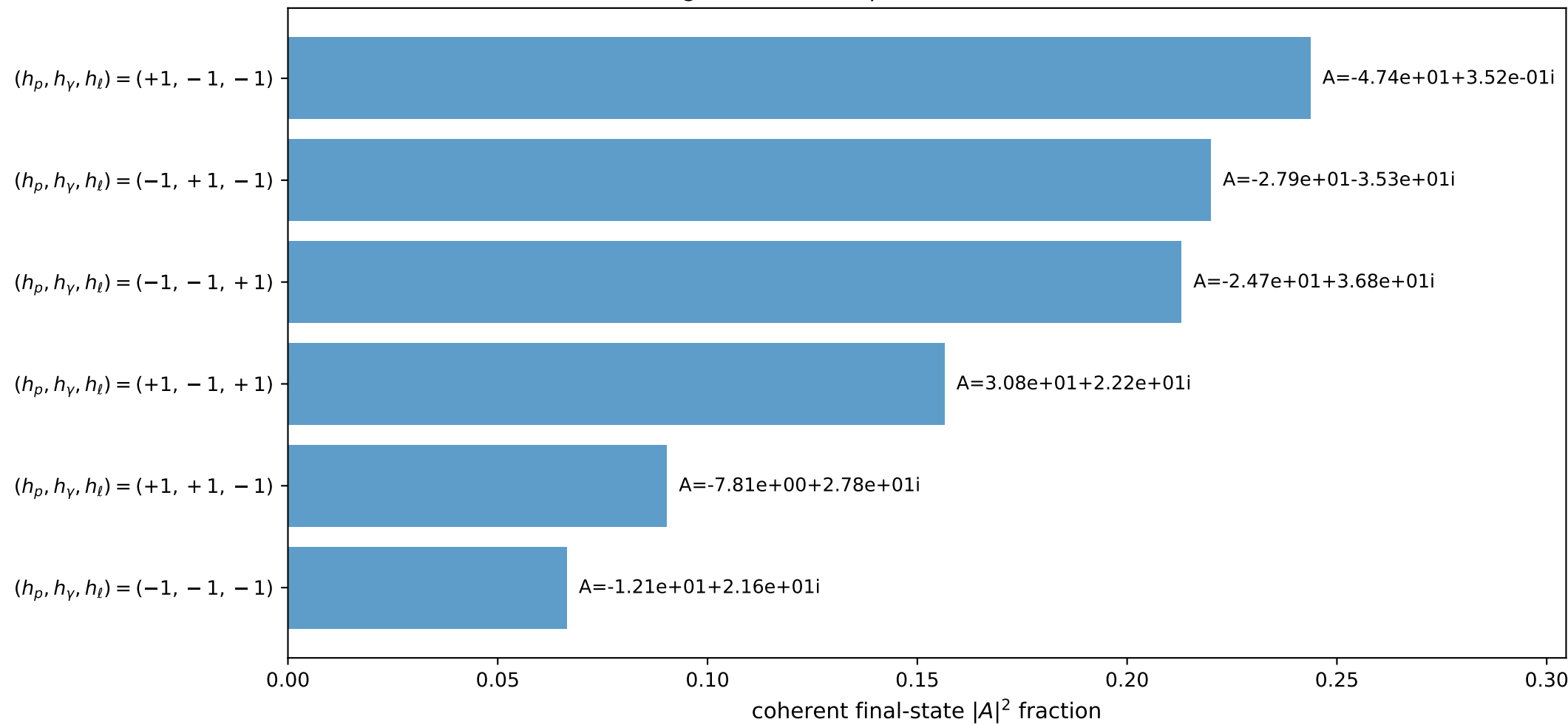
3D momenta



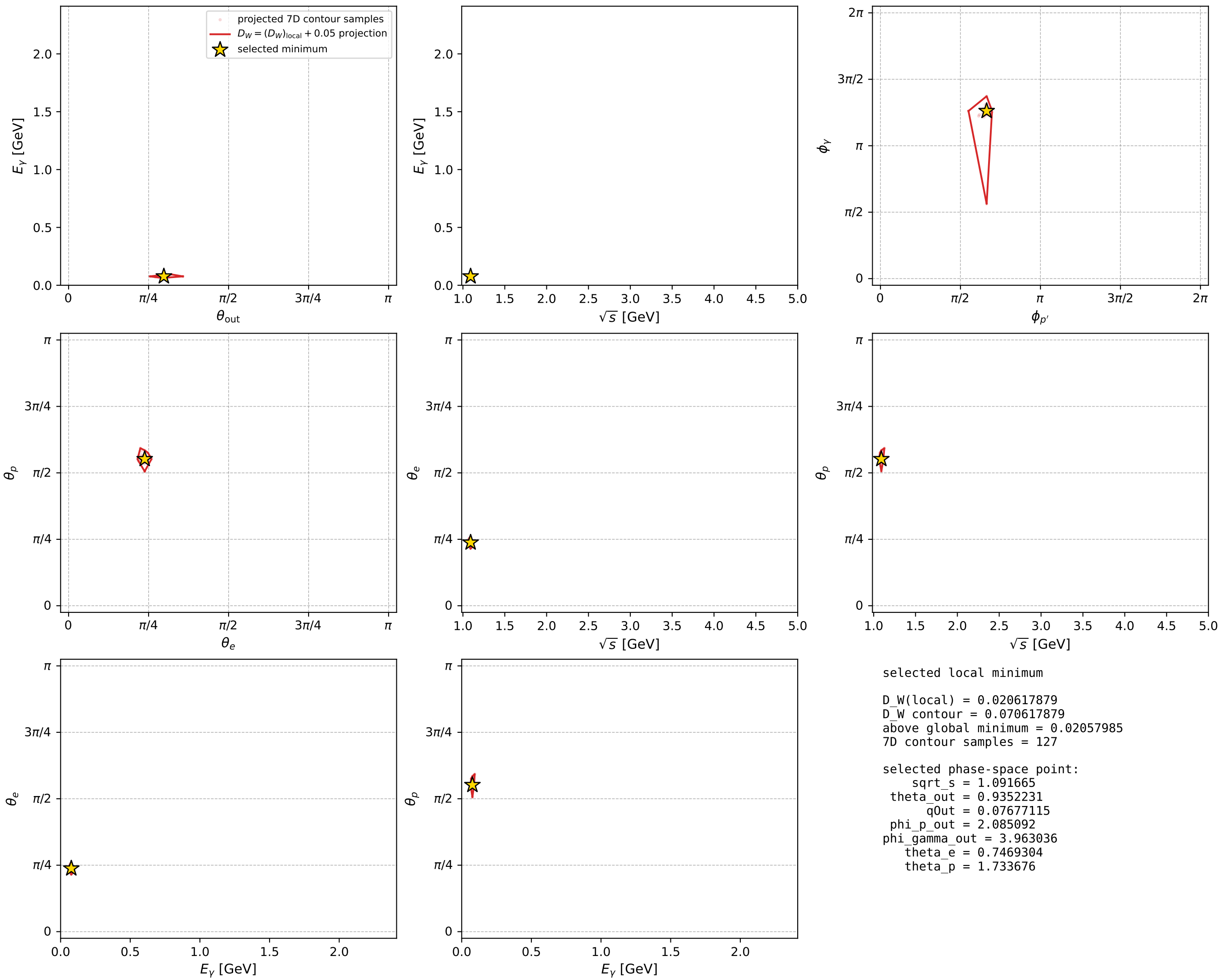
Transverse plane



Leading final-state amplitudes (N=6, retained=99.0%)



electron: polarization cluster P3, local minimum 692; pairwise projections of the 7D $D_W = (D_W)_{\text{local}} + 0.05$ contour

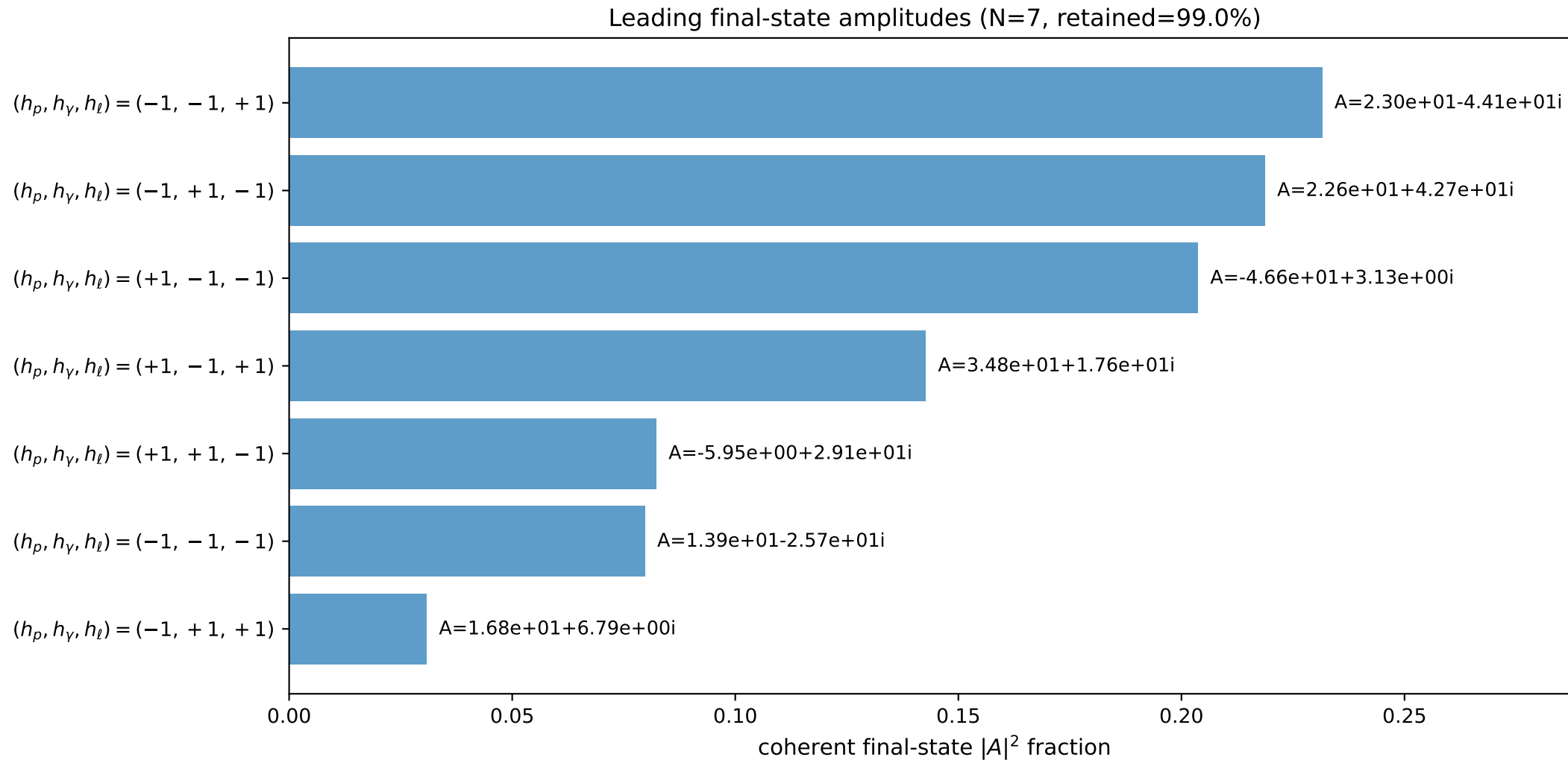
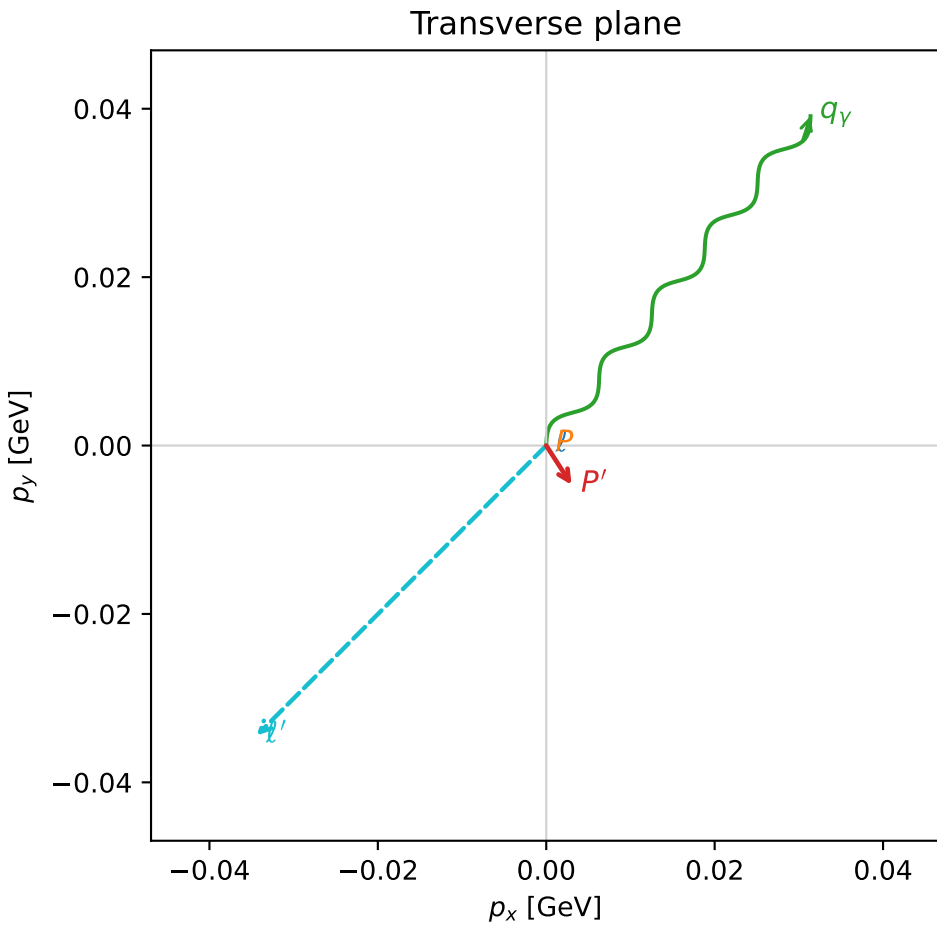
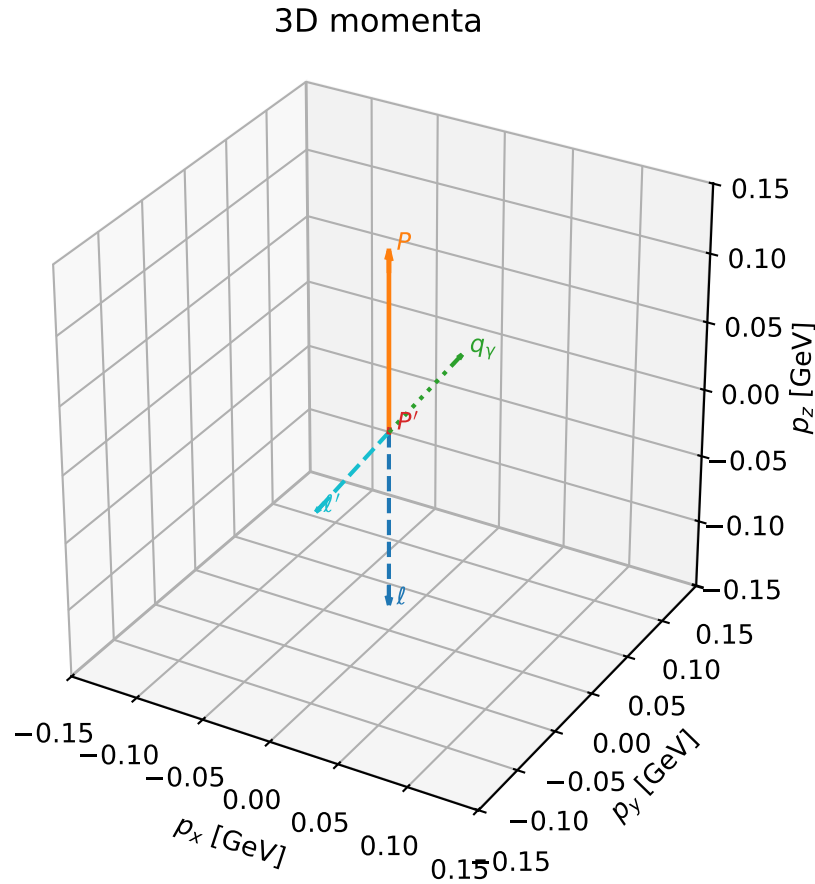


coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

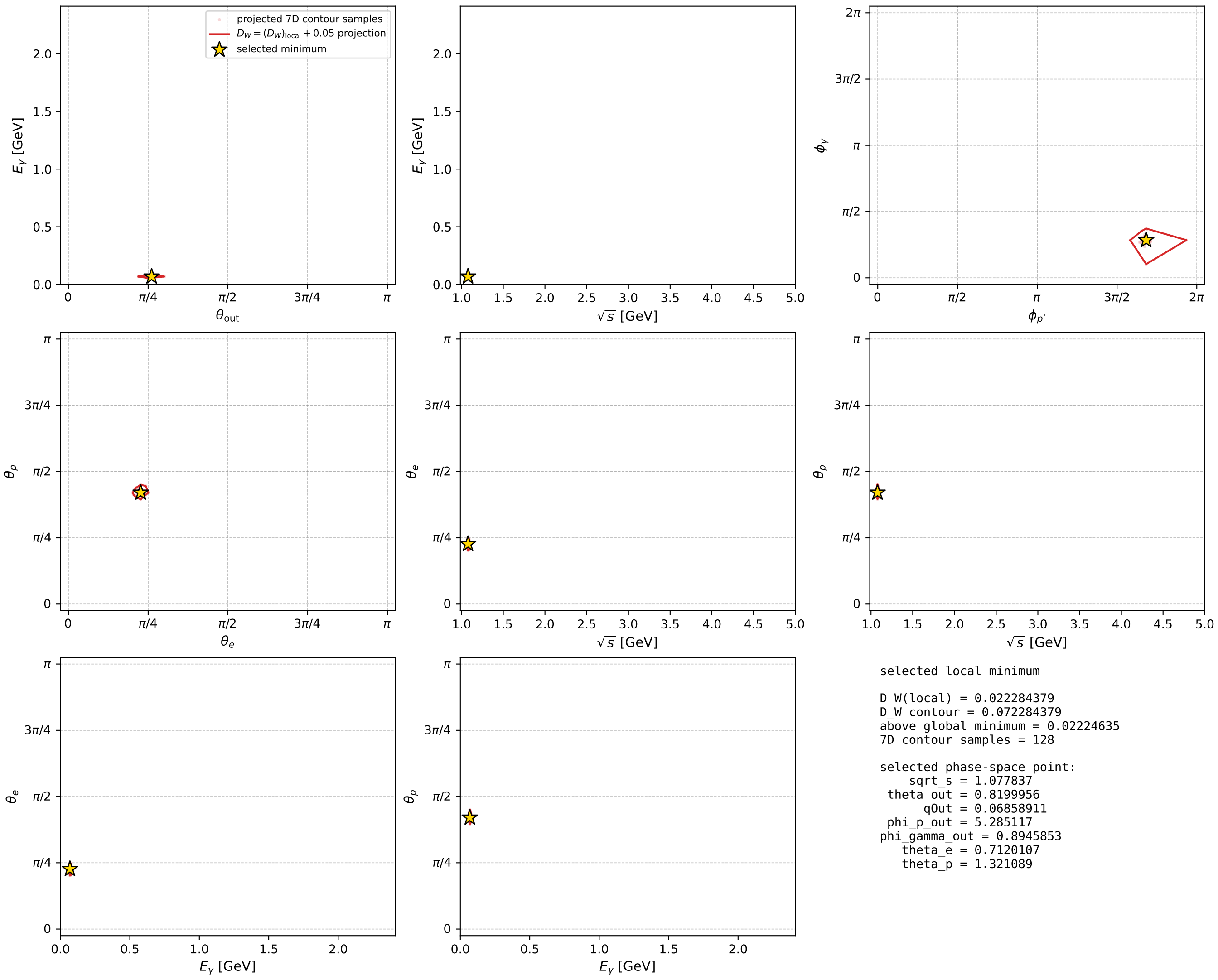
dw_mixing_angles_polarization_03_local_minimum_700 D_W=0.0222844
 region: polarization_cluster_03_local_minimum_700
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\theta_e=0.71201068$ rad
 incoming lepton: $0.75705|+\rangle +0.653357|-\rangle$
 proton mixing angle: $\theta_p=1.3210888$ rad
 incoming proton: $0.247121|+\rangle +0.968985|-\rangle$

$s=1.16173$, $\sqrt{s}=1.07784$, $|\vec{P}|=0.130765$, $|\vec{P}'|=0.00760736$
 $\theta_{\text{out}}=0.819996$, $\phi_{P'}=5.28512$, $\phi_Y=0.894585$
 $E_Y=0.0685891$, $\theta_{\ell'}=2.38904$, $\phi_{\ell'}=3.92756$
 $Q^2=0.00502982$, $x_B=0.0377053$, $t=-0.0157183$, $W^2=1.00821$, $y=0.473231$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1308$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1308$
 P $E= 0.9471$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1308$
 ℓ' $E= 0.0712$ $p_x= -0.0344$ $p_y= -0.0344$ $p_z= -0.0520$
 P' $E= 0.9380$ $p_x= 0.0030$ $p_y= -0.0047$ $p_z= 0.0052$
 q_Y $E= 0.0686$ $p_x= 0.0314$ $p_y= 0.0391$ $p_z= 0.0468$



electron: polarization cluster P3, local minimum 700; pairwise projections of the 7D $D_W = (D_W)_{\text{local}} + 0.05$ contour

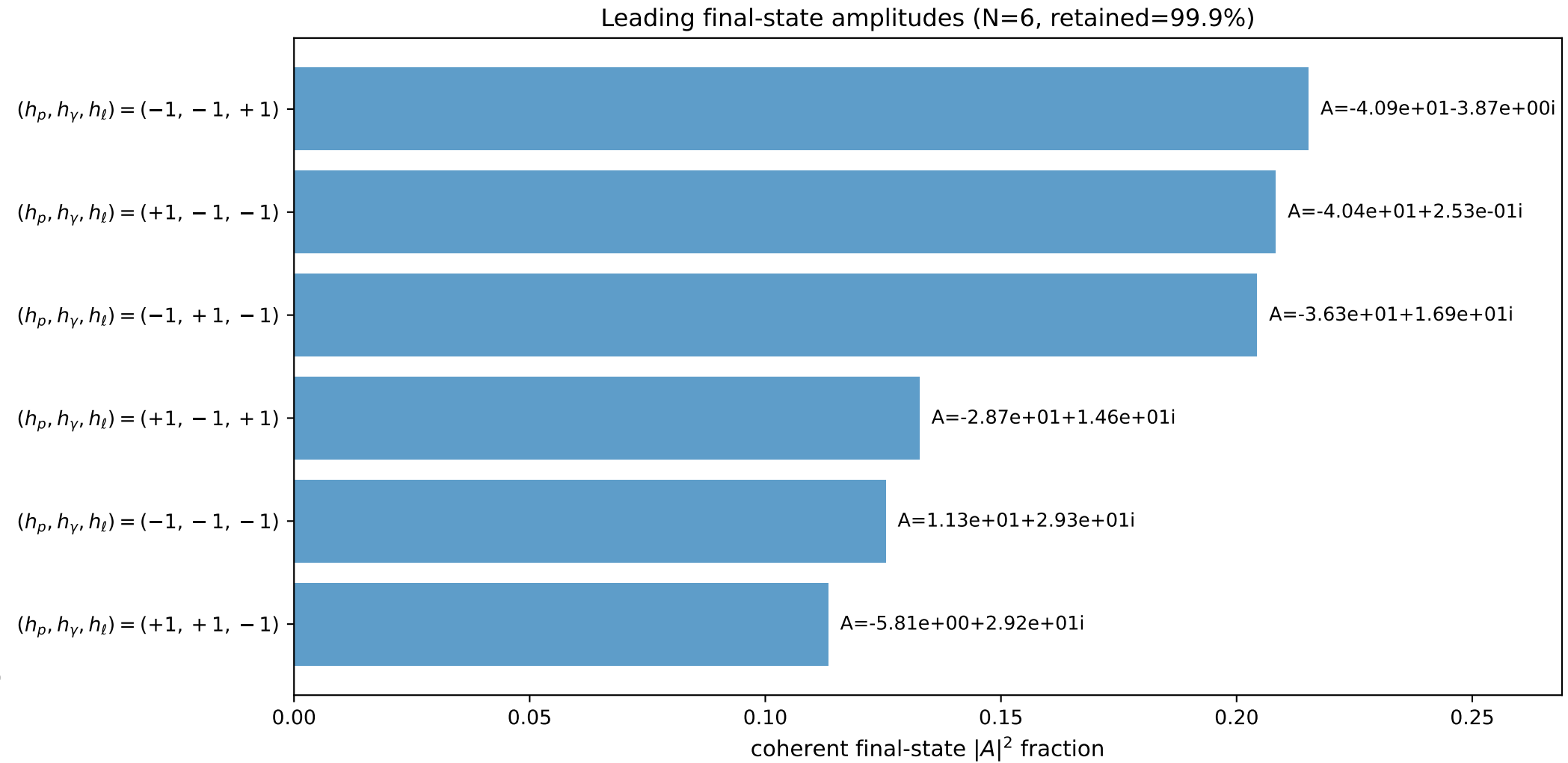
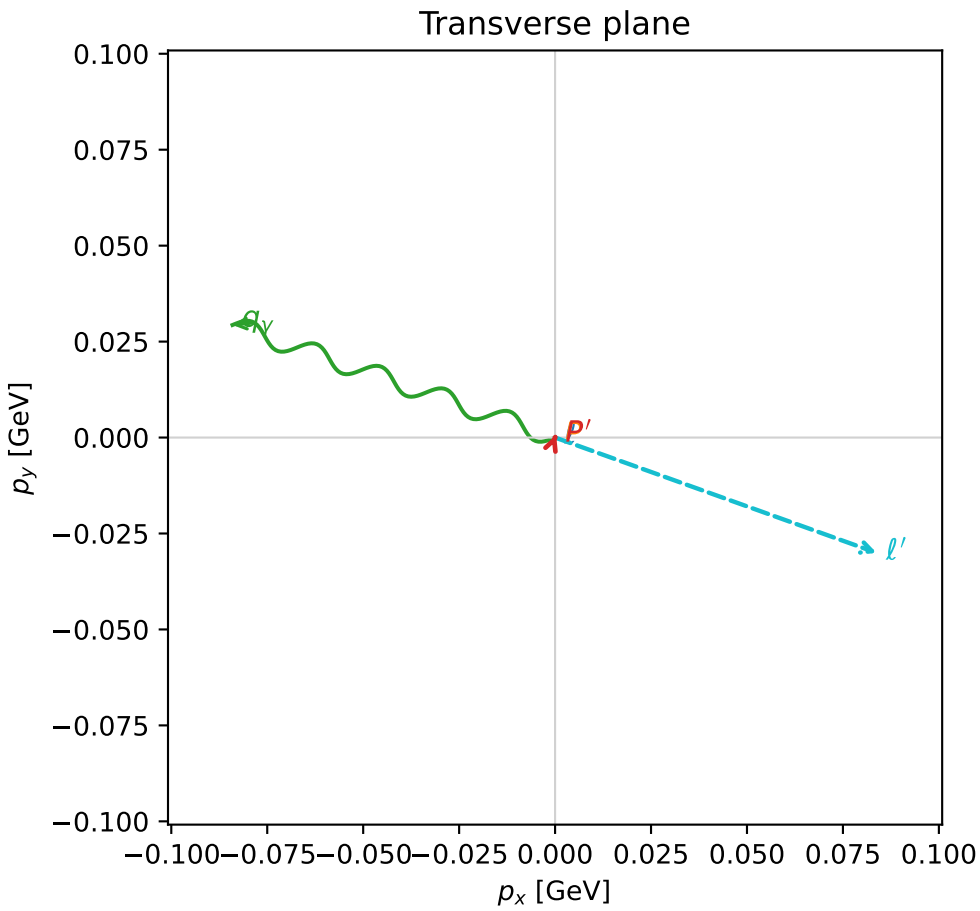
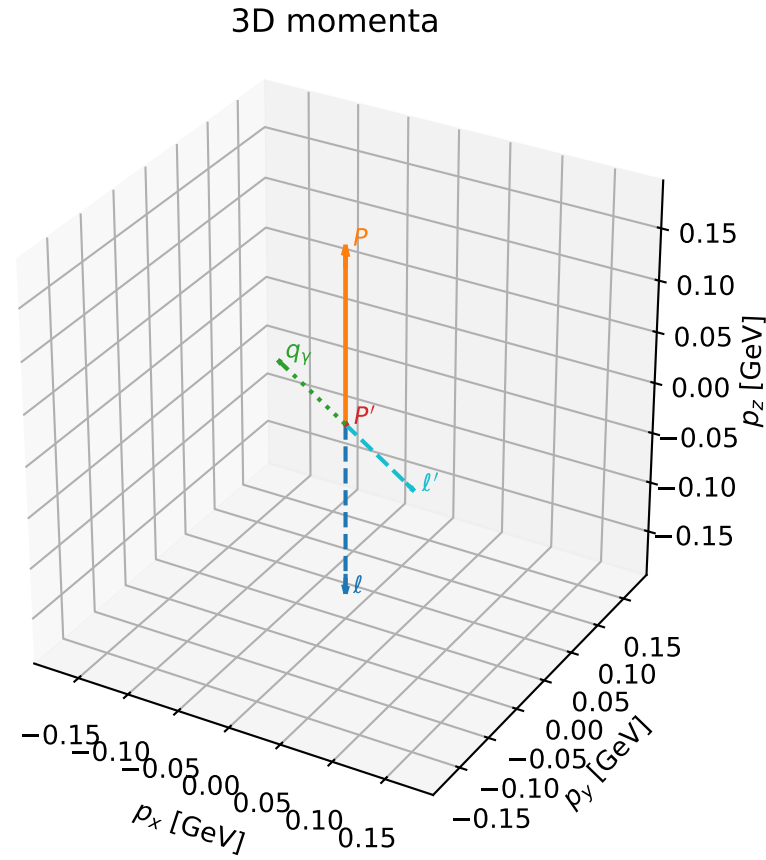


coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

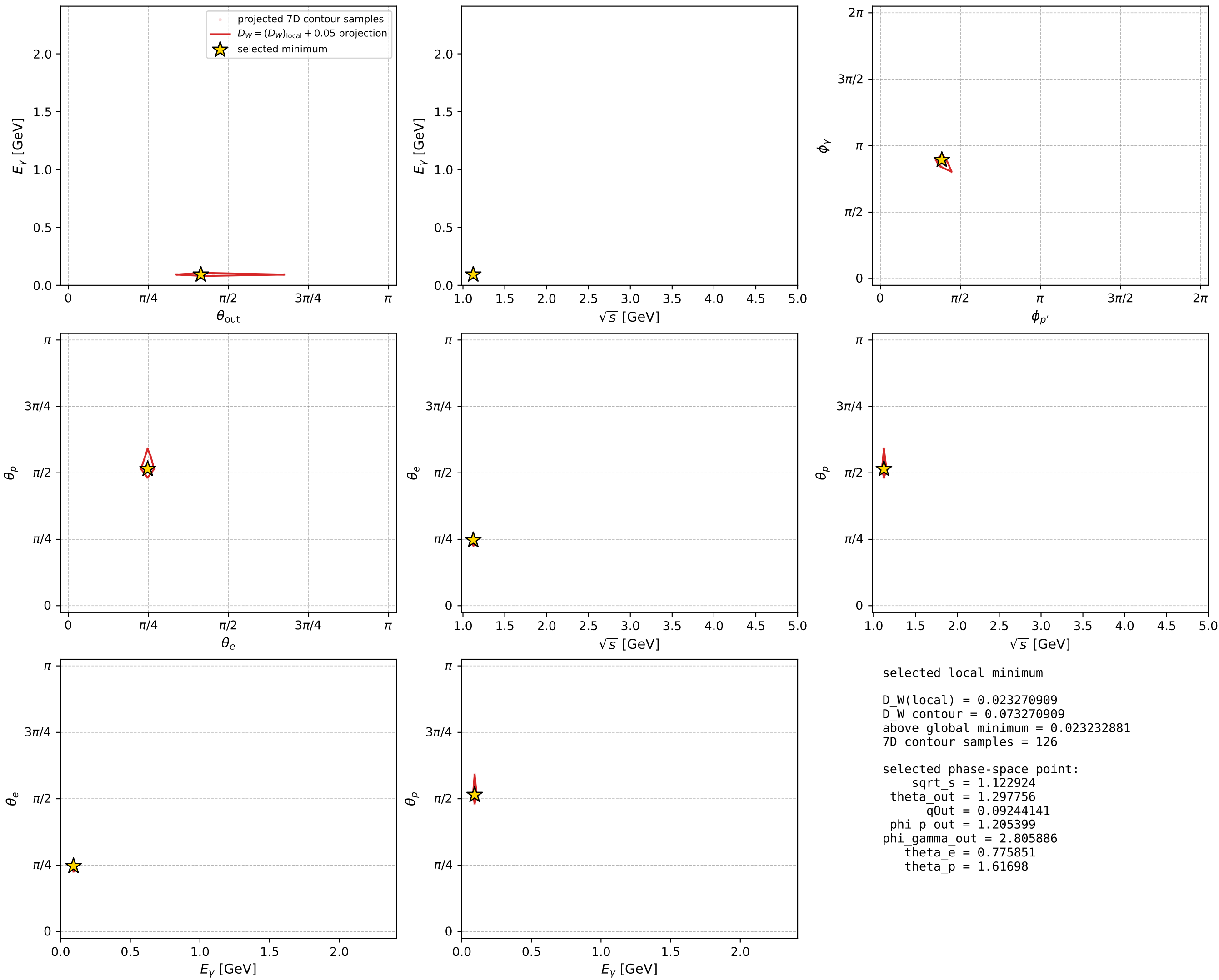
dw_mixing_angles_polarization_03_local_minimum_707 D_W=0.0232709
 region: polarization_cluster_03_local_minimum_707
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\theta_e=0.77585104$ rad
 incoming lepton: $0.713825|+\rangle +0.700324|-\rangle$
 proton mixing angle: $\theta_p=1.6169801$ rad
 incoming proton: $-0.0461674|+\rangle +0.998934|-\rangle$

$s=1.26096$, $\sqrt{s}=1.12292$, $|\vec{P}|=0.169696$, $|\vec{P}'|=0.000787812$
 $\theta_{\text{out}}=1.29776$, $\phi_{p'}=1.2054$, $\phi_\gamma=2.80589$
 $E_\gamma=0.0924414$, $\theta_{l'}=1.8461$, $\phi_{l'}=5.93896$
 $Q^2=0.0228548$, $x_B=0.116447$, $t=-0.0284935$, $W^2=1.05326$, $y=0.514987$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 0.1697$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1697$
 P $E= 0.9532$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1697$
 l' $E= 0.0925$ $p_x= 0.0838$ $p_y= -0.0300$ $p_z= -0.0251$
 P' $E= 0.9380$ $p_x= 0.0003$ $p_y= 0.0007$ $p_z= 0.0002$
 q_γ $E= 0.0924$ $p_x= -0.0840$ $p_y= 0.0293$ $p_z= 0.0249$



electron: polarization cluster P3, local minimum 707; pairwise projections of the 7D $D_W = (D_W)_{\text{local}} + 0.05$ contour

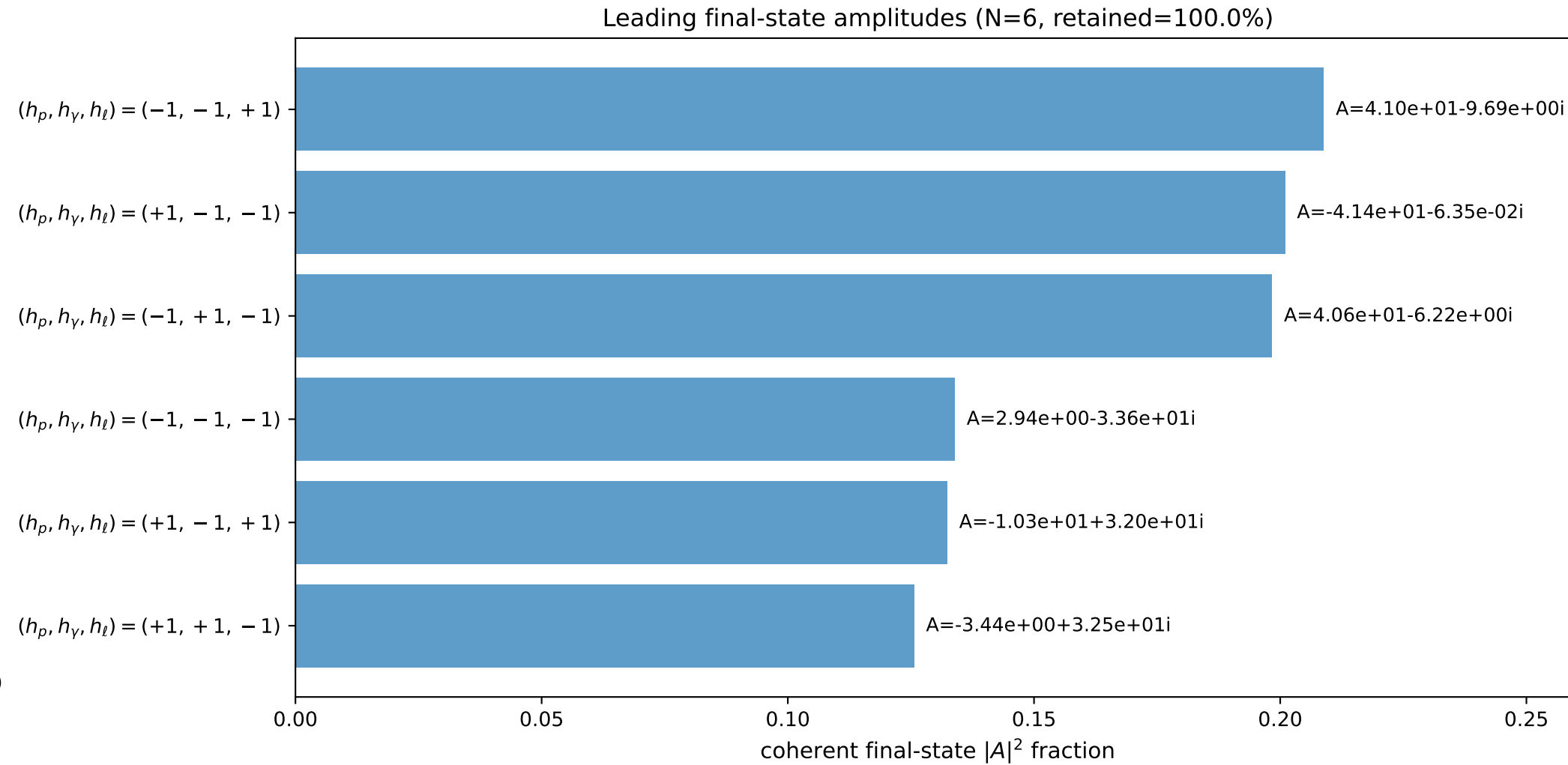
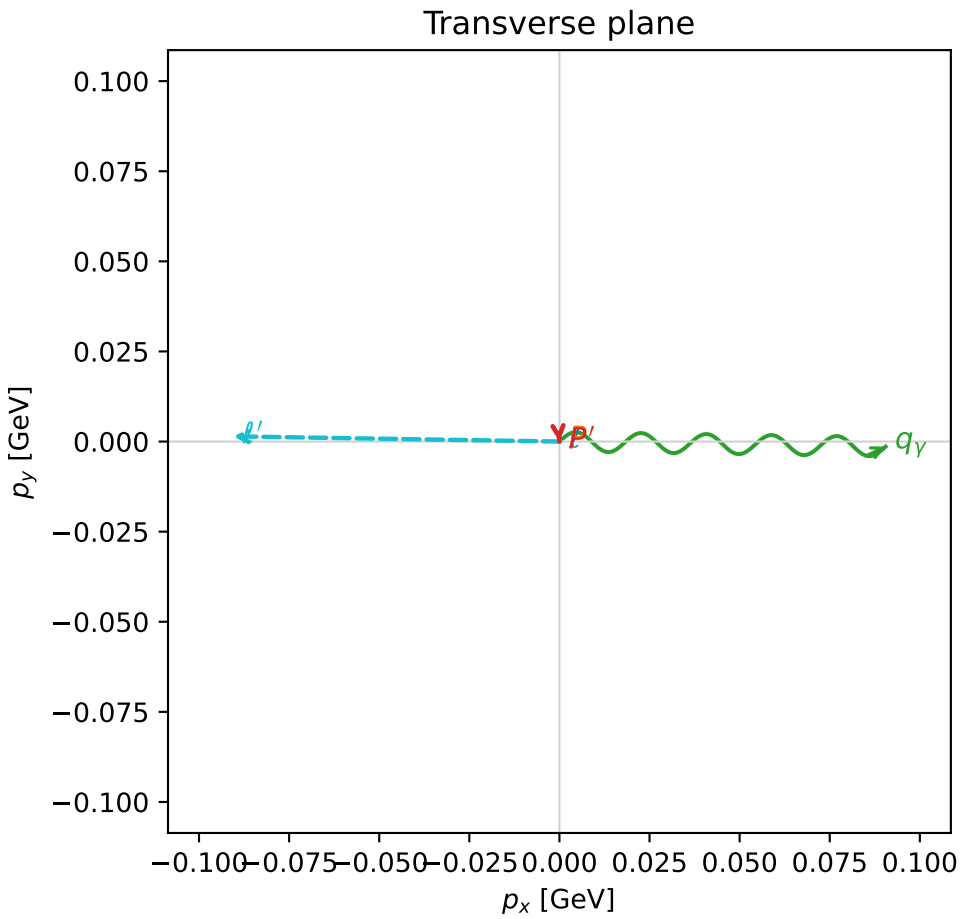
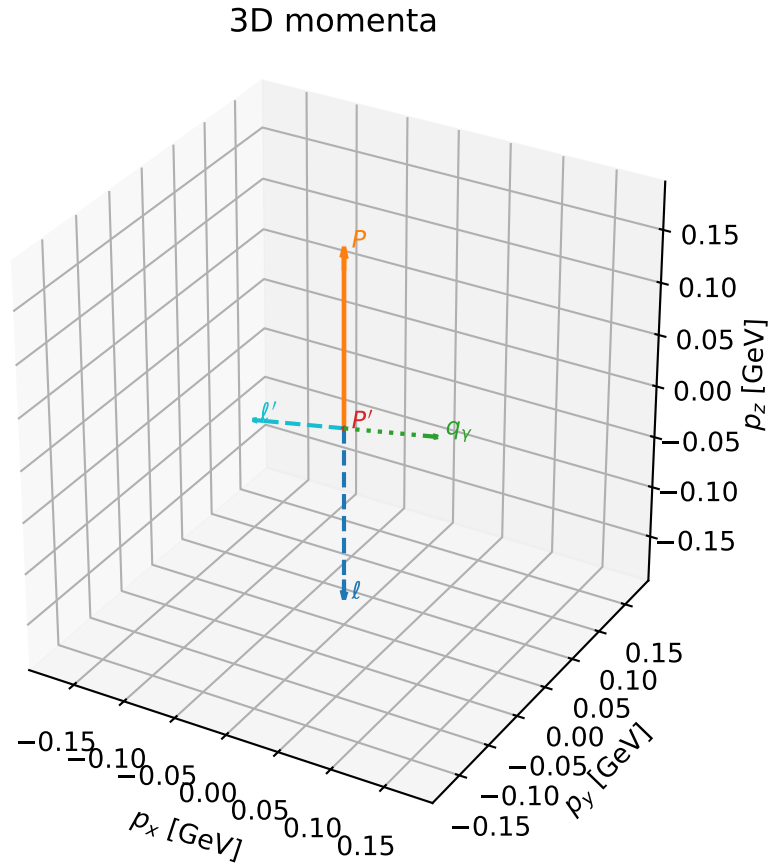


coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

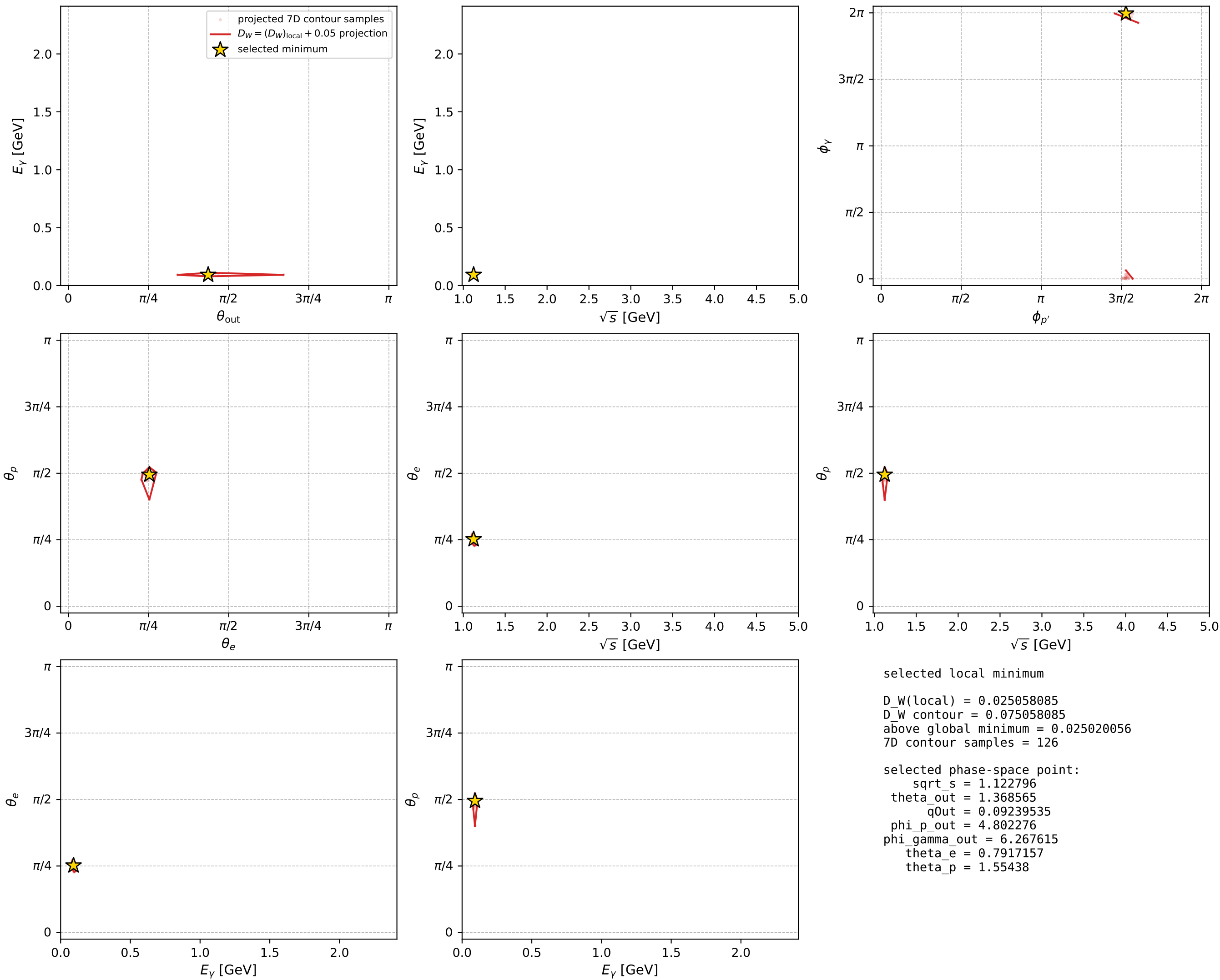
dw_mixing_angles_polarization_03_local_minimum_712 D_W=0.0250581
 region: polarization_cluster_03_local_minimum_712
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\theta_e=0.79171574$ rad
 incoming lepton: $0.702626|+\rangle +0.71156|-\rangle$
 proton mixing angle: $\theta_p=1.5543796$ rad
 incoming proton: $0.016416|+\rangle +0.999865|-\rangle$

$s=1.26067$, $\sqrt{s}=1.1228$, $|\vec{P}|=0.169588$, $|\vec{P}'|=2.97012e-05$
 $\theta_{\text{out}}=1.36856$, $\phi_{P'}=4.80228$, $\phi_Y=6.26762$
 $E_Y=0.0923953$, $\theta_{\ell'}=1.77308$, $\phi_{\ell'}=3.1257$
 $Q^2=0.0250433$, $x_B=0.126242$, $t=-0.0285269$, $W^2=1.05318$, $y=0.520909$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1696$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1696$
 P $E= 0.9532$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1696$
 ℓ' $E= 0.0924$ $p_x= -0.0905$ $p_y= 0.0014$ $p_z= -0.0186$
 P' $E= 0.9380$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 0.0000$
 q_Y $E= 0.0924$ $p_x= 0.0905$ $p_y= -0.0014$ $p_z= 0.0186$



electron: polarization cluster P3, local minimum 712; pairwise projections of the 7D $D_W = (D_W)_{\text{local}} + 0.05$ contour



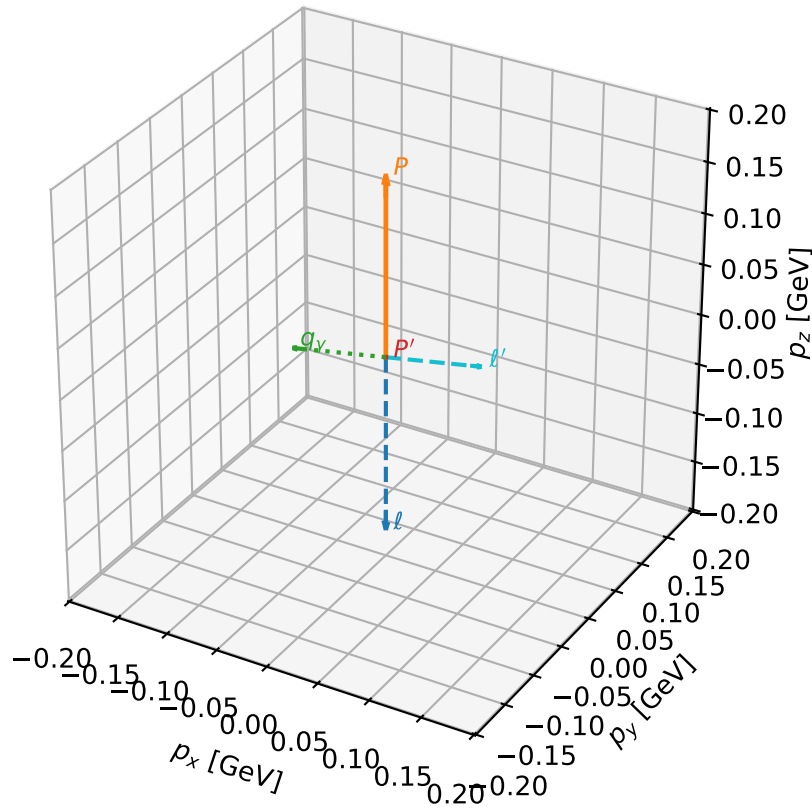
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

dw_mixing_angles_polarization_03_local_minimum_714 D_W=0.0252306
 region: polarization_cluster_03_local_minimum_714
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\theta_e=0.78063145$ rad
 incoming lepton: $0.710469|+\rangle + 0.703728|-\rangle$
 proton mixing angle: $\theta_p=1.578225$ rad
 incoming proton: $-0.0074286|+\rangle + 0.999972|-\rangle$

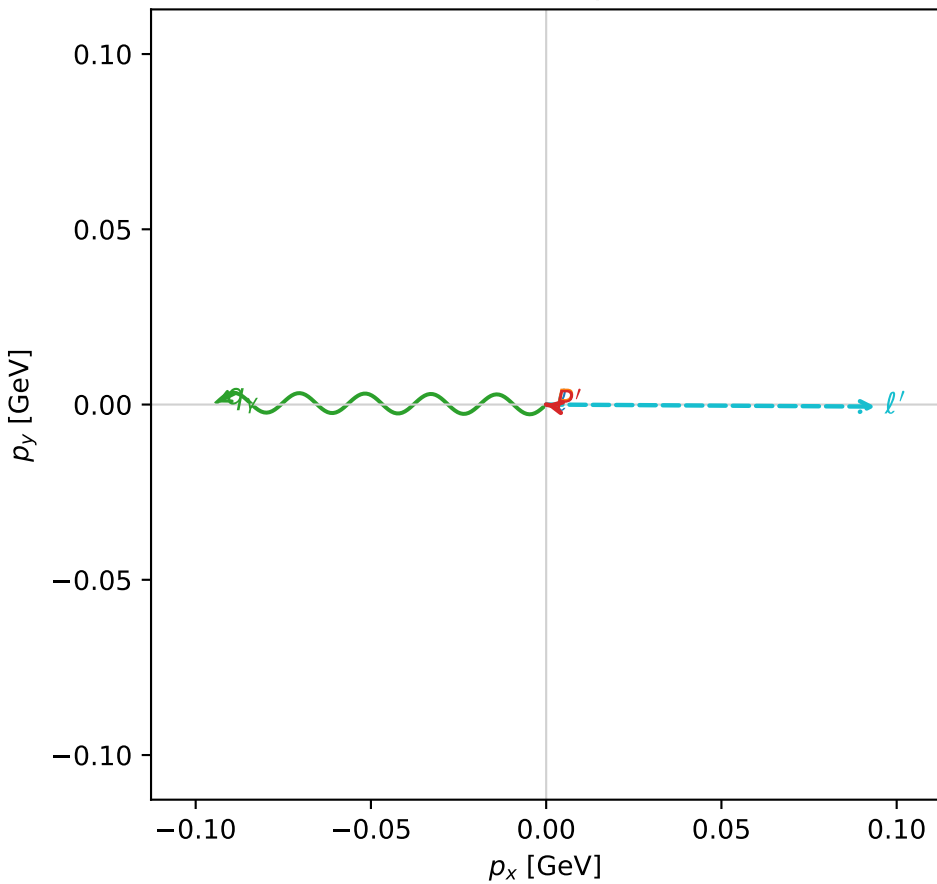
$s=1.27542$, $\sqrt{s}=1.12934$, $|\vec{P}|=0.175134$, $|\vec{P}'|=3.18134e-07$
 $\theta_{out}=1.7609$, $\phi_{P'}=2.95135$, $\phi_\gamma=3.1353$
 $E_\gamma=0.0956712$, $\theta_{\ell'}=1.38069$, $\phi_{\ell'}=6.27689$
 $Q^2=0.039843$, $x_B=0.181664$, $t=-0.0304091$, $W^2=1.05932$, $y=0.55444$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1751$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1751$
 P $E= 0.9542$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1751$
 ℓ' $E= 0.0957$ $p_x= 0.0939$ $p_y= -0.0006$ $p_z= 0.0181$
 P' $E= 0.9380$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -0.0000$
 q_γ $E= 0.0957$ $p_x= -0.0939$ $p_y= 0.0006$ $p_z= -0.0181$

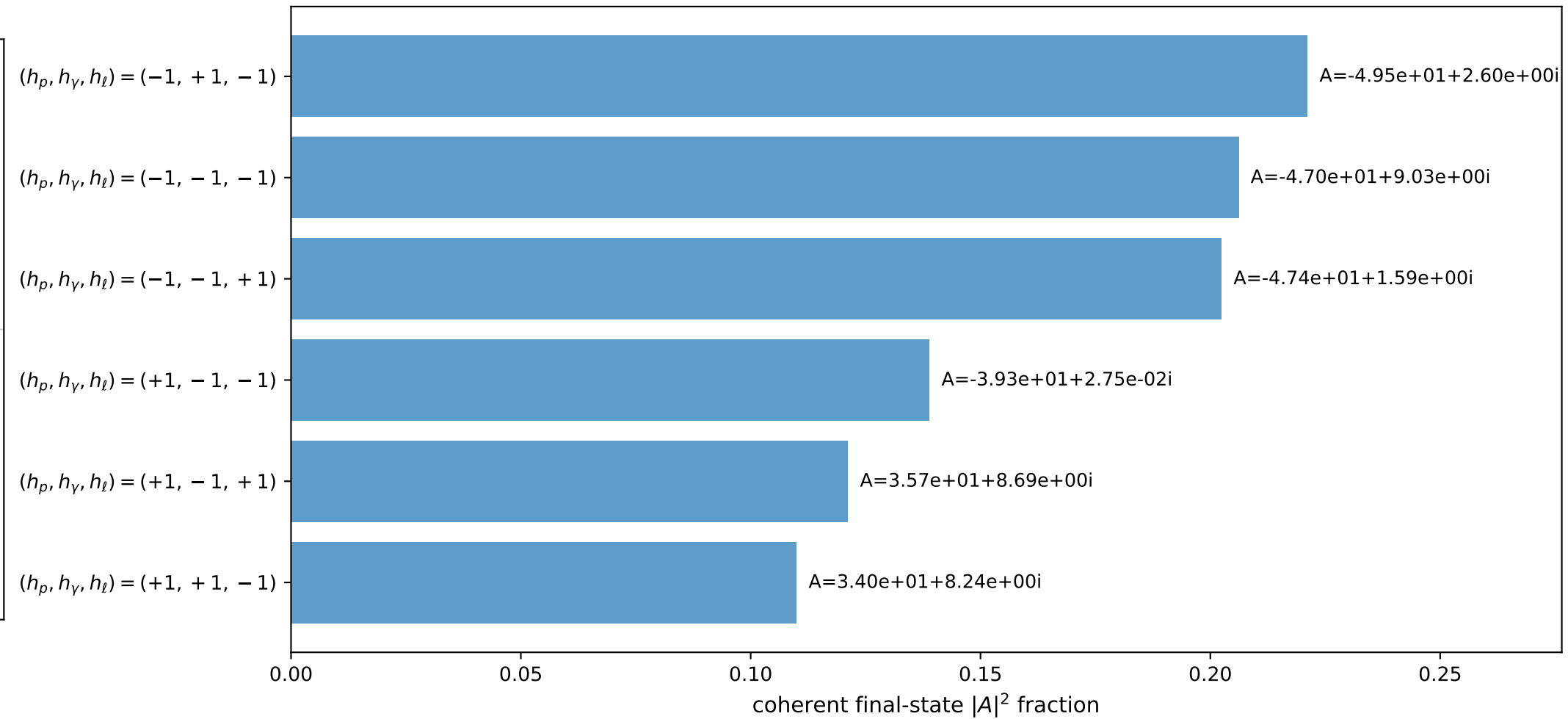
3D momenta



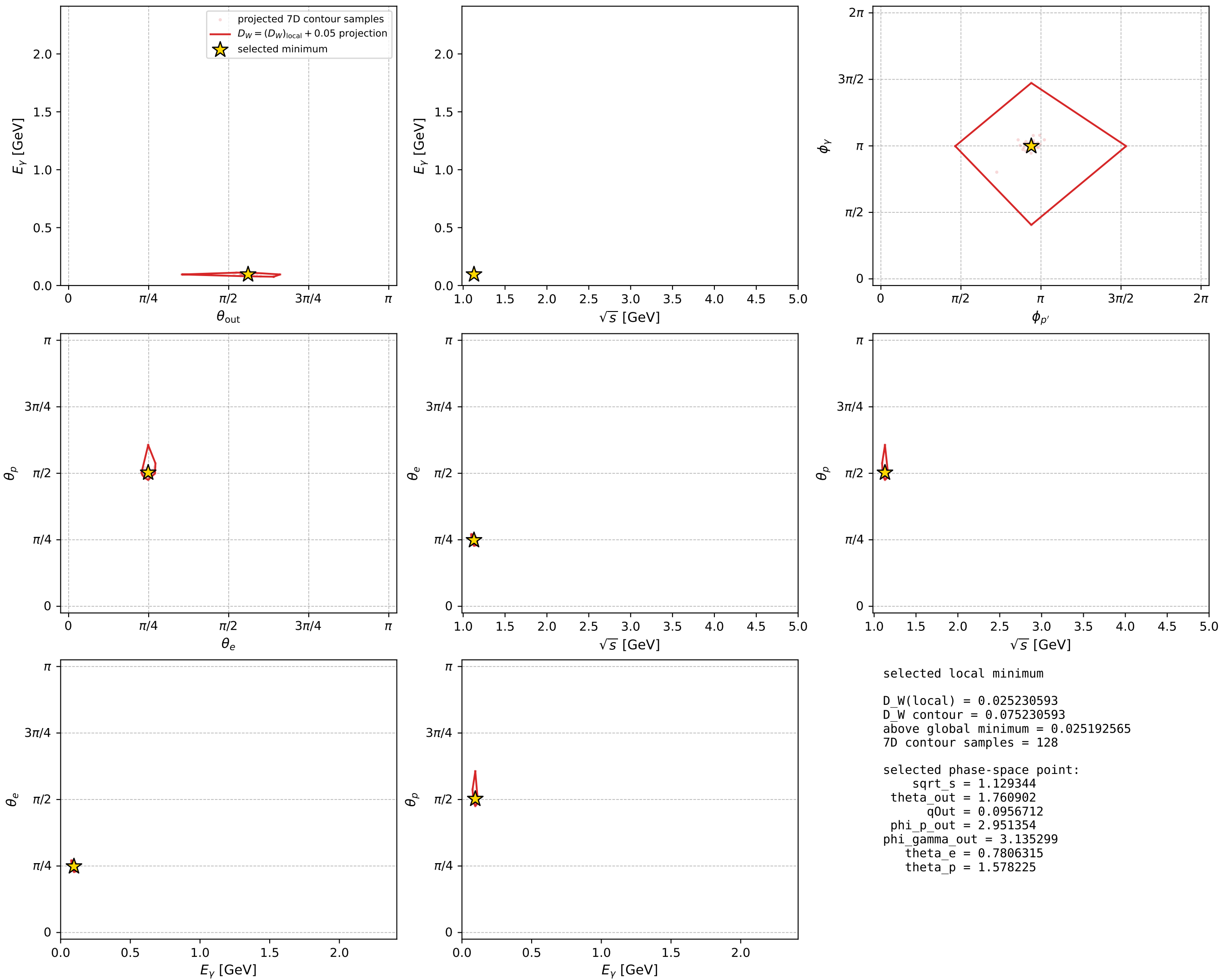
Transverse plane



Leading final-state amplitudes (N=6, retained=100.0%)



electron: polarization cluster P3, local minimum 714; pairwise projections of the 7D $D_W = (D_W)_{\text{local}} + 0.05$ contour

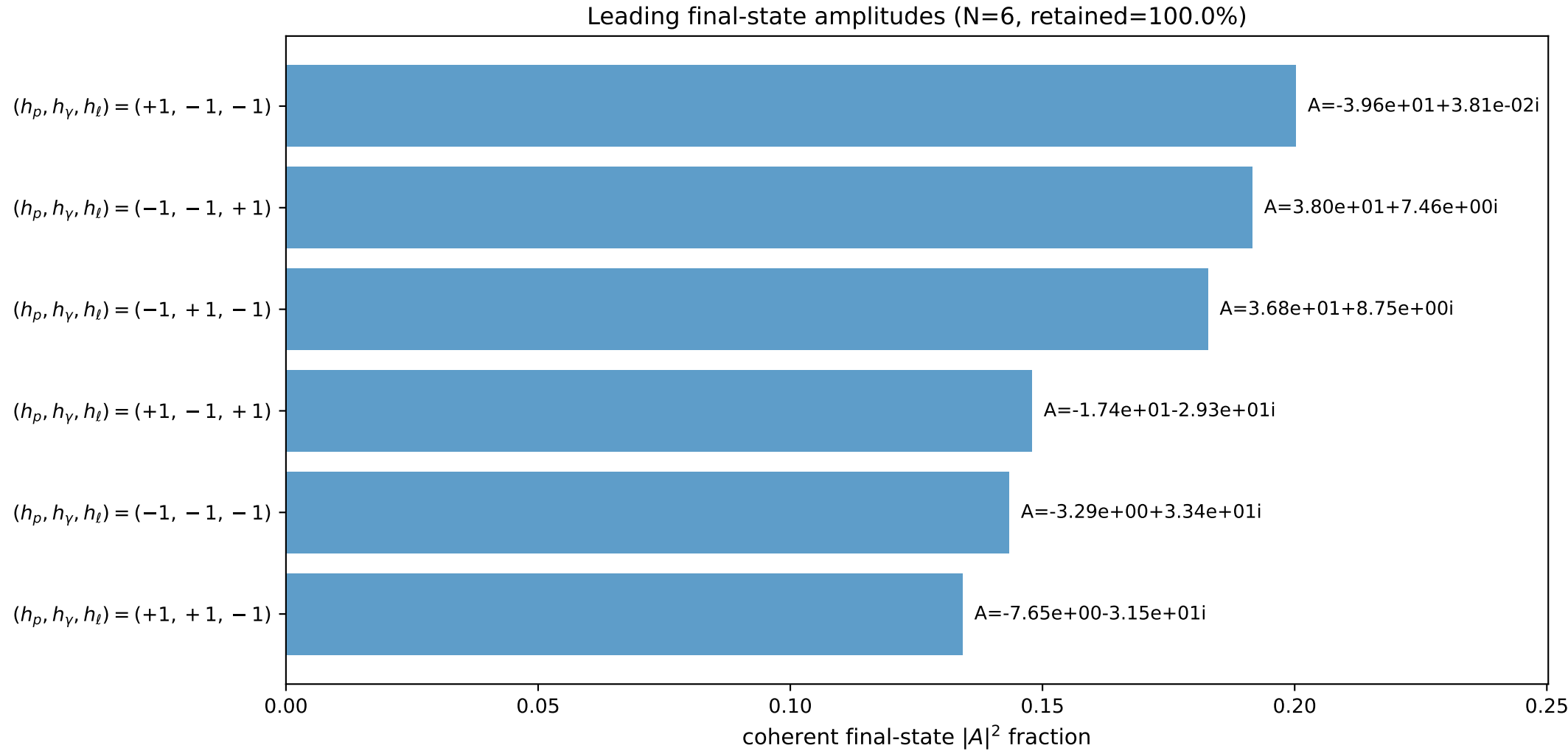
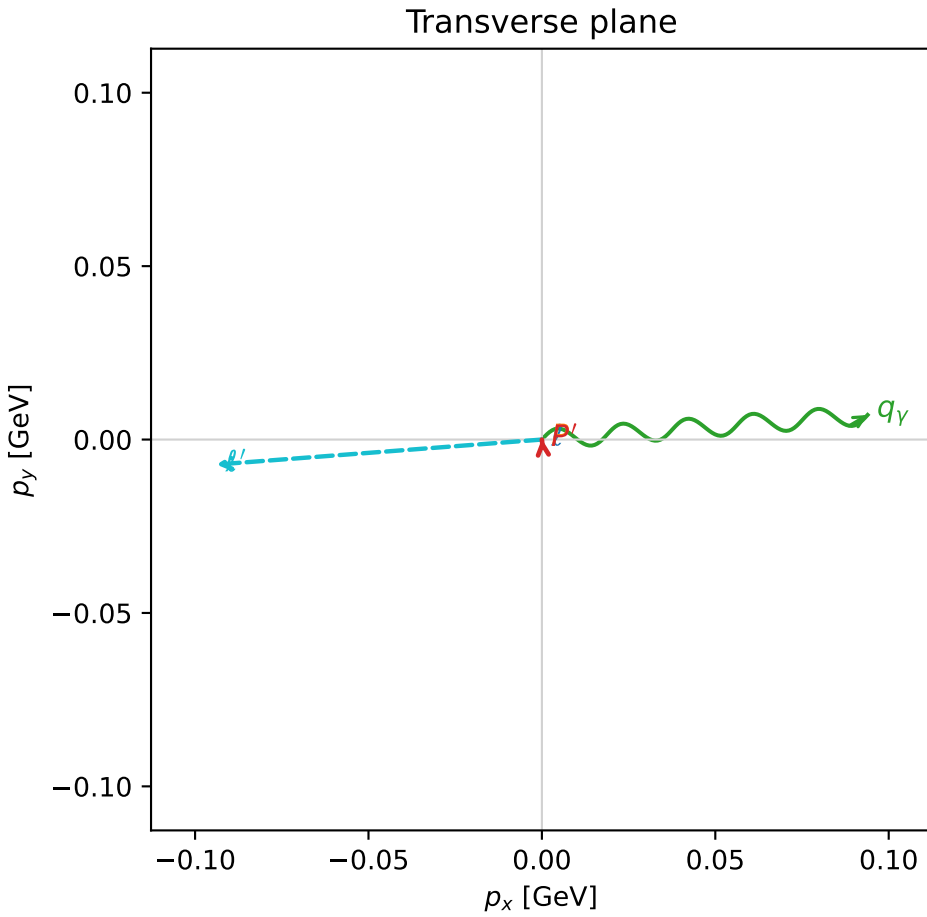
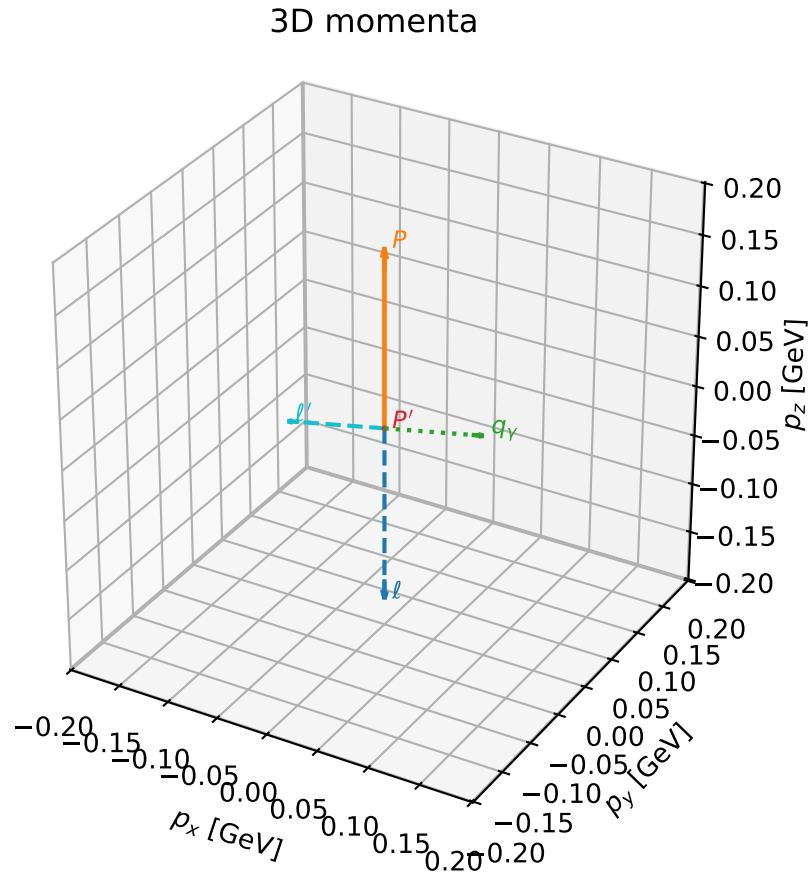


coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

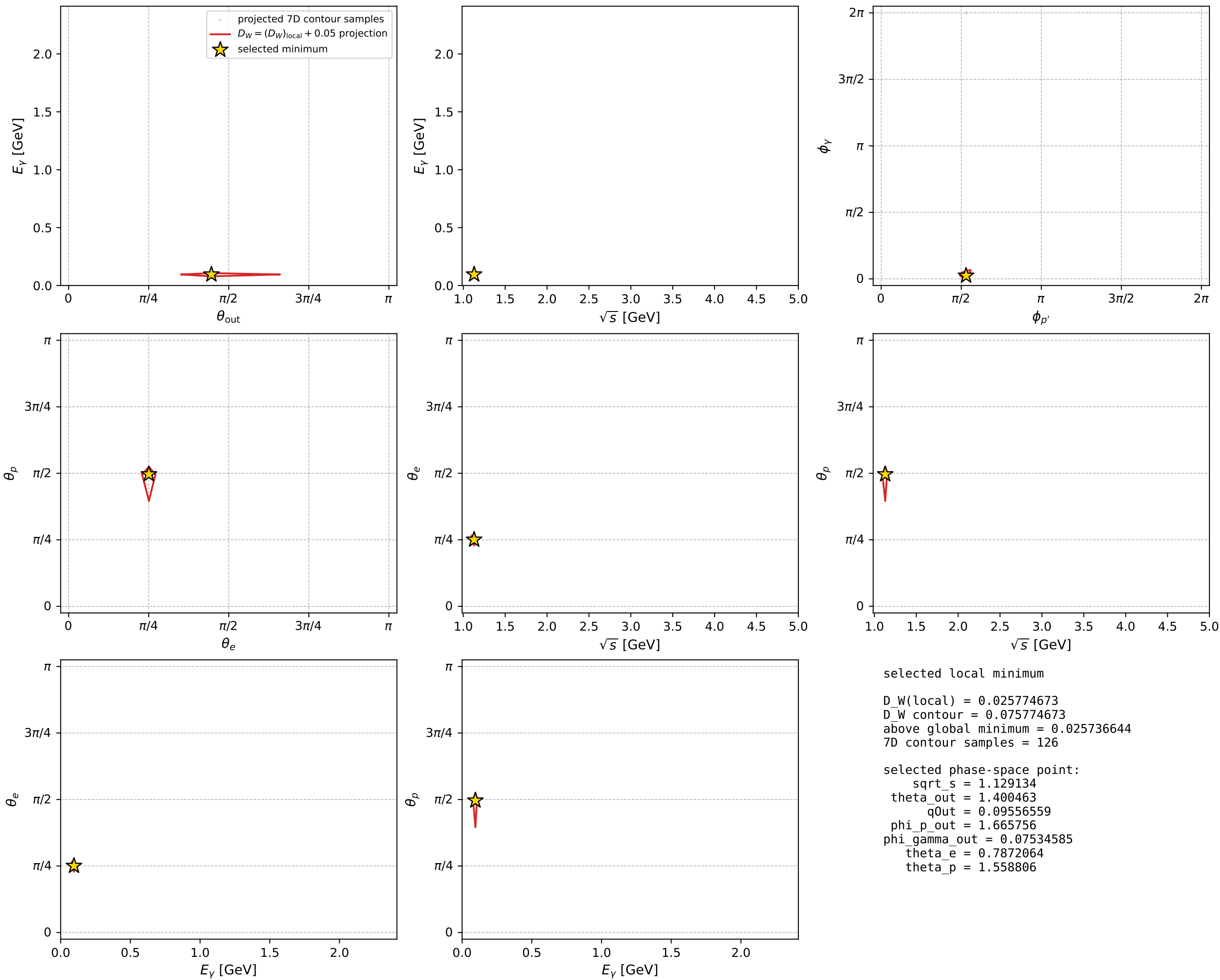
dw_mixing_angles_polarization_03_local_minimum_716 D_W=0.0257747
 region: polarization_cluster_03_local_minimum_716
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\theta_e=0.78720644$ rad
 incoming lepton: $0.705827|+\rangle + 0.708384|-\rangle$
 proton mixing angle: $\theta_p=1.5588059$ rad
 incoming proton: $0.0119901|+\rangle + 0.999928|-\rangle$

$s=1.27494$, $\sqrt{s}=1.12913$, $|\vec{P}|=0.174956$, $|\vec{P}'|=0.000124234$
 $\theta_{\text{out}}=1.40046$, $\phi_{P'}=1.66576$, $\phi_Y=0.0753458$
 $E_Y=0.0955656$, $\theta_{l'}=1.74135$, $\phi_{l'}=3.21824$
 $Q^2=0.0277644$, $x_B=0.134098$, $t=-0.0303406$, $W^2=1.05912$, $y=0.524034$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 0.1750$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1750$
 P $E= 0.9542$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1750$
 l' $E= 0.0956$ $p_x= -0.0939$ $p_y= -0.0072$ $p_z= -0.0162$
 P' $E= 0.9380$ $p_x= -0.0000$ $p_y= 0.0001$ $p_z= 0.0000$
 q_Y $E= 0.0956$ $p_x= 0.0939$ $p_y= 0.0071$ $p_z= 0.0162$



electron: polarization cluster P3, local minimum 716; pairwise projections of the 7D $D_W = (D_W)_{\text{local}} + 0.05$ contour

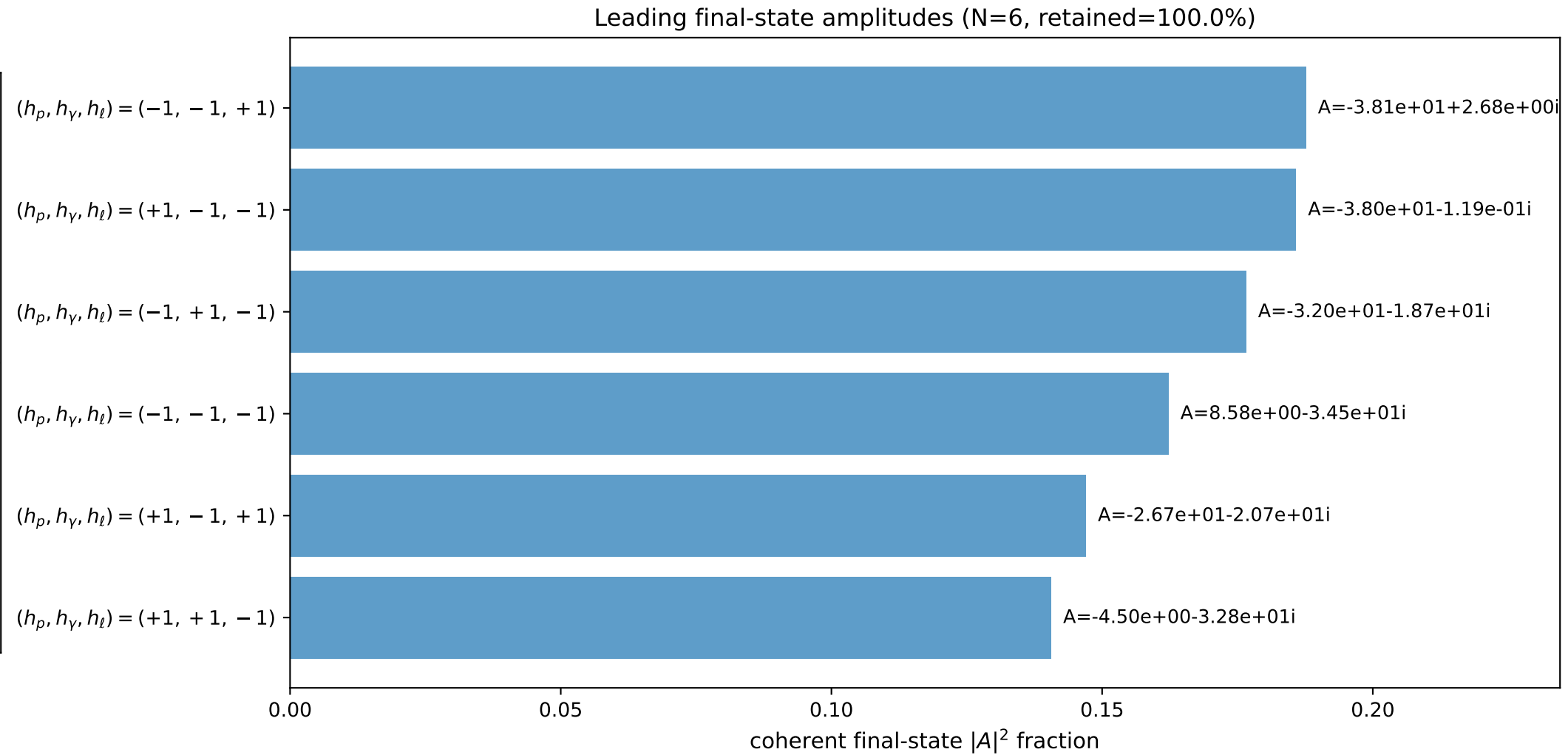
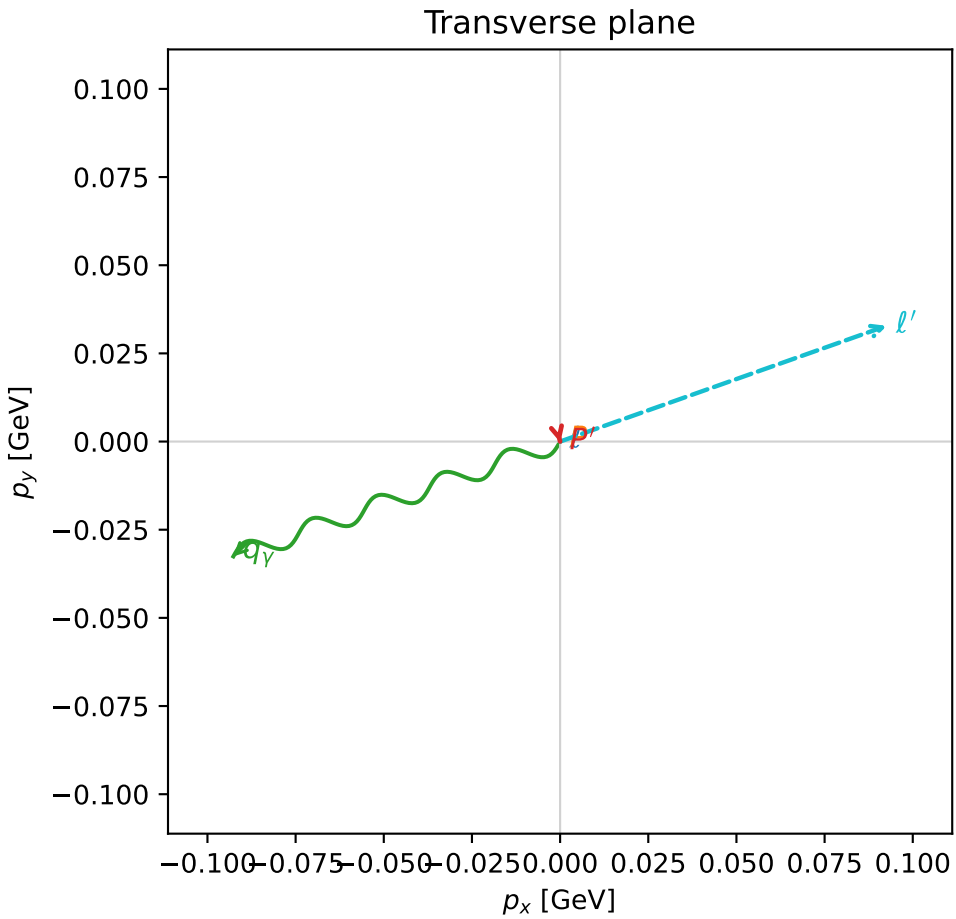
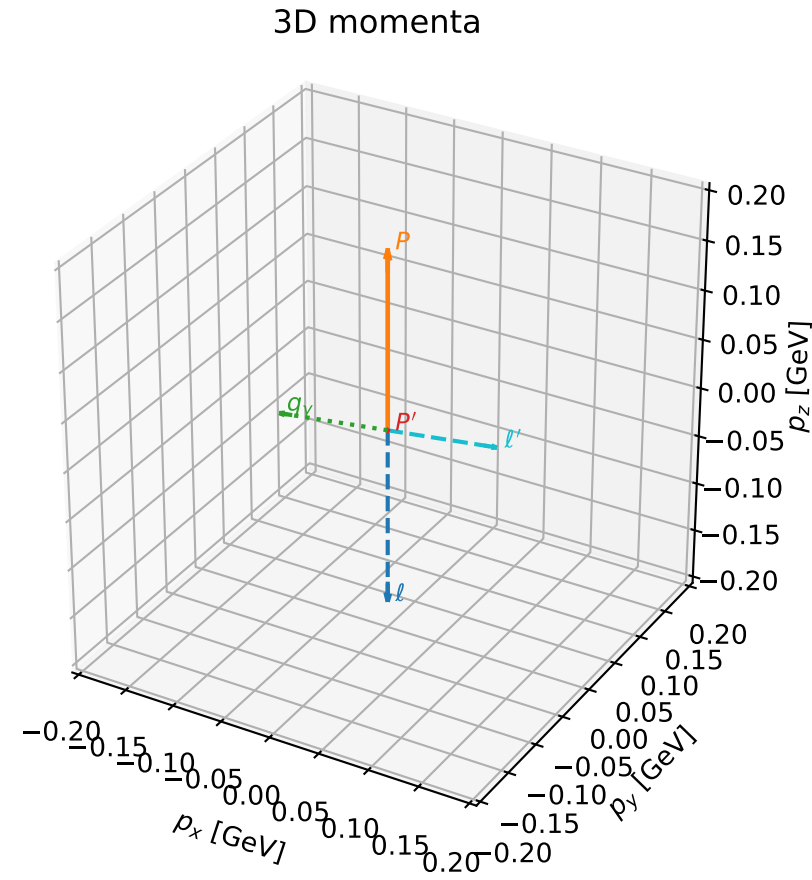


coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

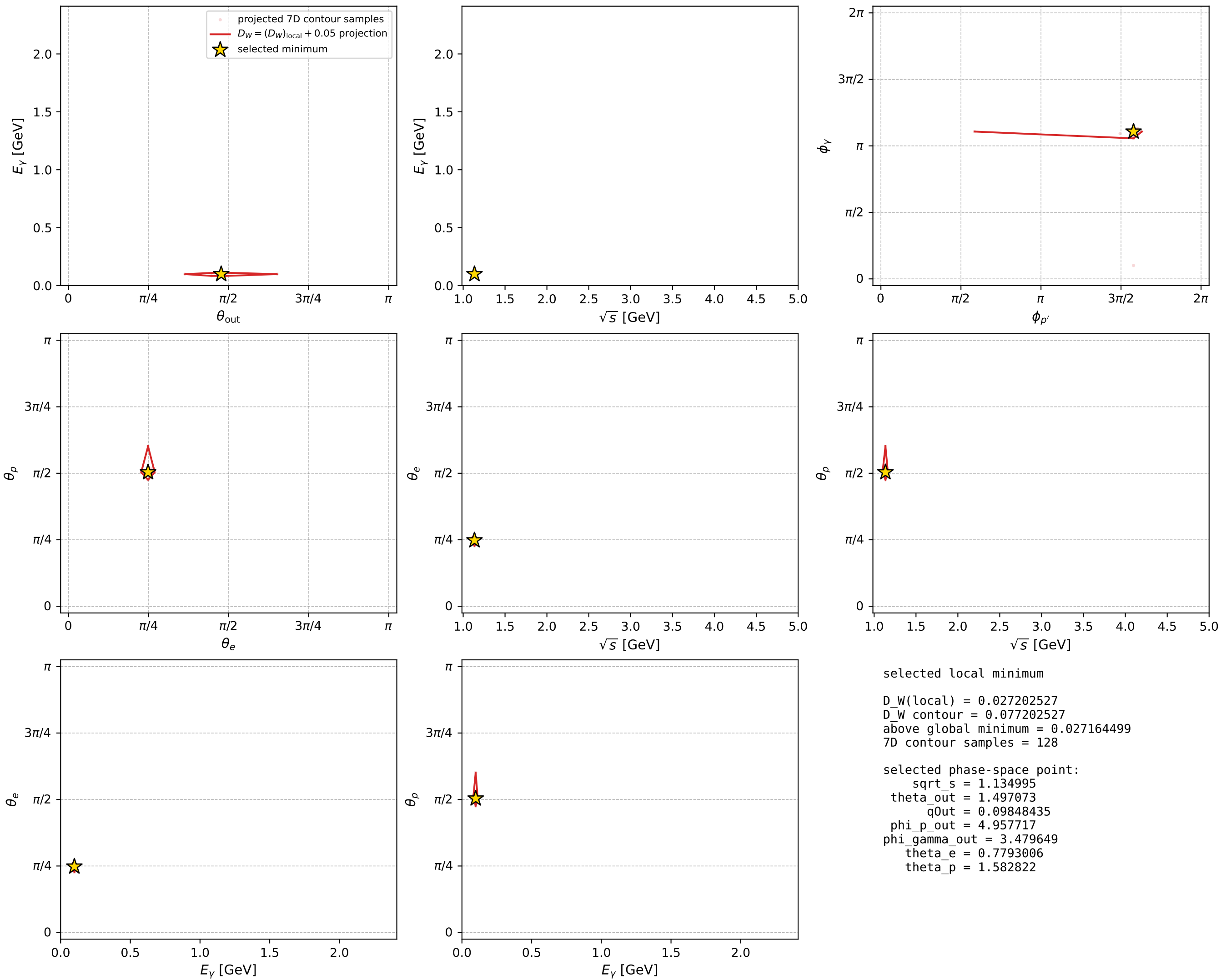
dw_mixing_angles_polarization_03_local_minimum_718 D_W=0.0272025
 region: polarization_cluster_03_local_minimum_718
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\theta_e=0.77930061$ rad
 incoming lepton: $0.711405|+> +0.702782|->$
 proton mixing angle: $\theta_p=1.5828223$ rad
 incoming proton: $-0.0120256|+> +0.999928|->$

$s=1.28821$, $\sqrt{s}=1.13499$, $|\vec{P}|=0.179898$, $|\vec{P}'|=0.000249079$
 $\theta_{\text{out}}=1.49707$, $\phi_{P'}=4.95772$, $\phi_Y=3.47965$
 $E_Y=0.0984843$, $\theta_{\ell'}=1.64469$, $\phi_{\ell'}=0.340574$
 $Q^2=0.0328267$, $x_B=0.150873$, $t=-0.0320646$, $W^2=1.0646$, $y=0.5328$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1799$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1799$
 P $E= 0.9551$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1799$
 ℓ' $E= 0.0985$ $p_x= 0.0926$ $p_y= 0.0328$ $p_z= -0.0073$
 P' $E= 0.9380$ $p_x= 0.0001$ $p_y= -0.0002$ $p_z= 0.0000$
 q_Y $E= 0.0985$ $p_x= -0.0927$ $p_y= -0.0326$ $p_z= 0.0073$



electron: polarization cluster P3, local minimum 718; pairwise projections of the 7D $D_W = (D_W)_{\text{local}} + 0.05$ contour



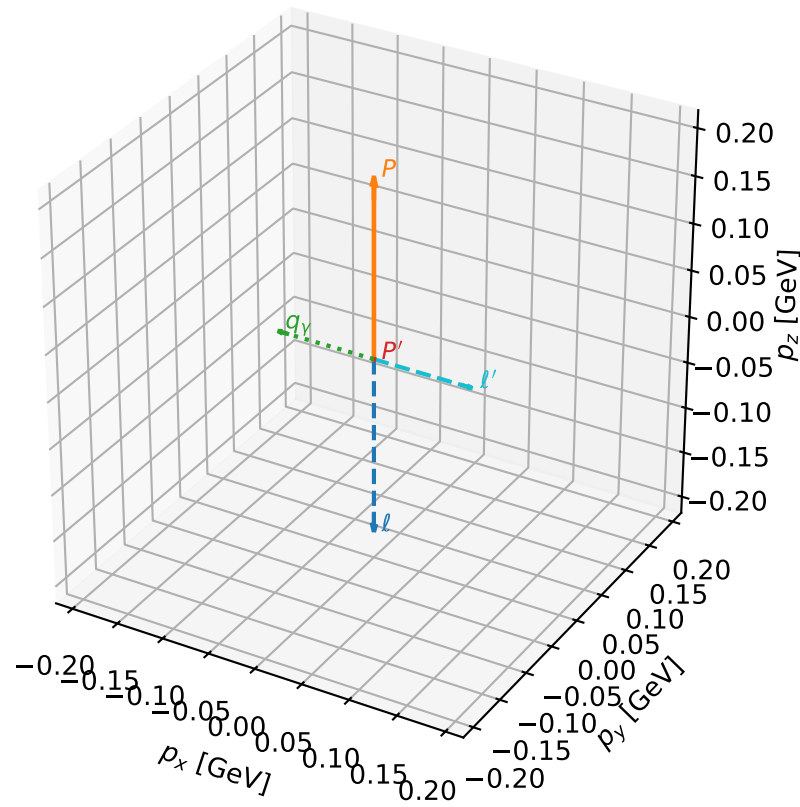
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

dw_mixing_angles_polarization_03_local_minimum_731 D_W=0.0393579
 region: polarization_cluster_03_local_minimum_731
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\theta_e=0.76881889$ rad
 incoming lepton: $0.718732|+\rangle + 0.695287|-\rangle$
 proton mixing angle: $\theta_p=1.5776148$ rad
 incoming proton: $-0.00681839|+\rangle + 0.999977|-\rangle$

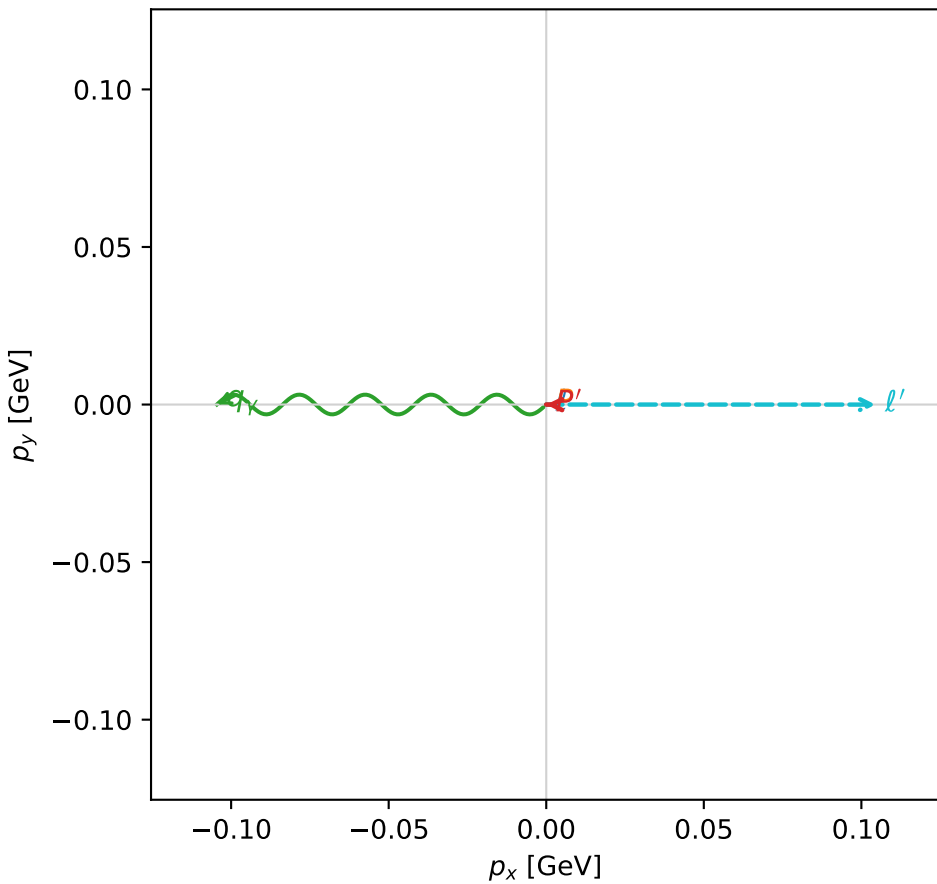
$s=1.31567$, $\sqrt{s}=1.14703$, $|\vec{P}|=0.18998$, $|\vec{P}'|=7.38927\text{e-}07$
 $\theta_{\text{out}}=1.56558$, $\phi_{P'}=3.13093$, $\phi_\gamma=3.14143$
 $E_\gamma=0.104512$, $\theta_{\ell'}=1.57601$, $\phi_{\ell'}=6.28302$
 $Q^2=0.0395035$, $x_B=0.167695$, $t=-0.0357295$, $W^2=1.07591$, $y=0.540511$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1900$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1900$
 P $E= 0.9570$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1900$
 ℓ' $E= 0.1045$ $p_x= 0.1045$ $p_y= -0.0000$ $p_z= -0.0005$
 P' $E= 0.9380$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 0.0000$
 q_γ $E= 0.1045$ $p_x= -0.1045$ $p_y= 0.0000$ $p_z= 0.0005$

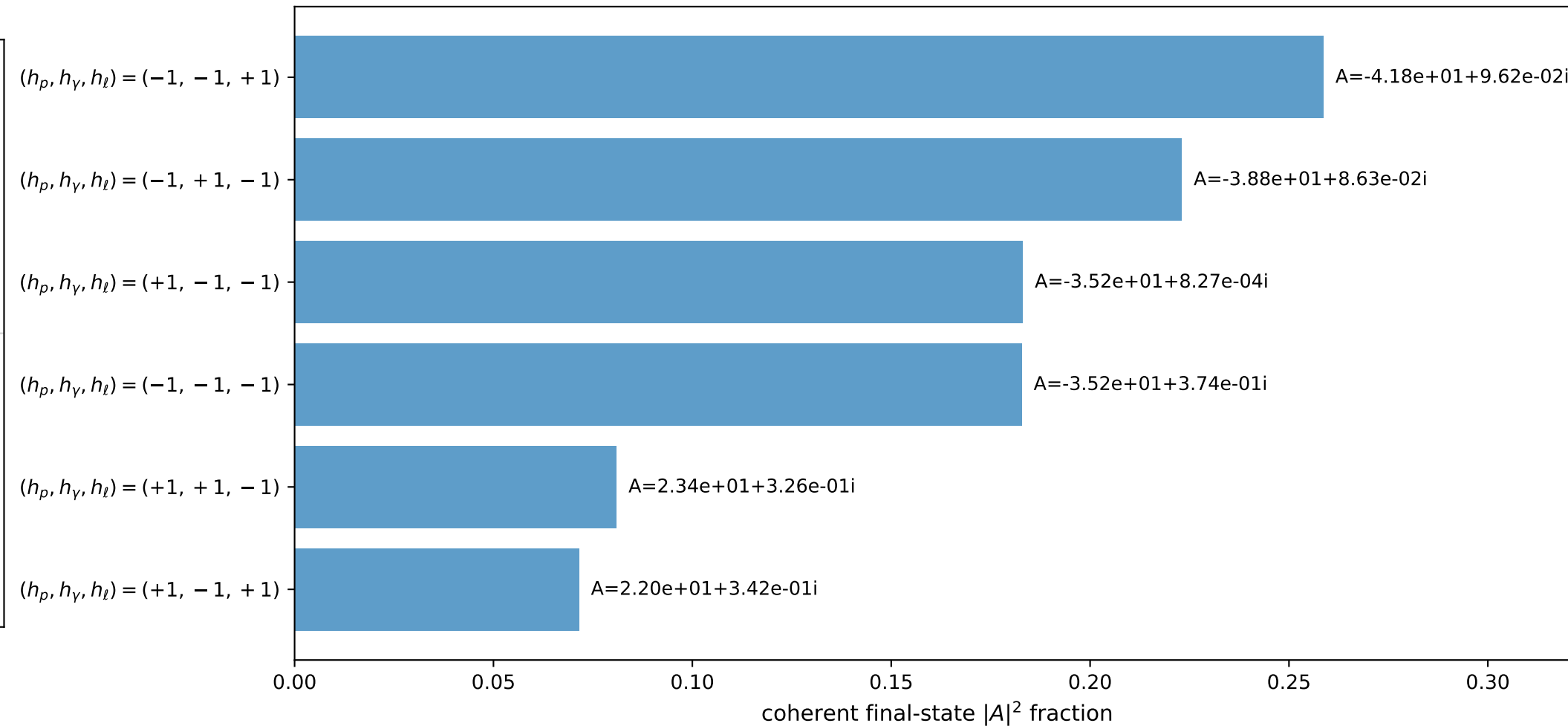
3D momenta



Transverse plane



Leading final-state amplitudes (N=6, retained=100.0%)



electron: polarization cluster P3, local minimum 731; pairwise projections of the 7D $D_W = (D_W)_{\text{local}} + 0.05$ contour

